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CELEBRATING 50 YEARS

Visualization of Data, Understanding and Interpretation

Unit 1 : Foundations of Visualization, Data Interpretation

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

Dr.Prajwala TR (Course Instructor)

Dept. of CSE, PES University, Bangalore

Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.



Visualization of Data, Understanding and Interpretation

Syllabus



Unit 1: Foundations of Visualization, Data Interpretation

Introduction to data visualization, Role of Visualization in AI-driven Analytics, Visual encoding (marks & channels), Human perception & cognition, Gestalt principles, Color theory and design, Static vs Interactive Visualizations, Data preprocessing & Exploratory Data Analysis (EDA), Identifying trends, correlations and patterns, Multivariate visualization, Statistical thinking, Perceptual effectiveness of charts, Data-ink ratio and chart junk, Misleading visualizations and ethical concerns.

(14 Hours)

Unit 2: Advanced Visualization & Story Telling

Advanced visualizations: Networks, node-link diagrams, treemaps, saliency visualization. Dimensionality reduction visualization: PCA, t-SNE, UMAP. Geospatial visualization, Feature importance visualization, Uncertainty visualization, Interactive dashboards, Coordinated multiple views. Storytelling with data, Narrative visualization, AI-generated reports and summaries, Audience-centric design, Annotation & highlighting, Ethical visualization and accessibility.

(14 Hours)

Visualization of Data, Understanding and Interpretation

Syllabus



Unit 3: Multimedia and Interactive Visualization

Multimedia visualization concepts, Image, video and audio visualization, Scientific visualization fundamentals, Scalar and vector field visualization, Volume visualization, Medical and simulation data visualization, 3D visualization and rendering basics, Animation and motion visualization, Temporal visualization, Graph drawing and network layout techniques (force-directed and hierarchical layouts), Visualization pipelines and rendering process, Interactive visualization techniques: brushing, linking, filtering, zooming and details-on-demand, Visual analytics and sensemaking, Visualization evaluation methods, Usability testing and cognitive load analysis, Dashboard design principles and accessibility in visualization systems. **(14 Hours)**

Unit 4: AI-Assisted Visualization

Automated visualization tools: Auto chart recommendation systems, Natural Language to Visualization (NL2Vis), conversational visualization systems, text-to-chart generation, Automated generation of dashboards, Data summarization and interpretation using LLMs for EDA insights, Introduction to human-in-the-loop AI in visualization, AI-assisted pattern recognition, Anomaly detection, Bias in AI-driven insights, Model drift visualization, Uncertainty visualization in AI models, Visualization of ML outputs: predictions, probabilities and confidence measures, Explainable AI (XAI) visualizations, Monitoring AI systems using visual analytics, Misuse and limitations of automated insights. **(14 Hours)**

Visualization of Data, Understanding and Interpretation

Links

[Course Information](#)

[Text Book](#)

[Project Guidelines](#)



Visualization of Data, Understanding and Interpretation

Evaluation Policy

Activity	Marks		Remarks
ISA 1	20	40	CBT/Hybrid Mode
ISA 2	20		CBT/Hybrid Mode
Experiential Learning	4	10	Hands-on Submissions (Hands-on activities related to each unit will be demonstrated to the students, and they will execute and submit similar tasks.)
	6		Project (Title & Literature Survey + Implementaion & Demo + IEEE Draft)
ESA	50		Final ESA Exam

Visualization of Data, Understanding and Interpretation

Teaching Assistants (TAs)



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Visualization of Data, Understanding and Interpretation

Project Ideas



1. AI-Powered Smart City Analytics Dashboard

Objective:

Develop an interactive dashboard for monitoring traffic, pollution, weather, energy consumption, and public transport data. The system should visualize trends, identify anomalies, and generate AI-assisted analytical reports for city administrators.

Tools:

Power BI, Python (Pandas, Plotly, Folium), Dash, OpenAI/Gemini API

Steps:

- Preprocess and analyze smart city datasets using EDA.
- Design interactive dashboards with geospatial and temporal visualizations.
- Develop linked and animated visualizations for traffic and environmental monitoring.
- Integrate AI to automatically generate insights, detect anomalies, and recommend actions.

Learning Outcomes:

Understanding EDA, interactive dashboards, geospatial & temporal visualization, and AI-assisted analytics.

Visualization of Data, Understanding and Interpretation

Project Ideas



2. Explainable Healthcare Analytics System

Objective:

Develop a healthcare analytics platform that visualizes patient records, predicts disease risk, and explains machine learning predictions using Explainable AI visualizations.

Tools:

Power BI, Python (Pandas, Plotly, Scikit-learn, SHAP), Streamlit, Open3D/3D Slicer, OpenAI/Gemini API

Steps:

- Perform EDA on healthcare datasets.
- Design dashboards showing patient trends and disease statistics.
- Visualize medical images and temporal patient records.
- Explain AI predictions using SHAP/LIME and generate AI-based medical summaries.

Learning Outcomes:

Understanding healthcare visualization, scientific visualization, explainable AI, and AI-assisted reporting.

Visualization of Data, Understanding and Interpretation

Project Ideas



3. Business Intelligence Dashboard with AI Insights

Objective:

Build an executive dashboard for sales, customer behavior, inventory, and financial analysis with AI-generated business reports and visualization recommendations.

Tools:

Power BI, Python (Pandas, Plotly), SQL, Dash, OpenAI/Gemini API

Steps:

- Perform exploratory analysis of business datasets.
- Create KPI dashboards with drill-down analysis.
- Develop interactive visualizations for customer behavior and sales trends.
- Generate AI-powered reports and actionable insights.

Learning Outcomes:

Understanding business intelligence, dashboard development, storytelling with data, and AI-assisted decision making.

Visualization of Data, Understanding and Interpretation

Project Ideas



4. Climate Change Visual Analytics Platform

Objective:

Develop an interactive platform to analyze climate change using temporal, geospatial, and AI-assisted visualizations for environmental decision making.

Tools:

Power BI, Python (Pandas, Plotly, Folium), Dash, OpenAI/Gemini API

Steps:

- Perform EDA on climate datasets.
- Visualize temperature, rainfall, and emissions using maps and dashboards.
- Develop animated temporal visualizations.
- Generate AI-powered reports and uncertainty visualizations.

Learning Outcomes:

Understanding environmental analytics, geospatial visualization, temporal visualization, and AI-assisted reporting.

Visualization of Data, Understanding and Interpretation

Project Ideas



5. Financial Market Analytics Dashboard

Objective:

Develop an interactive stock market analytics platform that visualizes historical trends, predicts market movements, and explains AI-generated forecasts.

Tools:

Power BI, Python (Pandas, Plotly, Scikit-learn, SHAP), Yahoo Finance API, OpenAI/Gemini API

Steps:

- Analyze stock market data using EDA.
- Develop interactive dashboards for market trends.
- Build temporal visualizations for stock performance.
- Integrate AI models with explainable visualizations.

Learning Outcomes:

Understanding financial visualization, predictive analytics, explainable AI, and dashboard development.

Visualization of Data, Understanding and Interpretation

Project Ideas



6. Cybersecurity Threat Visualization Platform

Objective:

Create a cybersecurity dashboard to monitor network traffic, detect anomalies, visualize attacks, and explain AI-based intrusion detection.

Tools:

Power BI, Python (Pandas, Plotly, NetworkX, Scikit-learn, SHAP), Streamlit, OpenAI/Gemini API

Steps:

- Analyze network traffic datasets.
- Create dashboards for attack monitoring.
- Develop interactive network and temporal visualizations.
- Integrate anomaly detection with explainable AI and automated reports.

Learning Outcomes:

Understanding network visualization, anomaly detection, interactive dashboards, and explainable AI.

Visualization of Data, Understanding and Interpretation

Project Ideas



7. E-Commerce Customer Analytics Dashboard

Objective:

Develop a customer analytics platform to analyze purchasing behavior, visualize customer journeys, and generate AI-assisted business recommendations.

Tools:

Power BI, Python (Pandas, Plotly, Scikit-learn), Dash, OpenAI/Gemini API

Steps:

- Perform EDA on customer transaction datasets.
- Design dashboards for customer segmentation and sales analysis.
- Develop interactive customer behavior visualizations.
- Generate AI-based recommendations and business reports.

Learning Outcomes:

Understanding customer analytics, interactive dashboards, storytelling with data, and AI-assisted decision making.

Visualization of Data, Understanding and Interpretation

Project Ideas



8. AI-Assisted Visualization Recommendation System

Objective:

Develop an intelligent system that automatically recommends suitable visualizations for uploaded datasets, generates interactive dashboards, and produces AI-generated analytical reports.

Tools:

Power BI, Python (Pandas, Plotly, Dash, Scikit-learn), Open3D, OpenAI/Gemini API

Steps:

- Analyze uploaded datasets and perform automated EDA.
- Recommend appropriate visualization techniques.
- Generate interactive dashboards with filtering and drill-down.
- Produce AI-generated summaries and explain visualization choices.

Learning Outcomes:

Understanding automated visualization, dashboard generation, Natural Language-to-Visualization (NL2Vis), and AI-assisted visual analytics.



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Visualization of Data, Understanding and Interpretation

Lecture Overview

- The core purpose of data visualization in transforming complex, large-scale datasets into clear, recognizable patterns that drive rapid and efficient business decisions.
- A standard data-to-decision analytics workflow that bridges raw data collection and preprocessing with visual insights and informed action.
- The essential integration of visual diagnostics across the AI and machine learning pipeline, from initial data exploration and feature relationships to model training, evaluation, and explainability.
- The critical design principles separating good graphs from bad ones to ensure data is represented accurately and communicates its message without misleading clutter.

Visualization of Data, Understanding and Interpretation

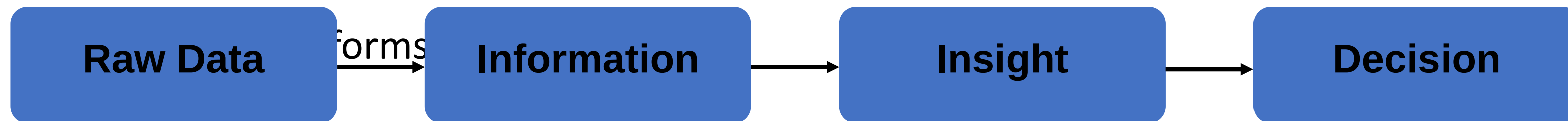
What is Data Visualization?

(TB 1, Ch1)

Data Visualization is the graphical representation of information and data using visual elements such as points, lines, bars, colors, shapes, and spatial positioning.

Rather than reading thousands of numerical values individually, visualization allows us to recognize meaningful patterns quickly.

Data visualization is the graphical representation and presentation of data to facilitate understanding.



Its primary objective is not to create attractive graphics but to communicate information accurately, efficiently, and effectively.

Visualization of Data, Understanding and Interpretation

What is Data Visualization?

Imagine you have just joined an e-commerce company as a Data Analyst.

On your first day, your manager gives you a spreadsheet containing more than one million customer transactions collected over the past five years.

Your task is to answer the following questions:

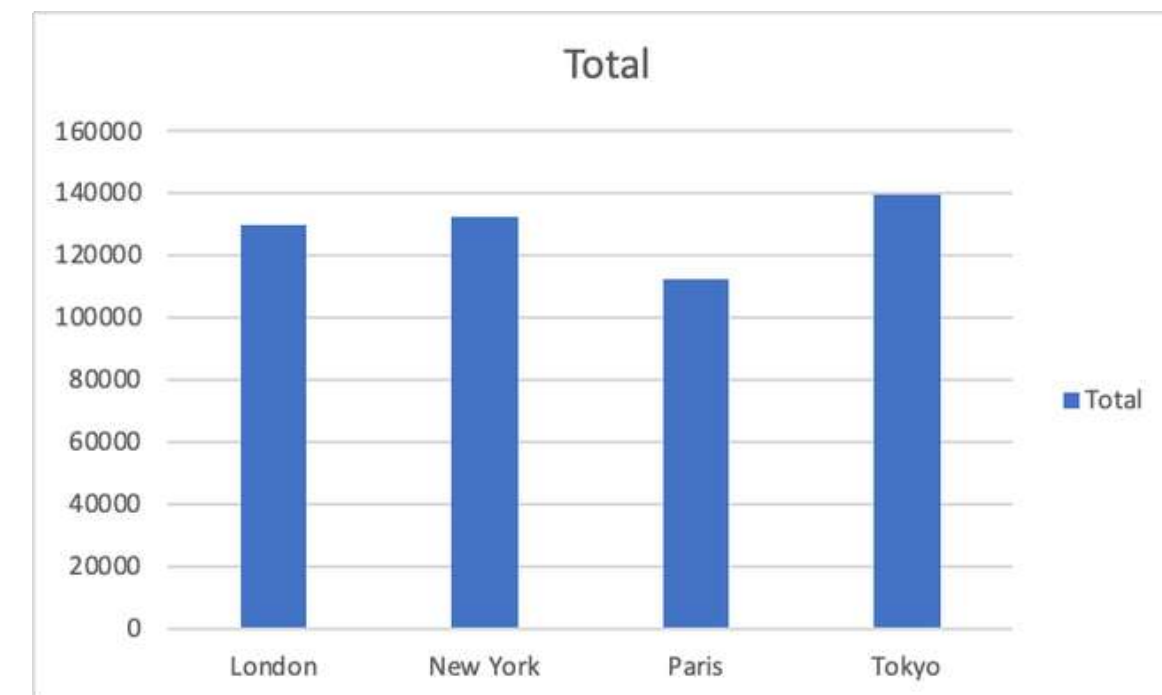
- Which region generated the highest revenue?
- During which month do sales peak?

Although all the information exists within the spreadsheet, identifying these insights manually would take hours. This is where data visualization becomes indispensable.

Visualization converts complex datasets into

CustomerID	OrderID	ProductInformation	TransactionAmount	PurchaseDate	Location
8814	890075	Product C	943.31	04/11/23	Tokyo
2188	176819	Product A	463.7	04/11/23	London
4608	340062	Product A	80.28	04/11/23	New York
2559	239145	Product A	221.29	04/11/23	London
9482	194545	Product A	739.56	04/11/23	Paris
8483	691194	Product C	375.23	04/11/23	Paris
8317	826847	Product B	272.56	04/11/23	New York
6911	963918	Product C	433.33	04/11/23	Tokyo
8993	112426	Product D	16.55	04/12/23	New York
3519	139726	Product C	464.63	04/12/23	New York
9005	972064	Product C	120.75	04/12/23	Paris
9409	597126	Product A	865.98	04/12/23	London
5670	995296	Product B	392.7	04/12/23	London
8724	739717	Product C	623.35	04/12/23	London
9148	629996	Product D	99.15	04/12/23	Tokyo
5837	387813	Product D	978.17	04/12/23	New York
2767	342397	Product D	299.57	04/12/23	Tokyo

CSV with Customer Transaction Data (Very Hard to Gather Insights)



Visualization of Sales by City

Visualization of Data, Understanding and Interpretation

Why Do We Need Data Visualization?

Organizations today generate enormous volumes of data from websites, sensors, mobile applications, financial transactions, social media, and IoT devices. While this data contains valuable information, extracting useful knowledge from millions of records is extremely difficult without visualization.

Visualization enables us to:

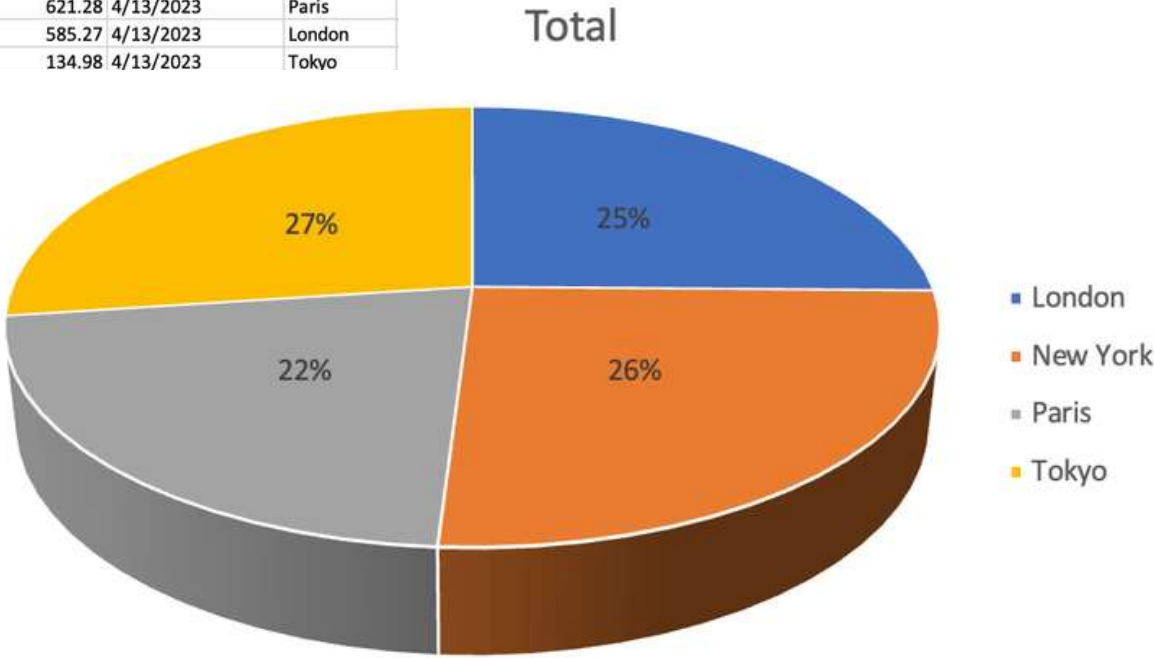
- Compare values across categories
- Identify trends over time
- Discover relationships between variables
- Detect anomalies and outliers
- Present analytical findings effectively

Visualization of Data, Understanding and Interpretation

Why Do We Need Data Visualization?

Without Visualization	With Visualization
Large Tables	Charts & Dashboards
Slow Analysis	Quick Understanding
Difficult Comparisons	Easy Comparisons
Hidden Trends	Visible Trends

CustomerID	OrderID	ProductInformation	TransactionAmount	PurchaseDate	Location
8814	890075	Product C	943.31	04/11/23	Tokyo
2188	176819	Product A	463.7	04/11/23	London
4608	340062	Product A	80.28	04/11/23	New York
2559	239145	Product A	221.29	04/11/23	London
9482	194545	Product A	739.56	04/11/23	Paris
8483	691194	Product C	375.23	04/11/23	Paris
8317	826847	Product B	272.56	04/11/23	New York
6911	963918	Product C	433.33	04/11/23	Tokyo
8993	112426	Product D	16.55	04/12/23	New York
3519	139726	Product C	464.63	04/12/23	New York
9005	972064	Product C	120.75	04/12/23	Paris
9409	597126	Product A	865.98	04/12/23	London
5670	995296	Product B	392.7	04/12/23	London
8724	739717	Product C	623.35	04/12/23	London
9148	629996	Product D	99.15	04/12/23	Tokyo
5837	387813	Product D	978.17	04/12/23	New York
2767	342397	Product D	299.57	04/12/23	Tokyo
2824	334053	Product B	744.13	04/12/23	London
9128	281201	Product D	388.9	04/12/23	Paris
1520	629903	Product B	102.76	04/12/23	New York
8719	313701	Product B	338.56	04/12/23	Paris
8529	562721	Product B	132.99	04/12/23	Tokyo
8705	466960	Product A	751.53	04/12/23	London
6347	614435	Product D	637.78	04/12/23	London
4161	859587	Product C	692.36	04/12/23	London
2228	777366	Product B	997.19	04/12/23	Tokyo
6638	812597	Product C	417.64	04/12/23	London
5533	768061	Product B	13.51	4/13/2023	Paris
9871	750817	Product C	849.53	4/13/2023	Paris
9864	761820	Product B	621.28	4/13/2023	Paris
1146	821846	Product D	585.27	4/13/2023	London
9115	205239	Product C	134.98	4/13/2023	Tokyo



Instead of asking people to read large tables of numbers, visualization leverages the human visual system to identify patterns almost instantly.

Visualization of Data, Understanding and Interpretation

From Data to Decisions

Data alone has limited value. The real value lies in the decisions made using that data.

A typical analytics workflow consists of the following stages:

1. Collect raw data from multiple sources.
2. Clean and preprocess the data.
3. Visualize the processed data.
4. Discover meaningful insights.
5. Make informed decisions.

Visualization serves as the bridge between raw data and actionable knowledge.



Visualization of Data, Understanding and Interpretation

Why is Data Visualization Important in AI and Machine Learning?

Visualization plays a critical role throughout the machine learning pipeline. Before training a model, data scientists rely on visualizations to understand data quality, detect missing values, identify outliers, and analyze feature distributions.

During model development, visualizations help monitor training progress through loss curves and accuracy plots. After training, evaluation metrics such as confusion matrices, ROC curves, and Precision–Recall curves provide insights into model performance.

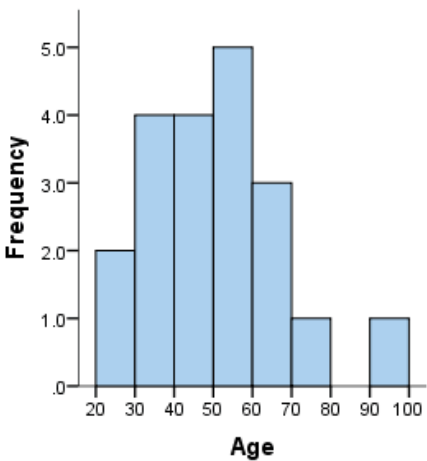
Modern Explainable AI (XAI) techniques also use visualizations to explain why machine learning models make specific predictions.

Visualization is therefore not an optional step—it is an essential component of every AI workflow.

Visualization of Data, Understanding and Interpretation

Visualization Throughout the AI Pipeline

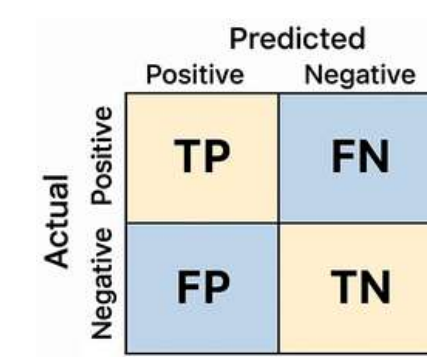
Stage	Visualization	Purpose
Data Exploration	Histogram	Understand distributions
Feature Relationships	Scatter Plot	Identify correlations
Correlation Analysis	Heatmap	Detect feature dependencies
Model Training	Loss Curve	Monitor learning progress
Model Evaluation	Confusion Matrix	Evaluate classification accuracy
Explainability	SHAP / Feature Importance	Interpret predictions



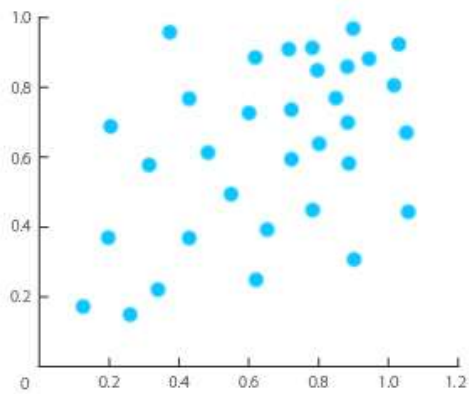
Histogram



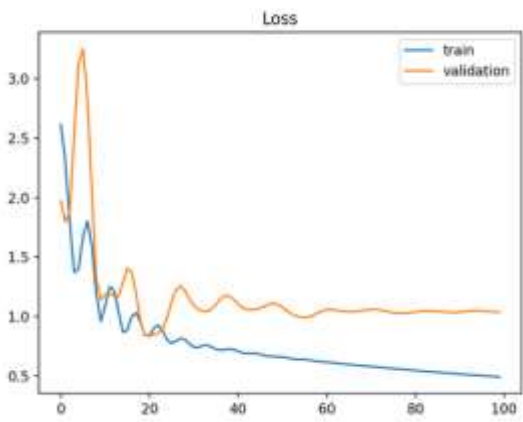
Heat Map



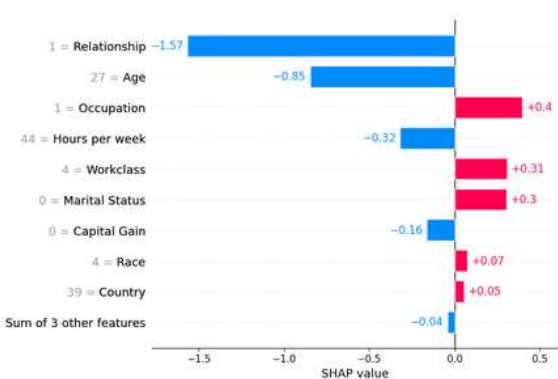
Confusion Matrix



Scatter Plot



Loss Curve



SHAP Feature Importance

Visualization of Data, Understanding and Interpretation

Good Graph vs Bad Graph

(TB 1, Ch1)



Not all visualizations communicate information equally well. Two charts can represent the same data, yet one may help the audience understand the message instantly, while the other creates confusion or even leads to incorrect conclusions.

A good graph emphasizes the data itself. It is simple, accurate, appropriately labeled, and uses the right chart type for the question being answered.

A bad graph distracts the audience with unnecessary design elements such as 3D effects, excessive colors, cluttered labels, or misleading scales, making the data harder to interpret.

Good Graph	Bad Graph
Simple and easy to read	Cluttered and distracting
Appropriate chart type	Inappropriate chart type
Clear labels and title	Missing or unclear labels
Accurate representation	Can distort the data
Highlights the message	Draws attention to decoration

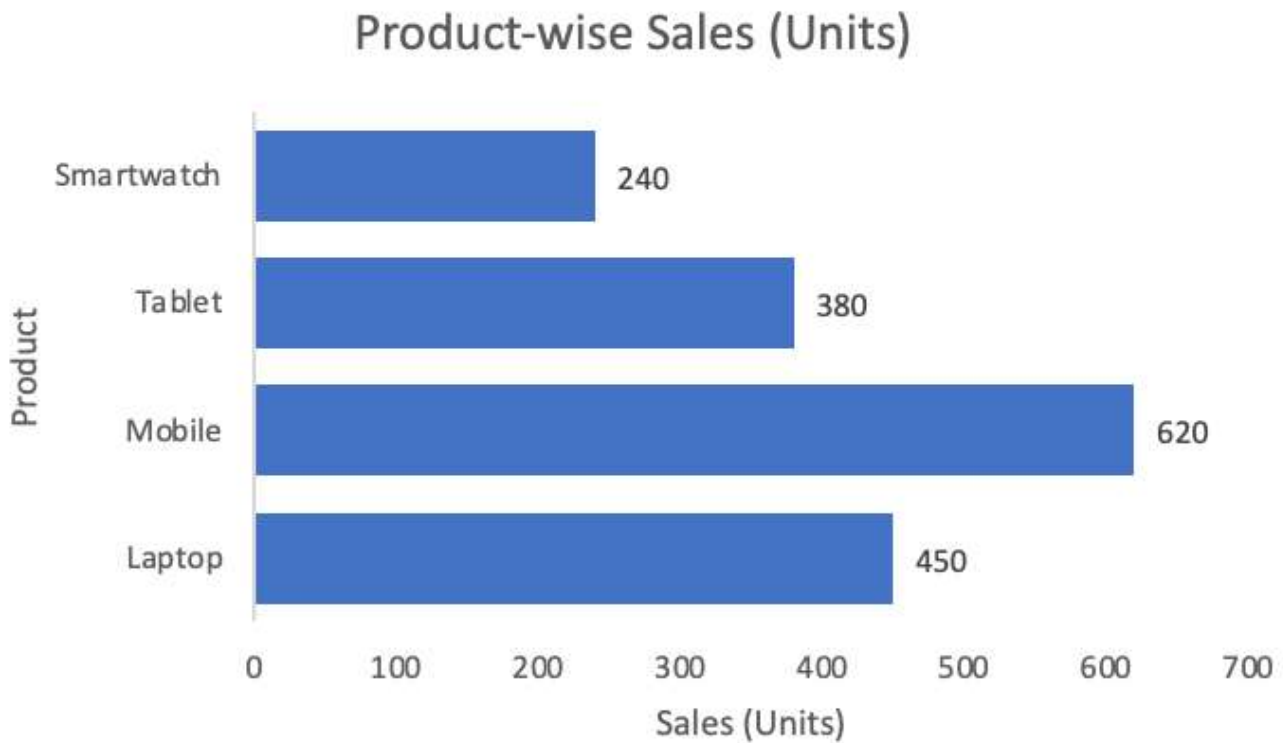
Visualization of Data, Understanding and Interpretation

Good Graph vs Bad Graph

(TB 1, Ch1)

Consider the following Data:

Product	Sales (Units)
Laptop	450
Mobile	620
Tablet	380
Smartwatch	240



Clear, simple, and easy to compare values.

A visualization should make insights easier to discover, not harder to see.



Poor design reduces clarity and may mislead interpretation.

Visualization of Data, Understanding and Interpretation

Human-in-the-Loop Visualization

- Many real-world analysis problems are ill-defined, where users do not know the exact questions to ask beforehand.
- Data exploration often involves discovering patterns, generating hypotheses, and refining questions rather than obtaining a single predefined answer.
- Visualization supports this exploratory process by combining human reasoning with computer computation.
- The goal of visualization is to augment human capabilities, not completely replace human decision-making.
- Humans contribute domain expertise, contextual understanding, and judgement, while computers efficiently process large datasets.

Visualization of Data, Understanding and Interpretation

Roles of Humans and Computers in Visualization



Visualization is most effective when human intelligence and computational power work together.

Humans	Computers
Interpret patterns	Process massive datasets
Apply domain knowledge	Perform rapid computations
Generate hypotheses	Execute algorithms
Validate insights	Detect statistical relationships
Make final decisions	Produce visual representations

Visualization of Data, Understanding and Interpretation

Why Vision Works Better than Tables



- Visualization exploits the human visual system, which provides a high-bandwidth channel for understanding information.
- Much of our visual processing occurs in parallel, allowing rapid recognition of patterns without consciously examining every value.
- Visual features such as color, position, and shape help important information stand out immediately (visual pop-out).
- Large numerical tables require sequential reading, making comparisons slower and increasing cognitive effort.
- Therefore, charts and graphs allow users to identify trends, clusters, and anomalies much faster than reading raw tables.

Visualization of Data, Understanding and Interpretation

MCQs

1. What is the primary purpose of data visualization?

- A. To make reports look more attractive
- B. To reduce the size of datasets
- C. To represent data graphically for easier understanding and decision-making
- D. To replace statistical analysis

Visualization of Data, Understanding and Interpretation

MCQs



1. What is the primary purpose of data visualization?

- A. To make reports look more attractive
- B. To reduce the size of datasets
- C. To represent data graphically for easier understanding and decision-making
- D. To replace statistical analysis

Answer: C

Visualization of Data, Understanding and Interpretation

MCQs



2. Which statement best describes the role of data visualization in AI and Machine Learning?

- A. It is used only after a model is deployed.
- B. It is useful only for presenting final results.
- C. It helps understand data, monitor models, and communicate insights throughout the AI workflow.
- D. It automatically improves the accuracy of machine learning models.

Visualization of Data, Understanding and Interpretation

MCQs



2. Which statement best describes the role of data visualization in AI and Machine Learning?

- A. It is used only after a model is deployed.
- B. It is useful only for presenting final results.
- C. It helps understand data, monitor models, and communicate insights throughout the AI workflow.
- D. It automatically improves the accuracy of machine learning models.

Answer: C

Visualization of Data, Understanding and Interpretation

MCQs

- 3. A company has collected five years of sales data containing millions of records. Why is visualization preferred over manually analyzing the spreadsheet?**
- A. Visualization reduces the amount of data collected.
 - B. Visualization replaces the need for data cleaning.
 - C. Visualization helps identify patterns, trends, and insights more quickly.
 - D. Visualization guarantees correct business decisions.

Visualization of Data, Understanding and Interpretation

MCQs

3. A company has collected five years of sales data containing millions of records. Why is visualization preferred over manually analyzing the spreadsheet?

- A. Visualization reduces the amount of data collected.
- B. Visualization replaces the need for data cleaning.
- C. Visualization helps identify patterns, trends, and insights more quickly.
- D. Visualization guarantees correct business decisions.

Answer: C

Visualization of Data, Understanding and Interpretation

MCQs

4. Which statement best describes the role of data visualization in AI and Machine Learning?

- A. It uses as many colors and effects as possible.
- B. It communicates information clearly and accurately with minimal clutter.
- C. It always uses 3D graphics to improve presentation.
- D. It focuses more on appearance than on the data.

Visualization of Data, Understanding and Interpretation

MCQs



4. Which statement best describes the role of data visualization in AI and Machine Learning?

- A. It uses as many colors and effects as possible.
- B. It communicates information clearly and accurately with minimal clutter.
- C. It always uses 3D graphics to improve presentation.
- D. It focuses more on appearance than on the data.

Answer: B

Visualization of Data, Understanding and Interpretation

MCQs

5. Two graphs display exactly the same dataset. One is clear and easy to interpret, while the other contains excessive colors, 3D effects, and clutter. Which statement is most appropriate?

- A. Both graphs are equally effective because the data is the same.
- B. The graph with more visual effects is always better.
- C. The way data is presented affects how easily it can be understood.
- D. Decorative elements improve the accuracy of the visualization.

Visualization of Data, Understanding and Interpretation

MCQs

5. Two graphs display exactly the same dataset. One is clear and easy to interpret, while the other contains excessive colors, 3D effects, and clutter. Which statement is most appropriate?

- A. Both graphs are equally effective because the data is the same.
- B. The graph with more visual effects is always better.
- C. The way data is presented affects how easily it can be understood.
- D. Decorative elements improve the accuracy of the visualization.

Answer: C

Visualization of Data, Understanding and Interpretation

MCQs

- 6. According to the design principles discussed in the lecture, why is a 3D exploded pie chart often categorized as a "bad graph"?**
- A. It is too simple and fails to represent complex categories.
 - B. It only supports the display of a single data column.
 - C. It can distort the data, reduce clarity, and draw attention to decoration rather than the message.
 - D. It lacks any option to label values or categories.

Visualization of Data, Understanding and Interpretation

MCQs

6. According to the design principles discussed in the lecture, why is a 3D exploded pie chart often categorized as a "bad graph"?

- A. It is too simple and fails to represent complex categories.
- B. It only supports the display of a single data column.
- C. It can distort the data, reduce clarity, and draw attention to decoration rather than the message.
- D. It lacks any option to label values or categories.

Answer: C

Visualization of Data, Understanding and Interpretation

Demo and activity



Data Visualization Demo:

A hands-on notebook that introduces key data visualization concepts using a gold price dataset, demonstrating various plots, data analysis techniques, and the role of visualization throughout the machine learning workflow.

Hands-on Exercise:

An interactive notebook where students analyze AI-related social media sentiment, build effective visualizations, compare graph quality, and transform **raw data** → **information** → **insights** → **business decisions** while exploring visualization techniques used in machine learning.



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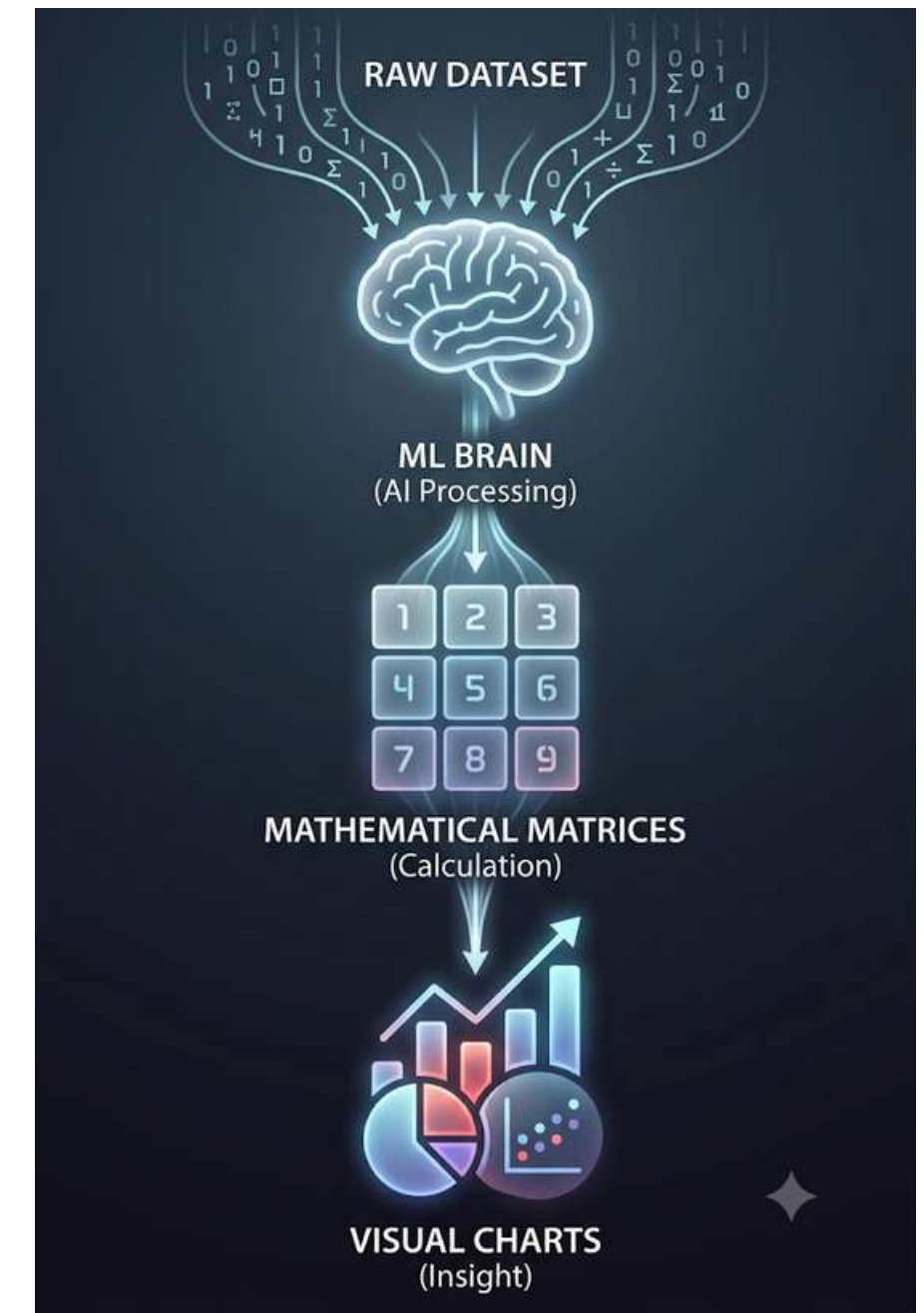
Lecture Overview

- Connecting AI models to human decision-makers by exploring how visualization serves as a crucial narrative bridge to translate abstract, multi-dimensional mathematical outputs into clear story arcs.
- Integrating visual interfaces across the modern machine learning lifecycle, showing how specific visual tools are applied during data collection, exploratory analysis, model training, performance evaluation, and deployment.
- Understand the Opening-Challenge-Action-Resolution (OCAR) model to structure data-driven stories.
- Avoiding cognitive overload and visual complexity by learning how to declutter multi-dimensional graphs, present progressive small multiples, and select appropriate presentation formats.
- Enhancing chart memorability and scale calibration through the use of isotype graphs containing scaled icons, varying chart types to prevent viewer fatigue, and adjusting figure counts for different

Visualization of Data, Understanding and Interpretation

AI-Driven Analytics (TB 1, Sec 1.1)

- AI-driven analytics represents the automatic process of extracting patterns and predictions from massive datasets using machine learning algorithms.
- Traditional analytics relies on human query formulation, whereas AI systems actively discover complex relationships without explicit programming.
- These automated analytics systems process highly dimensional mathematical spaces that lie far beyond simple human spreadsheets.
- The outputs of these models are often abstract probabilities, regression weights, or classification matrices that require transformation to be useful.
- Visualizing these multi-dimensional mathematical outputs represents the primary method of making automatic decisions understandable to human experts.



Visualization of Data, Understanding and Interpretation

Defining Data Visualization

(Sec 1.1)

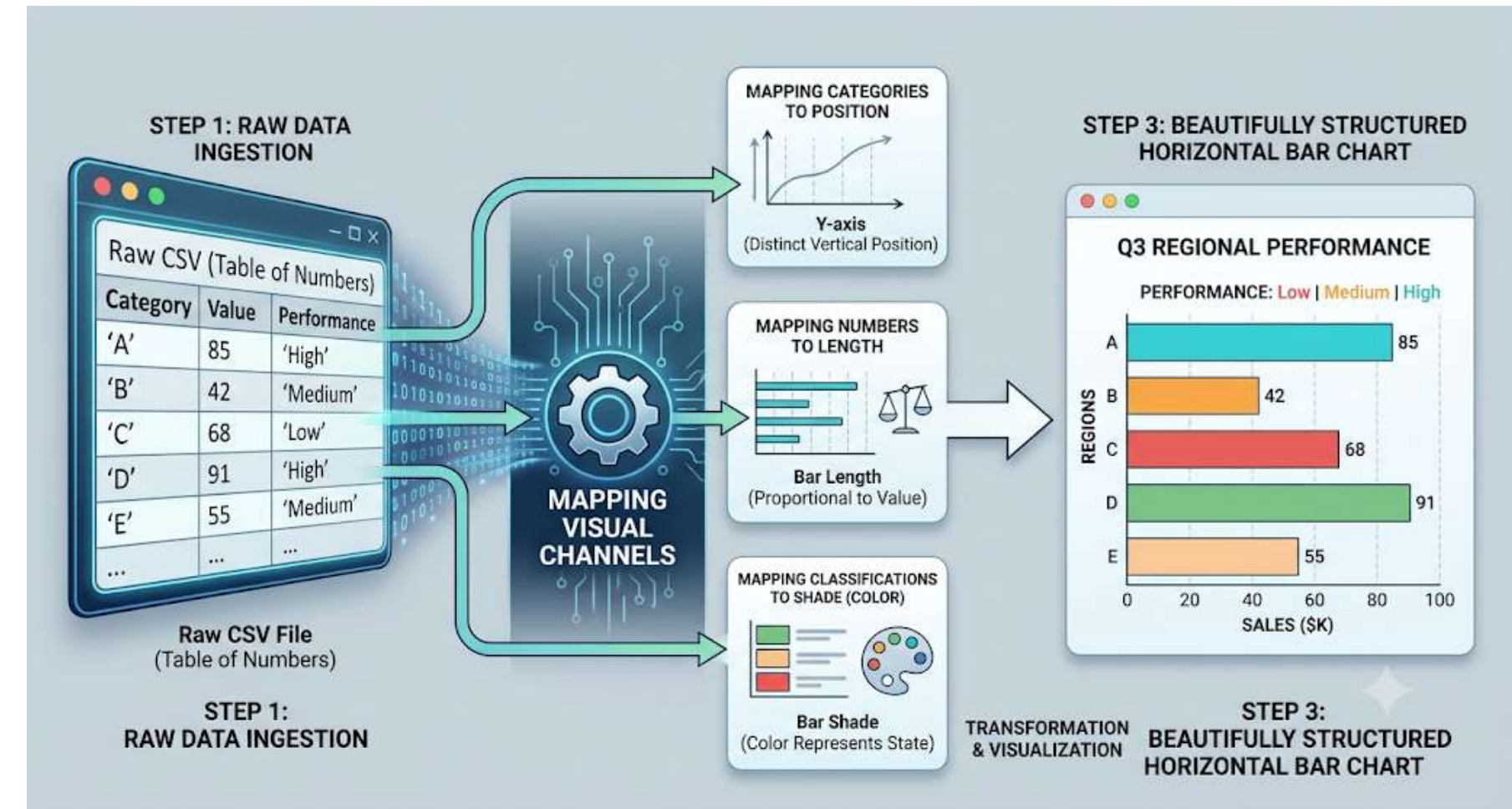
- Data visualization is the graphical representation and presentation of structured and unstructured information to facilitate deep cognitive understanding.

- We take complex data values and convert them in a systematic and logical way into distinct visual elements like points, lines, and solid shapes.

- These visual elements map quantitative variables onto spatial coordinates, color channels, and

- geometric sizes.
- The primary objective of data visualization is not to create attractive graphics but to communicate information accurately, efficiently, and effectively.

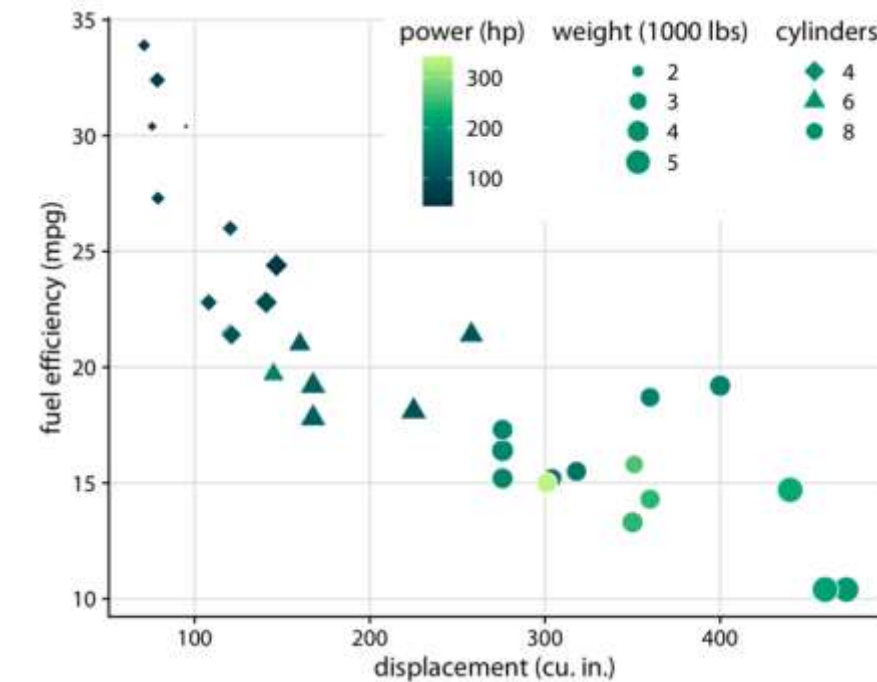
- A successful visualization maps unique variables to specific visual channels using one-to-one scales to ensure that the data remains completely unambiguous.



Visualization of Data, Understanding and Interpretation

The Symbiotic Relationship between AI and Visualization (TB 1, Ch29)

- Artificial intelligence and data visualization act as complementary forces in the modern data engineering and analytics workflow.
- AI models excel at processing billions of records to uncover complex mathematical correlations that are invisible to the naked human eye.
- Data visualization excels at presenting these model outputs so humans can evaluate their context, validity, and ethical implications.
- Without data visualization, artificial intelligence remains a silent box whose internal reasoning, feature weights, and predictions are inaccessible to decision-makers.



We utilize multi-scale mappings like the five-axis system in to let algorithms do the heavy computation while presenting the key dimensions visually.



Visualization of Data, Understanding and Interpretation

Why Visualization Bridges the AI-Decision Gap (Ch29)

- Machine learning models generate purely numerical predictions that lack the cognitive context needed to make final business and engineering choices.
- Data visualization acts as a narrative bridge by placing raw model predictions into a clear story arc that stakeholders can readily digest.
- We organize the visualization to lead the viewer from an initial opening state to a challenging discovery and finally to a model-driven resolution.
- Translating raw analytical variables into derived visual channels helps busy stakeholders grasp the core operational message immediately.
- Good visual design ensures that the audience does not have to guess

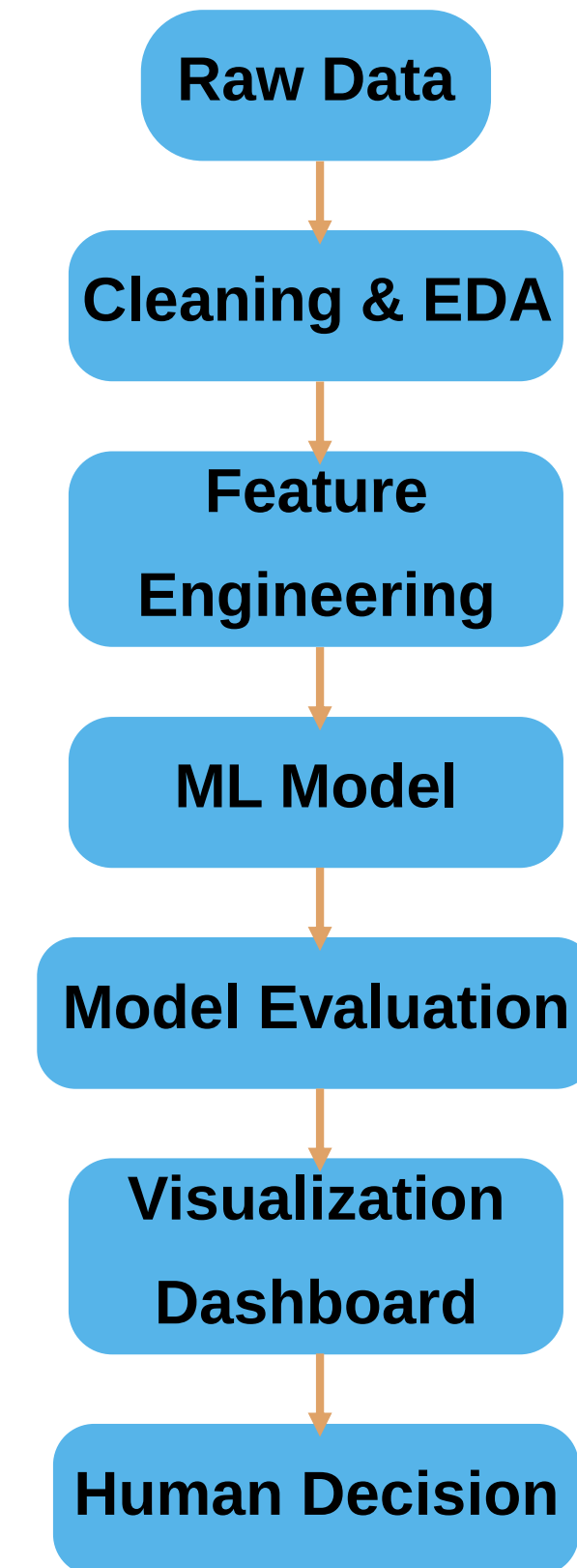


Visualization of Data, Understanding and Interpretation

The Modern AI Analytics Pipeline (TB 1, Ch1-

2)

- AI analytics is not a single algorithm but a complete pipeline from raw data to business decisions.
- Visualization plays a role at almost every stage, not just after model training.
- Analysts continuously inspect data quality, model behavior, and prediction reliability through visual interfaces.
- Human experts interact with visual dashboards to validate AI recommendations before deployment.
- Effective visualization shortens the gap between discovering insights and making informed decisions.



Visualization of Data, Understanding and Interpretation

Visualization Across the Machine Learning Lifecycle



Different visualizations answer different questions during AI development:

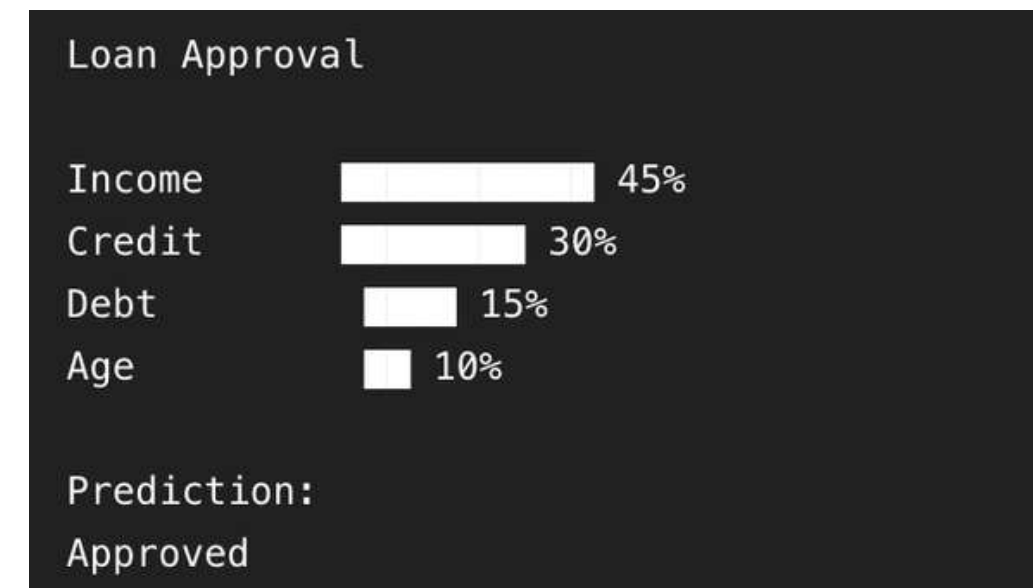
Stage	Typical Visualization	Purpose
Data Collection	Histograms, Missing-Value Heatmaps	Assess data quality
Exploratory Analysis	Scatter plots, Box plots	Discover patterns
Model Training	Loss Curves	Monitor learning
Model Evaluation	Confusion Matrix, ROC Curve	Measure performance
Model Deployment	Dashboards, Time-Series Plots	Monitor production systems

💡 *Different stages of AI require different visualization techniques, there is no universal chart*

Visualization of Data, Understanding and Interpretation

Explainable AI (XAI): Making AI Transparent

- Modern AI models often behave like black boxes, making their reasoning difficult to understand.
 - Explainable AI (XAI) uses visualization to reveal why a prediction was made.
 - Common visual explanations include:
 - Feature importance rankings
 - Saliency or attention maps
 - Decision trees
 - SHAP value plots
 - These visualizations improve:
 - Trust
 - Accountability
 - Regulatory compliance
- 💡 *AI should not only provide answers, it should explain them visually*
- Human oversight

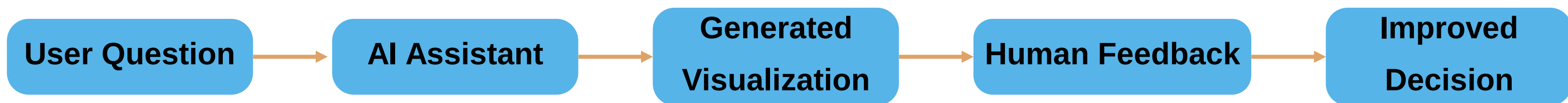


Visualization of Data, Understanding and Interpretation

Future Trends in AI Visualization

The future of AI analytics is moving beyond traditional dashboards:

- *Natural Language to Visualization (NL2Vis)*: Users ask questions in plain English and AI generates charts automatically.
- *Conversational Analytics*: AI assistants explain trends and answer follow-up questions.
- *Automated Insight Generation*: Systems detect anomalies, correlations, and important trends without manual exploration.
- *Interactive AI Dashboards*: Real-time filtering, drill-down analysis, and personalized recommendations.
- *Human-in-the-Loop Visualization*: Experts collaborate with AI instead of replacing human decision-making.



Visualization of Data, Understanding and Interpretation

MCQs

- 1. What is the primary role of data visualization when bridging the "AI-Decision Gap"?**
- A. To replace machine learning models with human-directed heuristics.
 - B. To serve as a narrative bridge that translates raw numerical predictions into a structured story arc that stakeholders can readily digest.
 - C. To maximize the aesthetic complexity and density of data plots.
 - D. To automatically clean and filter raw databases without human input.

Visualization of Data, Understanding and Interpretation

MCQs

1. What is the primary role of data visualization when bridging the "AI-Decision Gap"?

- A. To replace machine learning models with human-directed heuristics.
- B. To serve as a narrative bridge that translates raw numerical predictions into a structured story arc that stakeholders can readily digest.
- C. To maximize the aesthetic complexity and density of data plots.
- D. To automatically clean and filter raw databases without human input.

Answer: B

Visualization of Data, Understanding and Interpretation

MCQs

2. Why is making a figure for "the generals" highly recommended?

- A. Generals prefer highly complex plots
- B. Busy decision-makers need to grasp core insights instantly
- C. It reduces the need for data cleaning
- D. It automatically improves accuracy

Visualization of Data, Understanding and Interpretation

MCQs



Demo Notebook:

A hands-on demonstration of the end-to-end analytics workflow, covering data validation, exploratory analysis, machine learning, and business decision-making using a gold price dataset. The notebook highlights how AI models support real-world business decisions.

Student Activity:

A scenario-based exercise where students analyze gold price data, create effective visualizations, apply storytelling techniques, and transform analytical findings into clear business recommendations.

Visualization of Data, Understanding and Interpretation

MCQs

3. What does the Opening-Challenge-Action-Resolution format provide?

- A. A method for debugging model scripts
- B. A structured narrative arc for visual storytelling
- C. A database query optimizer
- D. An image file compression format

Visualization of Data, Understanding and Interpretation

MCQs

3. What does the Opening-Challenge-Action-Resolution format provide?

- A. A method for debugging model scripts
- B. A structured narrative arc for visual storytelling
- C. A database query optimizer
- D. An image file compression format

Answer: B

Visualization of Data, Understanding and Interpretation

MCQs

4. What is a key benefit of using small multiples progressively?

- A. It eliminates the need for legends
- B. It helps build up audience comprehension of complex visual grids
- C. It reduces the image file size
- D. It increases color-blind accessibility

Visualization of Data, Understanding and Interpretation

MCQs

4. What is a key benefit of using small multiples progressively?

- A. It eliminates the need for legends
- B. It helps build up audience comprehension of complex visual grids
- C. It reduces the image file size
- D. It increases color-blind accessibility

Answer: B

Visualization of Data, Understanding and Interpretation

MCQs

5. How does an Isotype graph enhance visual memorability compared to a generic, solid bar chart?

- A. It replaces traditional solid bars with repeated copies of a highly recognizable topic icon, where each icon represents a defined, scaled mathematical quantity.
- B. It utilizes a complex grid of multi-panel small multiples to display hundreds of subplots.
- C. It converts 2D coordinates into highly abstract 3D tensor-based visualizations.
- D. It introduces highly decorative clip art randomly around the axes to fill empty space.

Visualization of Data, Understanding and Interpretation

MCQs



5. How does an Isotype graph enhance visual memorability compared to a generic, solid bar chart?

- A. It replaces traditional solid bars with repeated copies of a highly recognizable topic icon, where each icon represents a defined, scaled mathematical quantity.
- B. It utilizes a complex grid of multi-panel small multiples to display hundreds of subplots.
- C. It converts 2D coordinates into highly abstract 3D tensor-based visualizations.
- D. It introduces highly decorative clip art randomly around the axes to fill empty space.

Answer: A

Visualization of Data, Understanding and Interpretation

Demo and Activity



5. How does an Isotype graph enhance visual memorability compared to a generic, solid bar chart?

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- B. It utilizes a complex grid of multi-panel small multiples to display hundreds of subplots.
- C. It converts 2D coordinates into highly abstract 3D tensor-based visualizations.
- D. It introduces highly decorative clip art randomly around the axes to fill empty space.

Answer: A



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Visualization of Data, Understanding and Interpretation

Unit 1 : Foundations of Visualization, Data Interpretation

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

Dr.Prajwala TR (Course Instructor)

Dept. of CSE, PES University, Bangalore

Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.

Visualization of Data, Understanding and Interpretation

Lecture Overview

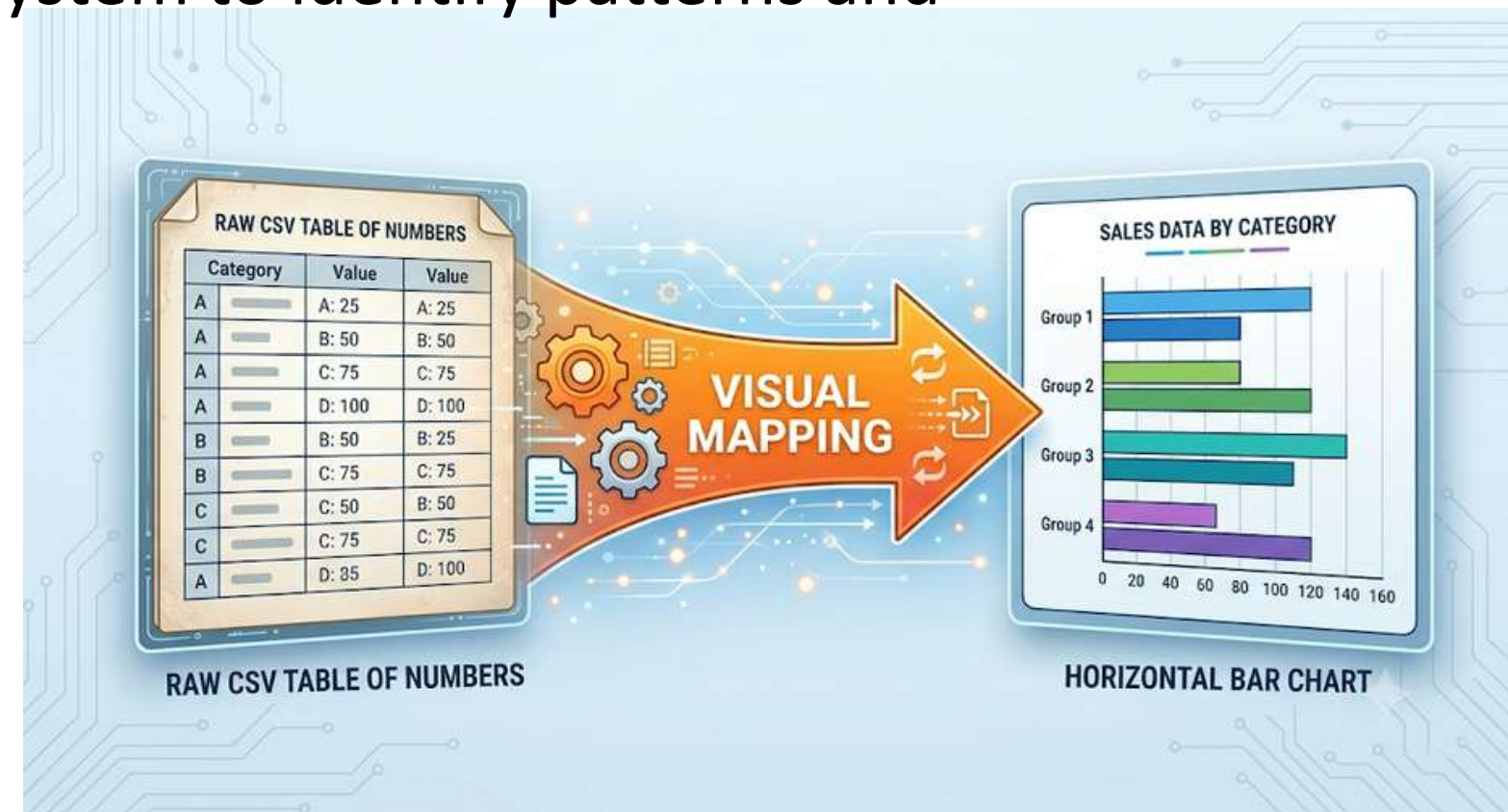
- Foundations of visual encoding and mapping, exploring how raw database variables are translated into visual marks and aesthetic channels.
- Mathematical principles of visual scales, highlighting the necessity of bijective, one-to-one mappings to prevent visualization ambiguity and ensure accurate data reconstruction.
- Rules of graphical integrity and proportion, focusing on the Principle of Proportional Ink and why area fills and bar plots require a strict zero-baseline to avoid distorting data.
- Multi-variable chart configurations and trade-offs, examining grouped and stacked bar charts, small multiples, and how scatter plots combine with bubble or shape scales to display multi-dimensional data.

Visualization of Data, Understanding and Interpretation

Introduction to Visual Encoding

(TB 1, Sec 2.1)

- Visual translation is the process of converting raw quantitative and qualitative values into distinct visual elements on screen.
- Raw databases consist of abstract numbers and factors that require mapping to physical visual properties to be processed by human vision.
- Structured representation maps each individual data value to a unique aesthetic value to prevent visual ambiguity.
- Cognitive interpretation leverages the human visual system to identify patterns and trends without reading raw database spreadsheets.

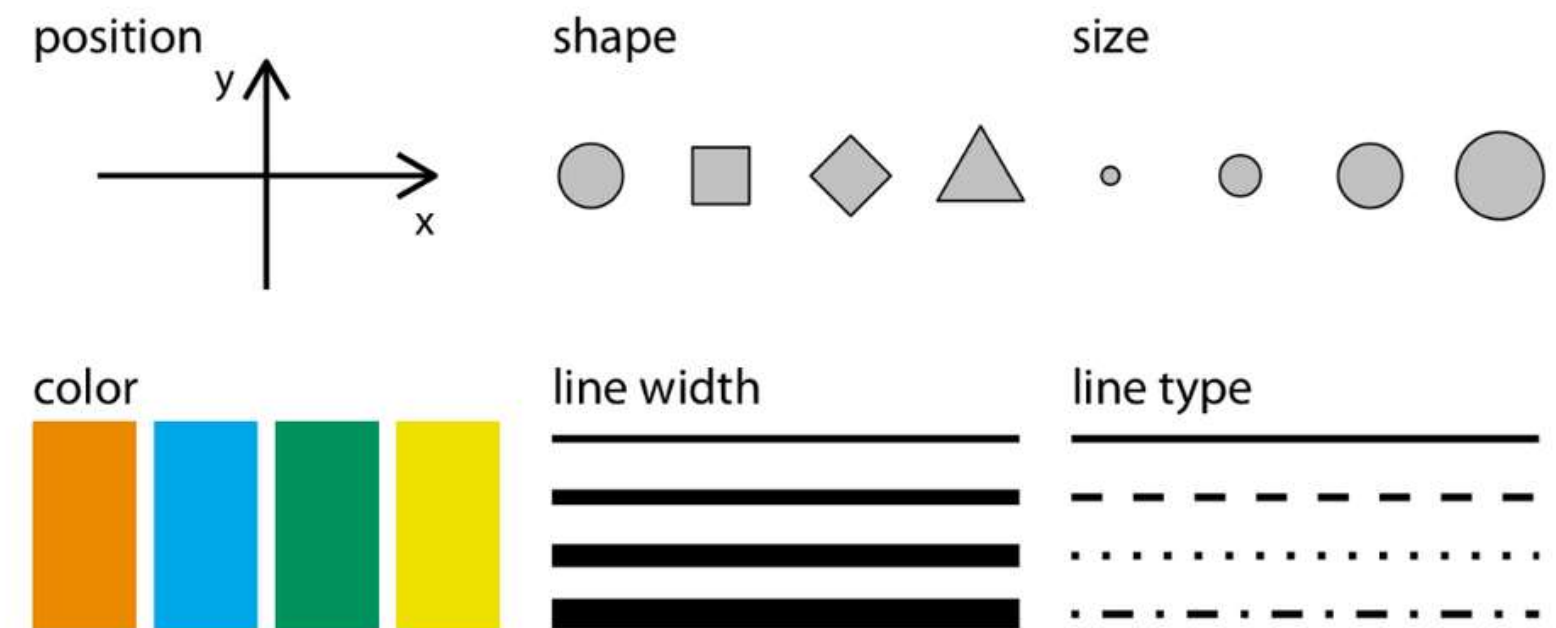


Visualization of Data, Understanding and Interpretation

Visual Primitives: Marks vs. Channels

(TB 1, Sec 2.1)

- Marks represent the basic geometric elements that anchor data points, including zero-dimensional points, one-dimensional lines, and two-dimensional shapes.
- Aesthetic channels represent the visual properties of a mark that can be varied to represent actual data dimensions.
- Position and scale are the most dominant aesthetic channels, defining where elements are located in standard two-dimensional graphics.
- Properties of marks such as shape, size, color, line width, and line type can all represent independent variables from the dataset.



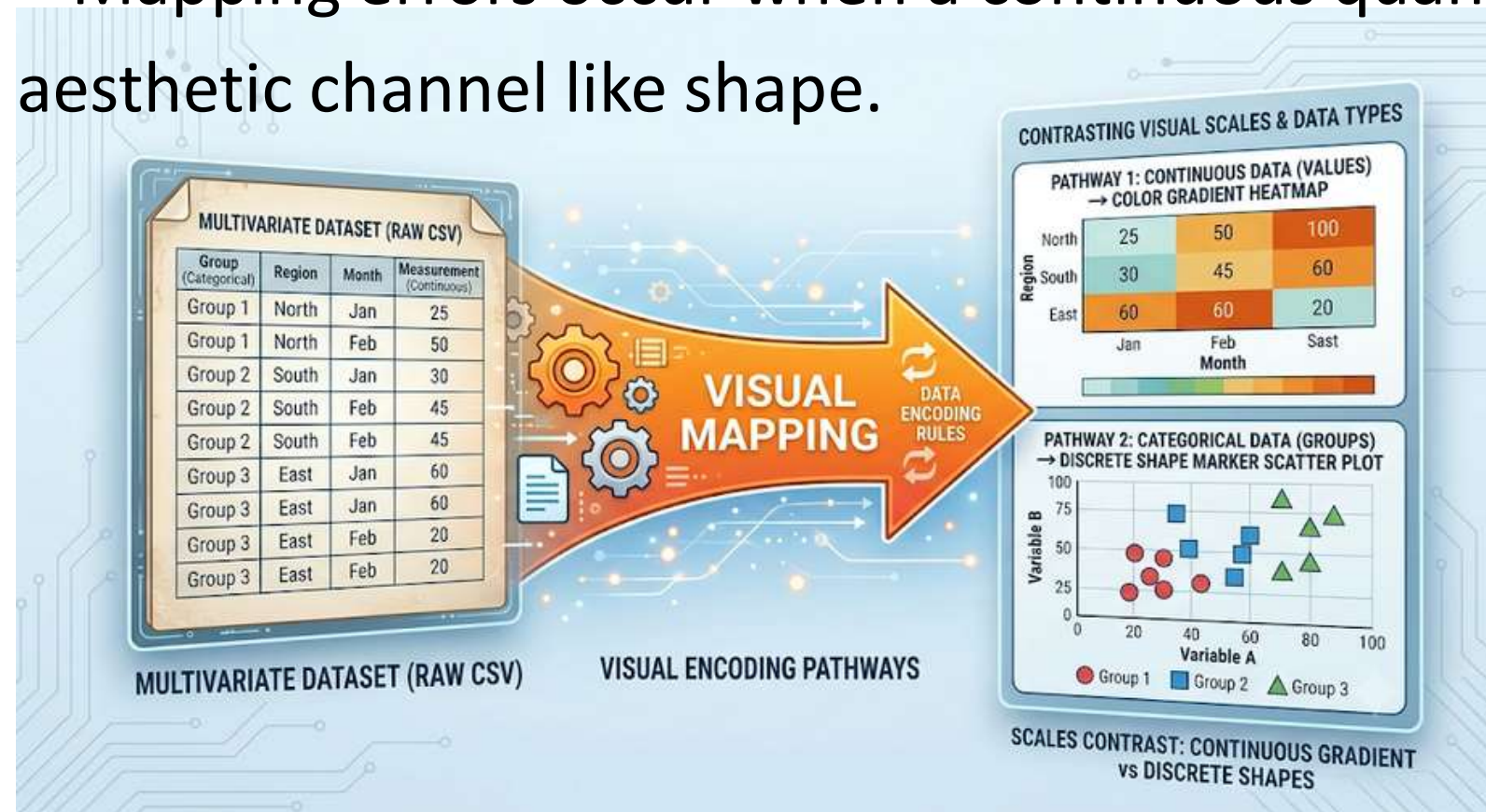
Commonly used aesthetics in data visualization: position, shape, size, color, line width, line type.

Visualization of Data, Understanding and Interpretation

Continuous vs. Discrete Aesthetics

(TB 1, Sec 2.1)

- Aesthetic channels are classified by their mathematical capability to represent continuous variables or only discrete variables.
- Continuous aesthetics include spatial position, geometric size, continuous color scales, and line width, which accept fine intermediate values.
- Discrete-only aesthetics include geometric shapes and line dash patterns, which cannot convey continuous quantities without causing confusion.
- Mapping errors occur when a continuous quantitative variable is assigned to a discrete aesthetic channel like shape.



Continuous vs. Discrete Visual Mapping

Visualization of Data, Understanding and Interpretation

Understanding Data Variables and Types (TB 1, Sec 2.1)

- Continuous quantitative variables consist of arbitrary real numbers such as weights, heights, temperatures, or time durations.
- Discrete quantitative variables represent numerical counts in discrete units, most commonly represented by integer values.
- Unordered qualitative variables consist of categories without any intrinsic order, such as animal species, product types, or regions.
- Ordered qualitative variables represent categorical factors with a clear, logical sequence such as ratings from poor to excellent.

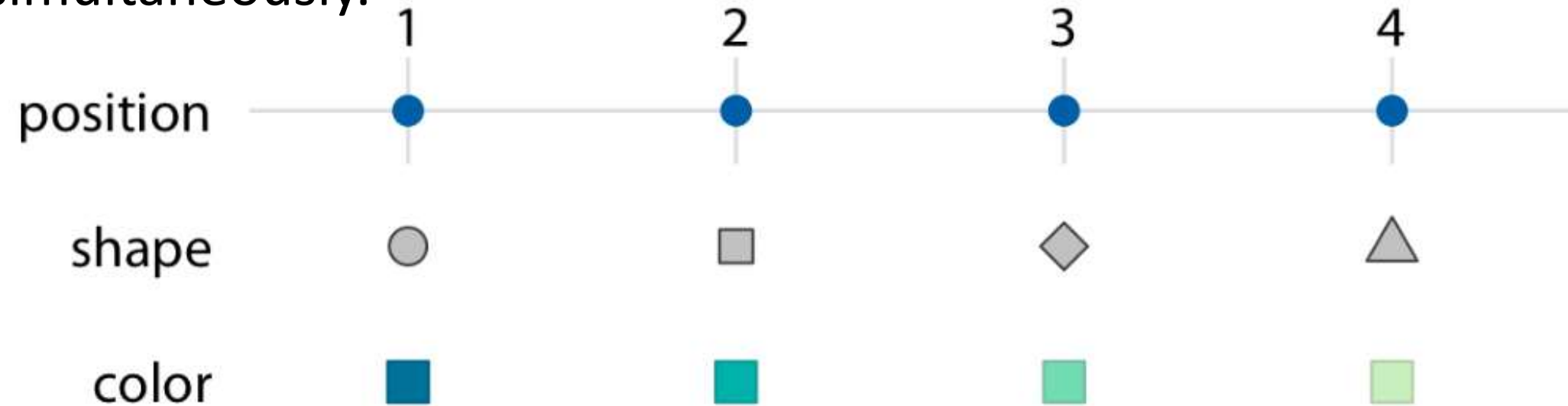
Type of variable	Examples	Appropriate scale	Description
quantitative/numerical continuous	1.3, 5.7, 83, 1.5×10^{-2}	continuous	Arbitrary numerical values. These can be integers, rational numbers, or real numbers.
quantitative/numerical discrete	1, 2, 3, 4	discrete	Numbers in discrete units. These are most commonly but not necessarily integers. For example, the numbers 0.5, 1.0, 1.5 could also be treated as discrete if intermediate values cannot exist in the given dataset.
qualitative/categorical unordered	dog, cat, fish	discrete	Categories without order. These are discrete and unique categories that have no inherent order. These variables are also called <i>factors</i> .
qualitative/categorical ordered	good, fair, poor	discrete	Categories with order. These are discrete and unique categories with an order. For example, "fair" always lies between "good" and "poor". These variables are also called <i>ordered factors</i> .
date or time	Jan. 5 2018, 8:03am	continuous or discrete	Specific days and/or times. Also generic dates, such as July 4 or Dec. 25 (without year).
text	The quick brown fox jumps over the lazy dog.	none, or discrete	Free-form text. Can be treated as categorical if needed.

Visualization of Data, Understanding and Interpretation

Scales: The Mathematical Bridge

(TB 1, Sec 2.1)

- Aesthetic scales define the functional mapping between data values and the physical properties of a visual element.
- Mathematical mapping must be one-to-one, meaning each unique data value corresponds to exactly one unique aesthetic value.
- Bijective scales prevent visualization ambiguity, ensuring that the reader can correctly reconstruct the raw data values from the visual graphics.
- Multiple scales can be combined on a single visual graphic to display five separate data variables simultaneously.

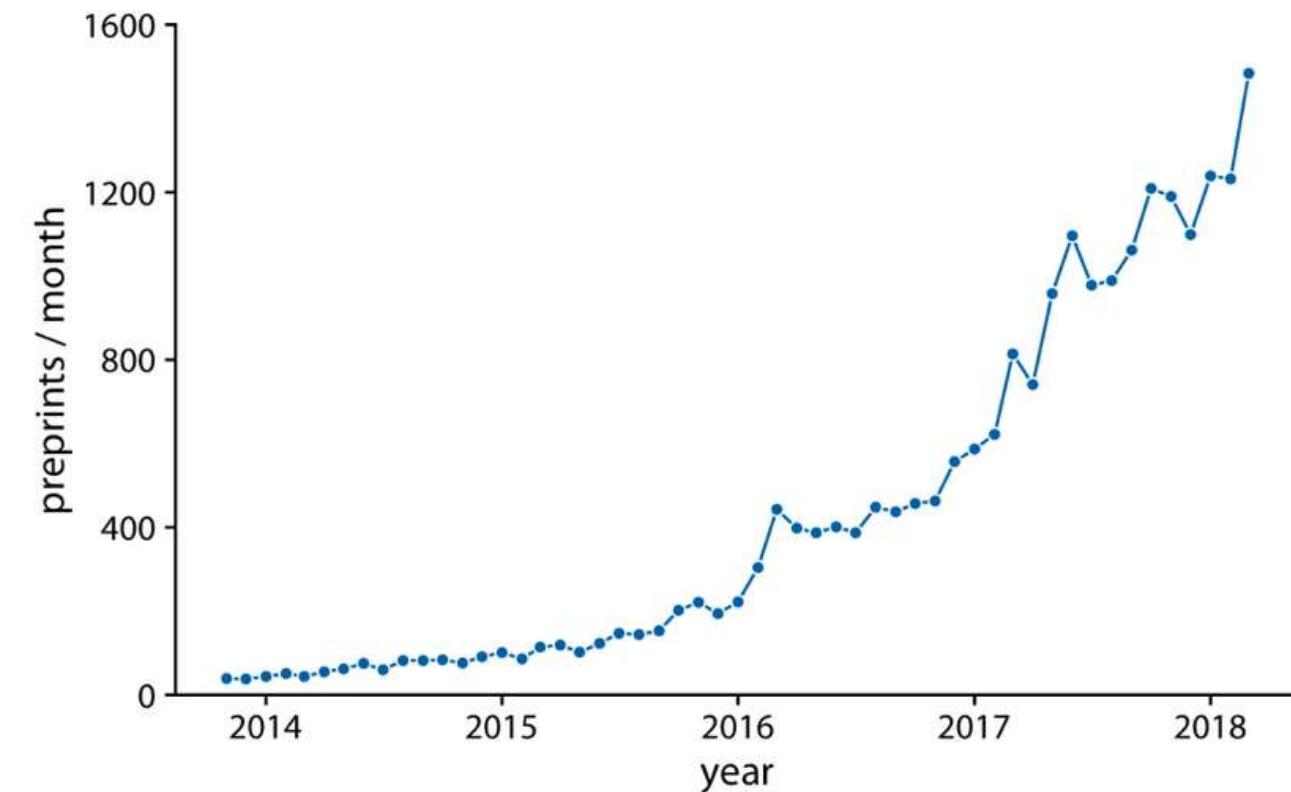


Scales link data values to aesthetics. Here, the numbers 1 through 4 have been mapped onto a position scale, a shape scale, and a color scale.

Visualization of Data, Understanding and Interpretation

Lines and Paths as Temporal Guides (TB1, Sec 13.1)

- Time-series visualization utilizes lines to emphasize temporal ordering and chronological sequence among data points.
- Connecting consecutive data points with lines guides the viewer's eye along a defined path from past to present.
- Continuous trends are emphasized when individual observation dots are omitted from dense time-series line graphs.
- Spacing density dictates whether dots are necessary, with dense series benefiting from simple, clean line plots.



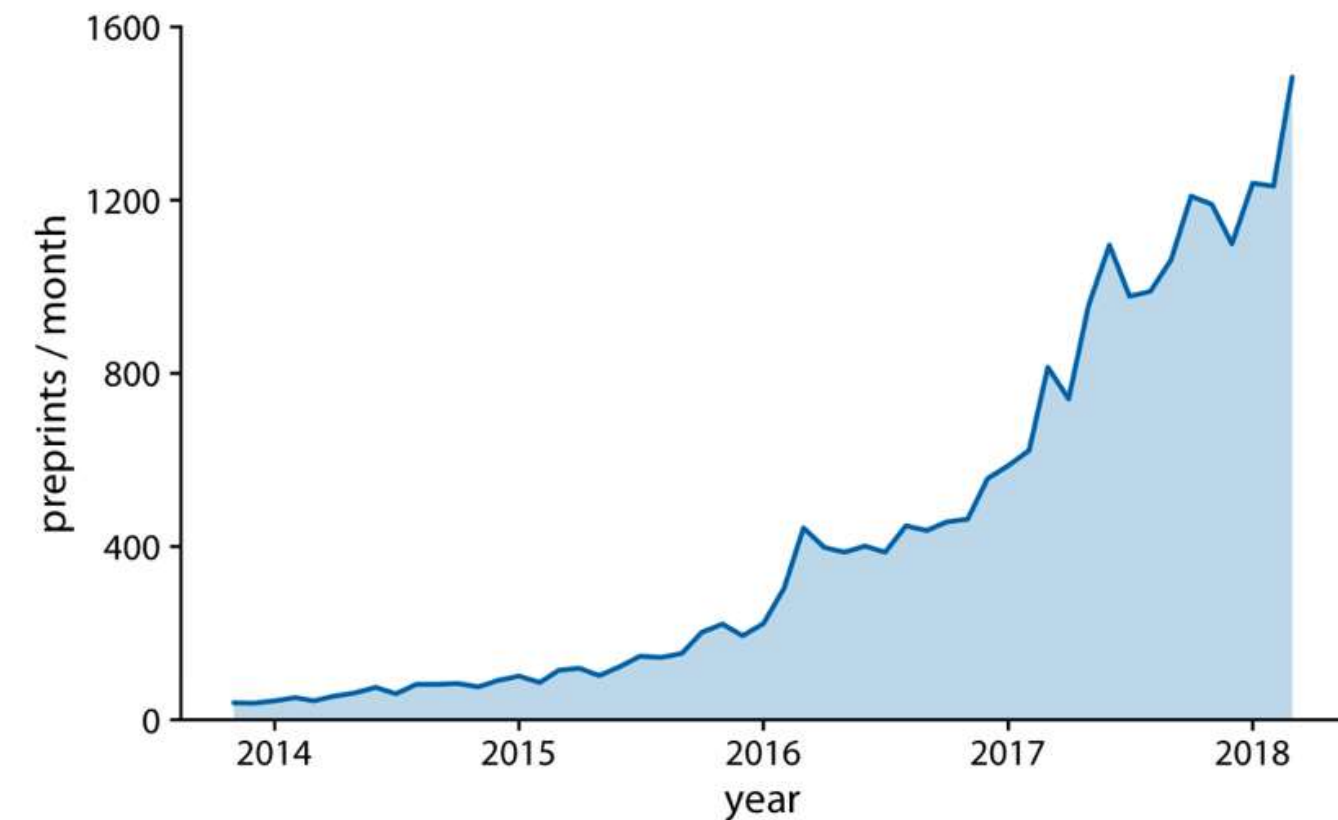
Monthly bioRxiv submissions with dots connected by thin guide lines.

Visualization of Data, Understanding and Interpretation

Noise Reduction and Area Fills

(TB1, Sec 13.1)

- Area fill graphs emphasize overall volume by shading the space underneath a time-series line with a solid color.
- Shaded area graphics are only mathematically valid if the vertical position scale starts strictly at zero.
- The Principle of Proportional Ink states that the shaded visual area must correspond exactly to the underlying numerical value.
- Arbitrary non-zero baselines distort visual volume and represent a fundamental charting error in area fill graphs.



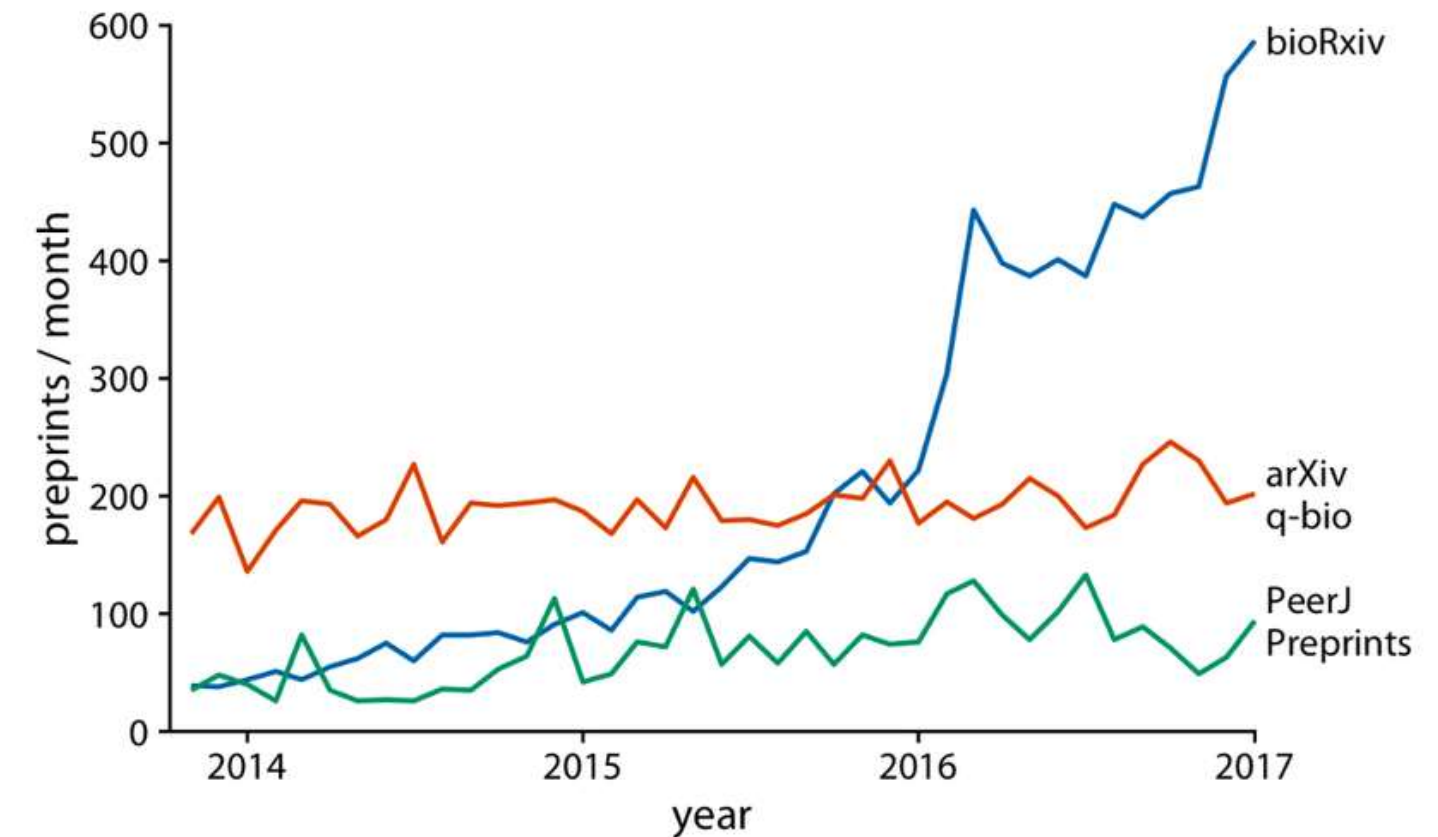
bioRxiv monthly preprint submissions as an area fill graph starting at zero.

Visualization of Data, Understanding and Interpretation

Direct Line Labeling vs. Distant Legends

(TB1, Sec 13.2)

- Distant legends create unnecessary cognitive load by forcing readers to scan back and forth to match colors.
- Direct labeling places text identifiers immediately adjacent to the ending of each corresponding line.
- Color-coded text labels enhance readability by matching the hue of the text to the hue of the visual line.
- Streamlined layouts are achieved by removing unnecessary legends and consolidating multiple shapes into a single line chart.



Multiple preprint server submission rates using direct color-coded text labels.

Visualization of Data, Understanding and Interpretation

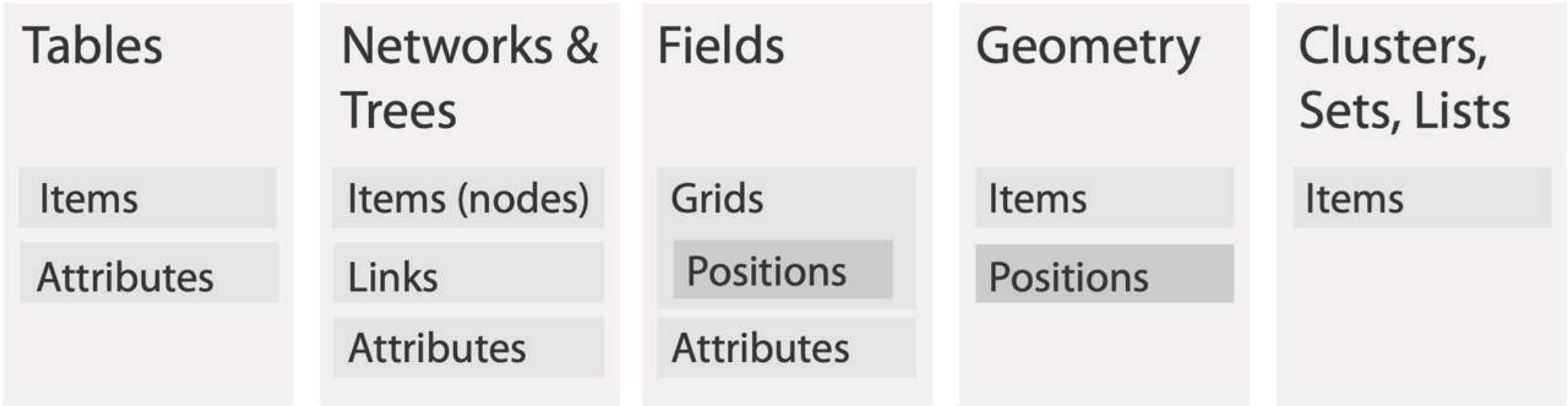
Data Types and Dataset Types

Before choosing a visualization, we must understand what kind of data we are working with:

- Data Types: the basic building blocks of information.
- Dataset Types: how these data elements are organized into larger structures.
- Choosing an appropriate visualization depends on correctly identifying both the data type and the dataset type.
- The same visualization technique may not work equally well for different datasets.

💡 *Understanding the structure of data is the first step toward selecting an effective visualization*

→ Data and Dataset Types



Visualization of Data, Understanding and Interpretation

Basic Dataset Types



We can identify **four primary dataset types**:

Dataset Type	Description	Example
Tables	Organized into rows (items) and columns (attributes).	Student records, sales data
Networks	Items connected through relationships (links).	Social networks, road maps
Fields	Continuous values measured over space or time.	Temperature maps, elevation models
Geometry	Represents spatial shapes and positions.	Country boundaries, CAD models

Additional collections include: clusters, sets, lists
These often appear as combinations of the four basic dataset types.

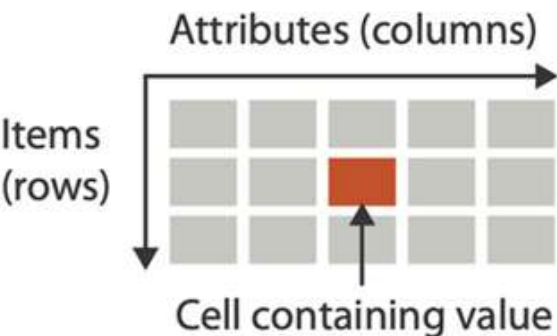
Source: Visualization Analysis & Design by Tamara Munzner, Ch 2 Sec 2.4

Visualization of Data, Understanding and Interpretation

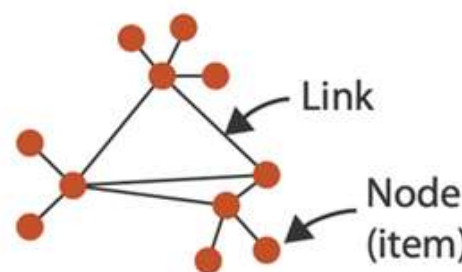
Basic Dataset Types

➔ Dataset Types

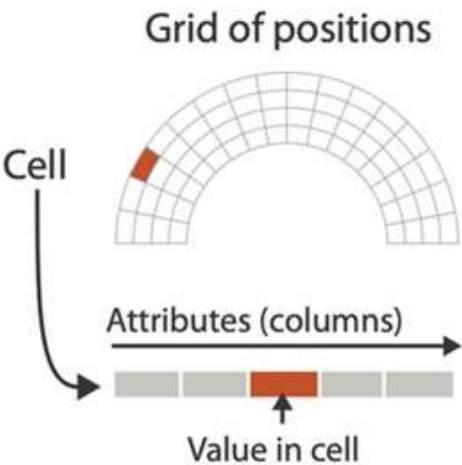
➔ Tables



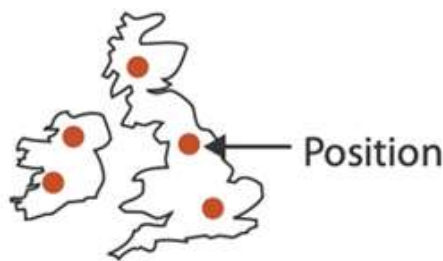
➔ Networks



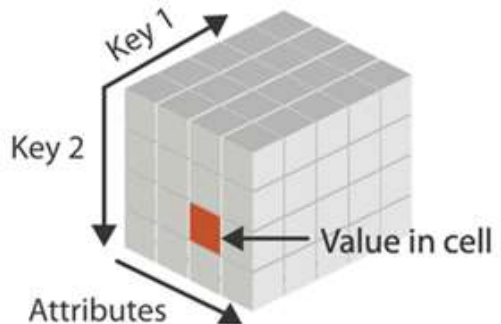
➔ Fields (Continuous)



➔ Geometry (Spatial)



➔ Multidimensional Table



➔ Trees

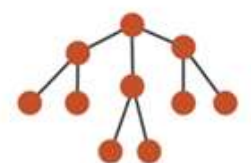


Figure 2.4. The detailed structure of the four basic dataset types.

Visualization of Data, Understanding and Interpretation

Attribute Types

An **attribute** is a measurable property associated with an item.

We can classify attributes into:

1. Categorical

- Represent labels or groups.
- No inherent ordering.
- Examples:

- Gender
- Country

➔ Attribute Types
➔ Categorical



➔ Ordered

➔ Ordinal



➔ Quantitative



1. Ordered

- Values have a meaningful order and are divided into:
 - Ordinal
 - Ordered categories
 - Differences are not numerically meaningful.
 - Example: Small < Medium < Large
 - Quantitative
 - Numerical values.
 - Arithmetic operations are meaningful.
 - Example: Height, Temperature, Sales

Visualization of Data, Understanding and Interpretation

Ordering Direction of Attributes

For ordered attributes, values may follow different ordering patterns:

Sequential

- Progresses continuously from low to high.
- Example:
 - Income
 - Height
 - Temperature

Diverging

- Values move away from a meaningful central point.
- Example:
 - Profit / Loss
 - Elevation relative to sea level
 - Positive / Negative sentiment

Cyclic

- Values repeat after completing a cycle.
- Example:
 - Hours of the day
 - Months of the year
 - Wind direction

Why it matters:

The ordering direction influences:

- Color scale selection
- Axis design
- Appropriate visualization techniques

→ Ordering Direction

→ Sequential



→ Diverging



→ Cyclic



Visualization of Data, Understanding and Interpretation

Dataset Availability

- Static datasets: Complete dataset available before analysis.
- Dynamic datasets: Data arrives continuously as a stream.
- Streaming dashboards, sensor networks, and stock markets are examples of dynamic datasets.
- Dynamic data requires continuous updates and interactive visualization.

➔ Dataset Availability

➔ Static



➔ Dynamic



Dataset availability can be either static or dynamic, for any dataset type.

Visualization of Data, Understanding and Interpretation

Bar Plot

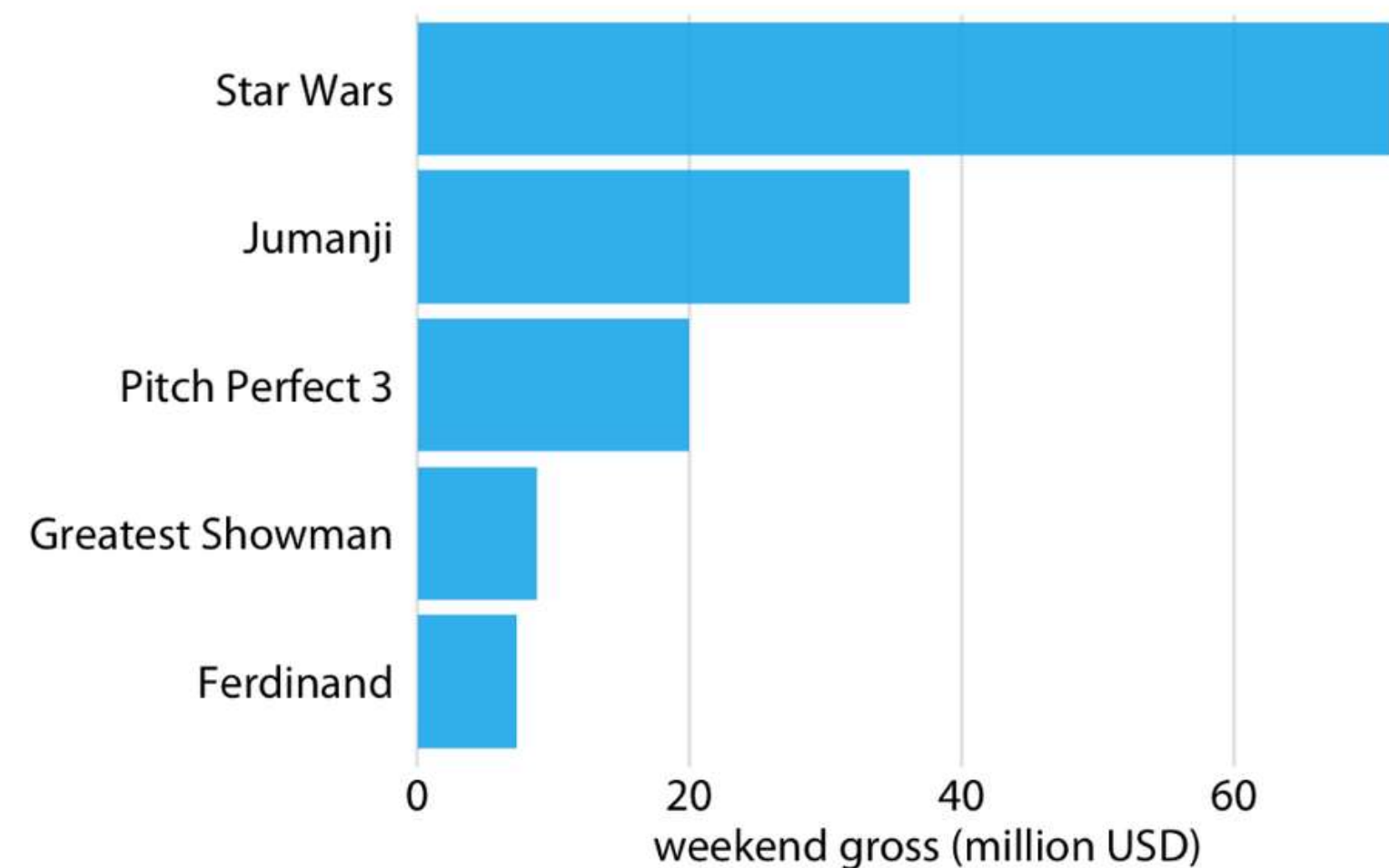
(TB1, Sec 6.1)

- Bar plots represent numerical amounts of categorical variables using rectangular shapes of varying heights or lengths.

- The zero-baseline rule requires that bars always start at zero on linear scales to preserve proportional area.

- Area-to-value ratios must remain perfectly constant across bars to strictly satisfy the Principle of Proportional Ink.

- Horizontal bar configurations provide a compact layout that prevents long category labels from overlapping or rotating.



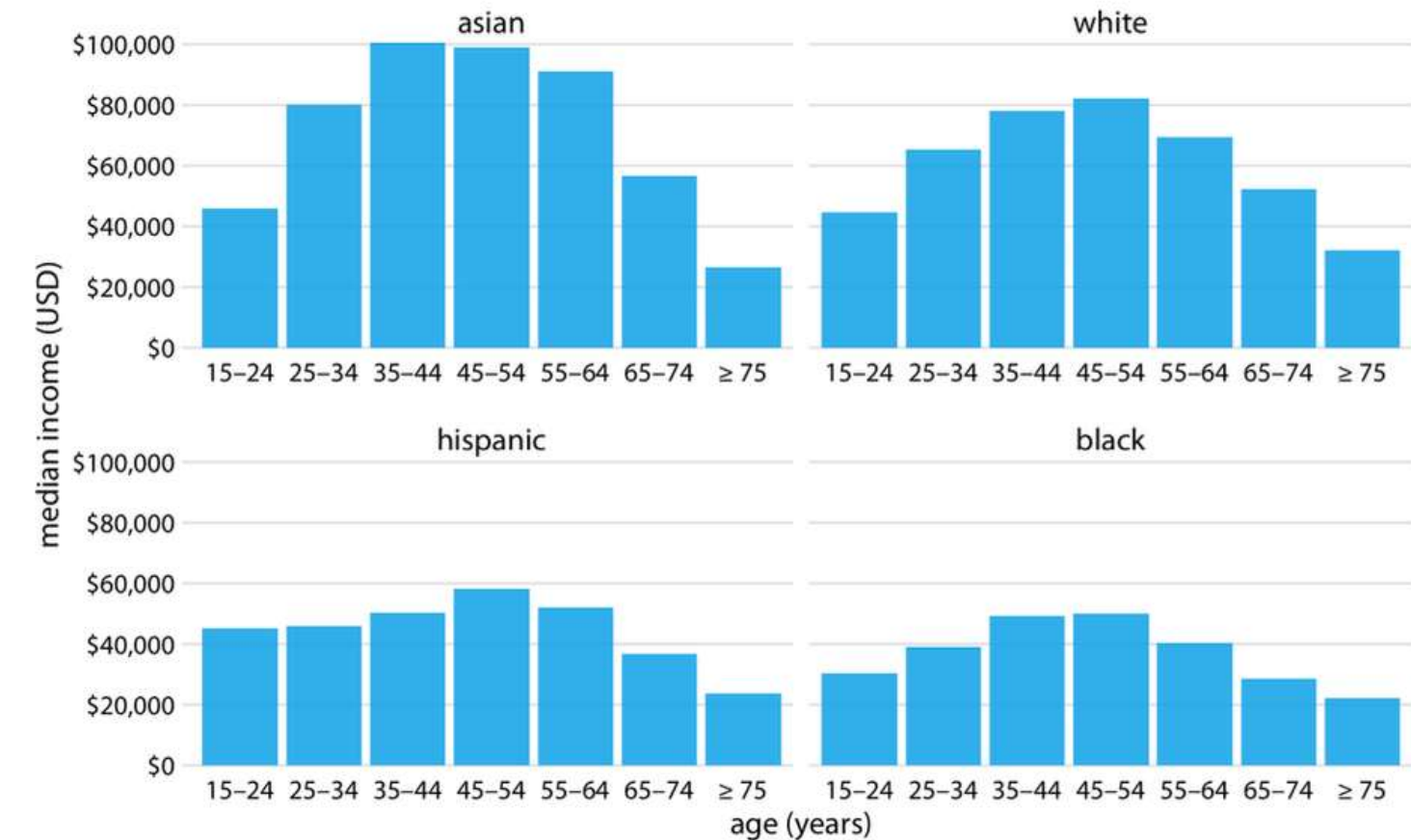
Displaying gross ticket sales for movies using a clean, sorted horizontal bar plot.

Visualization of Data, Understanding and Interpretation

Grouped vs. Stacked Bar Charts

(TB1, Sec 6.1)

- Grouped bar charts place colored rectangles side-by-side to facilitate comparison within multiple categories.
- Stacked bar charts place rectangles on top of each other to visualize proportions of a combined whole.
- Baseline shifting in stacked charts makes it difficult to compare the heights of individual internal bars.
- Small multiples offer a robust alternative to stacked charts by showing separate regular bar plots without color clutter.

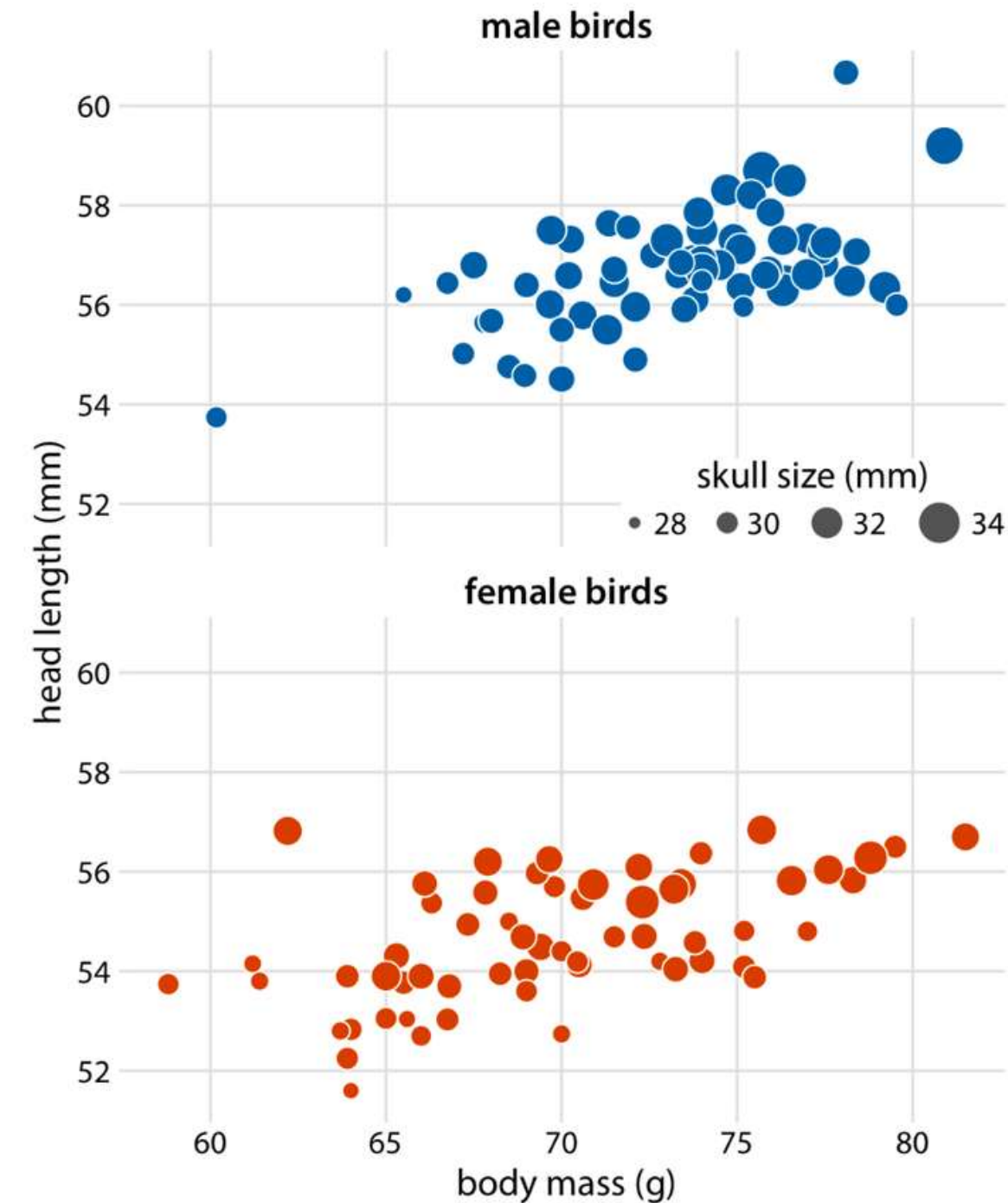


Median household incomes across demographics using four separate regular bar plots.

Visualization of Data, Understanding and Interpretation

Points as Marks: Scatter Plots and Bubble Charts (TB1, Sec 12.1)

- Scatter plots represent the standard visual mapping for examining associations between two continuous quantitative variables.
- Bubble charts introduce a third continuous variable by mapping numerical values to the physical size of individual points.
- Spatial positions must be mapped to orthogonal axes using identical metric units for paired datasets.
- Shape scales can be combined with point sizes to distinguish categorical factors from continuous variables.



Head length versus body mass for blue jays with point sizes representing skull size.

Visualization of Data, Understanding and Interpretation

MCQs

- 1. Which of the following represents a common visual mapping error in data visualization?**
- A. Mapping an unordered qualitative variable to geometric shapes.
 - B. Mapping a continuous quantitative variable to spatial position.
 - C. Mapping a continuous quantitative variable to a discrete channel like geometric shape.
 - D. Mapping an ordered qualitative variable to a continuous color scale.

Visualization of Data, Understanding and Interpretation

MCQs

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- C. Mapping a continuous quantitative variable to a discrete channel like geometric shape.
- D. Mapping an ordered qualitative variable to a continuous color scale.

Answer: C

Visualization of Data, Understanding and Interpretation

MCQs

2. According to the Principle of Proportional Ink, what is a fundamental requirement for area fill graphs and linear scale bar plots to be mathematically valid?

- A. They must use a non-zero baseline to emphasize minor fluctuations.
- B. The vertical position scale must start strictly at zero.
- C. They must feature color-coded text labels rather than solid fills.
- D. They must represent at least five independent variables simultaneously.

Visualization of Data, Understanding and Interpretation

MCQs



2. According to the Principle of Proportional Ink, what is a fundamental requirement for area fill graphs and linear scale bar plots to be mathematically valid?

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- B. The vertical position scale must start strictly at zero.
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- D. They must represent at least five independent variables simultaneously.

Answer: B

Visualization of Data, Understanding and Interpretation

MCQs

3. Why is mapping a continuous quantitative variable onto a line dash pattern or geometric shape considered a critical visual encoding error?

- A. These channels are mathematically and perceptually limited to discrete categories and cannot represent continuous intermediate values
- B. These channels automatically convert the numerical data into logarithmic coordinates
- C. These channels increase the physical file size of the rendered vector graphic
- D. These channels force the background grid lines to disappear

Visualization of Data, Understanding and Interpretation

MCQs



3. Why is mapping a continuous quantitative variable onto a line dash pattern or geometric shape considered a critical visual encoding error?

- A. These channels are mathematically and perceptually limited to discrete categories and cannot represent continuous intermediate values
- B. These channels automatically convert the numerical data into logarithmic coordinates
- C. These channels increase the physical file size of the rendered vector graphic
- D. These channels force the background grid lines to disappear

Answer: A

Visualization of Data, Understanding and Interpretation

MCQs



- 4. What specific limitation of stacked bar charts makes grouped bar charts or small multiples a preferred alternative for detailed comparisons?**
- A. Stacked bar charts cannot display combined categorical totals.
 - B. Stacked bar charts are mathematically invalid on linear scales.
 - C. Baseline shifting in stacked charts makes it difficult to compare the heights of individual internal bars.
 - D. Stacked bar charts require the use of continuous color scales.

Visualization of Data, Understanding and Interpretation

MCQs



4. What specific limitation of stacked bar charts makes grouped bar charts or small multiples a preferred alternative for detailed comparisons?

- A. Stacked bar charts cannot display combined categorical totals.
- B. Stacked bar charts are mathematically invalid on linear scales.
- C. Baseline shifting in stacked charts makes it difficult to compare the heights of individual internal bars.
- D. Stacked bar charts require the use of continuous color scales.

Answer: C

Visualization of Data, Understanding and Interpretation

demo and activity



Demo Notebook:

A hands-on demonstration of visual encoding principles, covering data types, chart selection, marks and channels, scales, and visualization best practices to effectively communicate insights from data.

Student Activity:

A time-based visualization exercise that challenges students to choose suitable charts, evaluate visualization quality, interpret data, and present meaningful insights using effective visual communication techniques.



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Visualization of Data, Understanding and Interpretation

Unit 1 : Foundations of Visualization, Data Interpretation

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

Dr.Prajwala TR (Course Instructor)

Dept. of CSE, PES University, Bangalore

Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.



Visualization of Data, Understanding and Interpretation

Lecture Overview

- Examine the biological mechanics of human vision and visual acuity.
- Explore how the brain perceives depth, motion, and three-dimensional surfaces.
- Analyze luminance, brightness, and lightness, illustrating how contrast helps direct attention and why common visual illusions.
- Study cognitive load and Edward Tufte's data-ink ratio, explaining how to reduce mental effort, eliminate unnecessary "chart junk", and maximize the signal-to-noise ratio in visualizations.

Visualization of Data, Understanding and Interpretation

Why Human Vision Matters in Data Visualization (TB 3, Ch2)

Data visualization is effective only when it *aligns with how people naturally perceive information*.

Our visual system is optimized to quickly recognize:

- Shapes
- Colors
- Movement
- Patterns
- Spatial relationships

Good visualizations take advantage of these abilities, allowing users to understand trends and relationships almost instantly.

Poor visual design increases cognitive effort and can even lead to incorrect conclusions.

Visualization of Data, Understanding and Interpretation

From Eye to Brain: How We See (Ch2)

Vision is not created by the eye alone. The process consists of three stages:

1. Light reflected from objects enters the eye.
2. The retina converts light into electrical signals.
3. The visual cortex interprets those signals to create meaningful perception.

💡 *We do not "see" the image on the retina, we see the brain's interpretation of it*

The retina contains two kinds of photoreceptors.

Cones

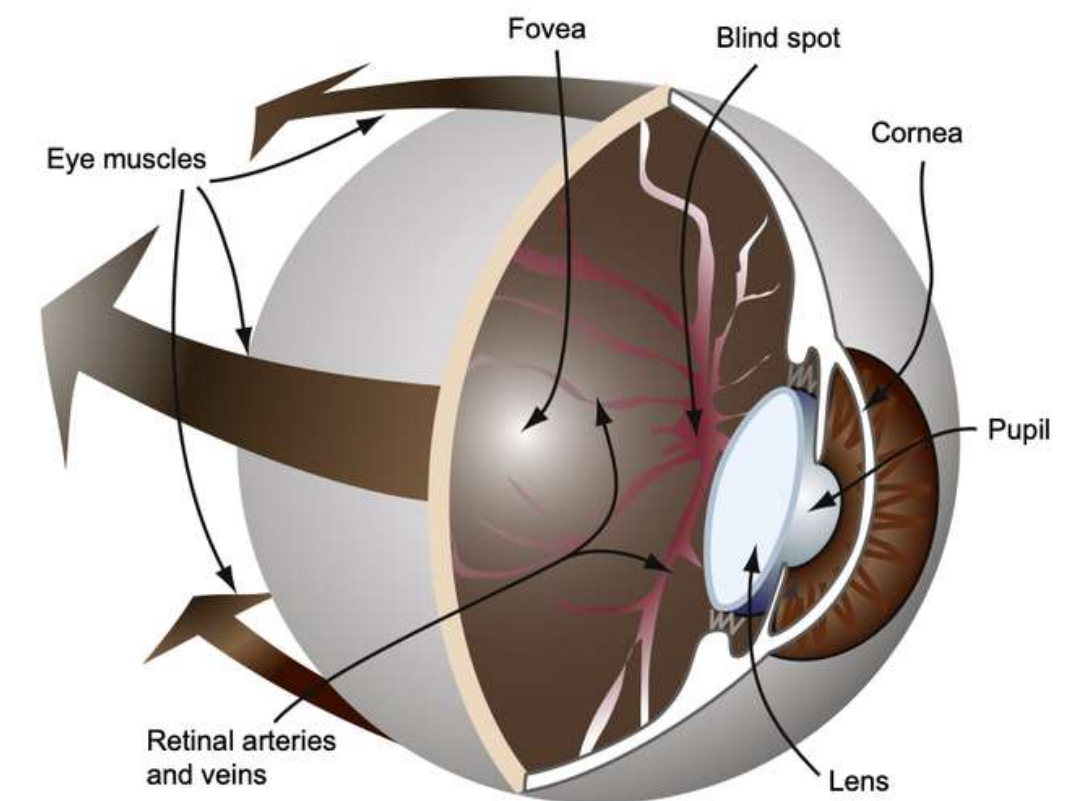
- Responsible for color vision
- High detail
- Concentrated in the fovea

Rods

- Sensitive in low light
- Detect motion
- Poor color perception

The *fovea* is a tiny central region where visual resolution is highest.

When reading charts, people constantly move their eyes so important information falls on the fovea.



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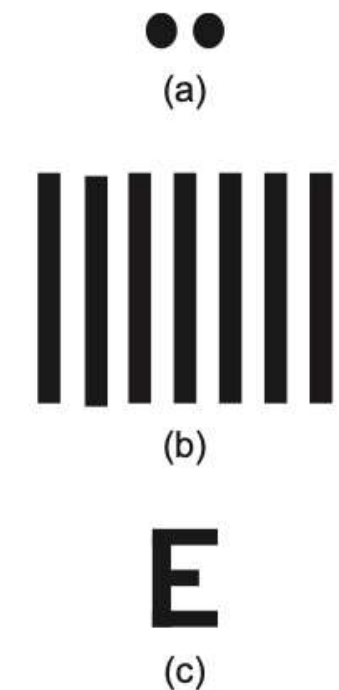
Visual Acuity: What Can We Actually See?

(Ch2)

Visual acuity refers to the eye's ability to distinguish fine detail.

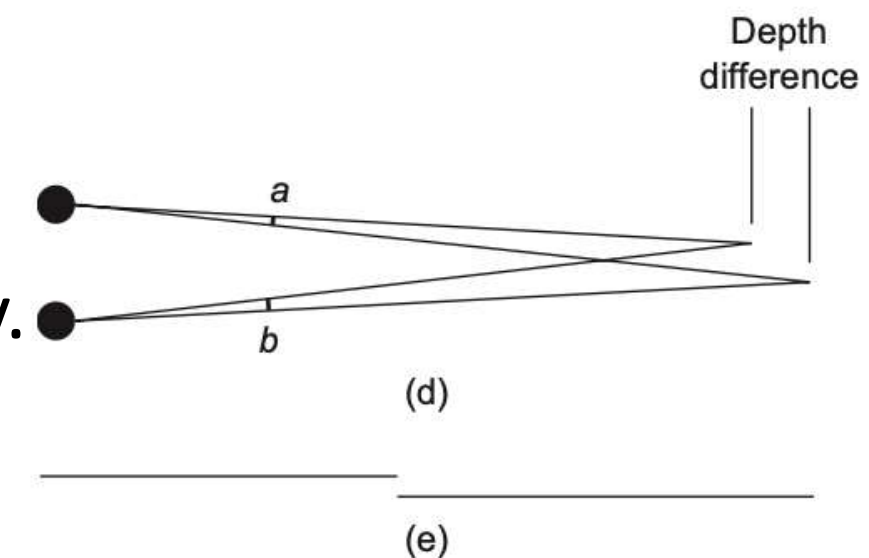
Different types include:

- (a) Point acuity: Ability to distinguish two closely spaced points as separate.
- (b) Grating acuity: Ability to distinguish a pattern of bright and dark bars.
- (b) Letter acuity: Ability to recognize and identify letters of different sizes.
- (c) Stereo acuity: Ability to perceive small differences in depth using both eyes.
- (d) Vernier acuity: Ability to detect tiny misalignments between two line segments.



Effect on Visualization Design

- Small labels become difficult to read.
- Thin lines may not be distinguishable.
- Tiny symbols can appear identical.
- Use appropriate font sizes, line thickness, and spacing for better readability.



Visualization of Data, Understanding and Interpretation

What is Visual Perception? (Ch2)



Visual perception is the process by which the brain interprets information received from the eyes to create a meaningful understanding of the surrounding world.

Perception is more than seeing. While the eyes capture light, the brain:

- Recognizes objects, faces, and patterns
- Estimates size, shape, and distance
- Identifies motion and direction
- Separates important objects from the background

As a result, we perceive meaningful scenes rather than individual pixels or colors.

💡 *Vision begins in the eye, but perception happens in the brain*

Visualization of Data, Understanding and Interpretation

How the Brain Perceives Depth and Surfaces (Ch2)

The world around us is *three-dimensional*, but the image formed on the retina is *two-dimensional*. The brain reconstructs depth by *combining multiple visual cues*.

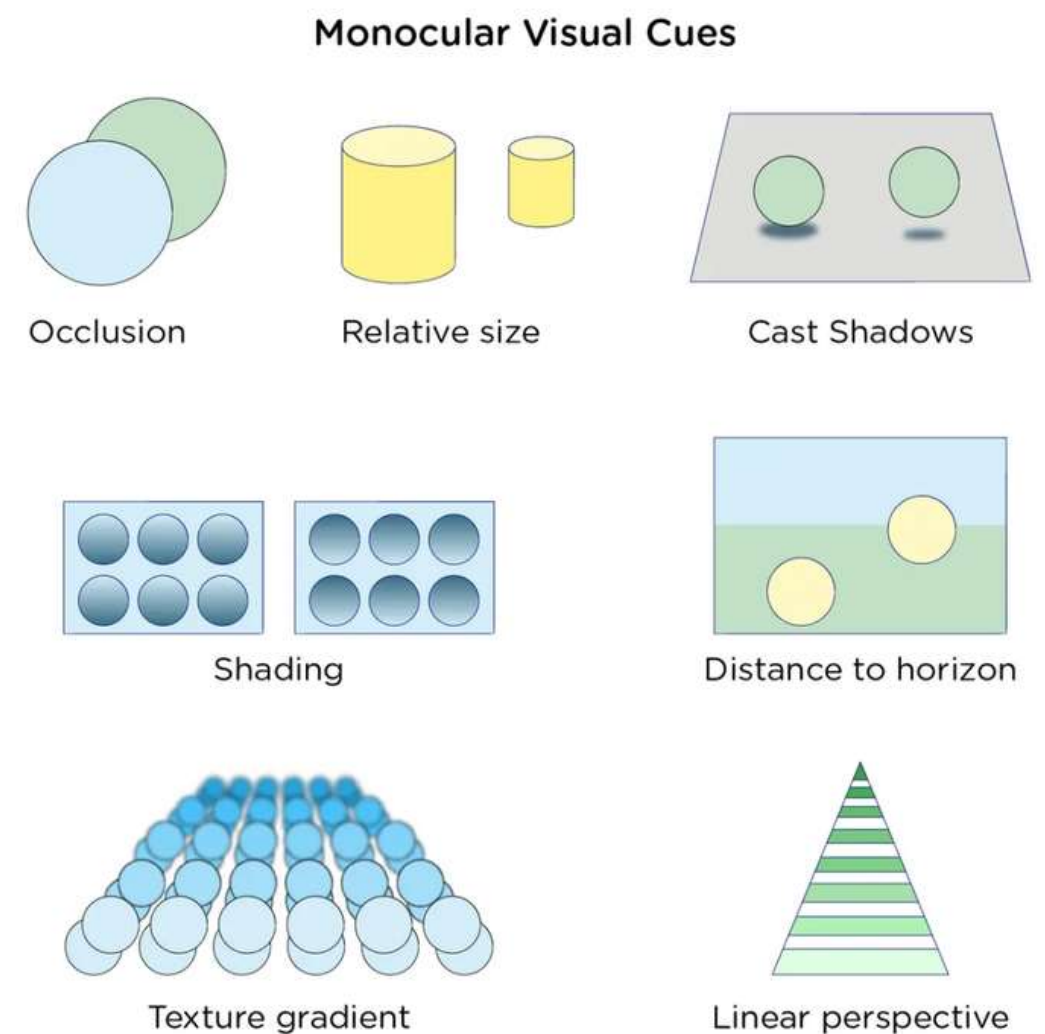
Important cues include:

- Texture gradients: Dense textures appear farther away.
- Shading: Light and shadow reveal the shape of objects.
- Cast shadows: Help estimate position and depth.
- Perspective: Parallel lines appear to converge with distance.
- Relative size: Smaller objects are usually perceived as farther away.

These cues allow us to quickly judge object shape, distance, and spatial relationships.

Why it matters for visualization?

Visualization designers use the same cues to make charts, maps, and 3D models easier to interpret. Example: terrain maps, heatmaps with hill shading, scientific surface plots, 3D scatter plots.



Jack Westin

Texture Gradients: Surfaces in the real world are rarely smooth-they contain textures that provide important information about shape, orientation, and distance.

As a textured surface extends away from us:

- Texture elements appear smaller and more densely packed.
- The brain interprets this gradual change as increasing distance.
- Texture gradients help estimate the orientation and curvature of surfaces.

Example: A tiled floor or brick road appears denser as it stretches into the distance.

Optical Flow: As we move through an environment, the image projected onto the retina continuously changes. This pattern of apparent motion is called optical flow.

The brain uses optical flow to:

- Determine the direction of movement.
- Estimate speed.
- Judge the distance of nearby objects.
- Detect potential obstacles.

Objects close to us move rapidly across our visual field, while distant objects appear to move much more slowly. *Example:* While driving, roadside trees seem to rush past, whereas distant mountains barely appear to move.

Visualization of Data, Understanding and Interpretation

Texture, Motion and Optical Flow (Ch2)

Why It Matters in Data Visualization?

Motion and texture can improve the understanding of visualizations by:

- Revealing surface shape in 3D plots.
- Showing movement and trends in animated visualizations.
- Enhancing terrain, flow maps, and scientific simulations.
- Helping users interpret spatial relationships more naturally.

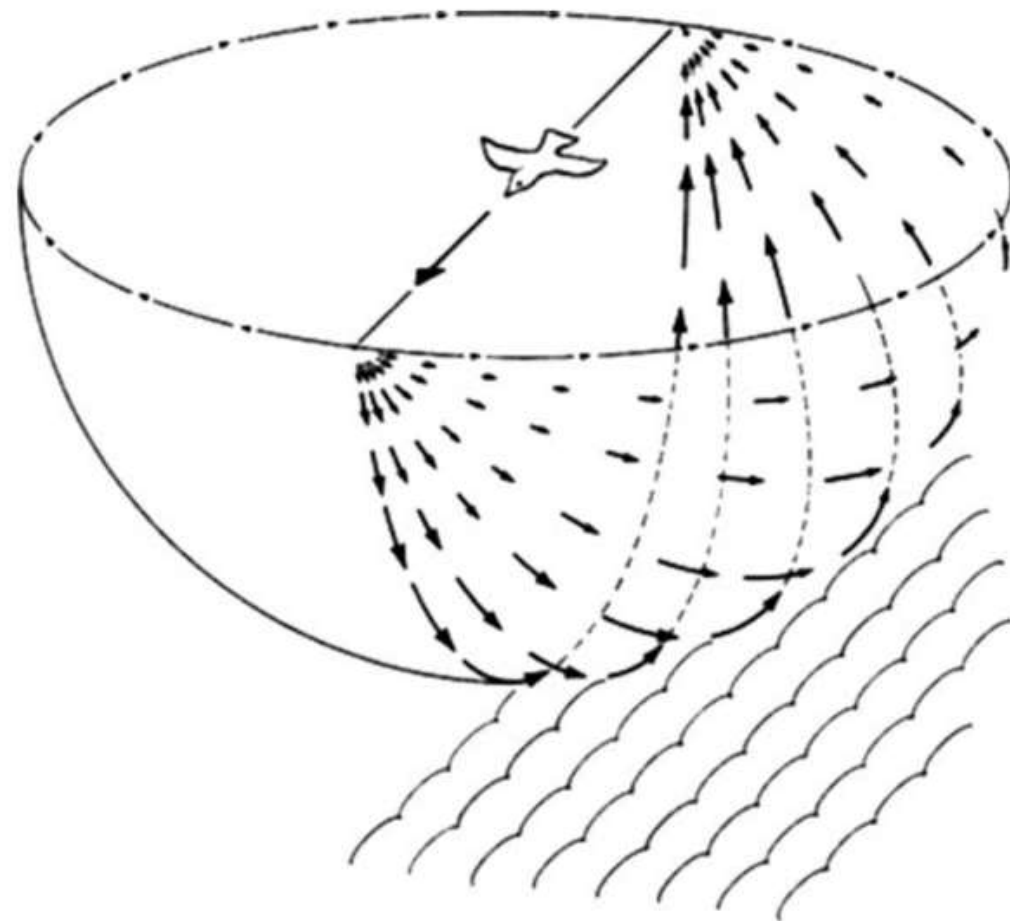
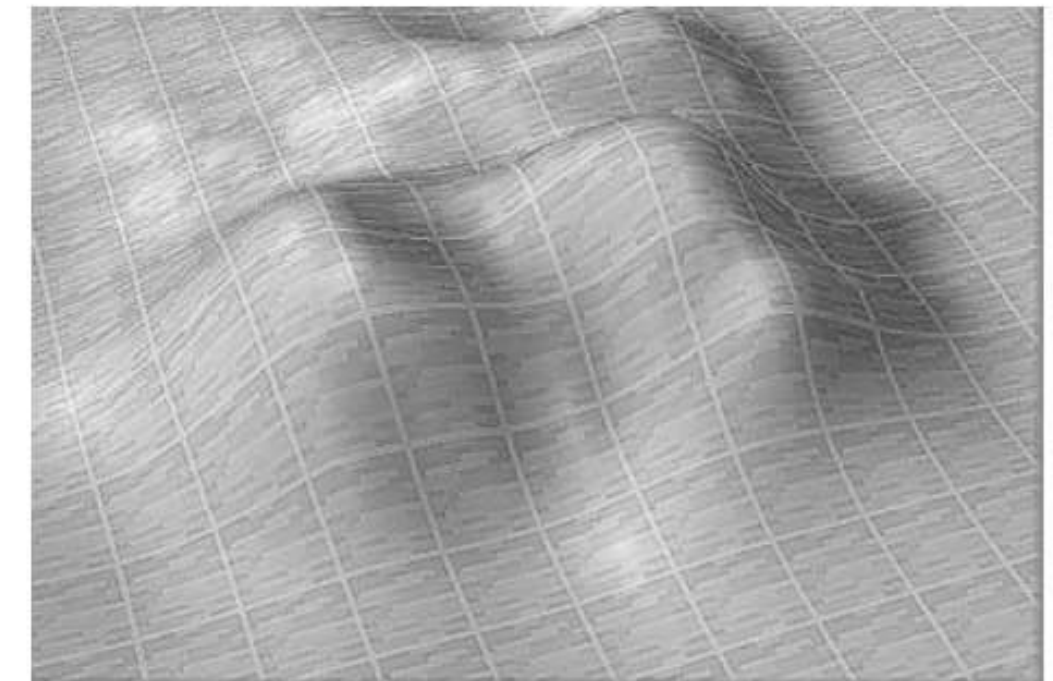


Figure 2.3 An expanding flow pattern of visual information is created as an observer moves while gazing in a forward direction. (From Gibson (1979). Reproduced with permission.)



(a)



(b)

Figure 2.4 An undulating surface with (a) and without (b) surface texture.

Visualization of Data, Understanding and Interpretation

Using Light and Shading in Visualization (Ch2)

Our visual system relies on light and shading to interpret the shape, depth, and orientation of objects. Without these visual cues, objects often appear flat and difficult to distinguish.

Different lighting effects reveal different aspects of an object's surface:

Lambertian Shading (Diffuse Lighting)

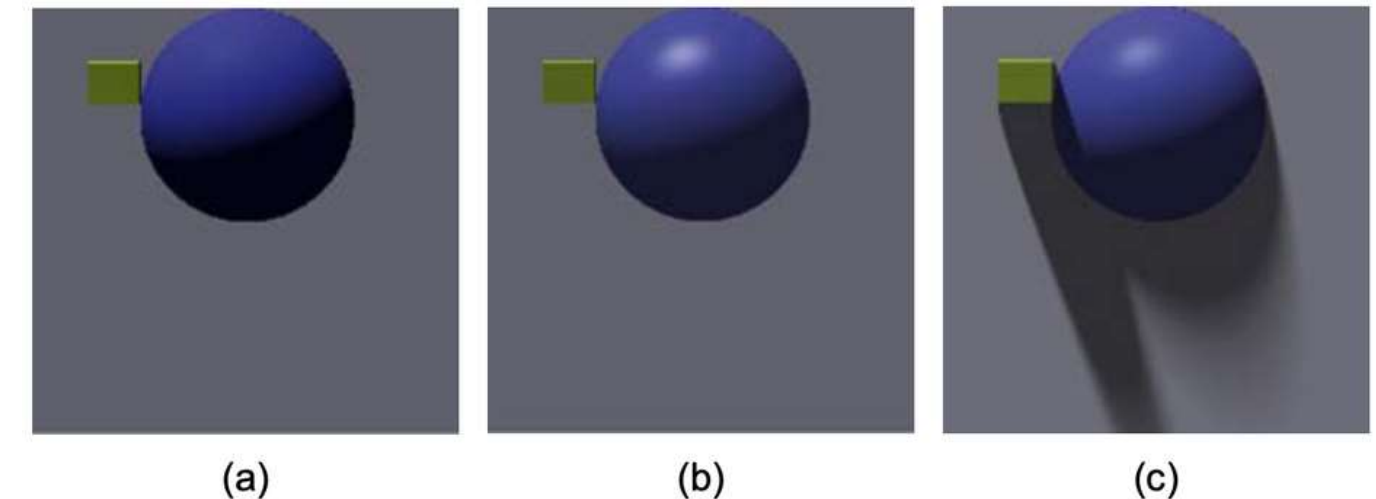
- Simulates light scattered evenly from a matte surface.
- Clearly reveals the overall shape and curvature of an object.
- Commonly used in scientific and engineering visualizations.

Specular Highlights

- Bright reflections produced on glossy surfaces.
- Emphasize fine surface details, roughness, and small imperfections.
- Help distinguish subtle changes in surface structure.

Cast Shadows

- Show how objects relate to surrounding surfaces.
- Improve the perception of depth, height, and spatial position.
- Make three-dimensional scenes easier to interpret.



(a) Lambertian shading only.

(b) Lambertian shading with specular and ambient shading.

(c) Lambertian shading with specular, ambient, and cast shadows.

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Display Design: What Designers Should Remember (Ch2)

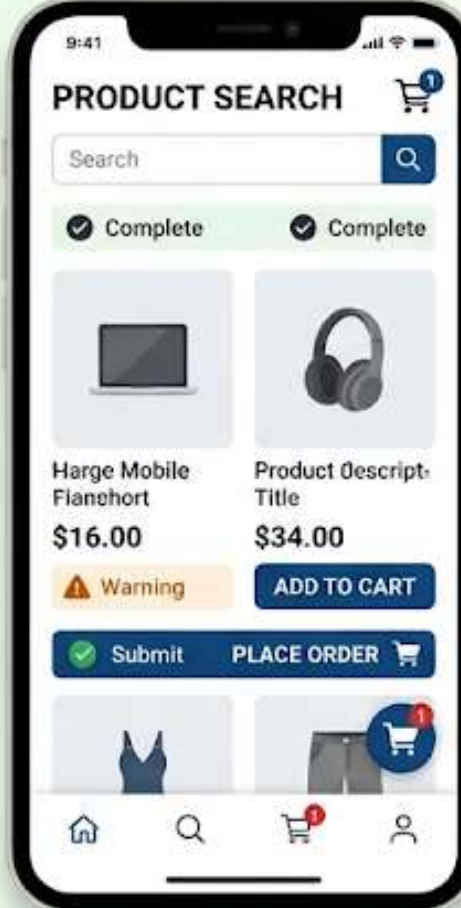
Human vision has practical limitations.

Effective displays should:

- Use sufficient contrast
- Avoid tiny text
- Avoid relying only on color
- Reduce unnecessary visual clutter
- Place important information near the visual focus

These principles improve readability and reduce cognitive effort.

COMPARING ACCESSIBLE UI: BAD DESIGN vs. GOOD DESIGN

BAD UI	GOOD UI
Disorganized, confusing visual	Clean, intuitive visual
	
NO CONTRAST Severe lack of contrast, making text unreadable	HIGH CONTRAST Black text on white background, clear readability
SMALL TEXT Tiny text descriptions, product prices are requiring zoom	READABLE TEXT All text: headings, descriptions, prices, navigation elements are minimum 14pt-16pt, bold
COLOR ONLY	SUCCESS
CLUTTER	MULTI-CHANNEL CUES
HIDDEN INFO	CLEAN DESIGN
	CLEAR FOCUS

Visualization of Data, Understanding and Interpretation

Human Perception, Cognition, Key Takeaways (Ch2)

The brain rapidly organizes visual information before conscious thinking.

Some important perceptual abilities are:

- Pattern recognition
- Figure-ground separation
- Similarity detection
- Proximity grouping
- Continuity
- Closure

These ideas form the foundation of *Gestalt principles*, which will be covered in the next lecture.

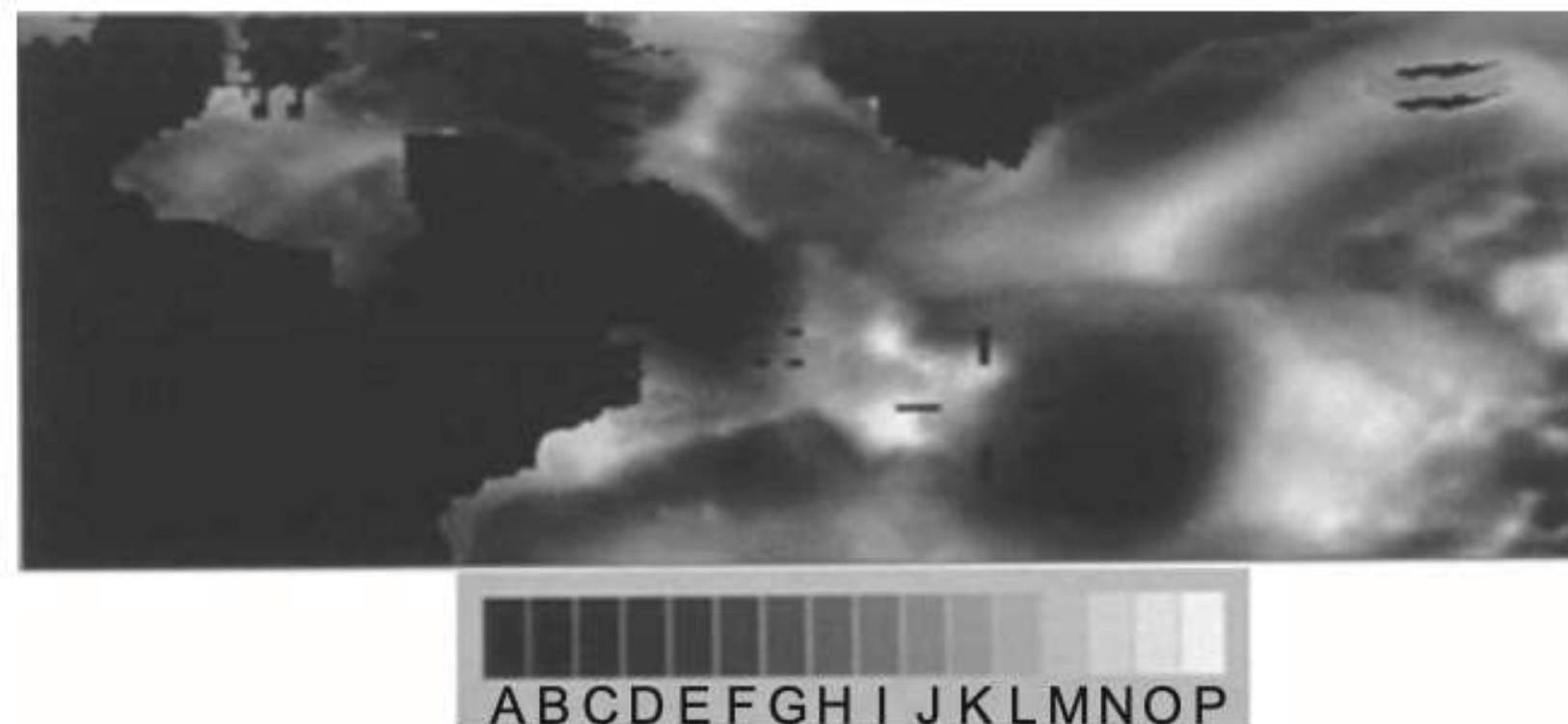
Good visualizations match the strengths of human perception.

- People notice patterns before numbers.
- Vision has limits in resolution and contrast.
- Texture, shading and motion improve understanding.
- Visual hierarchy guides attention.
- Design should reduce cognitive effort rather than increase it.

Visualization of Data, Understanding and Interpretation

Why Brightness Perception Matters (TB 3, Ch3)

- Visualizations often use *brightness* and *contrast* to **represent data values, emphasize information, and guide attention**.
- However, our brain does not perceive brightness as an absolute quantity.
- Instead, it compares neighboring regions and interprets relative differences, which can sometimes lead to visual illusions or misinterpretation.
- Understanding these perceptual mechanisms helps us create visualizations that are both accurate and easy to interpret.



Visualization of Data, Understanding and Interpretation

Luminance, Brightness and Lightness (Ch3)



Although often used interchangeably, these terms describe different concepts.

Luminance

- Physical amount of light emitted or reflected by a surface.
- Can be measured using scientific instruments.

Brightness

- The perceived intensity of light coming from a luminous source.
- Depends on human perception.

Lightness

- The perceived reflectance of a surface (how light or dark an object appears).
- Remains relatively stable under different lighting conditions.

💡 *Human perception depends more on brightness and lightness than on measured luminance*

Visualization of Data, Understanding and Interpretation

Brightness Illusions and Their Effect on Visualization (Ch3)

The visual system primarily responds to changes in brightness, not absolute light levels. This allows us to detect edges quickly, identify objects from their background and recognize patterns efficiently. Because the brain emphasizes differences, the same color or gray value can appear different depending on its surroundings. This property forms the basis of many visual illusions and influences how data visualizations are interpreted.

Visual illusions demonstrate that perception can differ from physical reality. Common brightness illusions include:

- *Hermann Grid*: Dark spots appear at intersections.
- *Mach Bands*: False bright or dark bands appear near edges.
- *Chevreul Illusion*: Uniform gray bands appear uneven because of adjacent shades.

These effects occur because the brain enhances edges and local contrast, which can distort the interpretation of charts and images.

Visualization of Data, Understanding and Interpretation

Brightness Illusions and Their Effect on Visualization (Ch3)

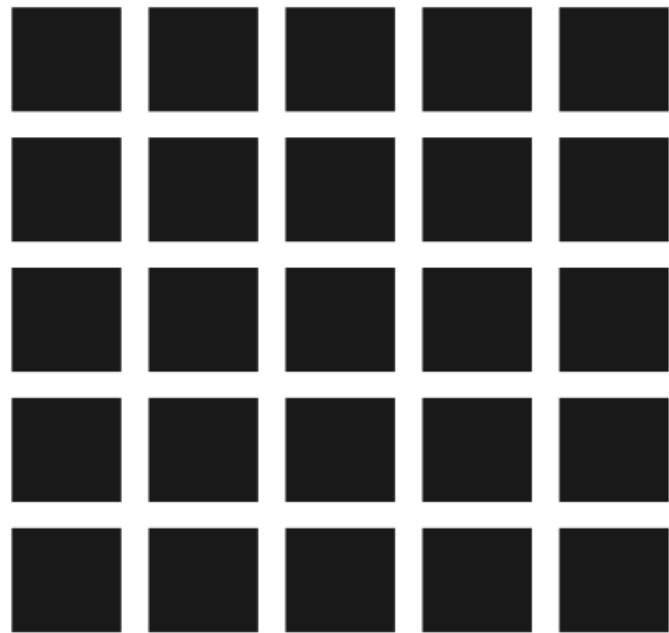


Figure 3.4 Hermann grid illusion. The black spots that are seen at the intersections of the lines are thought to result from the fact that there is more inhibition from the light in the periphery of the receptive field at position (a) than at position (b).

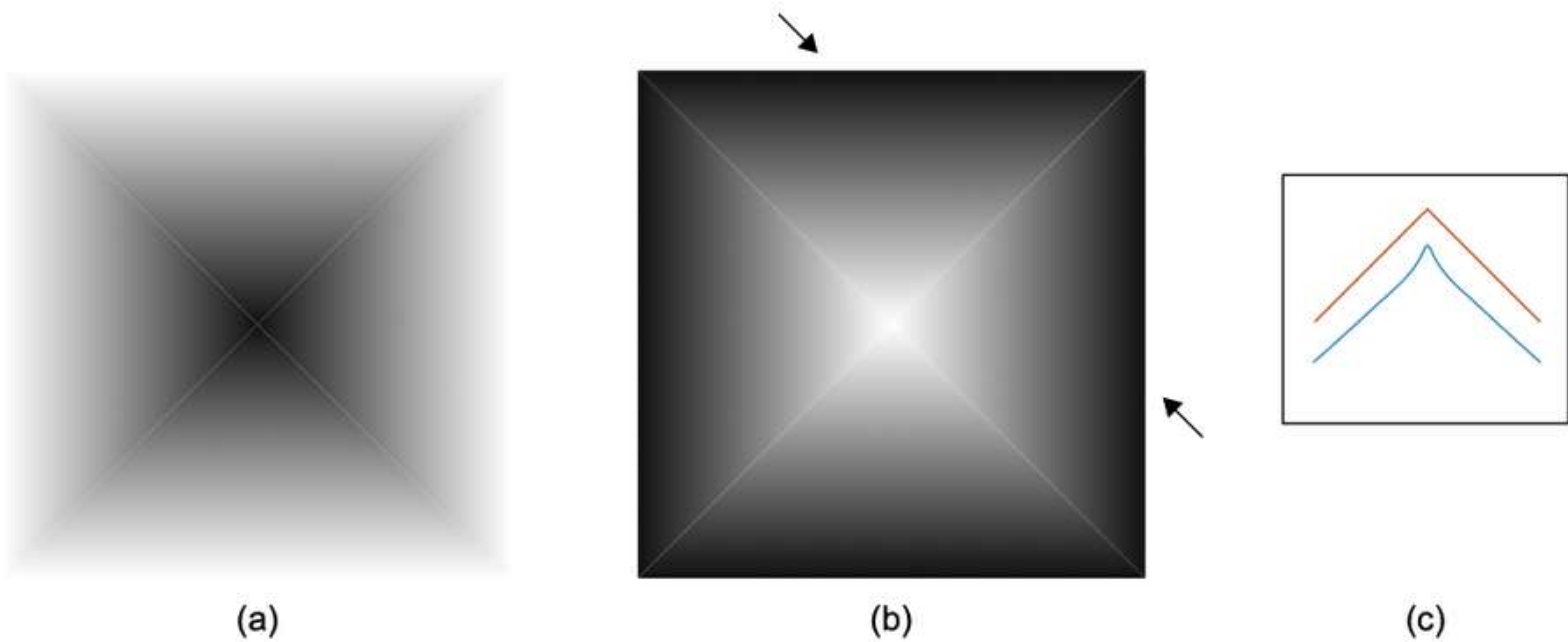
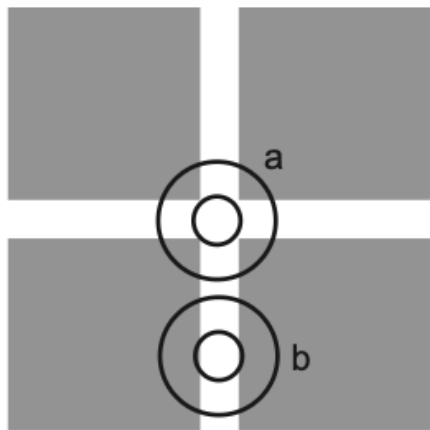


Figure 3.6 Illustration of Mach banding. (a, b) Dark and bright Mach bands are evident at the boundaries between the internal triangles. (c) The red curve shows the actual brightness profile between the two arrows. The blue curve shows how the application of a DoG filter models the bright bands that are seen.

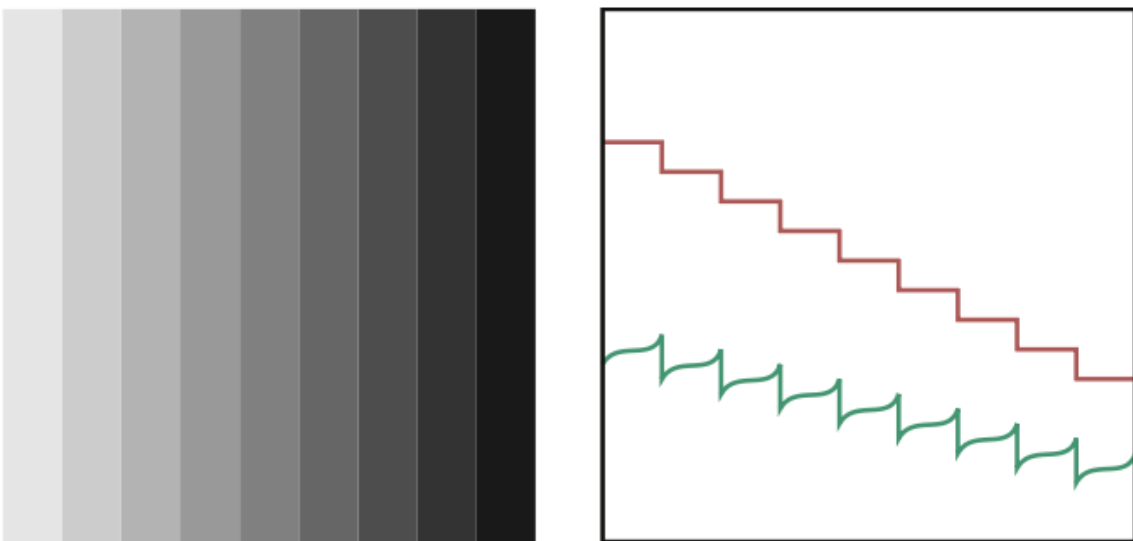


Figure 3.7 The Chevreul illusion. The measured lightness pattern is shown by the staircase pattern on the right. What is perceived can be closely approximated by a DoG model. The lower plot on the right shows the application of a DoG filter to the staircase pattern shown above.

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Gray Scales and Lightness Constancy (Ch3)

Gray scales are widely used to represent quantitative values, but they have limitations.

- Contrast illusions can distort perceived values.
- Equal gray steps are not always perceived equally.
- Reading exact values from gray scales is difficult.

For continuous quantitative data, carefully designed color scales are often more effective than grayscale alone.

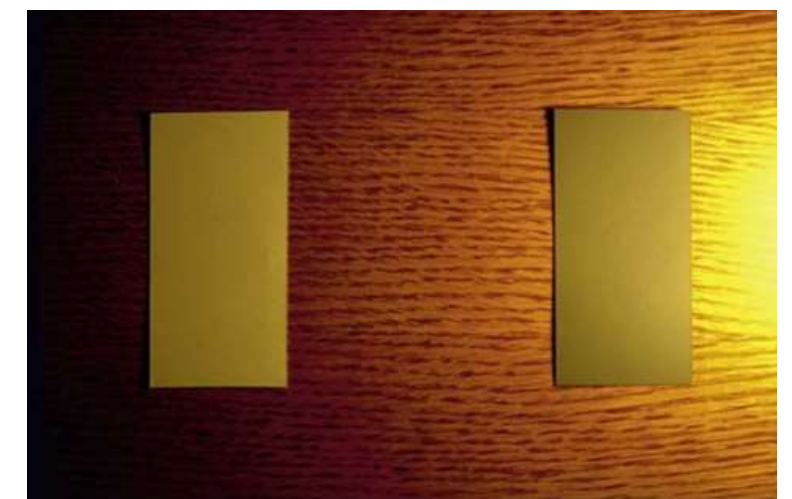
💡 *Avoid using grayscale to represent many distinct numerical values*

Although lighting conditions change dramatically throughout the day, we usually perceive an object's color and lightness as remaining constant. This ability is called **lightness constancy**.

The brain achieves this by considering:

- Surrounding objects
- Illumination
- Shadows
- Surface orientation
- Relative contrast

This allows us to recognize objects correctly even under different lighting conditions.



These two pieces of paper are illuminated by a desk lamp just to the right of the picture. This makes the amount of light reflected from the gray paper about the same as the light reflected from the white paper.

Visualization of Data, Understanding and Interpretation

Contrast, Practical Design Guidelines (Ch3)

Contrast helps direct attention and improve readability.

Effective use of contrast includes:

- Increasing contrast for important information.
- Reducing contrast for secondary elements.
- Maintaining sufficient contrast between text and background.
- Using contrast instead of excessive decoration.

Research also recommends a **minimum 3:1 luminance contrast ratio** for displaying fine details such as text and texture.

When designing visualizations:

- Use high luminance contrast for labels and fine details.
- Avoid relying only on grayscale for quantitative data.
- Use contrast to emphasize important information.
- Reduce unnecessary background clutter.
- Choose backgrounds that improve readability.

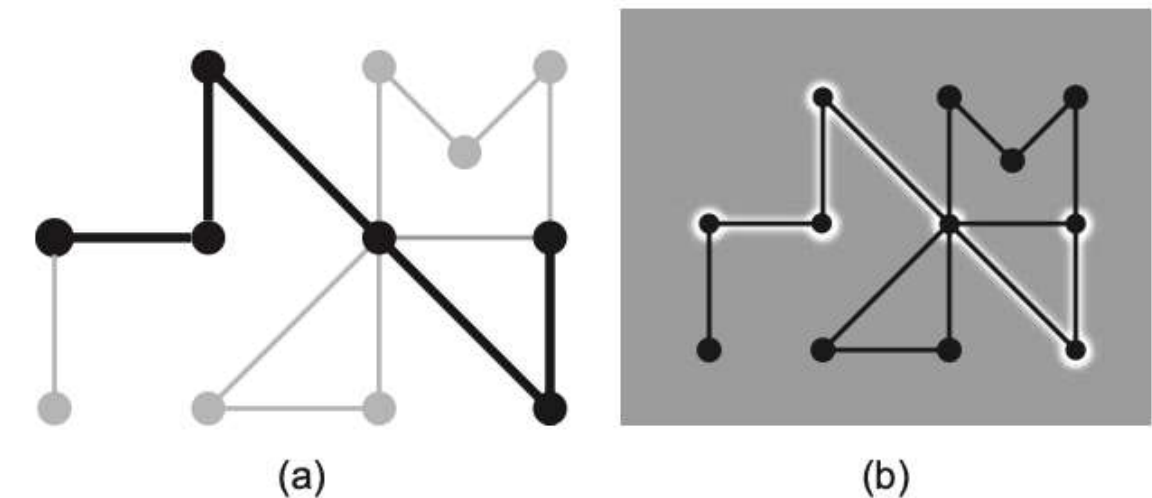


Figure 3.13 Two methods for highlighting a node-link diagram. (a) The contrast is reduced for the less important parts of the network. (b) The background contrast is increased using haloing to emphasize important parts.

💡 *Good visualization design aligns with how the human visual system naturally interprets brightness and contrast*

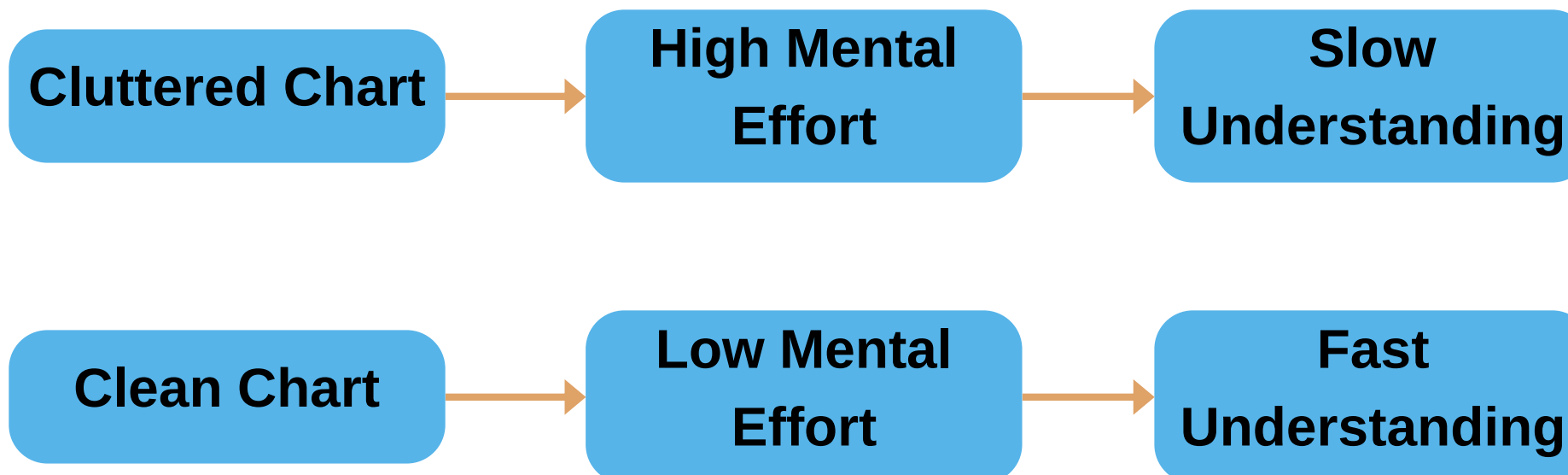
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Cognitive Load

(TB 1, Ch23)

- Cognitive load is the mental effort required to understand a visualization.
- Every visual element (lines, colors, labels, icons) demands the viewer's attention.
- Unnecessary visual elements increase effort without improving understanding.
- Good visualizations reduce mental effort so viewers can focus on the message, not decoding the chart.

💡 *Design for understanding, not decoration*



Visualization of Data, Understanding and Interpretation

Data-Ink Ratio & Chart Junk

(Ch23)



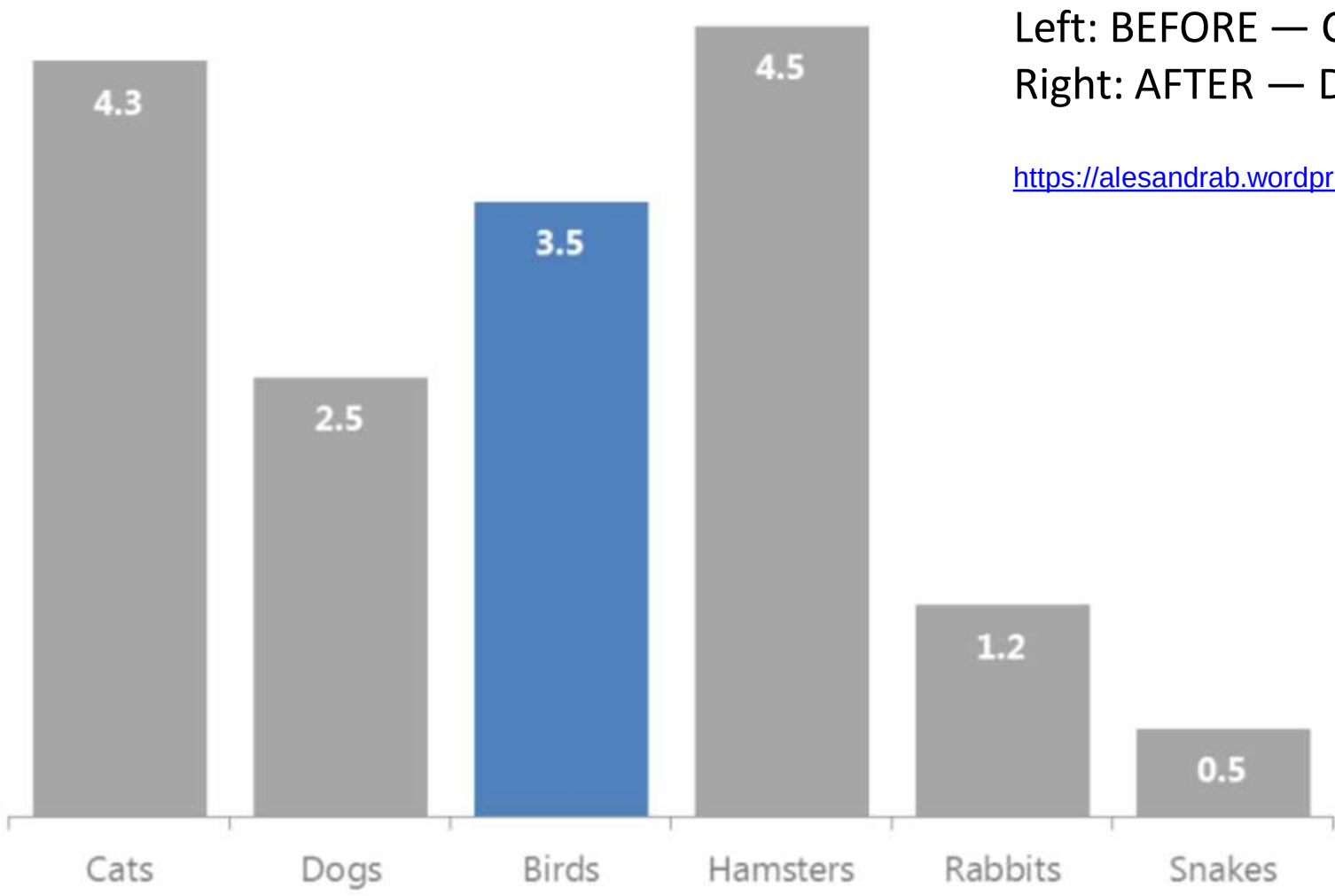
- *Edward Tufte's Principle: Data-Ink Ratio* is the proportion of ink (or pixels) in a visualization that directly represents the actual data.
- A high data-ink ratio means most visual elements communicate information, while unnecessary decorations are minimized.
- Remove non-essential ink such as excessive gridlines, heavy borders, 3D effects, gradients, shadows, and decorative backgrounds that do not improve understanding.
- The objective is not to remove all supporting elements: axes, labels, and legends should remain whenever they help interpretation.
- A clean visualization improves readability, directs attention to the data, and reduces cognitive load.

💡 Every visual element should have a purpose- if it doesn't communicate data or improve understanding, consider removing it

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Data-Ink Ratio & Chart Junk (Ch23)

Why less is more in data visualization:



Left: BEFORE — Chart Junk
Right: AFTER — Data-Ink Ratio Applied
<https://alesandrab.wordpress.com/2015/07/09/less-is-more/>

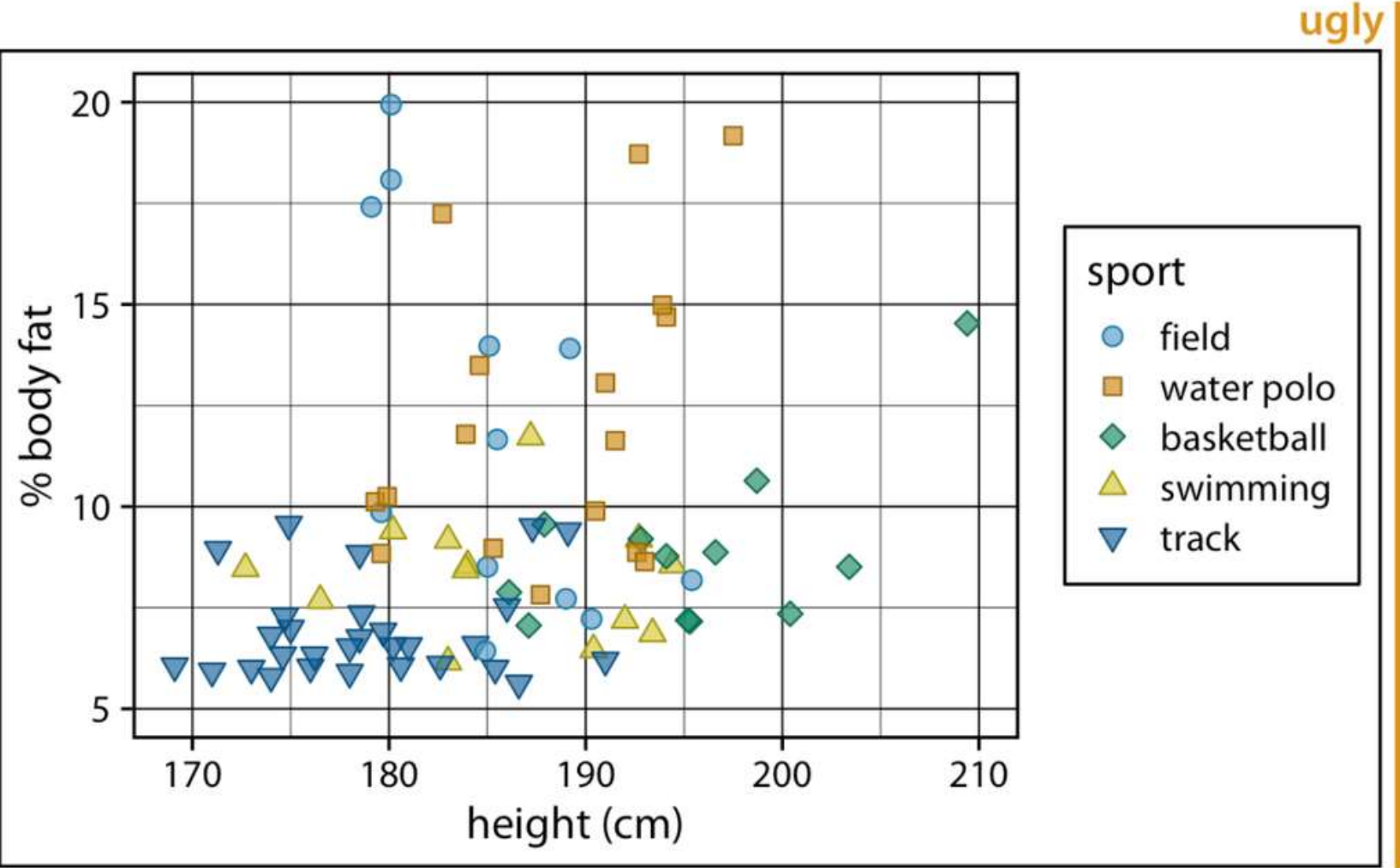
In the left graph, what exactly is the message behind this chart? What data is important to know? It turned out that the “bird” data was important. The category had increased by 60% in the last month. Looking at that chart can you tell that the “bird” category has improved? Can you even tell that the “bird” category is important?

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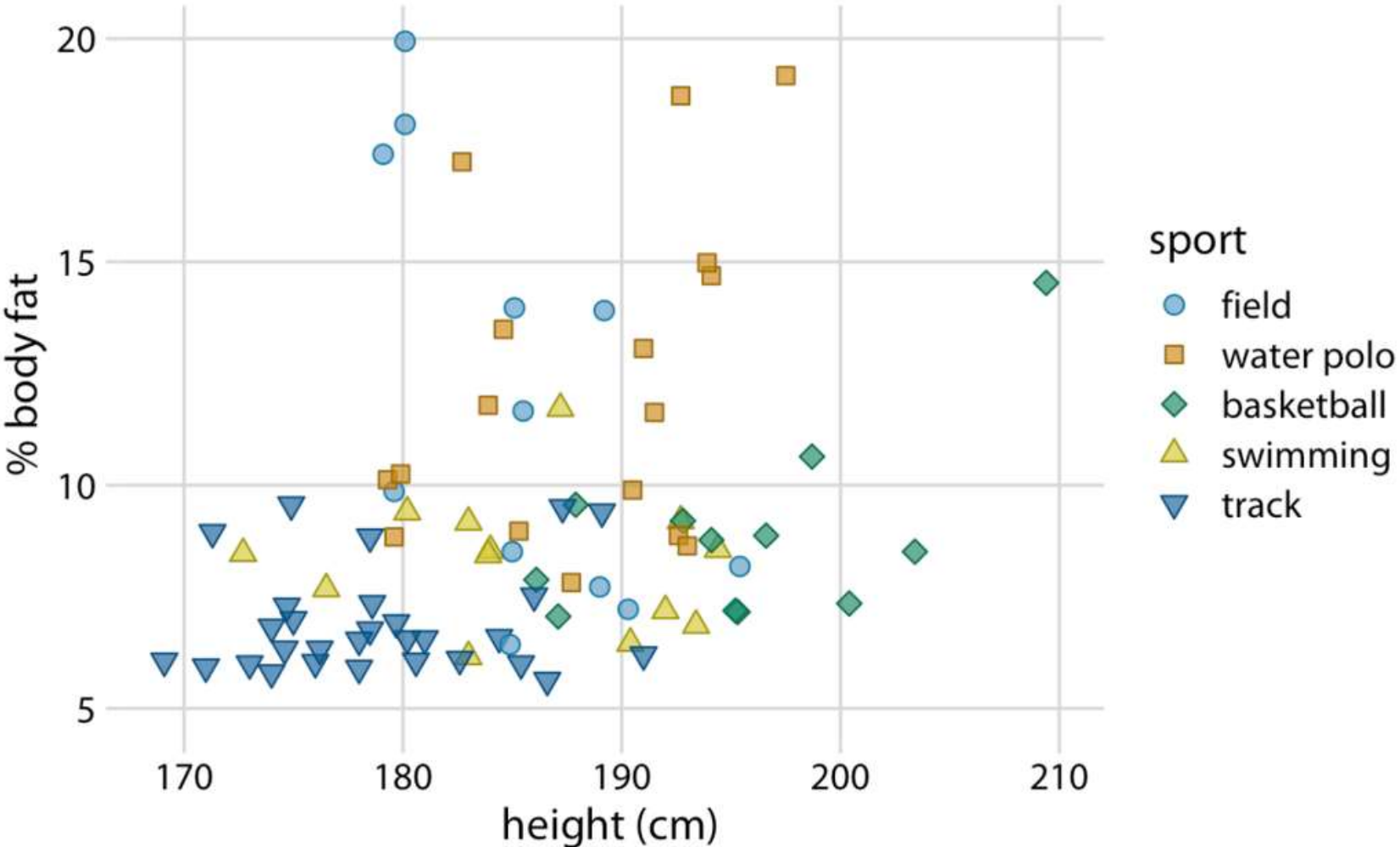
Data-Ink Ratio & Chart Junk

(Ch23)

Another example:



Old
version



Cleaned- up
version

Visualization of Data, Understanding and Interpretation

Signal vs Noise

(Ch23)



A good visualization maximizes the signal while minimizing the noise.

Signal	Noise
Data values	Decorative icons
Trends	Unnecessary colors
Comparisons	Shadows & gradients
Important labels	Redundant legends

💡If an element doesn't improve understanding, remove it.

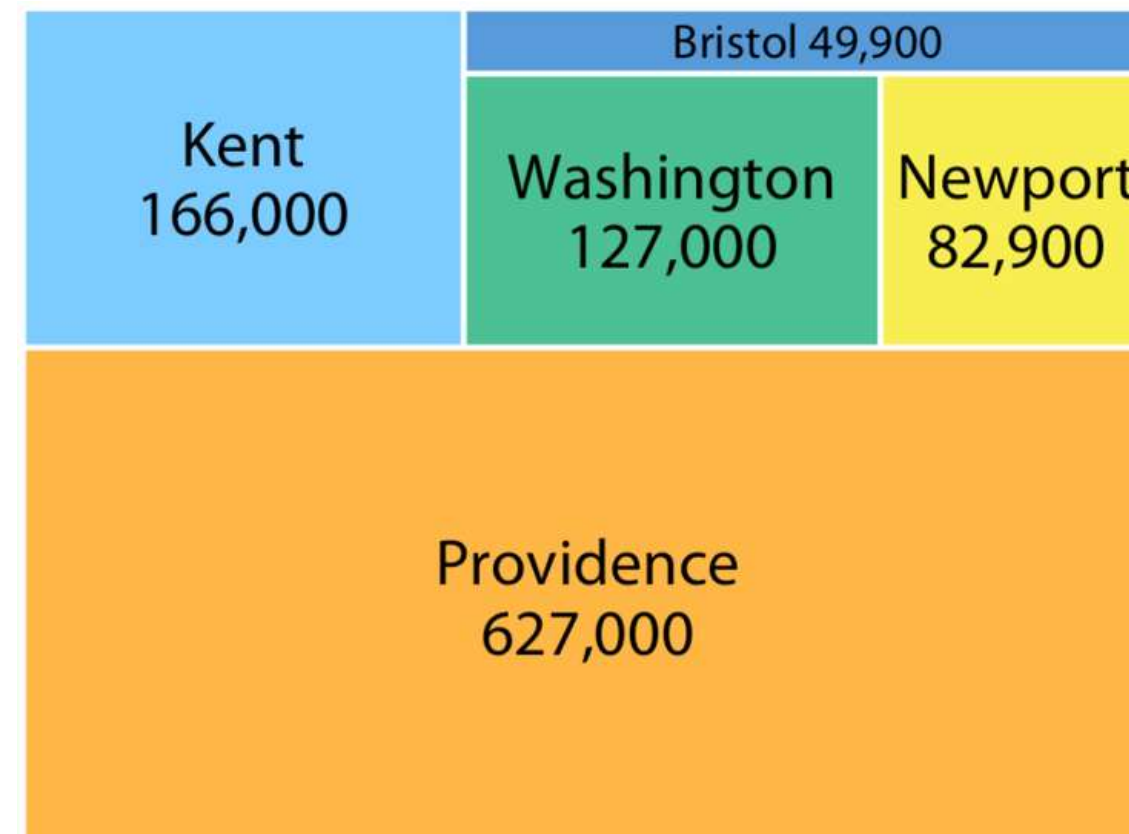
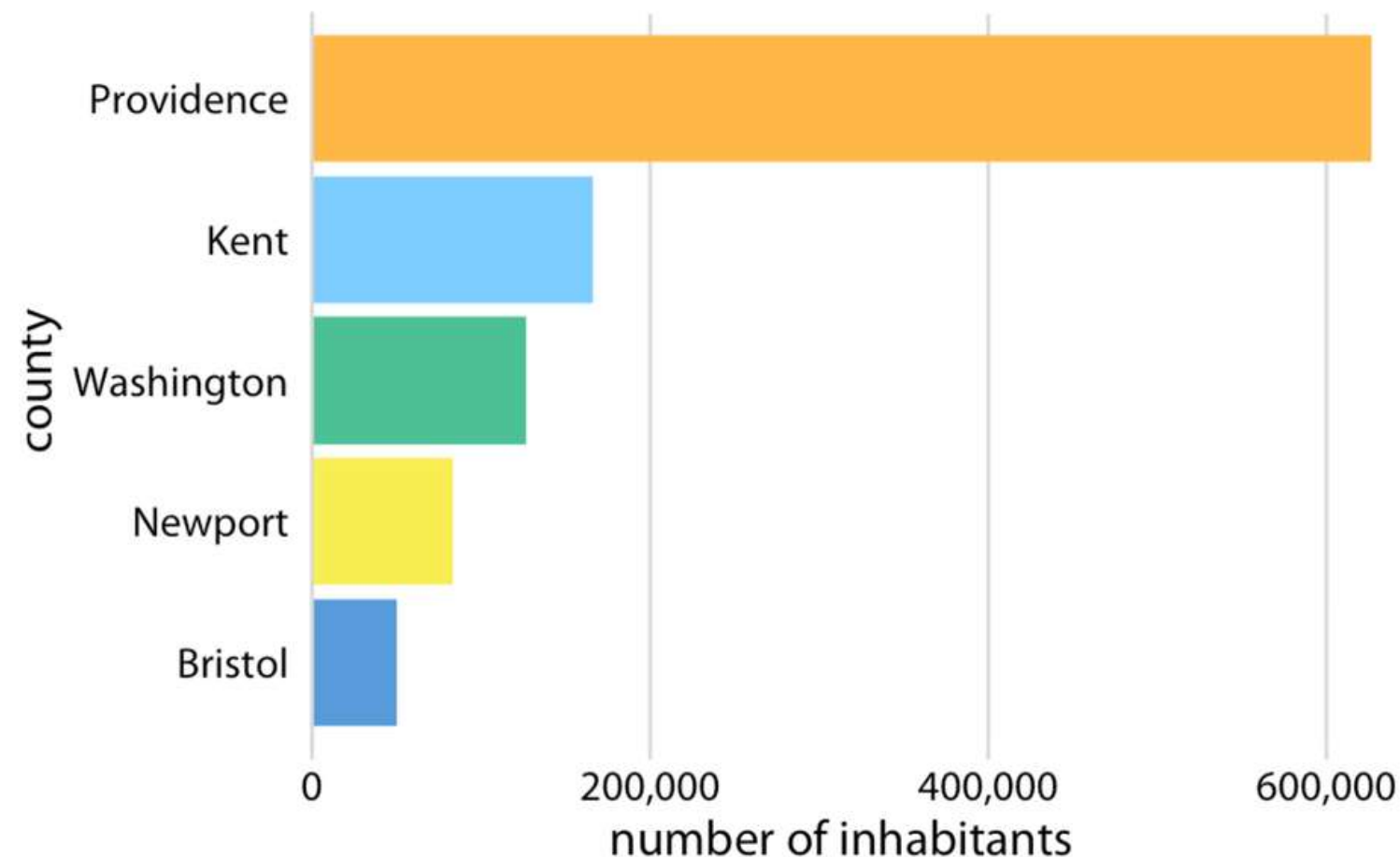
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Principle of Proportional Ink (Ch17)

Principle

- The size of a visual element should be proportional to the value it represents.
- Viewers naturally compare lengths, areas, and volumes when interpreting graphics.
- If the visual size grows faster or slower than the data, the visualization becomes misleading.
- A graphic should represent data honestly by keeping visual size proportional to the underlying values.

In the example below, one uses length whereas the other uses area:



Different visual encodings can represent the same data accurately. In bar charts, proportionality is shown through length, while in treemaps it is shown through area.

Visualization of Data, Understanding and Interpretation

Common Violations of Proportional Ink (Ch17)



Violations occur when:

- Bars do not start at zero, exaggerating differences.
- Areas are scaled incorrectly.
- Logarithmic axes are used with bars or shaded regions.
- Visual size no longer matches the underlying data.

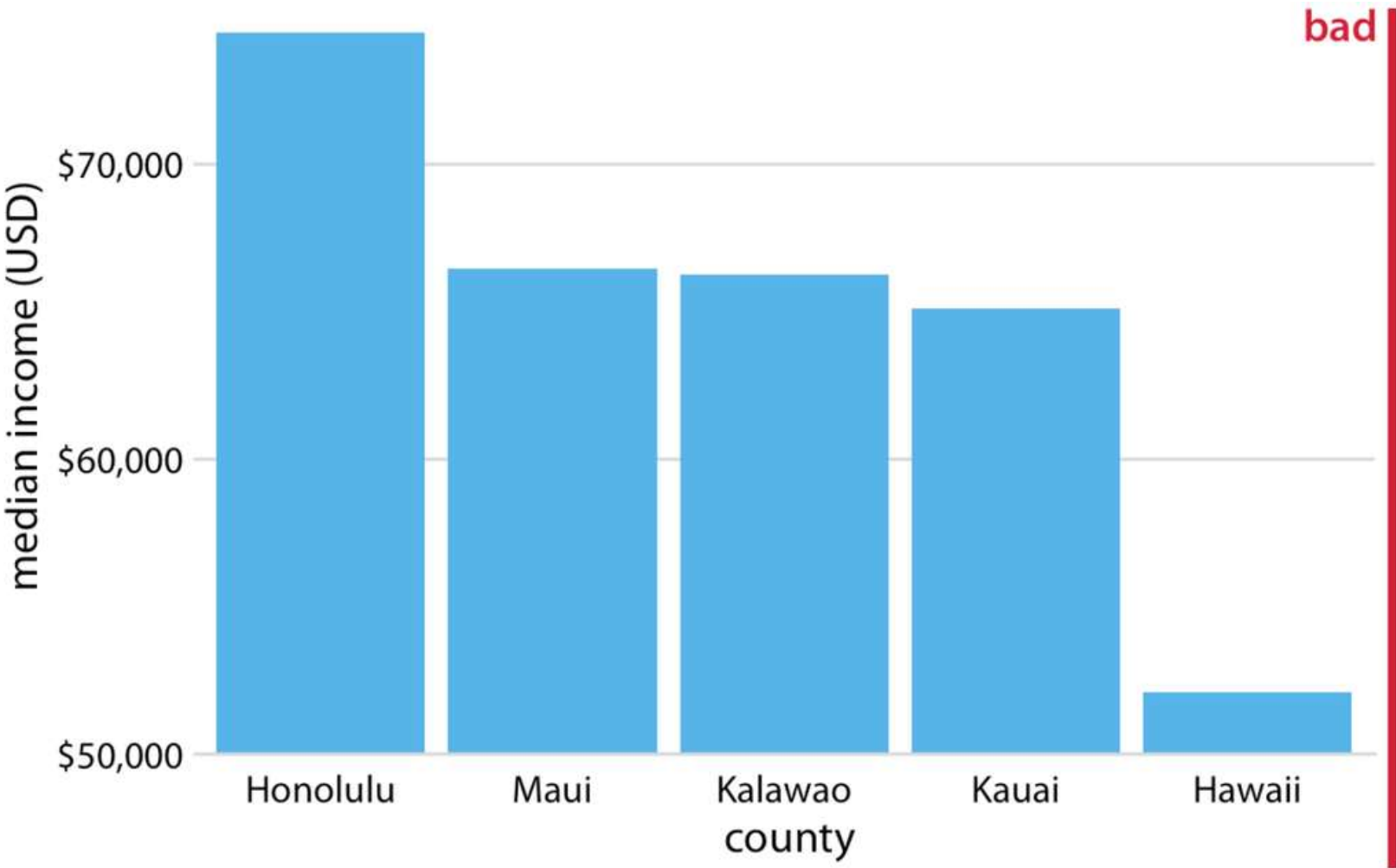
Best practices:

- Bars and filled areas on a linear scale should begin at 0
- For logarithmic scales, use points instead of bars
- Match the visual encoding to the type of scale being used.

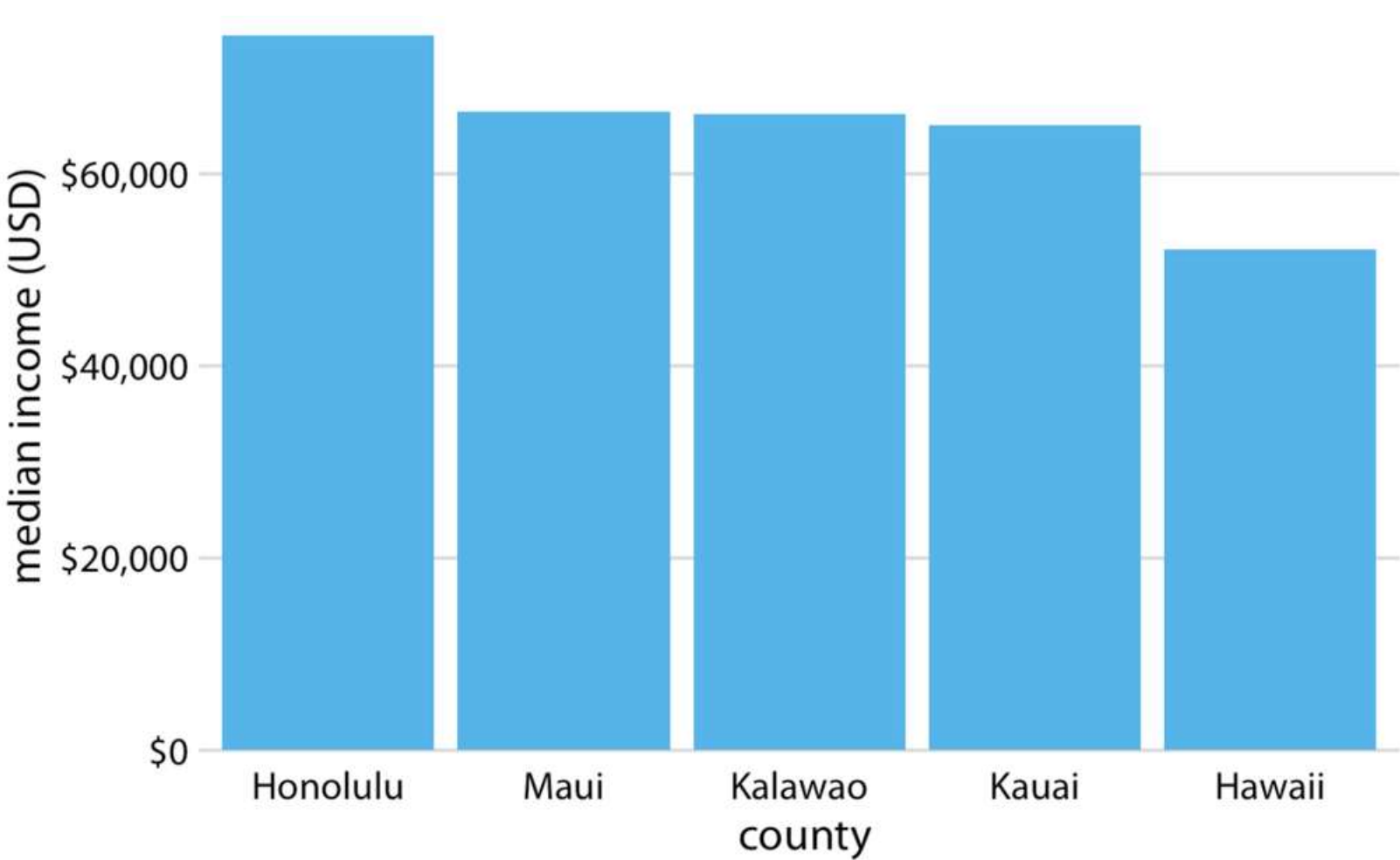
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Common Violations of Proportional Ink

(Ch17)



**BA
D**

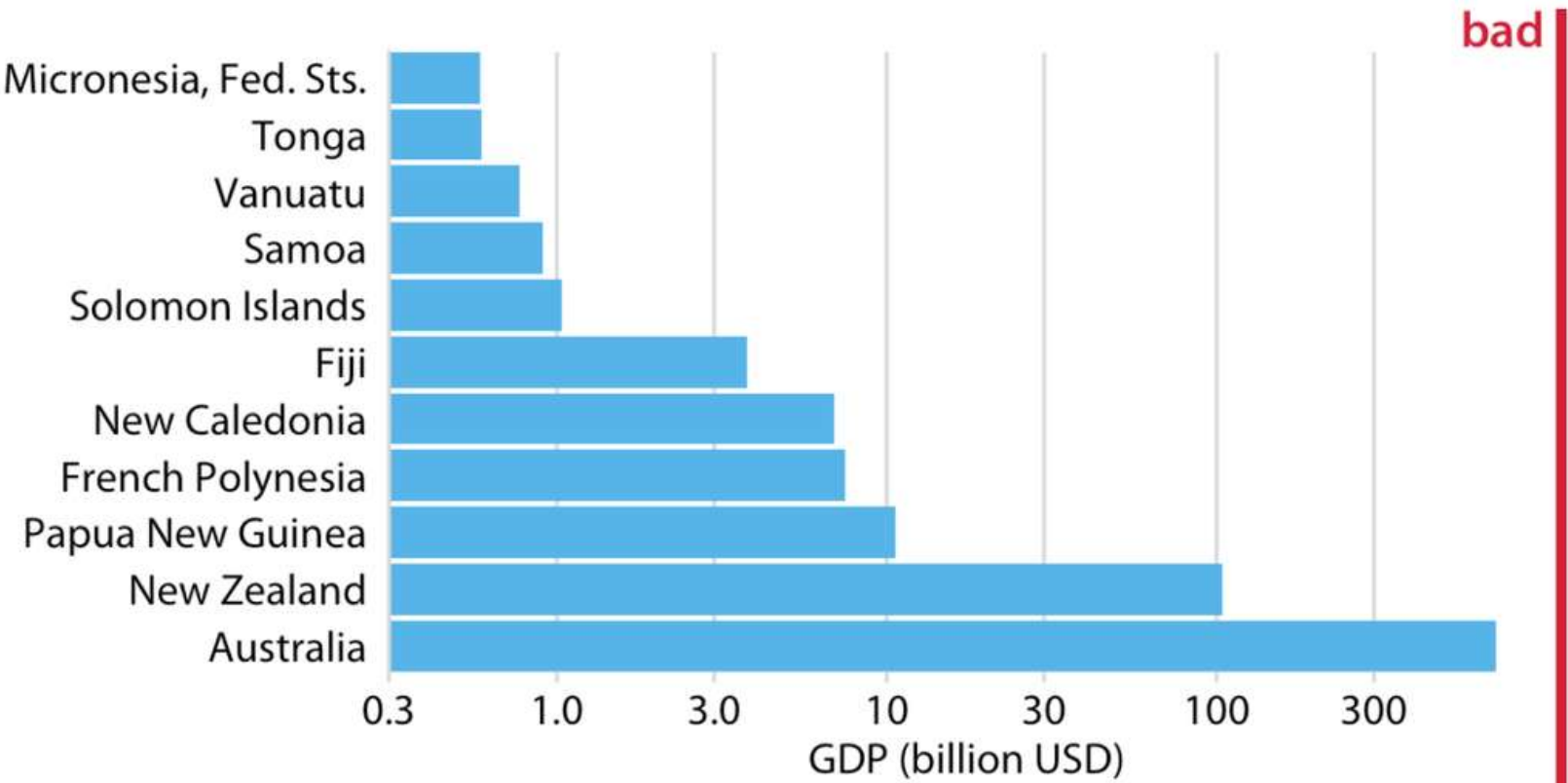


**GOO
D**

Bars should start at zero. Truncating the baseline exaggerates differences because the bar length no longer represents the data value.

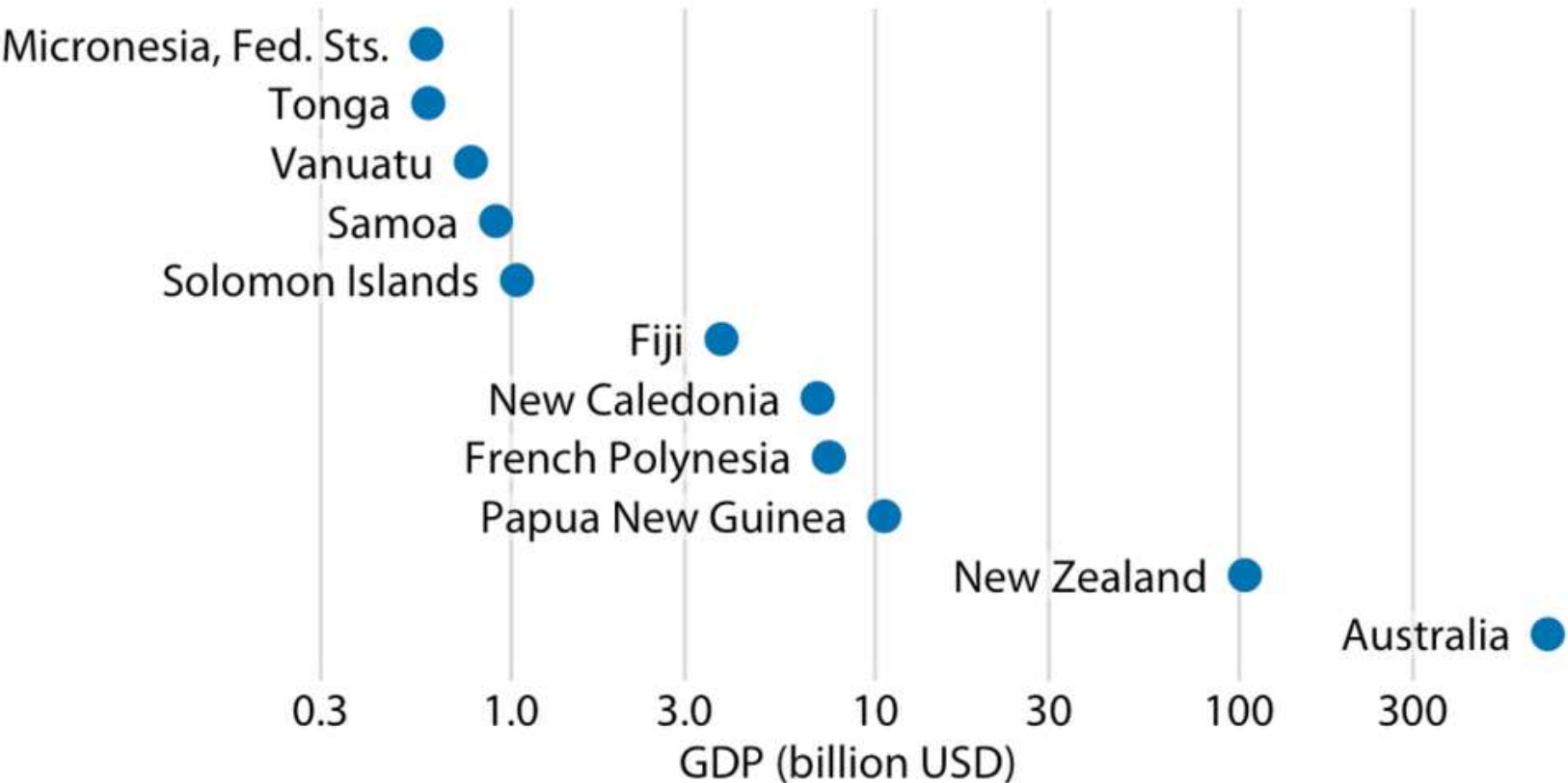
Visualization of Data, Understanding and Interpretation

Common Violations of Proportional Ink (Ch17)



**BA
D**

On logarithmic axes, bar lengths are no longer proportional to the values. Point plots avoid this problem because only position, not length is used to encode the data.



**GOO
D**



1. Which type of visual acuity refers to the eye's ability to detect tiny misalignments between two adjacent line segments?

- A. Point acuity
- B. Grating acuity
- C. Stereo acuity
- D. Vernier acuity

1. Which type of visual acuity refers to the eye's ability to detect tiny misalignments between two adjacent line segments?

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Answer: D



2. Which type of lighting effect simulates light scattered evenly from a matte surface and is commonly used in scientific and engineering visualizations to clearly reveal overall shape and curvature?

- A. Specular highlights
- B. Lambertian shading
- C. Cast shadows
- D. Ambient shading



2. Which type of lighting effect simulates light scattered evenly from a matte surface and is commonly used in scientific and engineering visualizations to clearly reveal overall shape and curvature?

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- B. Lambertian shading
- C. Cast shadows
- D. Ambient shading

Answer: B

3. To ensure readability and display fine details such as text and texture, research recommends maintaining what minimum luminance contrast ratio between the elements and the background?

- A. 2:1
- B. 3:1
- C. 4.5:1
- D. 7:1



3. To ensure readability and display fine details such as text and texture, research recommends maintaining what minimum luminance contrast ratio between the elements and the background?

- A. 2:1
- B. 3:1
- C. 4.5:1
- D. 7:1

Answer: B

4. According to Edward Tufte's principle of the Data-Ink Ratio, what should visual designers minimize to improve readability and reduce cognitive load?

- A. The use of point plots on logarithmic axes
- B. High-contrast text sizes
- C. Non-essential ink, such as excessive gridlines, heavy borders, and decorative backgrounds
- D. The overall size of a treemap's layout areas

4. According to Edward Tufte's principle of the Data-Ink Ratio, what should visual designers minimize to improve readability and reduce cognitive load?

- A. The use of point plots on logarithmic axes
- B. High-contrast text sizes
- C. Non-essential ink, such as excessive gridlines, heavy borders, and decorative backgrounds
- D. The overall size of a treemap's layout areas

Answer: C

Visualization of Data, Understanding and Interpretation

MCQs

5. Which design choice best improves the readability of fine details in a visualization?

- A. Increasing saturation only
- B. Using decorative backgrounds
- C. Maintaining sufficient luminance contrast between foreground and background
- D. Replacing all text with icons

Visualization of Data, Understanding and Interpretation

MCQs

5. Which design choice best improves the readability of fine details in a visualization?

- A. Increasing saturation only
- B. Using decorative backgrounds
- C. Maintaining sufficient luminance contrast between foreground and background
- D. Replacing all text with icons

Answer: C



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Thank You



Visualization of Data, Understanding and Interpretation

Unit 1 : Foundations of Visualization, Data Interpretation

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

Dr.Prajwala TR (Course Instructor)

Dept. of CSE, PES University, Bangalore

Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.



Visualization of Data, Understanding and Interpretation

Lecture Overview

- The fundamental psychological concepts of Gestalt theory and how our visual systems organize isolated elements into structured wholes.
- Spatial grouping mechanisms including proximity, similarity, connectedness, and enclosing boundaries.
- How visual continuity, symmetrical balance, closure of incomplete shapes, and figure-ground separation guide our eyes.
- Dynamic grouping via common fate and a complete framework for applying these principles to data visualization design.

Visualization of Data, Understanding and Interpretation

Gestalt Principles (TB 3, Ch6)

Gestalt psychology explains how people perceive organized wholes rather than isolated parts.

Instead of processing every object separately, the brain automatically groups objects using principles such as:

- Proximity
- Similarity
- Connectedness
- Continuity
- Symmetry
- Closure
- Common Region
- Figure & Ground
- Common Fate

These principles help viewers:

- identify relationships quickly
- reduce visual clutter
- understand visualizations with minimal effort



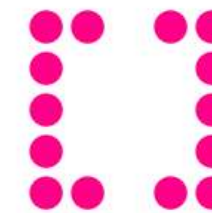
Prägnanz



Similarity



Proximity



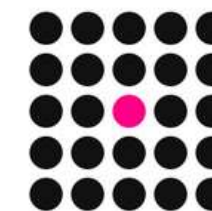
Closure & Symmetry



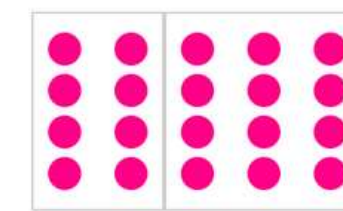
Continuity



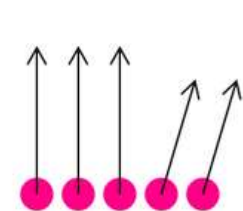
Figure/Ground



Focal point



Common Region



Common Fate

Principle: Objects positioned close together are perceived as belonging to the same group.

Design Implications:

- Use spacing to organize information.
- Separate unrelated elements with white space.
- Avoid unnecessary boxes if spacing is sufficient.

Examples:

- Items on a restaurant menu are grouped simply by spacing.
- Grouped bars appear related because they are placed close together.

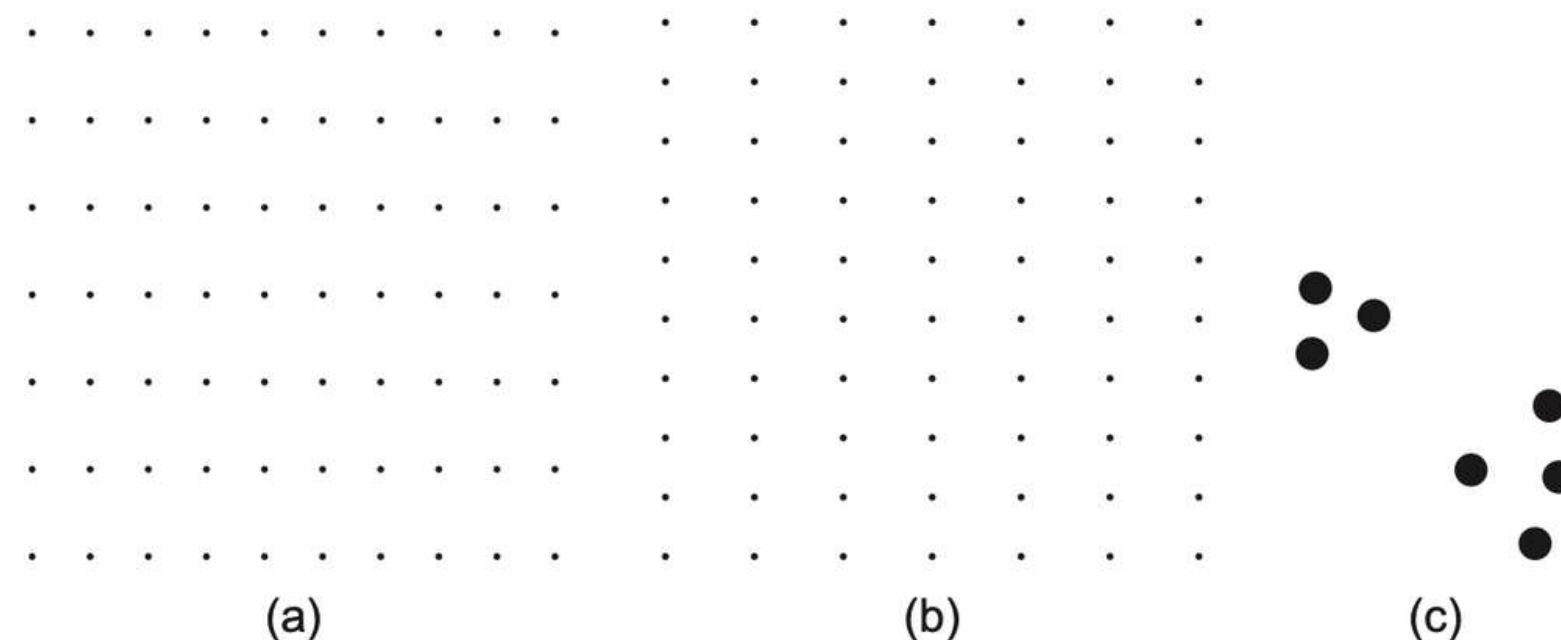


Figure 6.2 Spatial proximity is a powerful cue for perceptual organization. A matrix of dots is perceived as rows on the left (a) and columns on the right (b). In (c) we perceive two groups of dots because of proximity relationships.

- The Gestalt principle of *Proximity* states that objects positioned close to one another are perceived as a group, while objects spaced further apart are seen as separate.
- Spatial distance alone — with no borders, colors, or labels — is enough to communicate which elements belong together, making proximity one of the most efficient grouping tools available to designers.

Takeaways:

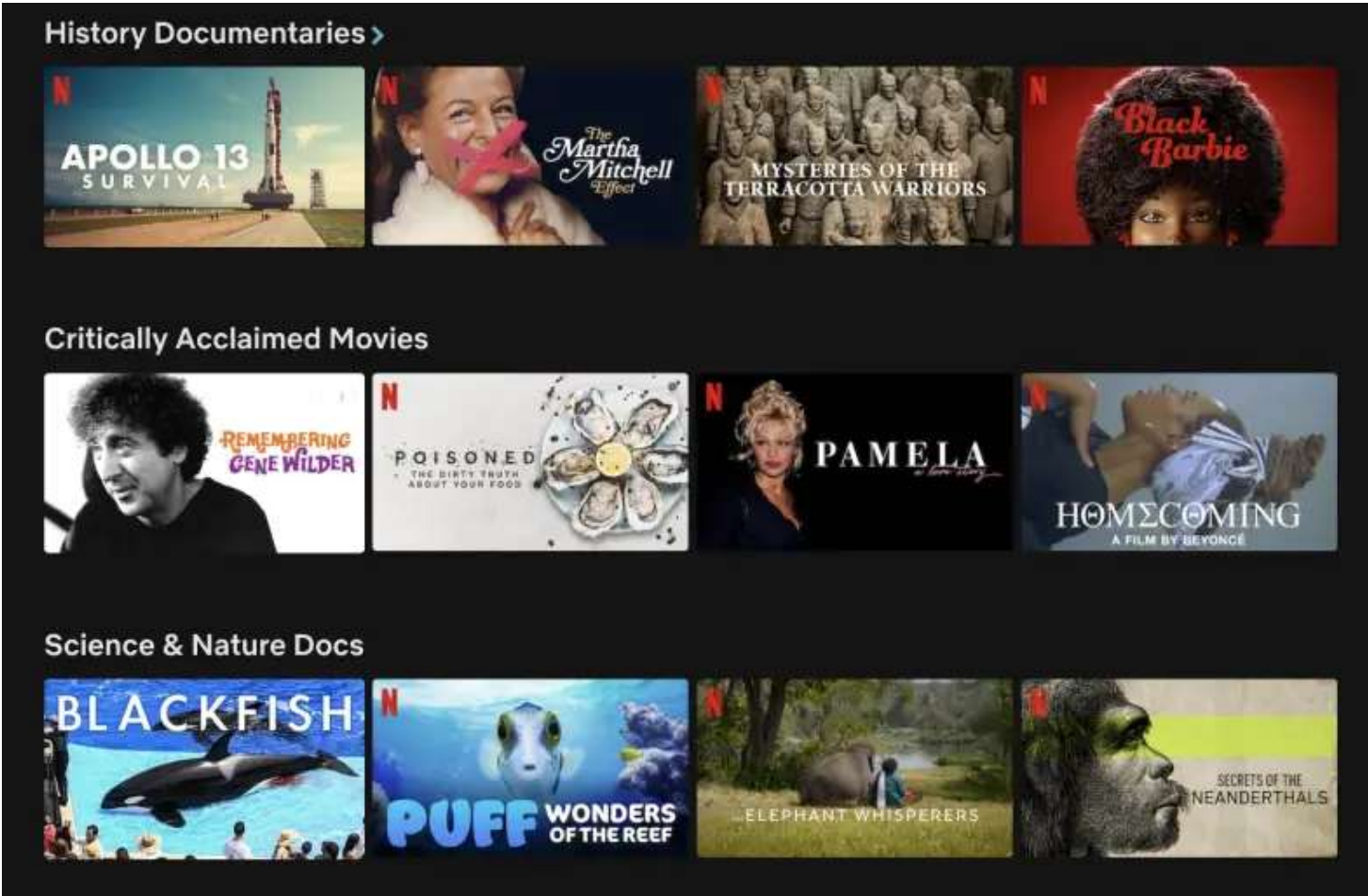
- Space Signals Connection: Distance between elements implies the strength of their connection
- Distance Hierarchy: Closer items have stronger perceived relationships, increase whitespace to signal a different relationship
- Modularity: Break content into digestible chunks using proximity to structure information.

Source: <https://www.gestaltprinciples.com/>

Visualization of Data, Understanding and Interpretation

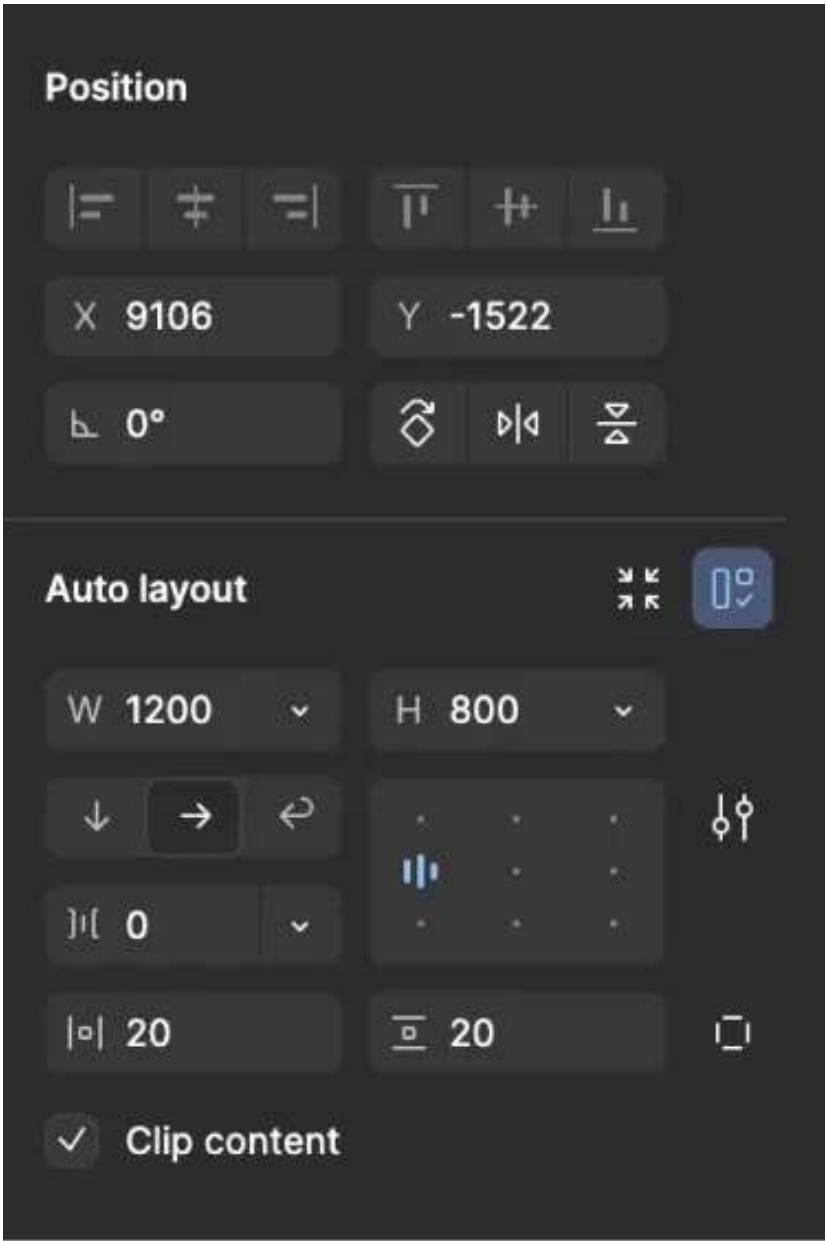
Proximity

(Ch6)



UI Design: Netflix Previews

Related content is closely grouped with the descriptive title it belongs to



UI Design: Figma
Properties Panel
Related properties are positioned close together for easy mental mapping

Visualization of Data, Understanding and Interpretation

Similarity

(Ch6)

Principle: Objects with similar color, shape, size, texture are perceived as belonging together.

Examples:

- Subway maps use the same color to represent one train line.
- Sports teams wear matching jerseys, making teammates instantly recognizable.
- Same-colored bars represent one category.

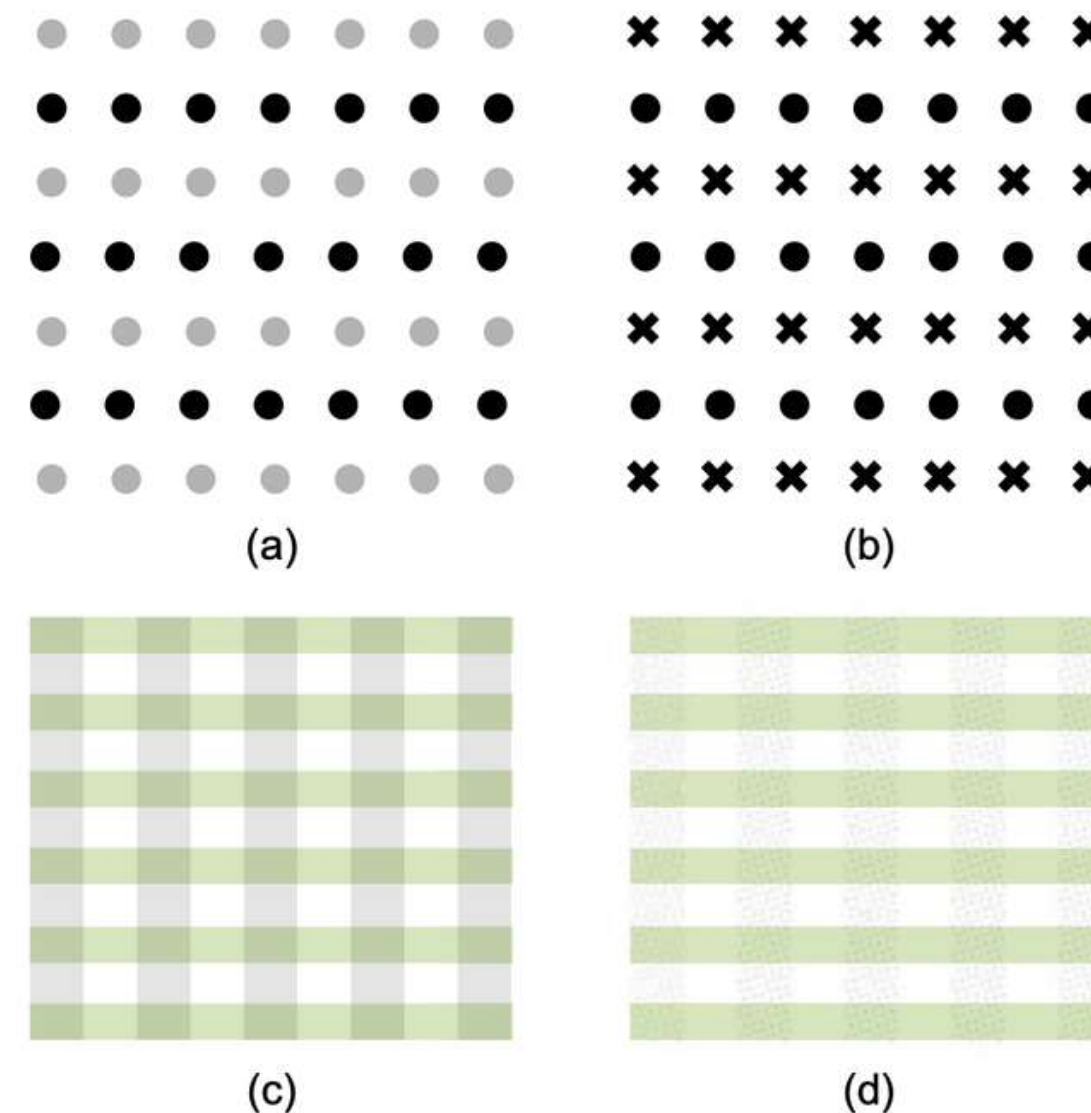


Figure 6.4 (a, b) According to the Gestalt psychologists, similarity between the elements in alternate rows causes the row percept to dominate. (c) Integral dimensions are used to delineate rows and columns. (d) When separable dimensions (color and texture) are used, it is easier to attend separately to either the rows or the columns.

Visualization of Data, Understanding and Interpretation

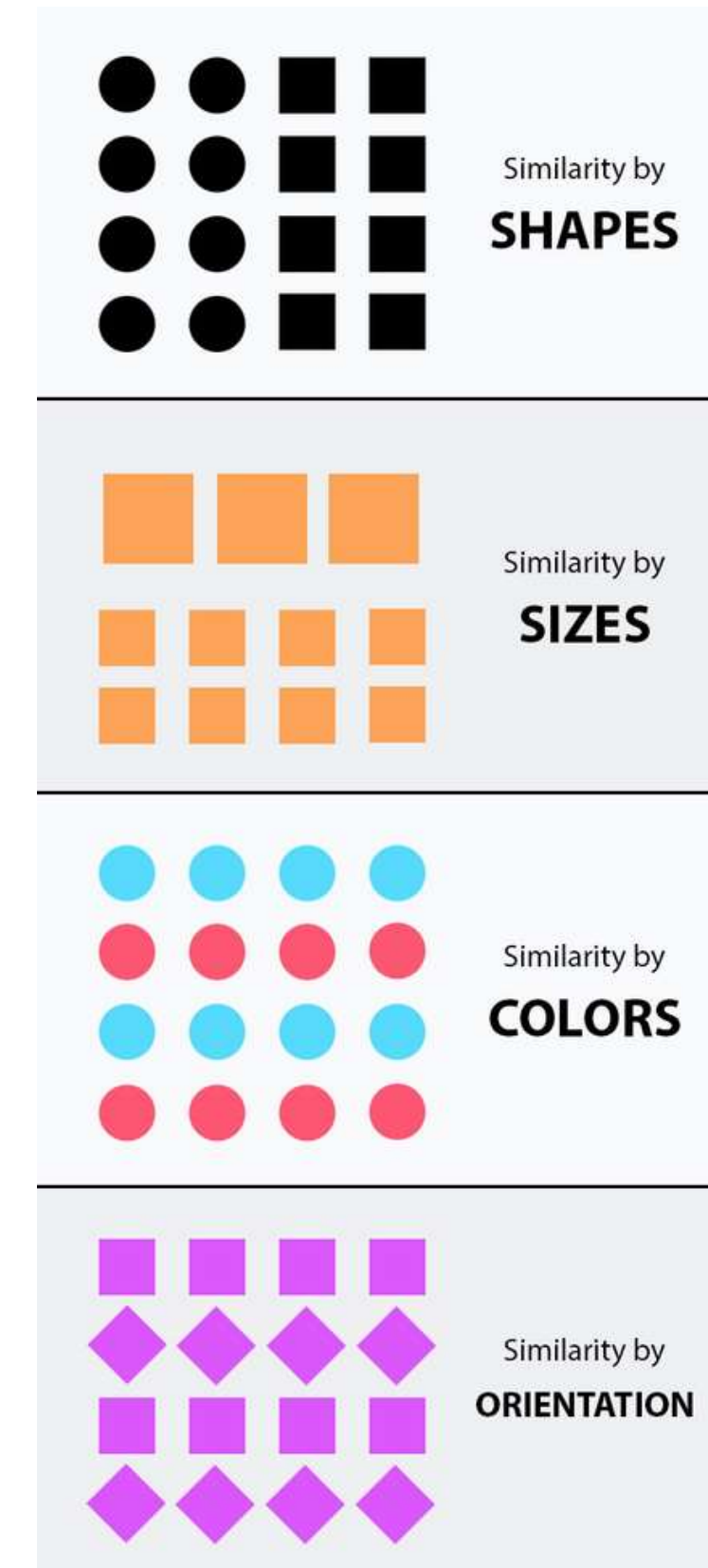
Similarity

(Ch6)

- The Gestalt principle of *Similarity* states that when visual elements share characteristics such as color, shape, size, orientation, or texture, our brains automatically perceive them as part of the same group.
- This principle explains why a row of identical icons reads as a single set, while one element styled differently instantly stands apart.

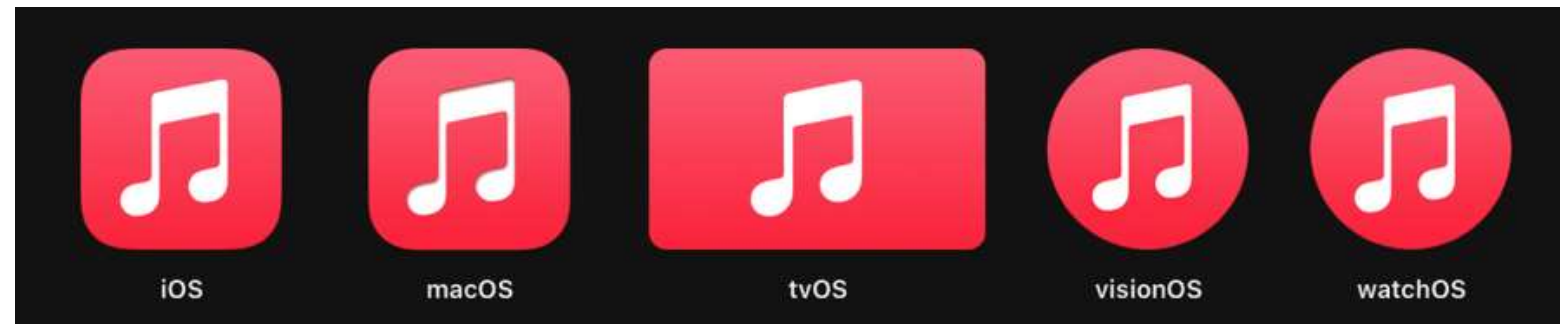
Takeaways:

- Pattern Recognition: Our brains automatically group elements that share visual properties
- Visual Properties: Color, shape, size, orientation, and texture can all establish similarity
- Relational Strength: Multiple shared properties create stronger grouping
- Avoid Accidental Similarity: Ensure unrelated elements are visually distinct to prevent confusion



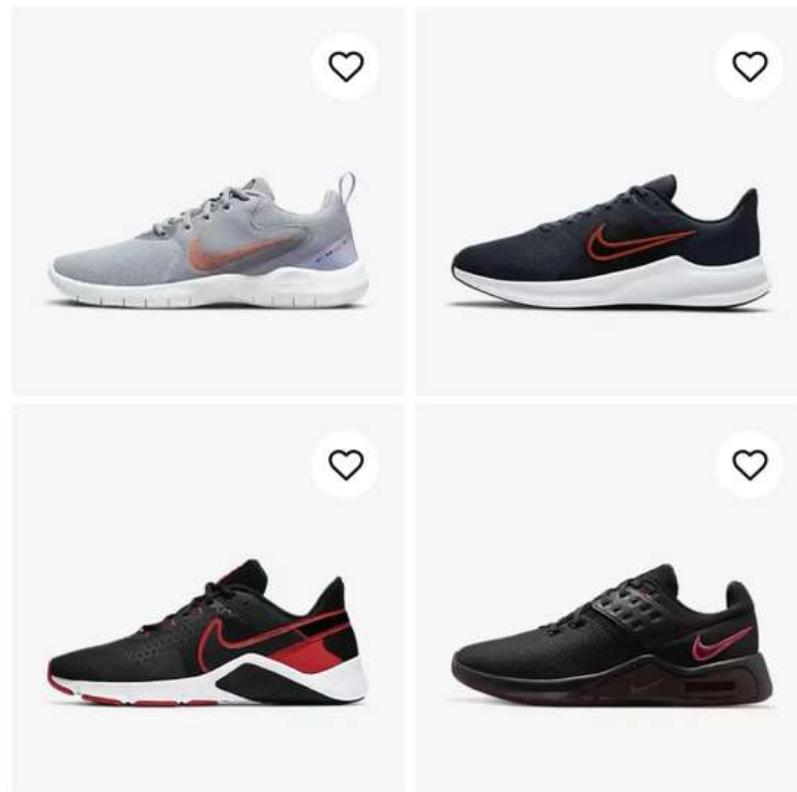
Visualization of Data, Understanding and Interpretation

Similarity (Ch6)



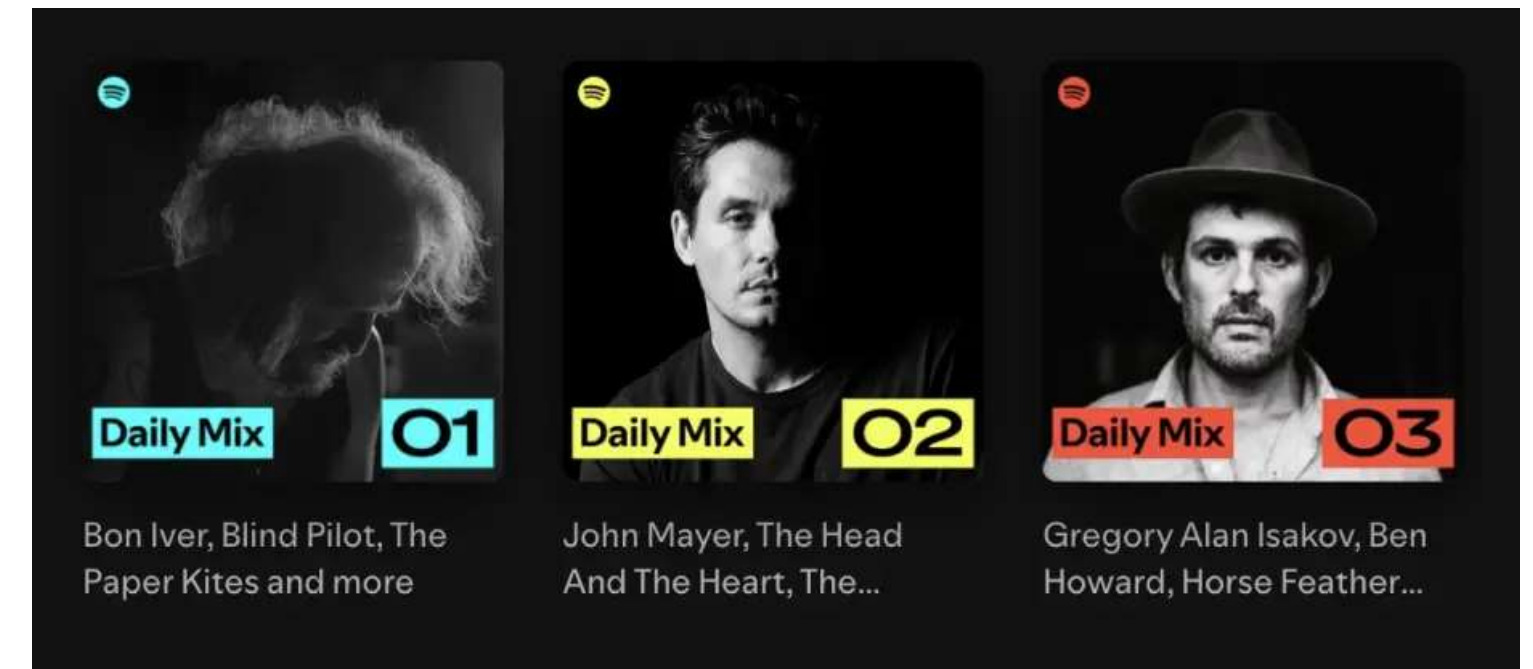
Iconography: Apple Music Logo

Icons share similar properties depending on the OS they belong to



Branding: Nike

Standardized product photography angles and lighting across their catalog



UI Design: Spotify's Playlist Cards

Consistent visual treatment across playlist cards creates a unified browsing experience

Visualization of Data, Understanding and Interpretation

Connectedness/Prägnanz (Ch6)

Principle: Objects joined by a line or edge are perceived as related, even more strongly than proximity or color.

Examples:

- Organization charts connect employees to managers.
- Network diagrams, family trees, flowcharts.

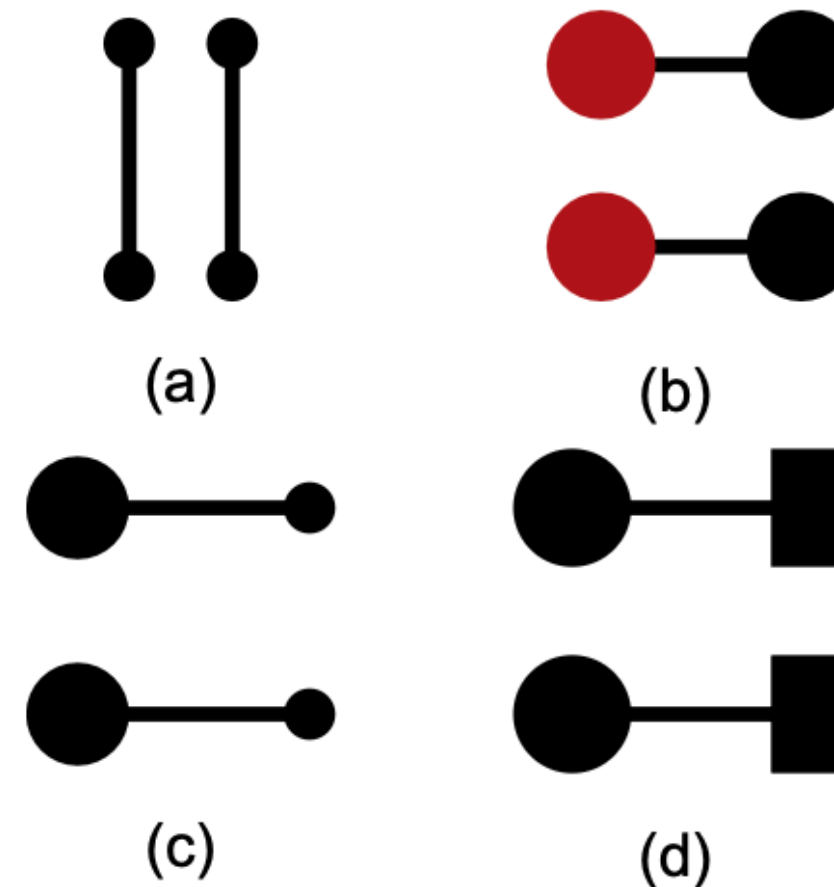


Figure 6.5 Connectedness is a powerful grouping principle that is stronger than (a) proximity, (b) color, (c) size, or (d) shape.

Visualization of Data, Understanding and Interpretation

Connectedness/Prägnanz

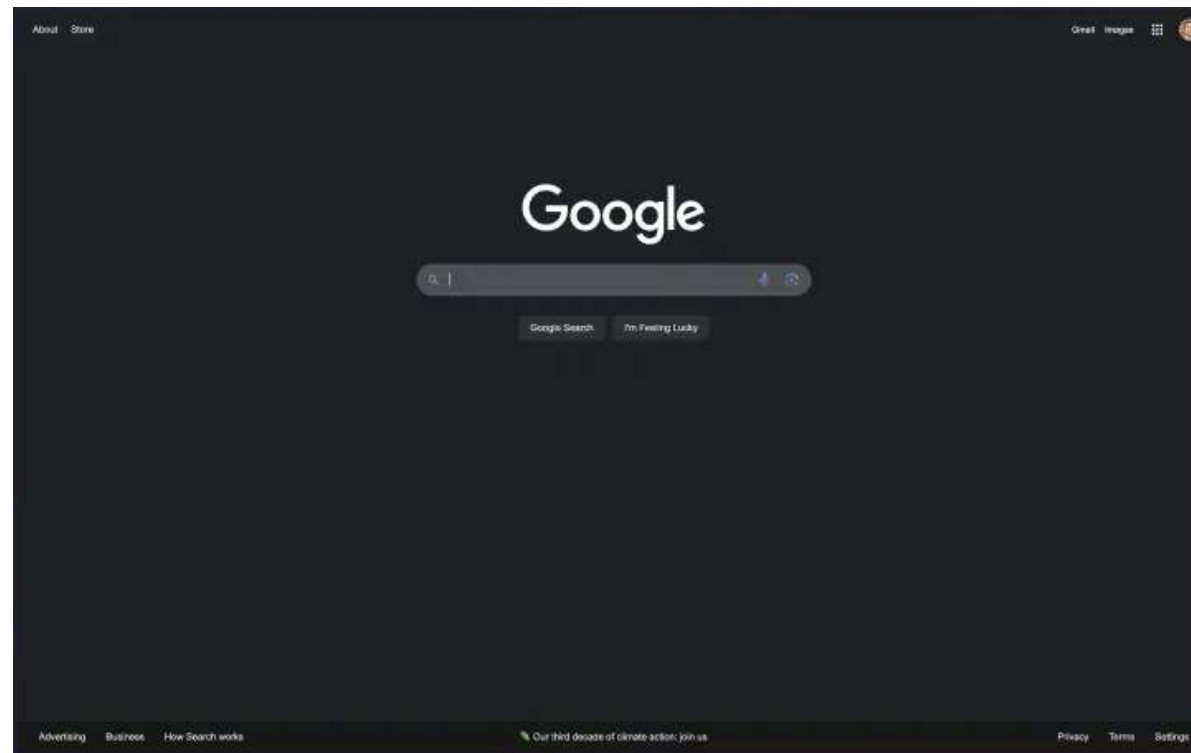
(Ch6)

- *Prägnanz* — German for 'conciseness' or 'pithiness' — is often considered the overarching law of Gestalt psychology.
- It states that when faced with complex or ambiguous visual information, the brain will interpret it in the simplest, most stable, and most regular form possible.
- Every other Gestalt principle can be seen as a specific way the mind pursues this drive toward simplicity.

Takeways:

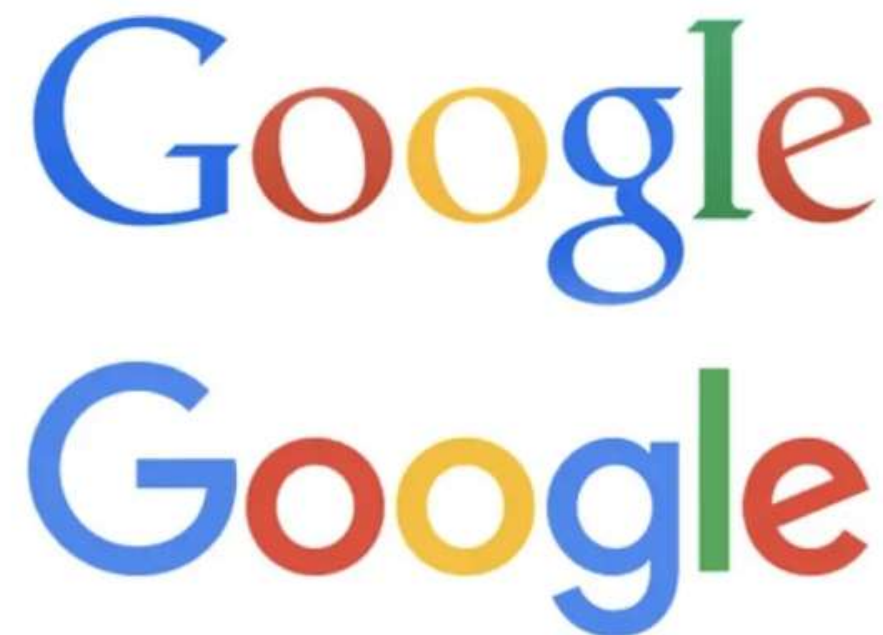
- Visual Simplicity: Reduction of visual complexity helps users process information faster
- Clear Structure: Clear visual hierarchies make complex information digestible
- Pattern Recognition: Simple, recognizable patterns reduce cognitive load
- Prioritize Key Actions: Highlight primary CTAs or content using size, color, or position

Visualization of Data, Understanding and Interpretation Connectedness/Prägnanz (Ch6)



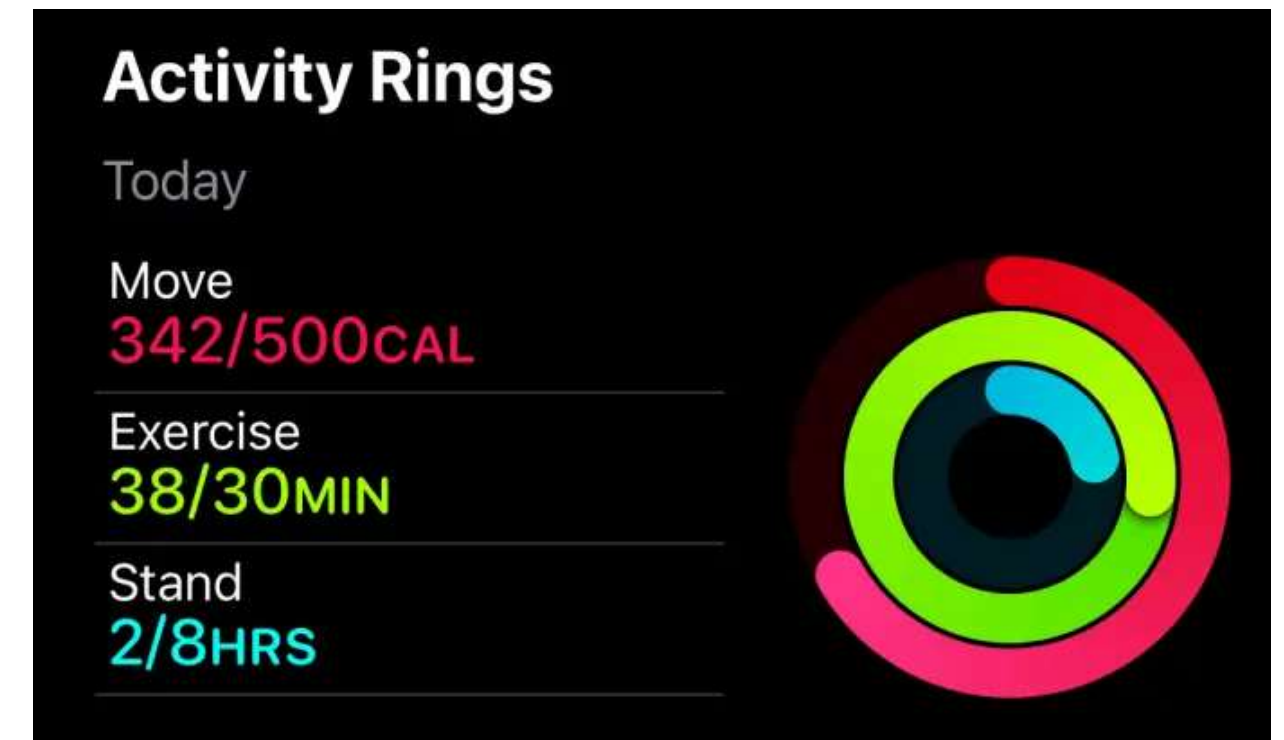
UI Design: Google Search Page

A complex interaction happens mostly behind the scenes as the interface is nearly as simple as it can be



Branding: Google Logo Redesign

Google's 2015 logo redesign simplified the letters to be more easily digestible



UI Design: Apple Activity Rings

The progress bars for your daily activities are simplified into closable rings, easily readable and transferrable across different devices

Visualization of Data, Understanding and Interpretation

Continuity (Ch6)

Principle: The eye naturally follows smooth, continuous paths rather than abrupt changes.

Design Implications:

- Smooth curves improve readability.
- Minimize unnecessary line crossings.
- Preserve visual flow in network diagrams.

Examples:

- Road markings guide drivers smoothly through intersections.
- Well-routed Sankey diagrams or node-link graphs.

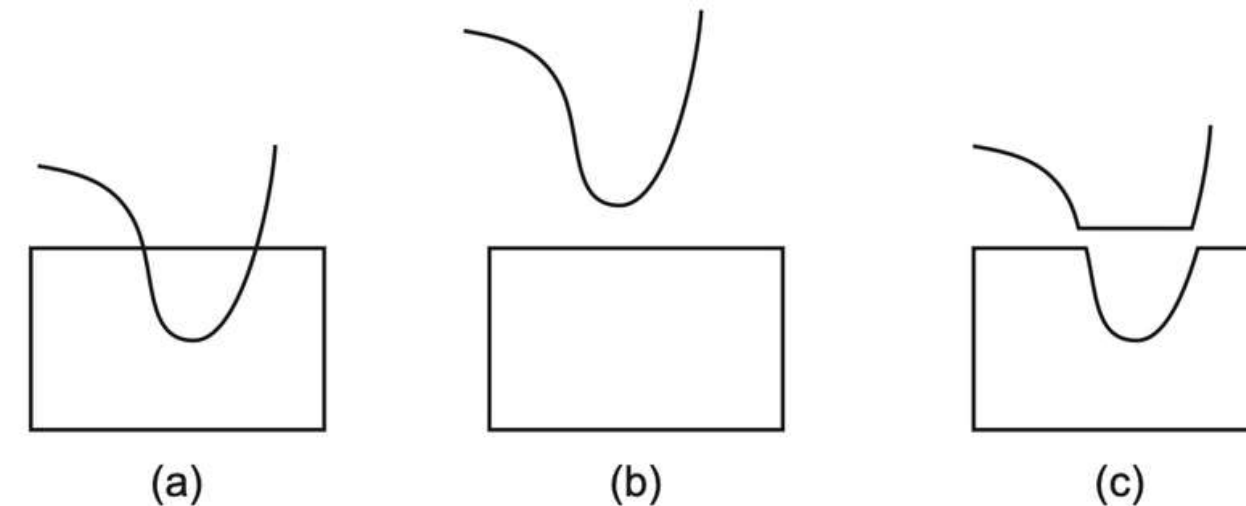


Figure 6.6 The pattern on the left (a) is perceived as a smoothly curved line overlapping a rectangle (b) rather than as the more angular components shown in (c).

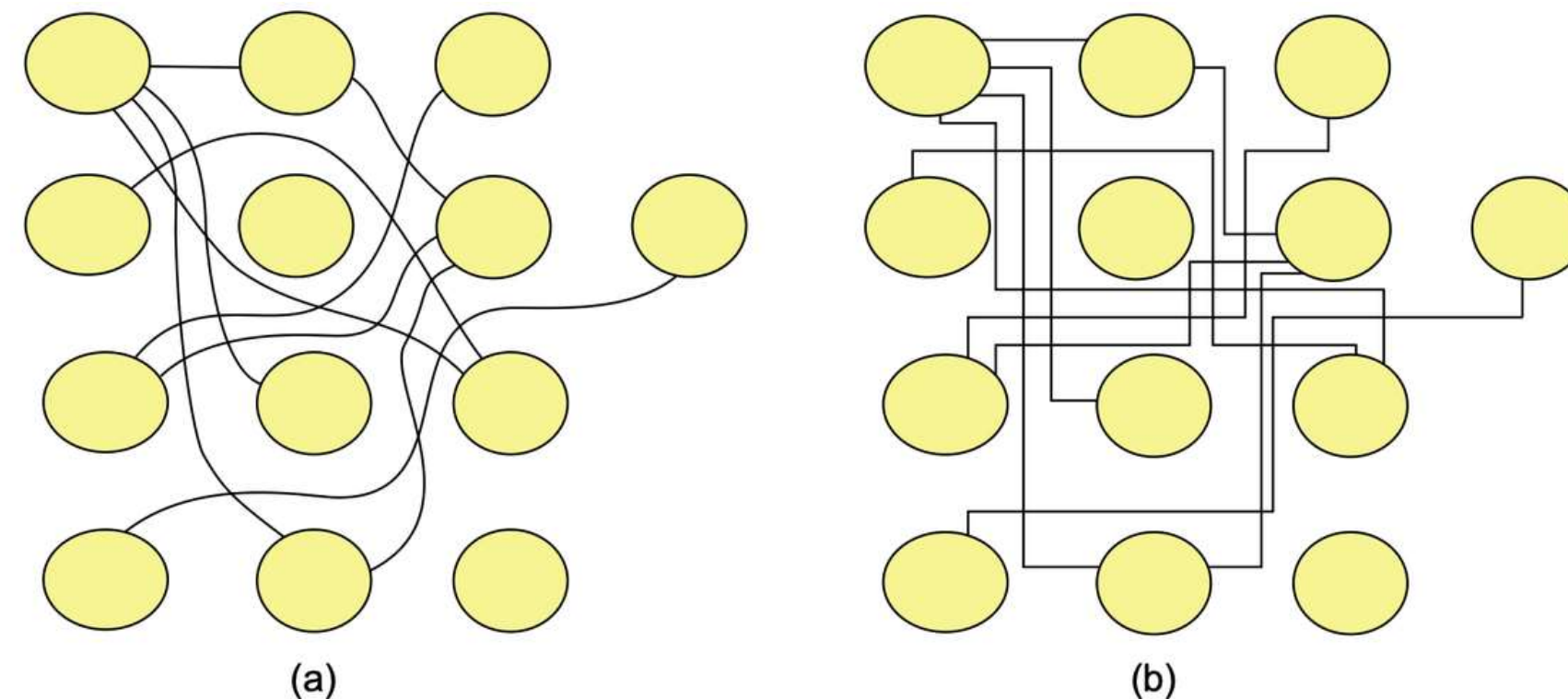


Figure 6.7 In (a), smooth continuous contours are used to connect nodes in the diagram; in (b), lines with abrupt changes in direction are used. It is much easier to perceive connections with the smooth contours.

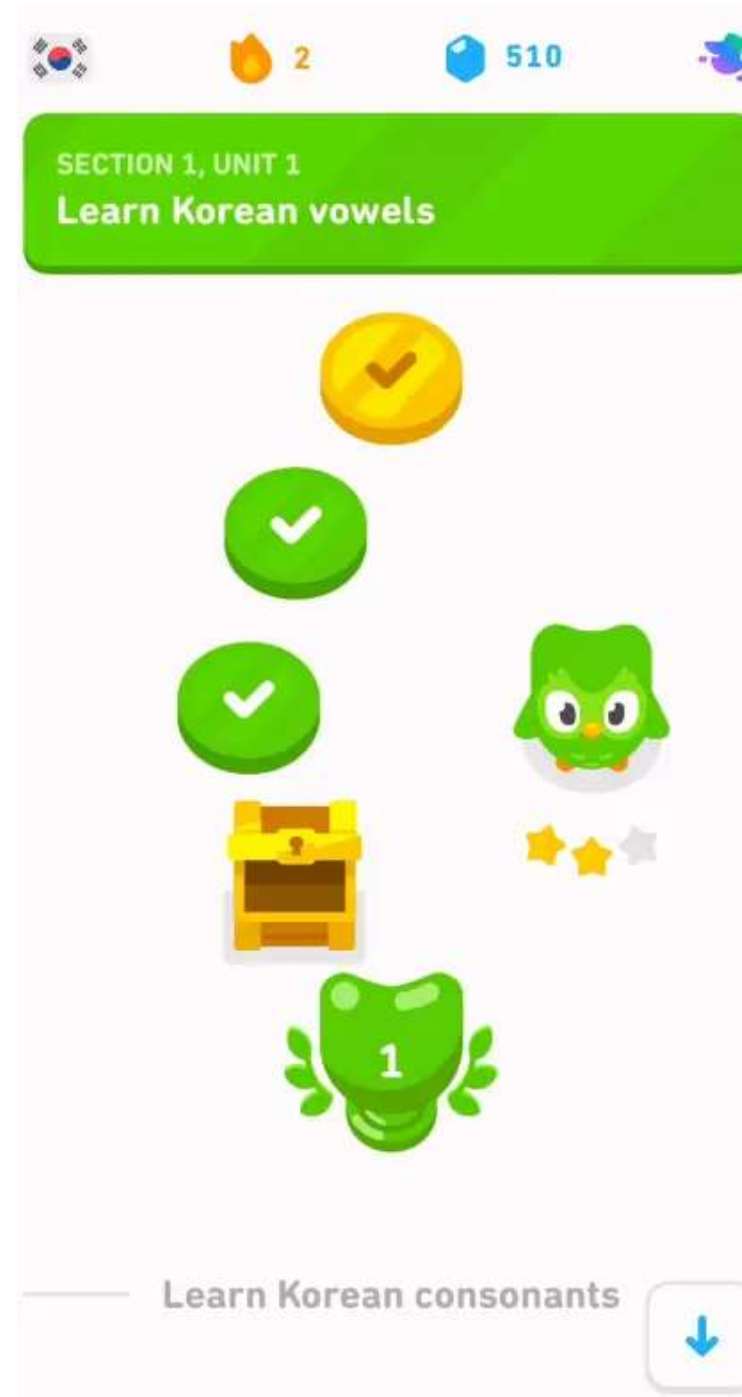
- The Gestalt principle of *Continuity* (also called continuation) describes our tendency to perceive elements arranged on a line or smooth curve as more related than elements that are not.
- Our eyes naturally follow the smoothest path, so when objects are aligned or flow in a consistent direction, the brain reads them as a connected sequence rather than separate parts.

Takeaways:

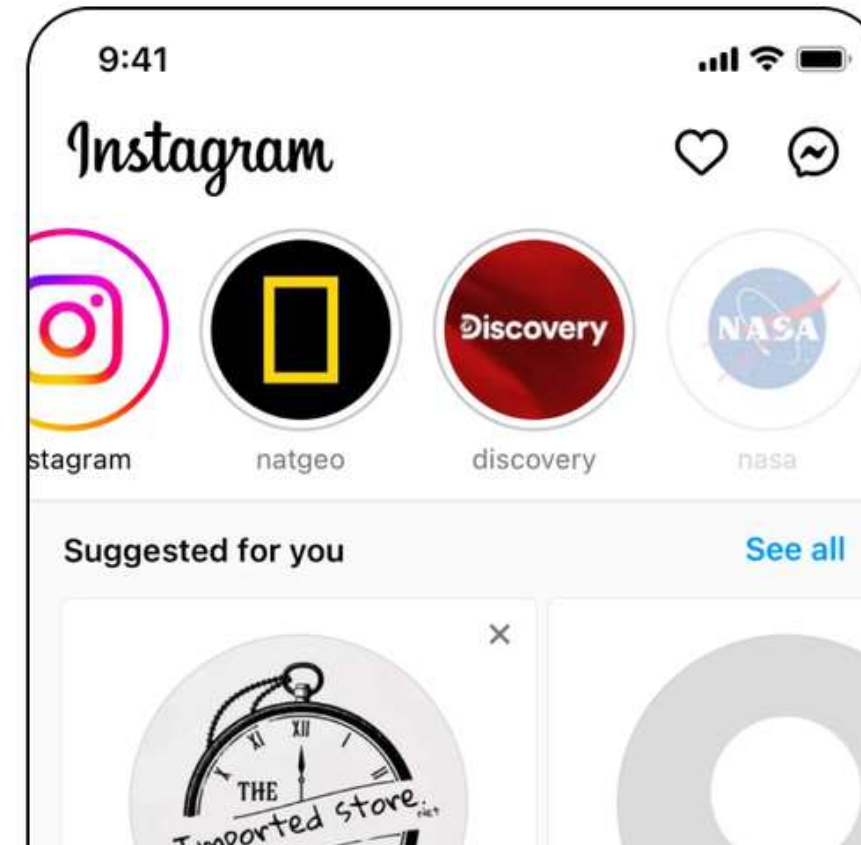
- Visual Flow: Eyes naturally follow the smoothest path between elements, creating visual connection
- Design for Scanning: Arrange content in predictable patterns (like F-shape or Z-shape layouts) for intuitive navigation
- Line Completion: The mind connects elements that form continuous lines or curves
- Directional Force: Elements aligned on a path appear related and sequential
- Smooth Transitions: Gradual transitions between states or elements maintain context and flow

Visualization of Data, Understanding and Interpretation

Continuity (Ch6)



Game Design:
Duolingo Progress Path
Curving path to lead you
to the goal while showing
progress



UI Design: Instagram
Stories
Natural line of elements
leading you from the left to
the right



Branding: Amazon
Logo
Line from A-Z implying
the breadth of products

Visualization of Data, Understanding and Interpretation

Symmetry

(Ch6)

Principle: Symmetrical objects are perceived as complete and belonging together.

Why it matters:

- improves comparison
- supports pattern recognition
- reduces visual effort

Examples:

- Company logos (Apple, Mercedes, Target) often use symmetry.
- Comparing two mirrored distributions or time series.

💡 *Use symmetry only when comparisons benefit from mirrored layouts*

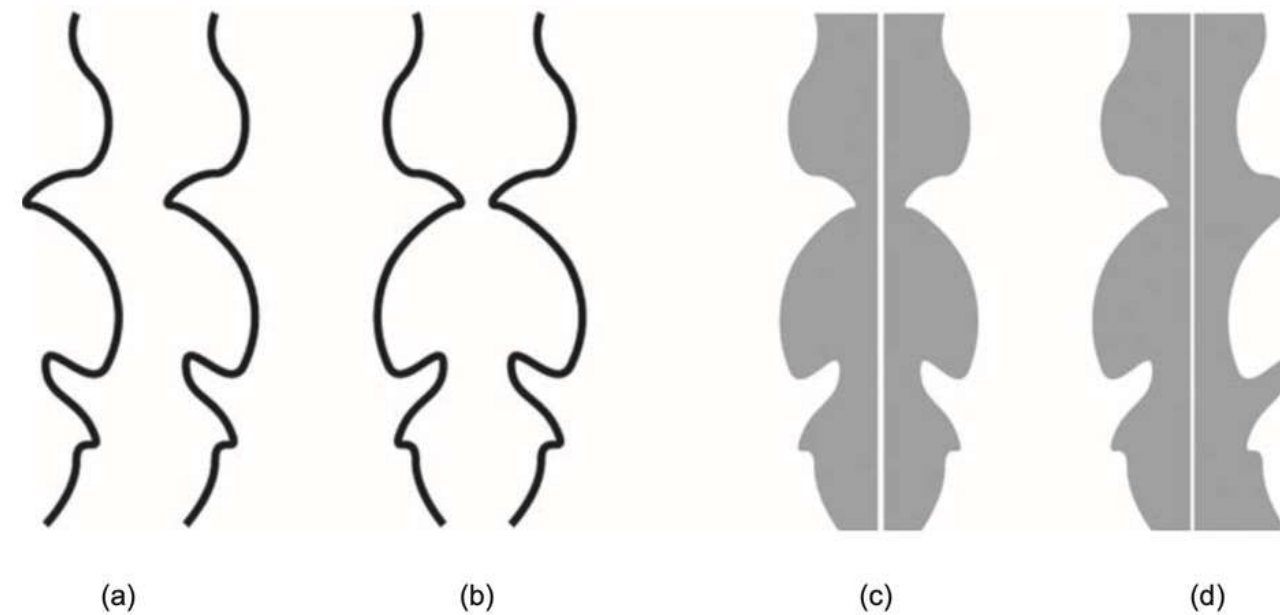


Figure 6.9 (a) The pattern on the left consists of two identical parallel contours. (b) Mirror symmetry provides more of a sense of a whole figure. In (d) the edge parallel symmetry and reversed edge polarity make it more difficult to see the symmetry respect to (c).

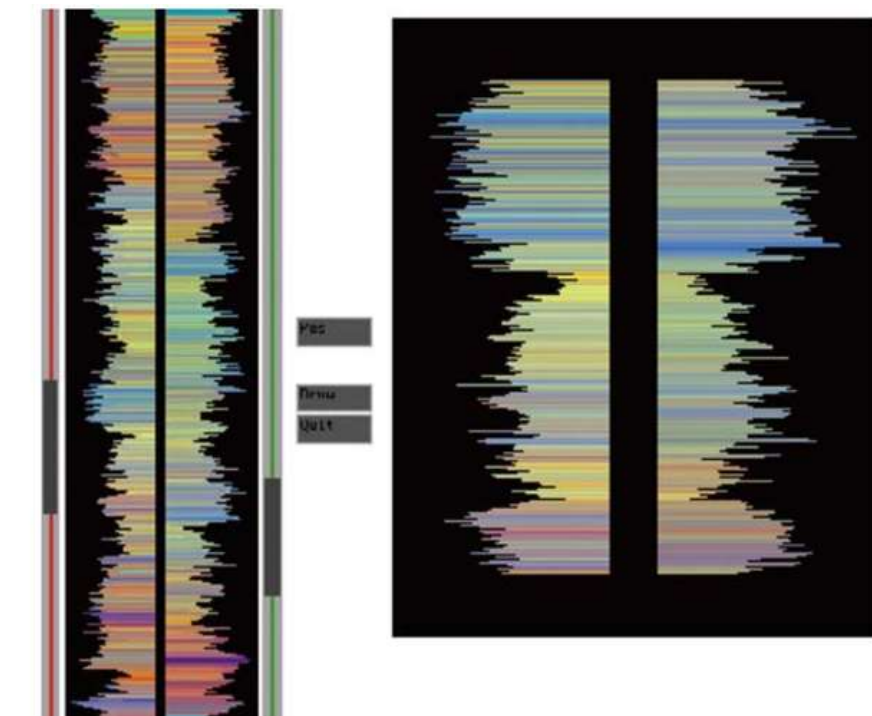


Figure 6.10 An application designed to allow users to recognize similar patterns in different time-series plots. The data represents a sequence of measurements made on deep ocean drilling cores. Two subsets of the extended sequences are shown on the right.

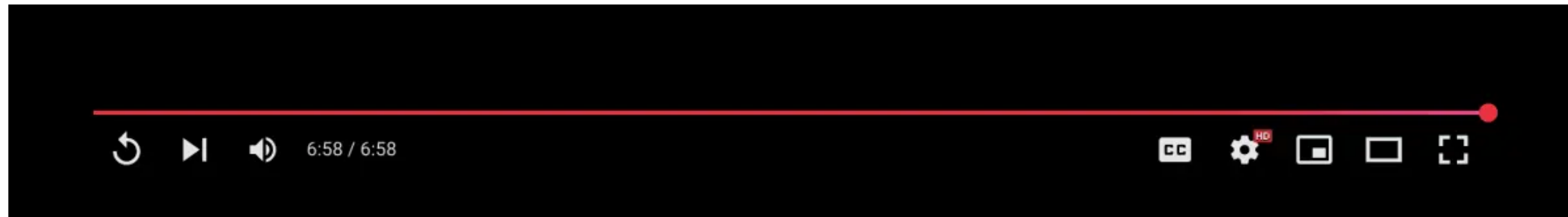
- The Gestalt principle of *Symmetry and Order* — closely tied to the law of Prägnanz — states that people perceive symmetrical, balanced arrangements as more unified, stable, and pleasing than asymmetrical ones.
- The brain treats symmetrical elements as belonging together and reads balanced compositions as orderly and trustworthy, often with less mental effort.

Takeaways:

- Consistent Alignment: Use grids and alignment tools to ensure symmetry and clean designs
- Circular Harmony: Radial balance creates focus and draws attention to central elements
- Visual Weight: Distribute elements evenly to avoid visual “heaviness” on one side

Visualization of Data, Understanding and Interpretation

Symmetry (Ch6)



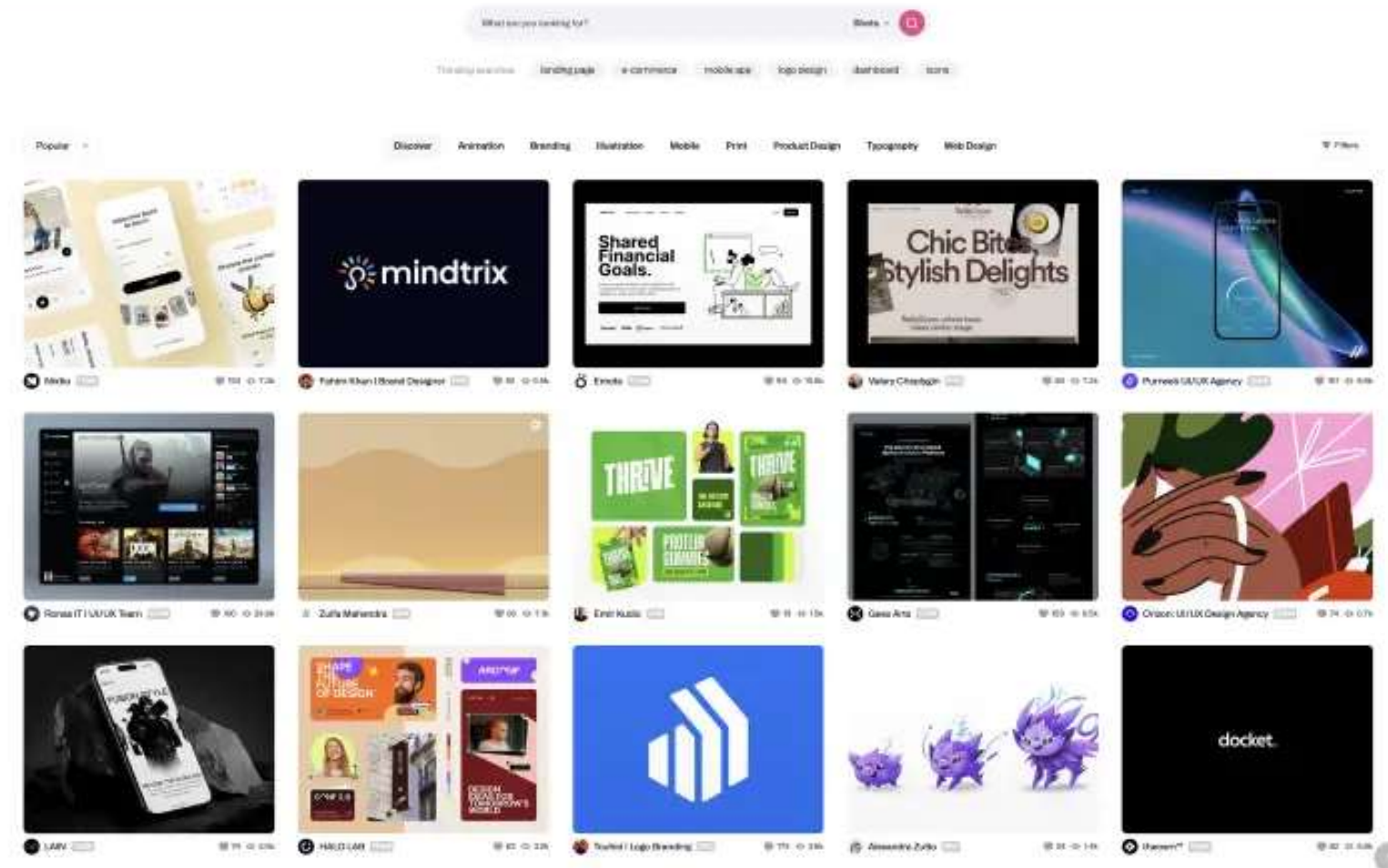
Design Systems: YouTube Player Controls

The separate groupings within the controls container are balanced and orderly



Branding: Airbnb Logo

The Airbnb logo is perfectly balanced and symmetrical



Web Design: Dribbble Grid Layout

The grid layout is a symmetrical repeatable pattern making it easy to scan

Visualization of Data, Understanding and Interpretation

Closure (Ch6)

Principle: The brain fills missing information to perceive complete objects.

Design Implication:

Use partial outlines or overlapping objects to simplify visuals without sacrificing recognition, but avoid removing so much information that the shape becomes ambiguous.

Examples:

- A broken or dashed circle is still perceived as a complete circle.
- The WWF panda logo is recognized even though much of the outline is missing.

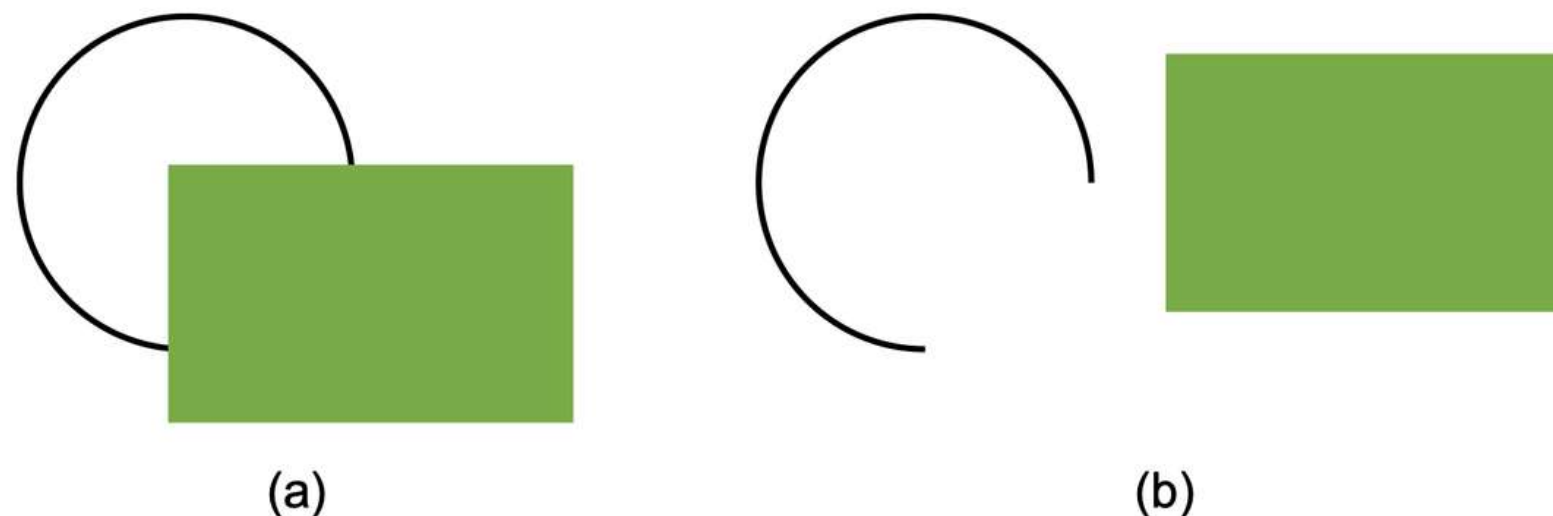


Figure 6.13 The Gestalt principle of closure holds that neural mechanisms operate to find perceptual solutions involving closed contours. In (a), we see a circle behind a rectangle, not a broken ring as in (b).

- The Gestalt principle of *Closure* describes the mind's tendency to complete incomplete shapes. When we see a form with gaps, our brain fills in the missing information to perceive a whole, recognizable object rather than disconnected fragments.
- This happens because the brain prefers complete, stable figures and works actively to resolve ambiguity.

Takeaways:

- Pattern Completion: Users naturally complete incomplete shapes based on familiar patterns
- Use Implied Shapes: Suggest shapes with partial lines or patterns to create intrigue
- Space Utilization: Negative space can create recognizable shapes and meaning
- Ensure Clarity: Make sure implied forms are easy to recognize, avoiding overly abstract gaps

Visualization of Data, Understanding and Interpretation

Closure

(Ch6)



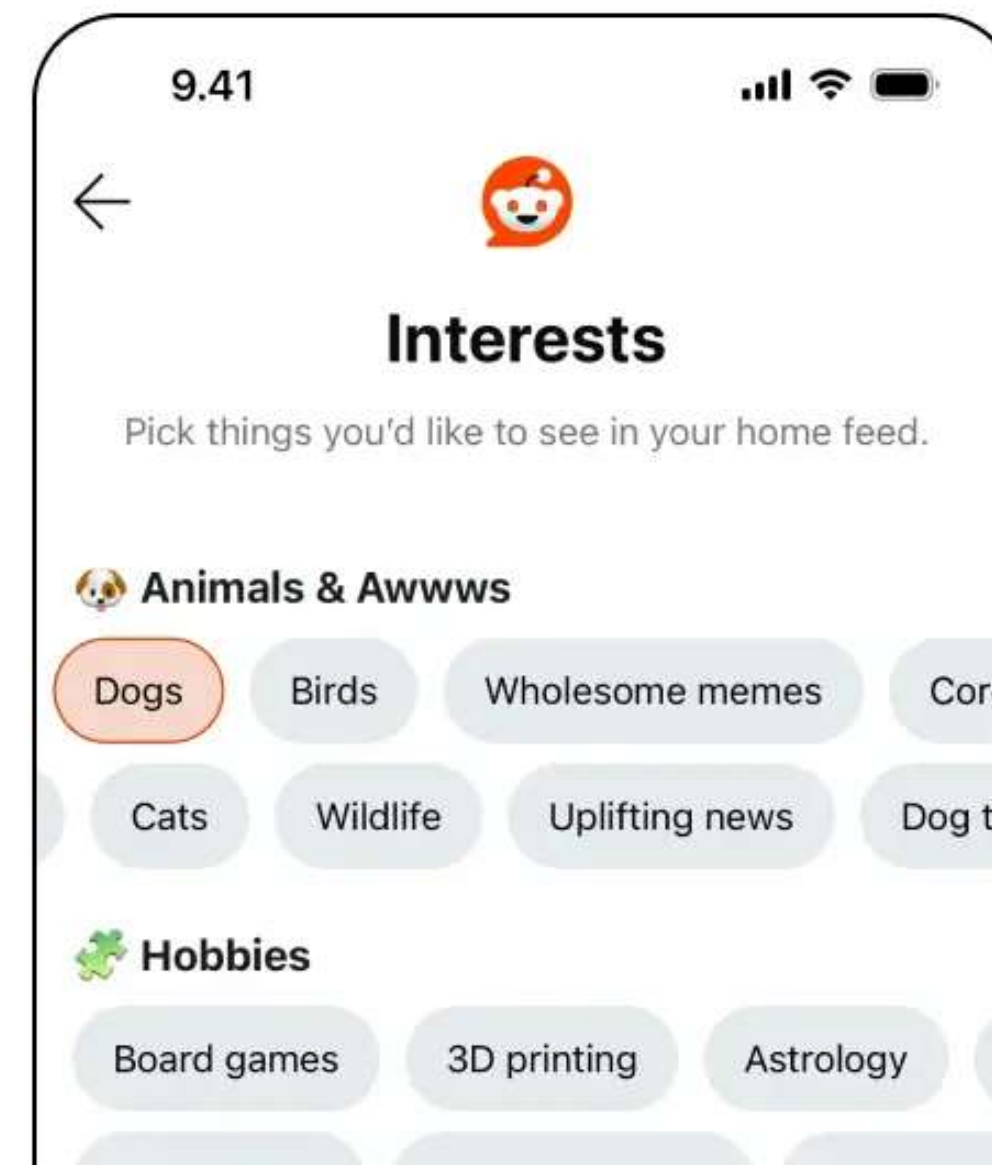
Iconography: Material UI Icons

Multiple icons are incomplete by design, but are still perceived as objects



Branding: IBM Logo

The IBM logo is not a complete shape, but we can still make out the letters



UI Design: Reddit Onboarding Pills

The recommended categories in the pill shape are partially off the screen, but we are still able to fill in the missing shape and content

Principle: Objects enclosed within the same boundary are perceived as belonging together.

Examples:

- dashboard cards
- panels
- grouped legends
- Venn/Euler diagrams

💡 *Design tip: a subtle border often groups information better than adding more labels.*

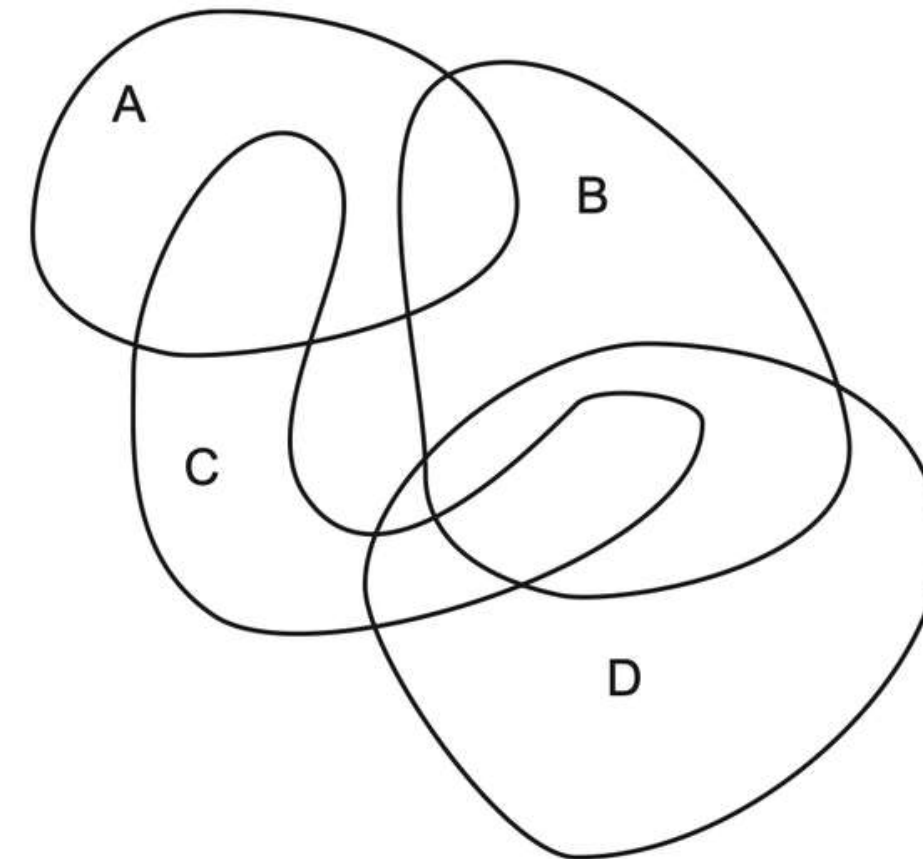


Figure 6.14 An Euler diagram. This diagram tells us (among other things) that entities can simultaneously be members of sets A and C but not of A, B, and C. Also, anything that is a member of both B and C is also a member of D. These rather difficult concepts are clearly expressed and understood by means of closed contours.

- The Gestalt principle of *Common Region* states that elements located within the same closed boundary or shared background are perceived as a group.
- A simple container — a card, a panel, a colored block, or even a thin dividing line — is enough to bind its contents together, and this grouping can be strong enough to override proximity and similarity.

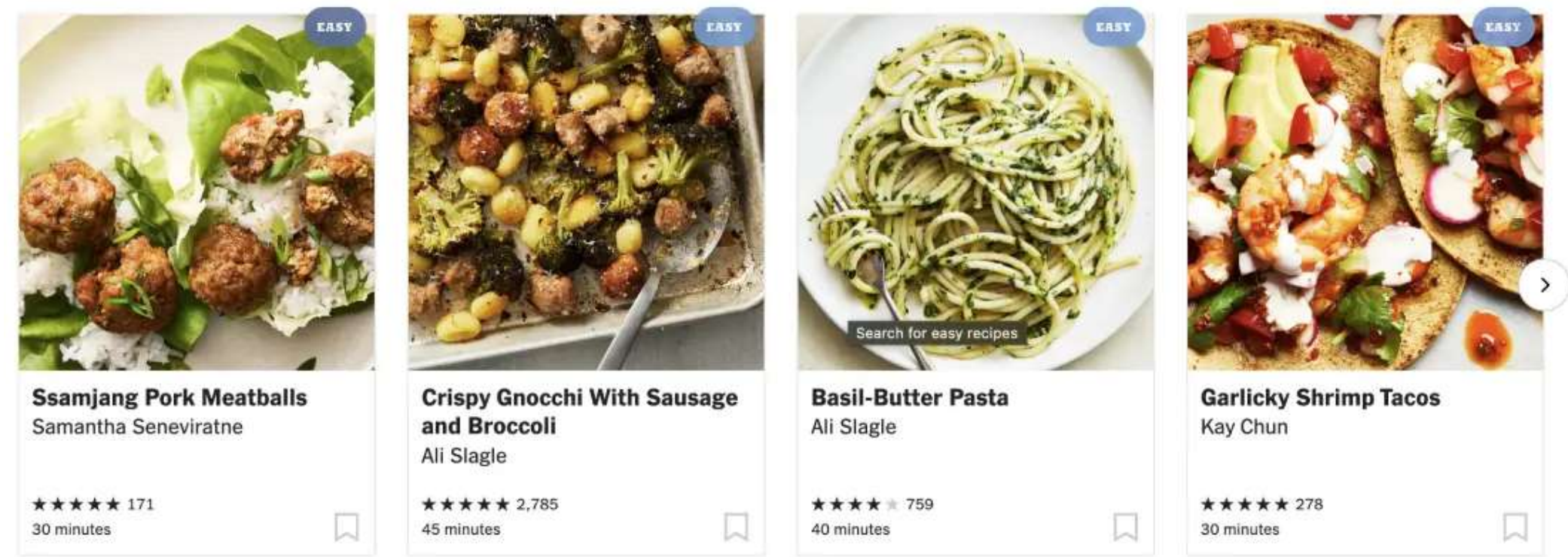
Takeaways:

- Boundary Effect: Elements within shared boundaries are automatically grouped
- Background Unity: Shared backgrounds create immediate association between elements
- Don't Abuse Borders: Don't overuse boundaries; whitespace can group elements effectively too
- Hierarchy Nesting: Nested regions effectively show content hierarchy and relationships

Visualization of Data, Understanding and Interpretation

Common Region

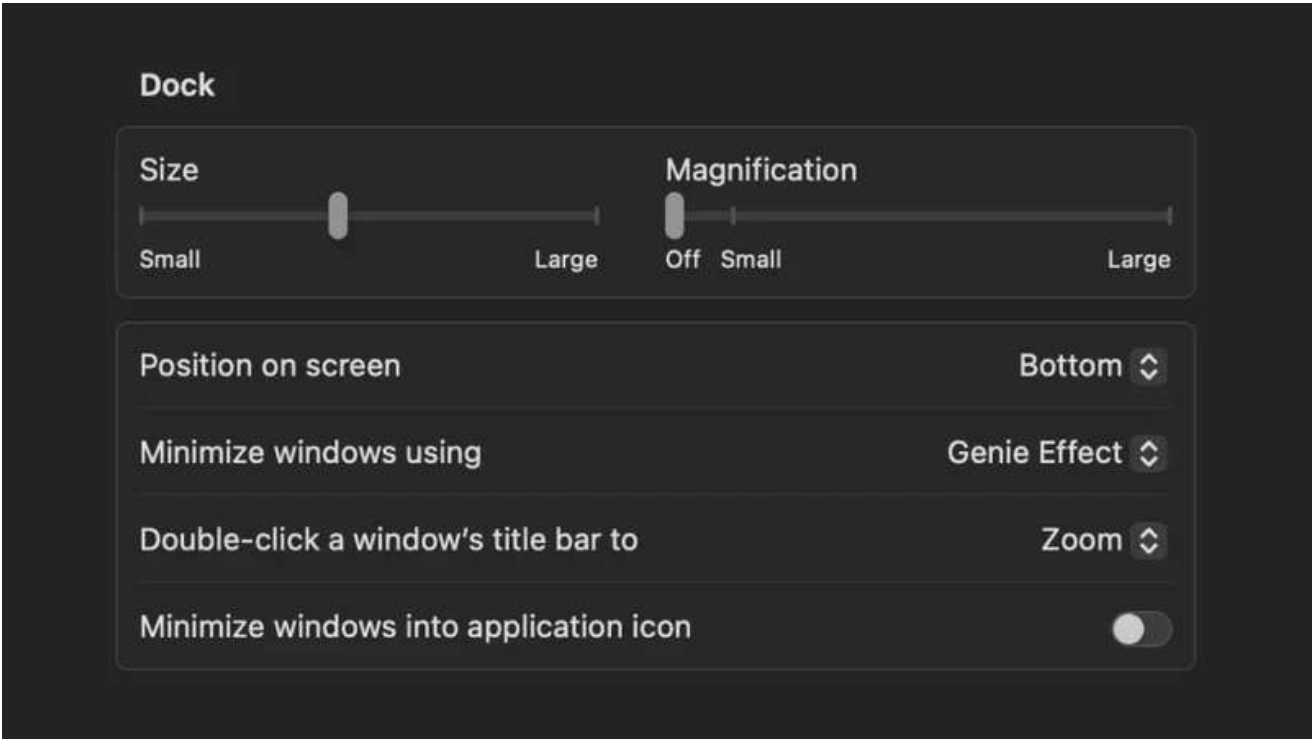
(Ch6)



UI Design: NYT Recipe Cards
Each recipe and actions are contained in a card with a border



UI Design: Apple Weather App
Different kinds of forecasts live within their own containers



Information Architecture: Apple Settings
Settings within the same category are grouped in separate containers with more closely related actions

Visualization of Data, Understanding and Interpretation

Figure & Ground

(Ch6)

Principle: The visual system separates a scene into:

- Figure → object of focus
- Ground → background

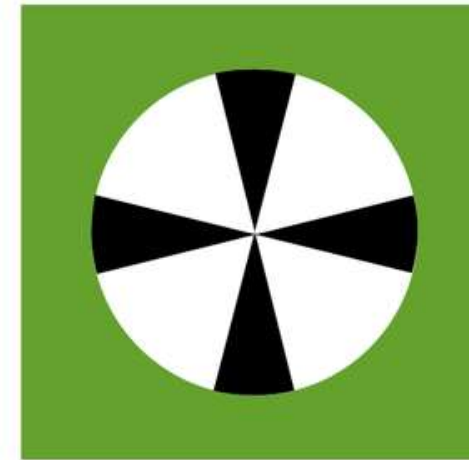
Good visualizations clearly distinguish the two.

Good Practices:

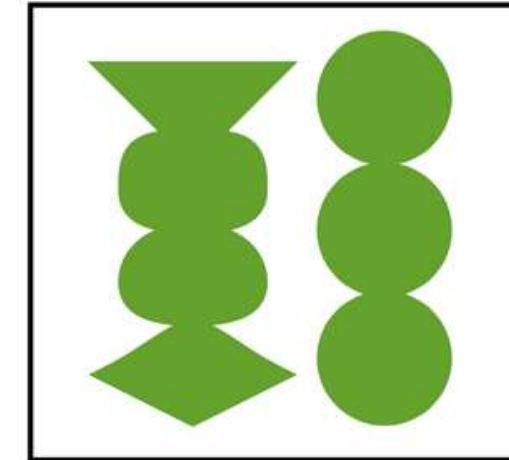
- strong contrast
- sufficient whitespace
- muted backgrounds

Examples:

- Reading black text on a white page
- Rubin's vase illusion (on the right)



(a)



(b)



(c)

Figure 6.19 (a) The black areas are smaller and therefore more likely to be perceived as an object. It is also easier to perceive patterns that are oriented horizontally and vertically as objects. (b) The green areas are seen as figures because of several Gestalt factors, including size and closed form. The area between the green shapes in (c) is generally not seen as a figure.



Figure 6.20 Rubin's Vase. The cues for figure and ground are roughly equally balanced, resulting in a bistable percept of either two faces or a vase.

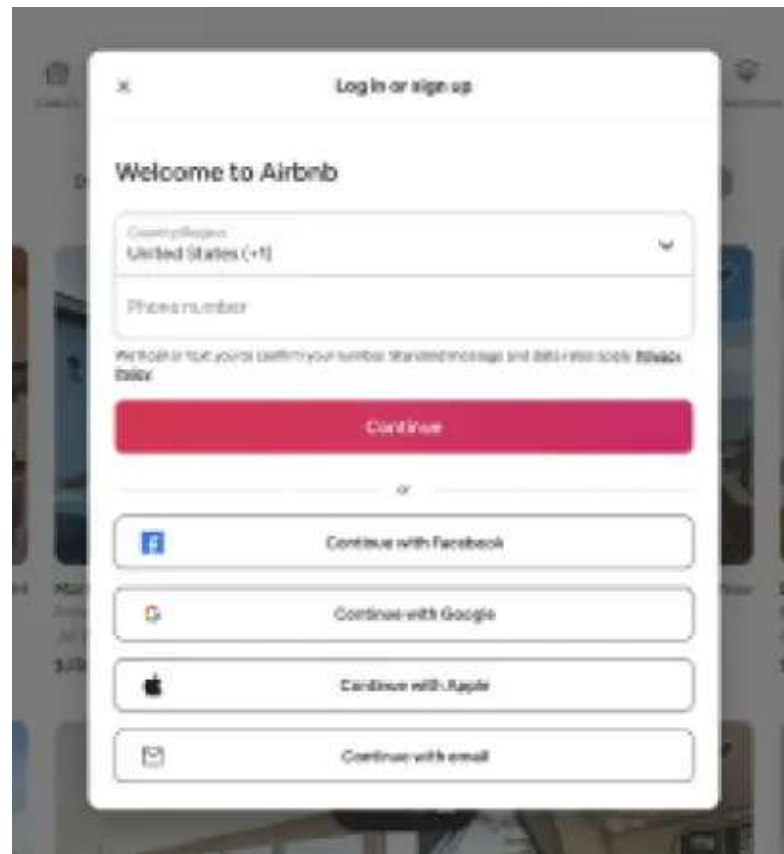
- The Gestalt principle of *Figure-Ground* describes how the human eye instinctively separates a scene into a focal object (the figure) and the surface behind it (the ground).
- This separation lets us focus on what matters while treating everything else as background, and it happens automatically whenever there is sufficient contrast in color, value, size, or depth.

Takeaways:

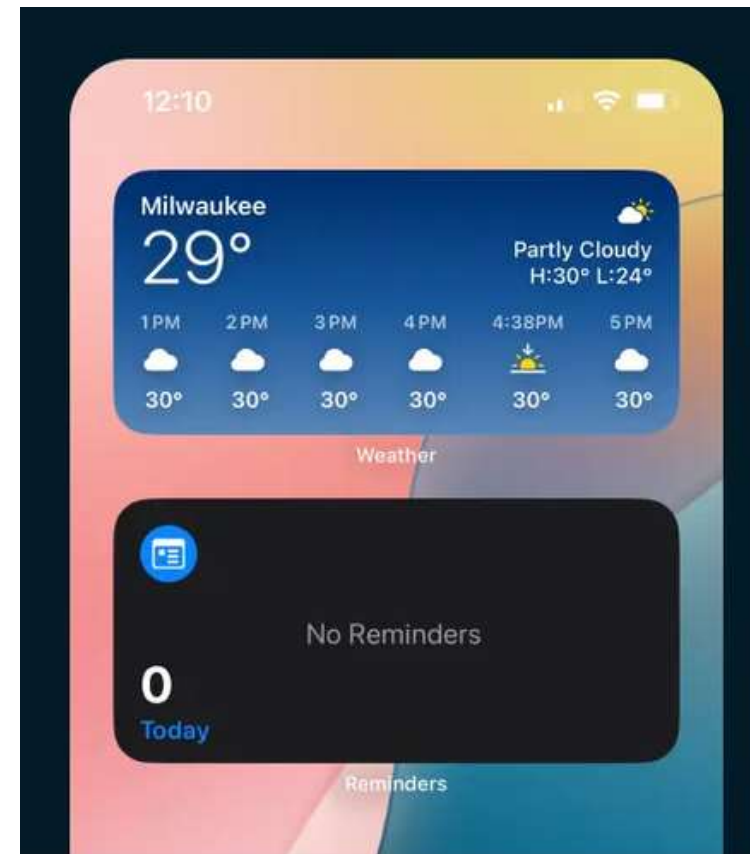
- Visual Contrast: Clear distinction between foreground and background elements guides attention
- Depth Perception: Use of shadows, blur, and opacity creates meaningful layers of content
- Ensure Clear Contrast: Use color, size, or depth to separate primary content (figure) from the background
- Test for Ambiguity: Make sure users can easily distinguish actionable elements (like buttons) from decorative ones

Visualization of Data, Understanding and Interpretation

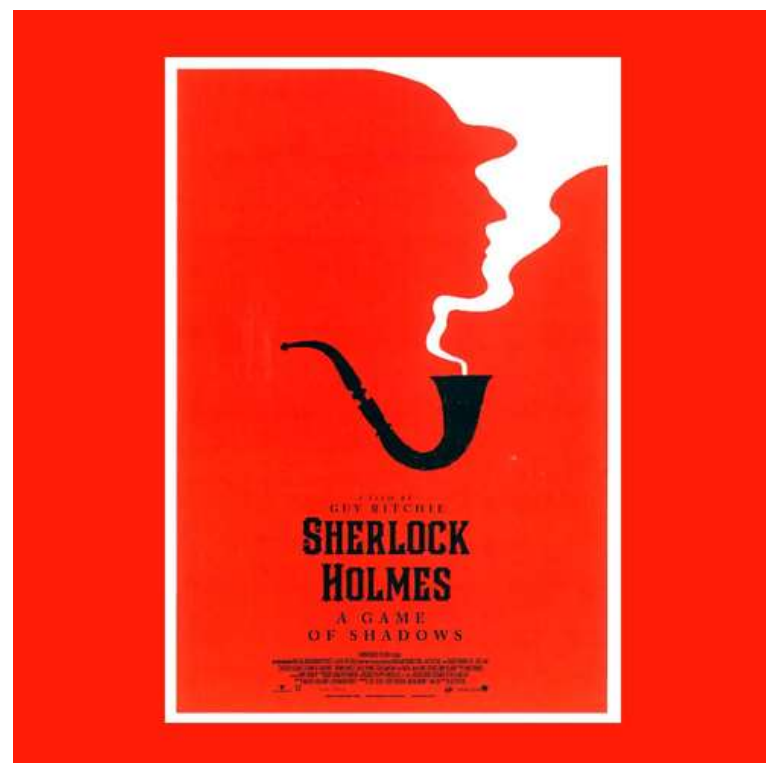
Figure & Ground (Ch6)



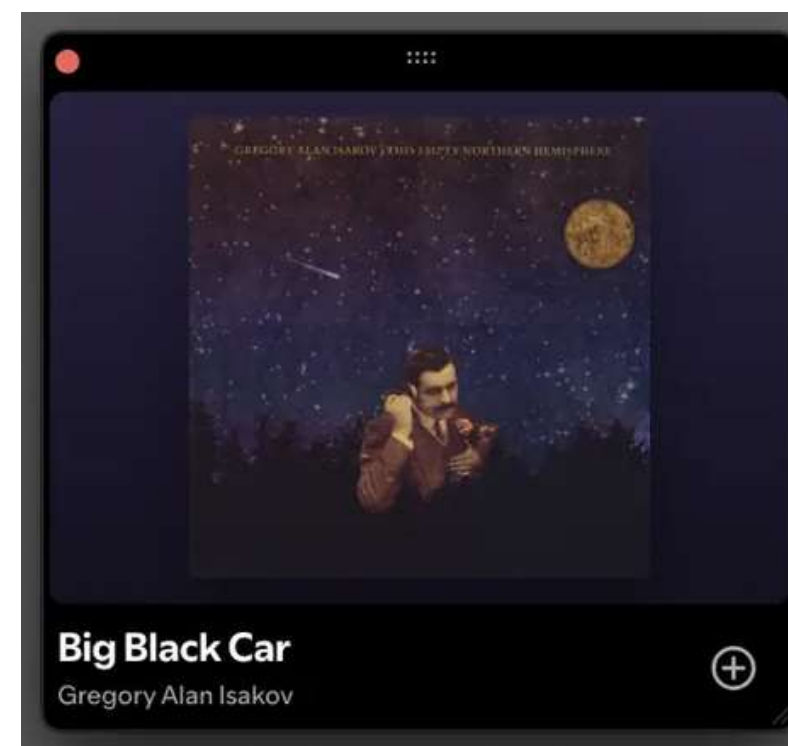
*UI Design: Airbnb
Login Modal*
Modal is focused in on
top of the background
content



UI Design: iOS Widgets
Widgets are based on top of
the background



*Graphic Design:
Sherlock Holmes
Poster*
Figure-ground
illustration that swaps
between smoke and a
face



*UI Design: Spotify
Miniplayer*
Miniplayer is moveable
above any background

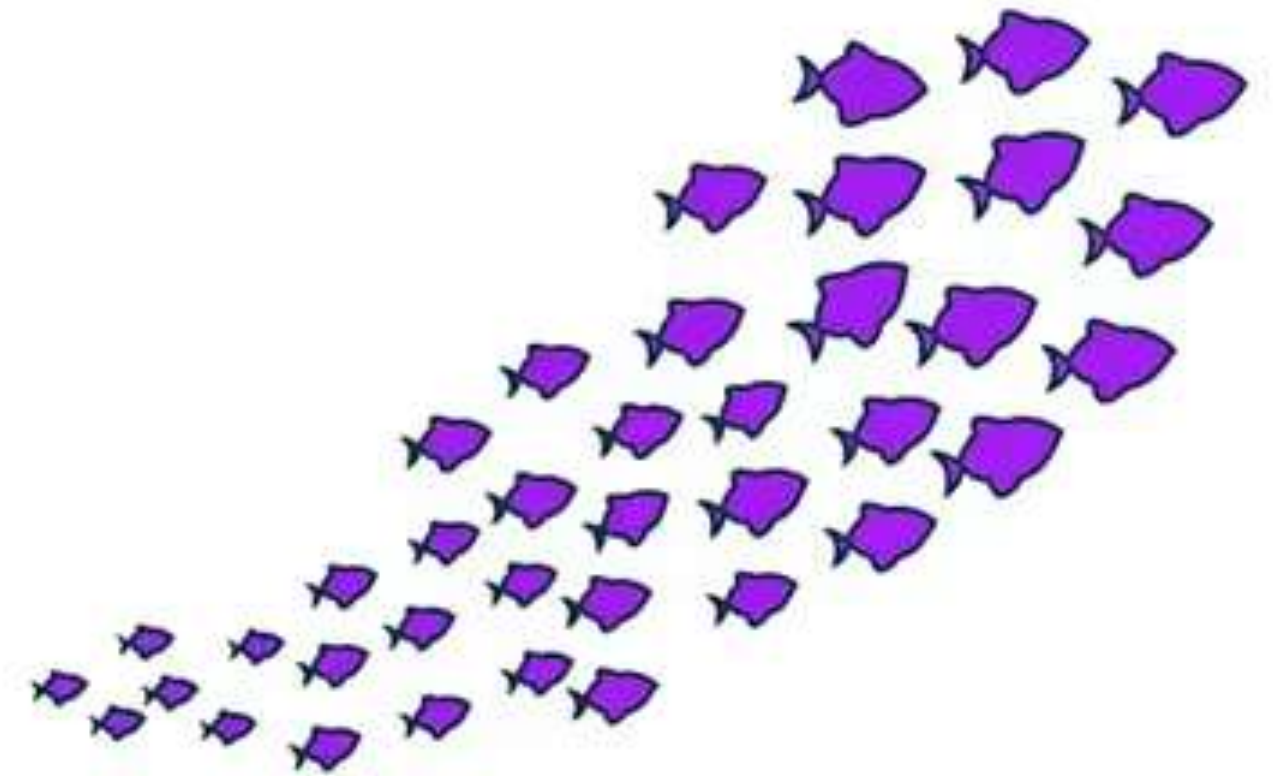
Principle: Objects moving in the same direction or changing together are perceived as belonging to one group.

Applications:

- Animated dashboards
- Motion tracking
- Interactive visualizations
- Traffic flow maps

Examples:

- A flock of birds appears as one group because they move together.
- Animated scatter plots where one category moves together over time.



- The Gestalt principle of *Common Fate* states that elements moving in the same direction, at the same speed, or transforming in the same way are perceived as a single group.
- Of all the Gestalt laws, common fate is the most dynamic — it relies on motion rather than static appearance, and it can override other grouping cues like color or proximity

Takeaways:

- Shared Movement: Elements moving together are seen as related
- Convey Hierarchy: Use animations to guide users through workflows or transitions
- Shared Transitions: Elements transforming in similar ways form natural groups
- Dynamic Relationships: Motion and transformation can create temporary groups

Check <https://www.gestaltprinciples.com/principles/common-fate> for live examples

Visualization of Data, Understanding and Interpretation

Applying Gestalt Principles to Data Visualization

(Ch6)



Principle	Design Guideline
Proximity	Group related information with spacing
Similarity	Use consistent colors and shapes
Connectedness	Connect related data with lines
Continuity	Preserve smooth visual flow
Symmetry	Use balanced layouts for comparison
Closure	Simplify incomplete shapes
Common Region	Enclose related elements
Figure & Ground	Maximize contrast
Common Fate	Animate related elements together

Visualization of Data, Understanding and Interpretation

MCQs

1. According to Gestalt principles, which visual cue creates the strongest perception of a relationship between objects, even overriding differences in proximity, color, size, or shape?

- A. Proximity
- B. Connectedness
- C. Similarity
- D. Common Region

Visualization of Data, Understanding and Interpretation

MCQs

1. According to Gestalt principles, which visual cue creates the strongest perception of a relationship between objects, even overriding differences in proximity, color, size, or shape?

- A. Proximity
- B. Connectedness
- C. Similarity
- D. Common Region

Answer: B

2. When designing a visualization using the principle of Closure, what is the primary design implication to keep in mind?

- A. Use highly saturated, contrasting colors for all background elements.
- B. Only use complete, unbroken borders to group legends.
- C. Use partial outlines to simplify visuals without making the shape ambiguous.
- D. Group unrelated variables together using spacing.

2. When designing a visualization using the principle of Closure, what is the primary design implication to keep in mind?

- A. Use highly saturated, contrasting colors for all background elements.
- B. Only use complete, unbroken borders to group legends.
- C. Use partial outlines to simplify visuals without making the shape ambiguous.
- D. Group unrelated variables together using spacing.

Answer: C



3. In the context of interactive dashboards and data visualization, which principle explains why elements that move in the same direction or change together over time are perceived as a single group?

- A. Continuity
- B. Symmetry
- C. Common Fate
- D. Proximity

3. In the context of interactive dashboards and data visualization, which principle explains why elements that move in the same direction or change together over time are perceived as a single group?

- A. Continuity
- B. Symmetry
- C. Common Fate
- D. Proximity

Answer: C

4. To effectively apply the Figure & Ground principle in a data presentation, which of the following practices is recommended?

- A. Use complex, highly textured patterns for the background.
- B. Keep the visual cues for background and foreground balanced so they are roughly equal.
- C. Maximize contrast by using a strong distinction between the object of focus and a muted background.
- D. Avoid using whitespace completely to prevent visual clutter.

4. To effectively apply the Figure & Ground principle in a data presentation, which of the following practices is recommended?

- A. Use complex, highly textured patterns for the background.
- B. Keep the visual cues for background and foreground balanced so they are roughly equal.
- C. Maximize contrast by using a strong distinction between the object of focus and a muted background.
- D. Avoid using whitespace completely to prevent visual clutter.

Answer: C

5. Which of the following correctly pairs a Gestalt principle with its recommended design guideline for data visualization?

- A. Proximity → Connect related data with lines
- B. Symmetry → Animate related elements together
- C. Common Region → Enclose related elements
- D. Similarity → Preserve smooth visual flow

5. Which of the following correctly pairs a Gestalt principle with its recommended design guideline for data visualization?

- A. Proximity → Connect related data with lines
- B. Symmetry → Animate related elements together
- C. Common Region → Enclose related elements
- D. Similarity → Preserve smooth visual flow

Answer: C



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Thank You



Visualization of Data, Understanding and Interpretation

Unit 1 : Foundations of Visualization, Data Interpretation

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

Dr.Prajwala TR (Course Instructor)

Dept. of CSE, PES University, Bangalore

Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.

Visualization of Data, Understanding and Interpretation

Lecture Overview

- The foundational role of color as an aesthetic channel in data visualization and how mathematical variable types determine the selection of functional color scales.
- The three core functional use cases of color in technical visualization: qualitative scales to distinguish groups, quantitative scales to represent continuous values, and accent scales to highlight key narratives.
- The design principles and constraints required to build perceptually sound qualitative, sequential, diverging, and accent color scales.

Visualization of Data, Understanding and Interpretation

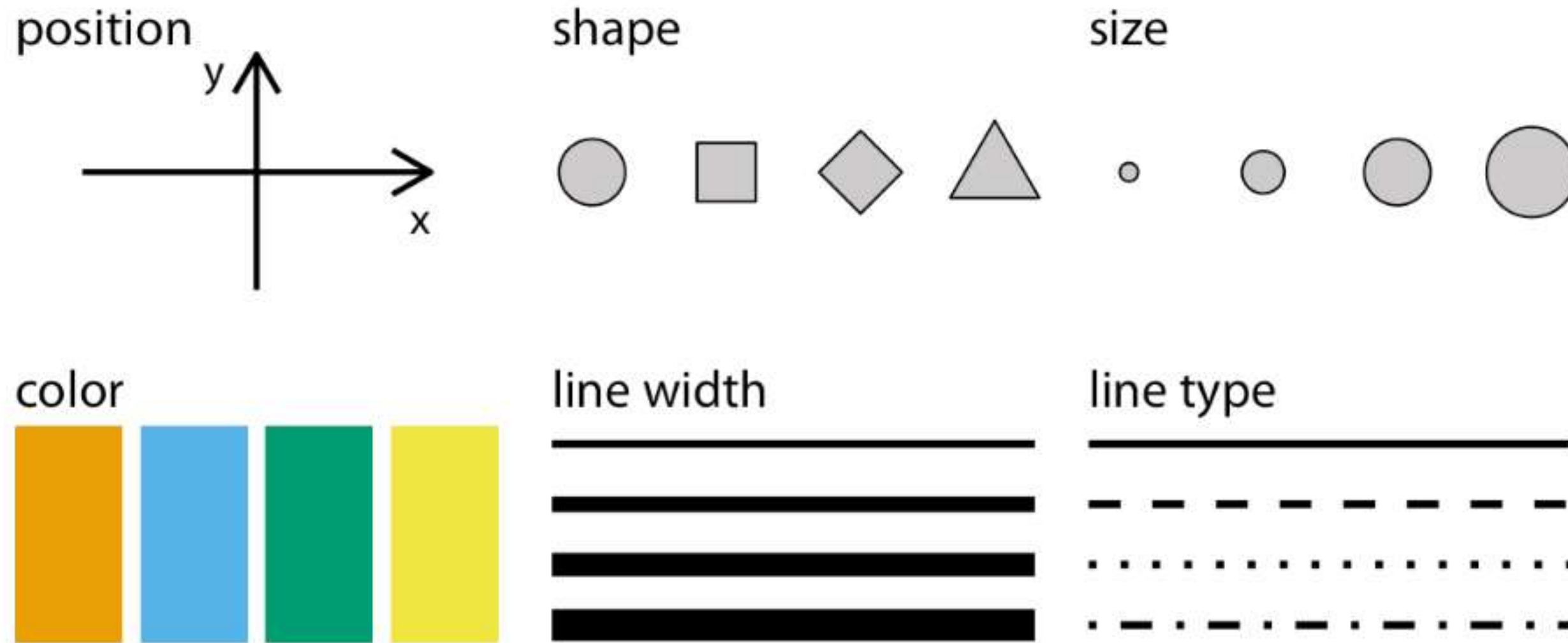
Color Fundamentals and Variable Mapping

- Color is a primary aesthetic dimension used to transform raw data values into quantifiable visual features. It belongs to a group of aesthetics, alongside position and size that can represent both continuous (numerical) and discrete (categorical) data.
- In technical visualization, color fundamentally serves three purposes: to distinguish groups, to represent data values, and to highlight specific information.
- A visualization must accurately convey data without misleading the viewer. If a color choice makes two distinct values look identical, or makes a small change appear massive, the figure is scientifically "wrong". Understanding the mathematical nature of your variables (quantitative vs. qualitative) is required to select the correct color scale.

How it Works: Variables are mapped to color via **scales**, which must define a unique, one-to-one mapping. For continuous variables like temperature, we use a gradient; for categorical variables like sensor IDs, we use distinct hues.

Visualization of Data, Understanding and Interpretation

Color Fundamentals and Variable Mapping



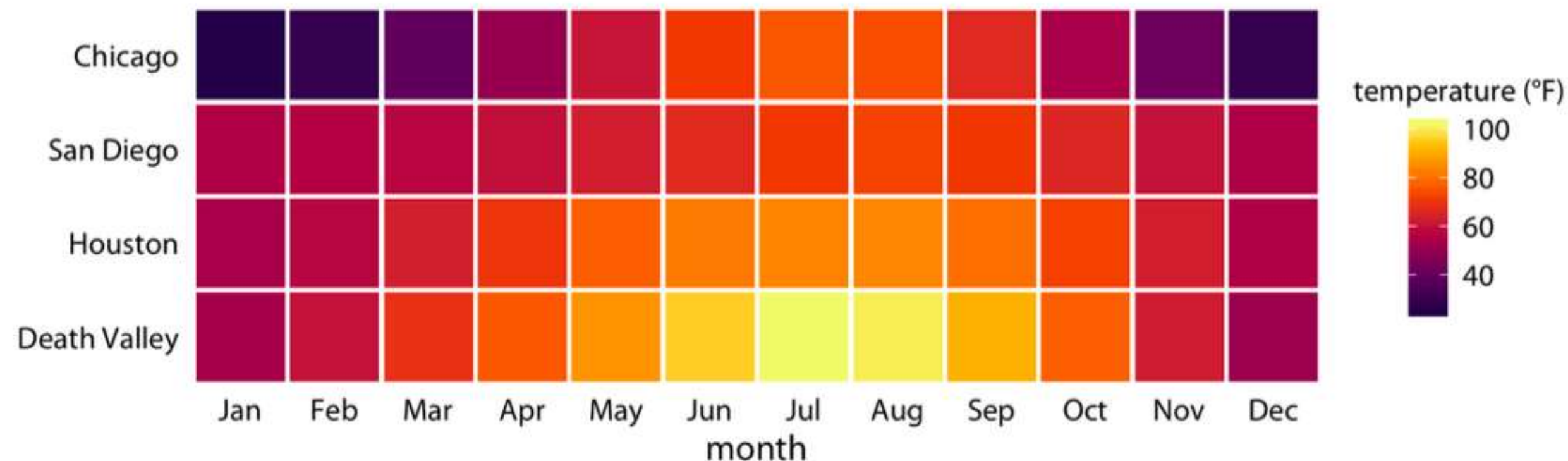
Commonly used aesthetics in data visualization: position, shape, size, color, line width, line type. Some of these aesthetics can represent both continuous and discrete data (position, size, line width, color) while others can only represent discrete data (shape, line type).

Visualization of Data, Understanding and Interpretation

Principles of Color Mapping

(TB 1, Sec 4.1)

- Color serves as a highly expressive aesthetic channel for encoding both continuous numerical gradients and discrete qualitative groups.
- Perception of color is highly non-linear, requiring color scales to be designed around the human visual system.
- Categorical mappings use color to indicate group membership for items that do not possess any mathematical order.
- Quantitative mappings use color gradients to represent continuous numerical magnitudes such as temperatures or density values.

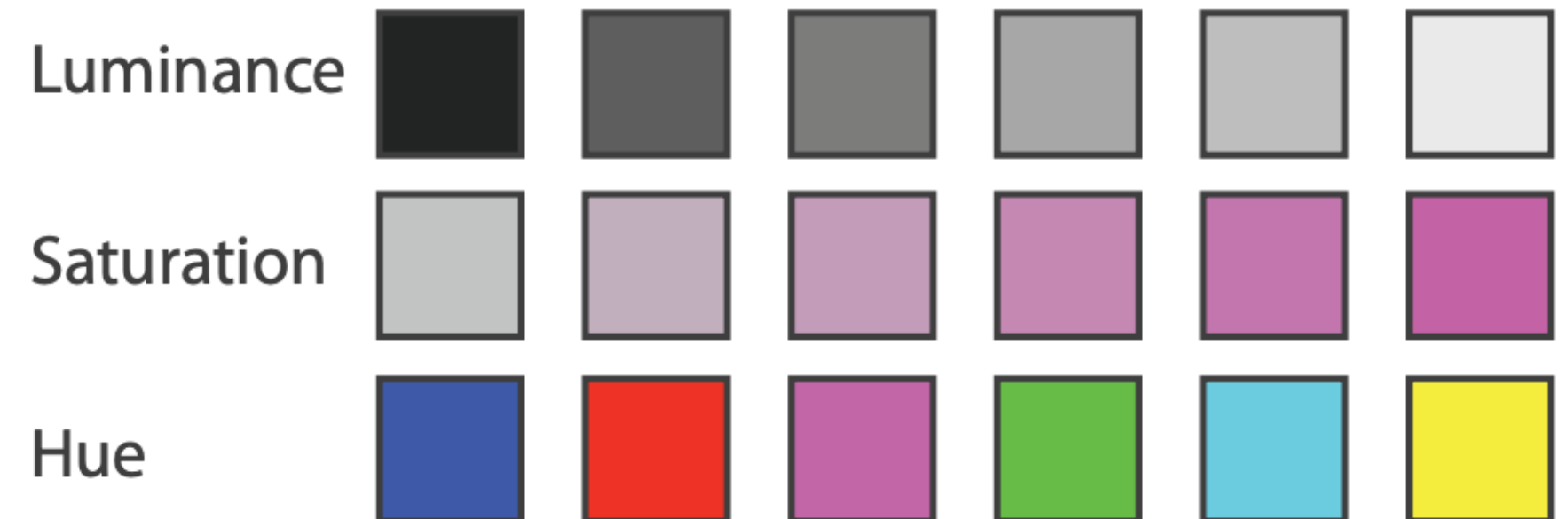


Average monthly temperatures for four locations encoded inside a colored matrix.

Visualization of Data, Understanding and Interpretation

Decomposing Color

- First rule of color: Do not (just) talk about color
 - Color is confusing if treated as monolithic
- Decompose into three channels:
 - Ordered can show magnitude
 - **Luminance**: how bright (B/W)
 - **Saturation**: how colorful
 - Categorical can show identity
 - **Hue**: what color
- Channels have different properties:
 - What they convey directly to perpetual system
 - How much they can convey
 - How many discriminable bins can we use

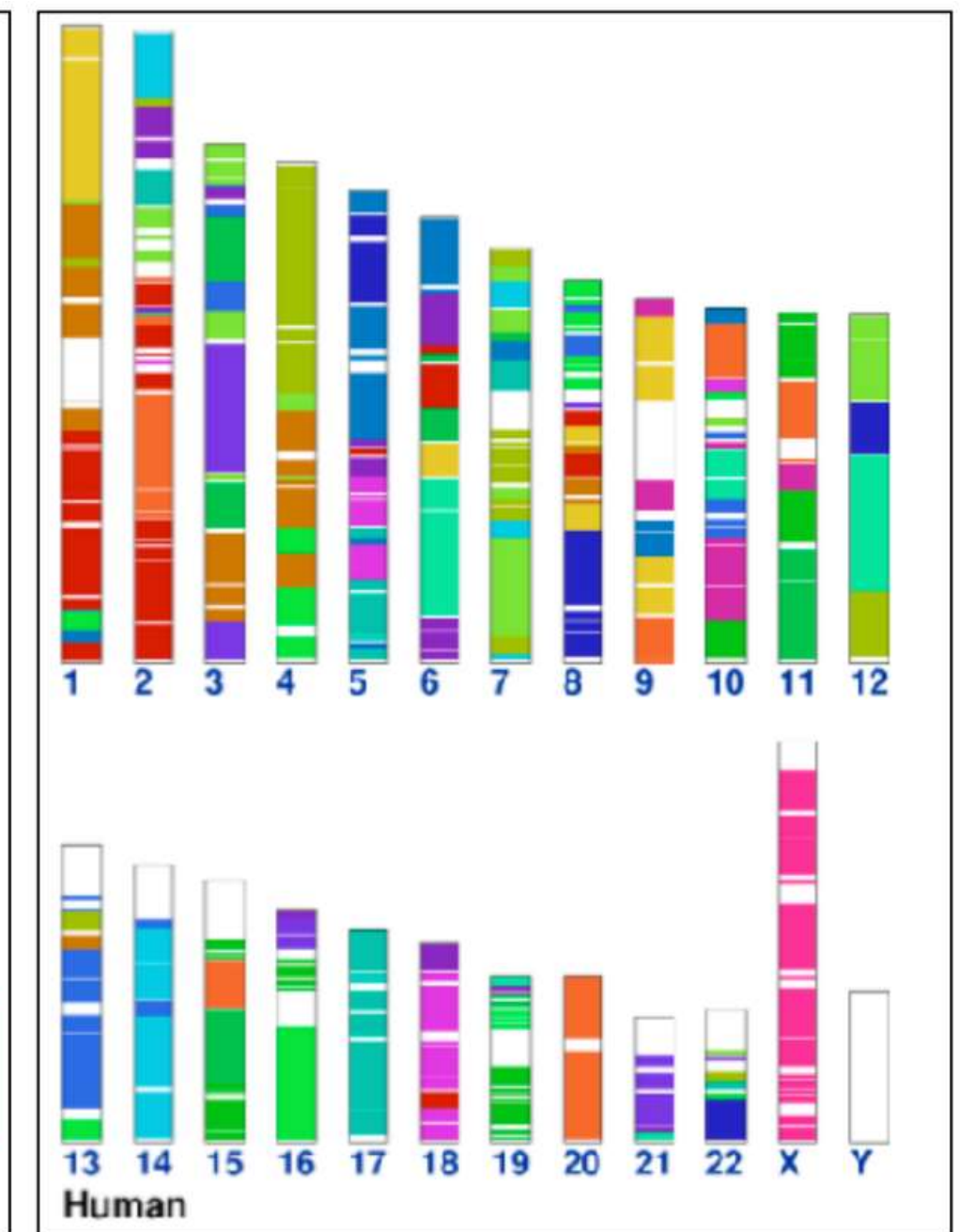
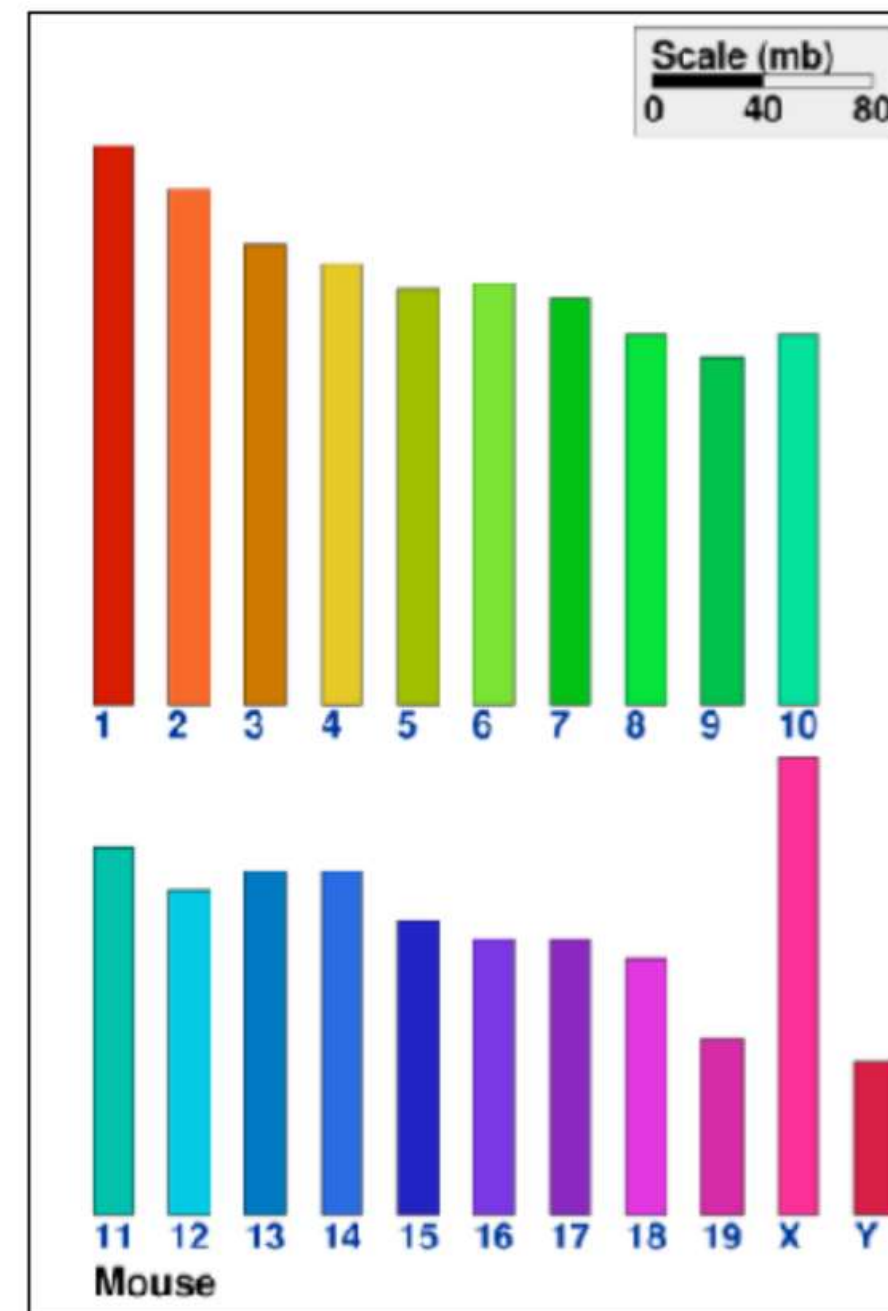


Visualization of Data, Understanding and Interpretation

Categorical Color

Limited number of discriminable bins

- Human perception built on relative comparisons
 - Great if color contiguous
 - Surprisingly bad for absolute comparisons
- Noncontiguous small regions of color
 - Fewer bins than you want
 - Rule of thumb: 6-12 bins, including background and highlights



Visualization of Data, Understanding and Interpretation

Three Functional Use Cases of Color (TB 1, Ch4)

1. To Distinguish

Qualitative Scales

Mapping: Discrete, Unordered Items

- Used to identify distinct categorical entities like countries, companies, or biological species.
 - Hues must look completely distinct but carry equivalent visual weight.
- Must never imply order or scale sequence, which corrupts perception.

2. To Represent Value

Sequential/Diverging Scales

Mapping: Quantitative Magnitudes

- Used to visualize continuous quantitative variables like income, density, or temperature.
- Sequential: Progression of dark-to-light hues mapped monotonically.
- Diverging: Two sequential scales stitched symmetrically at a light midpoint.

3. To Highlight

Accent Scales

Mapping: Narrative Focal Points

- Used in storytelling to direct the audience's gaze to crucial segments of a dataset.
- Composed of a highly drab, neutral background paired with bright, vivid anomalies.
- Enforces preattentive contrast, reducing cognitive load on complex figures.

Visualization of Data, Understanding and Interpretation

Color as a Tool to Distinguish: Qualitative Scales (TB1, Sec 4.1)

Defining Qualitative Color Scales

- Primary Function: Distinguish discrete categories or groupings that do not possess any inherent mathematical order (e.g., product types, departments, regions).
- Finite Set of Hues: Selected carefully to maximize perceived difference under the human visual system.
- Unordered Context: Items colored qualitatively represent completely equal concepts — no group is superior.

The Two Strict Constraints of Qualitative Design

A mathematically and perceptually sound qualitative scale must enforce:

- Equivalent Visual Weight: No single hue can stand out, be overly saturated, or look *louder* than the others. Equal luminance prevents false attention.
- No Implied Order: Palette must avoid sequential lightness transitions (e.g., pale blue to dark blue). Gradations force the brain to perceive a scale, corrupting categorical equality.

Visualization of Data, Understanding and Interpretation

Case Study - U.S. Population Growth Categorized by Region

Case Study:

- The Visual Setup: States are arranged along the Y-axis in order of their population growth percentage (2000-2010).
- Categorical Encoding: Rather than uniform gray bars, states are colored based on their geographic region (West, South, Midwest, Northeast).
- Equivalent Contrast: A proper qualitative palette is used where each region is clearly distinct yet equivalent in visual weight.

Cognitive Outcome of the Design

Result: Regional patterns emerge instantly

- Regional Clustering: The qualitative colors reveal a strong regional divide in population growth dynamics.
- Instant Pattern Recognition: The West (e.g., Nevada) and South dominate the high-growth end of the chart.
- Context Preservation: Midwest and Northeast states cluster at the lower end, showing very small increases or decreases.

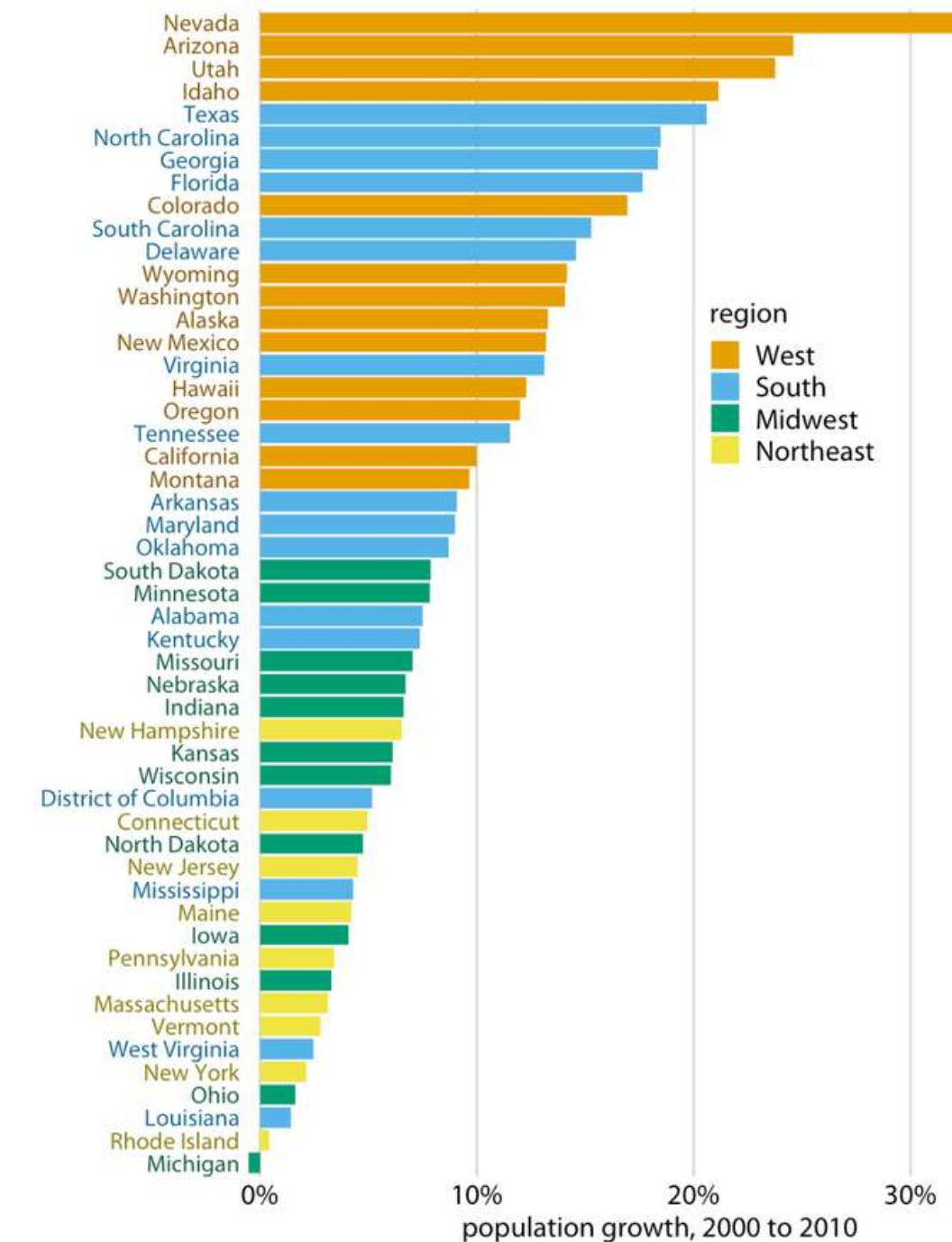


Figure 4.2 (TB 1, Sec 4.1)

Authoritative Qualitative Palettes

- Okabe Ito Scale: The academic standard. Designed specifically as a Color Universal Design (CUD) palette to be completely accessible to colorblind viewers.
- ColorBrewer Dark2: Optimized cartographic palette representing highly balanced, equivalent weight categorical attributes.
- ggplot2 Default Hue: Built-in scale spacing categories equally around a circular HSL color space — though less optimal for accessibility.

Palette Swatch Visualization

Okabe Ito Color Universal Design (CUD)

Scale:



ColorBrewer Dark2 Map Scale:



ggplot2 Hue:



Visualization of Data, Understanding and Interpretation

Quantitative Scales: Sequential and Diverging

(Sec 4.2)

- **Sequential Scales:** Map continuous numerical values from dark to light or vice-versa to represent magnitude or progression.
- **Diverging Scales:** Feature two sequential scales stitched together at a common neutral midpoint (usually a light color) to highlight deviation.
- **Perceptual Uniformity:** Scales must vary uniformly across their entire range so equal distance in data matches equal distance in color.
- **Symmetry Requirement:** Diverging scales must be balanced so that above-midpoint and below-midpoint values of equal magnitude have equal visual weight.

ColorBrewer Blues



Heat

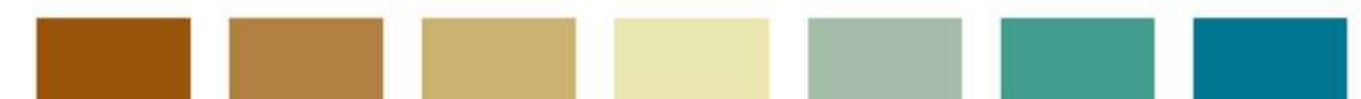


Viridis

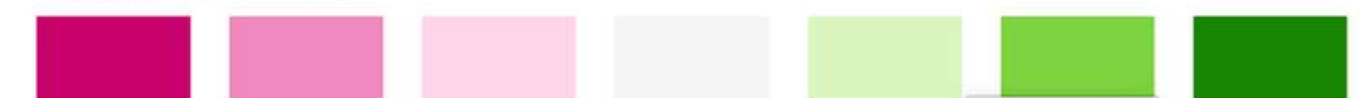


On the left are the sequential ColorBrewer Blues and Viridis scales. On the right are three diverging scales (brown to blue-green, pink to green, and blue to red).

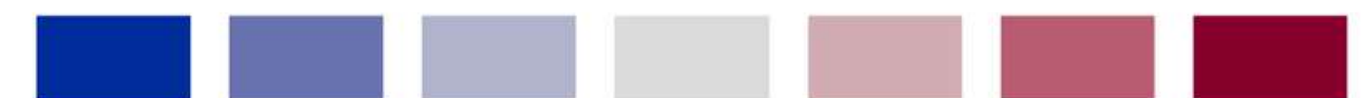
CARTO Earth



ColorBrewer PiYG



Blue-Red



Rotate Left

Visualization of Data, Understanding and Interpretation

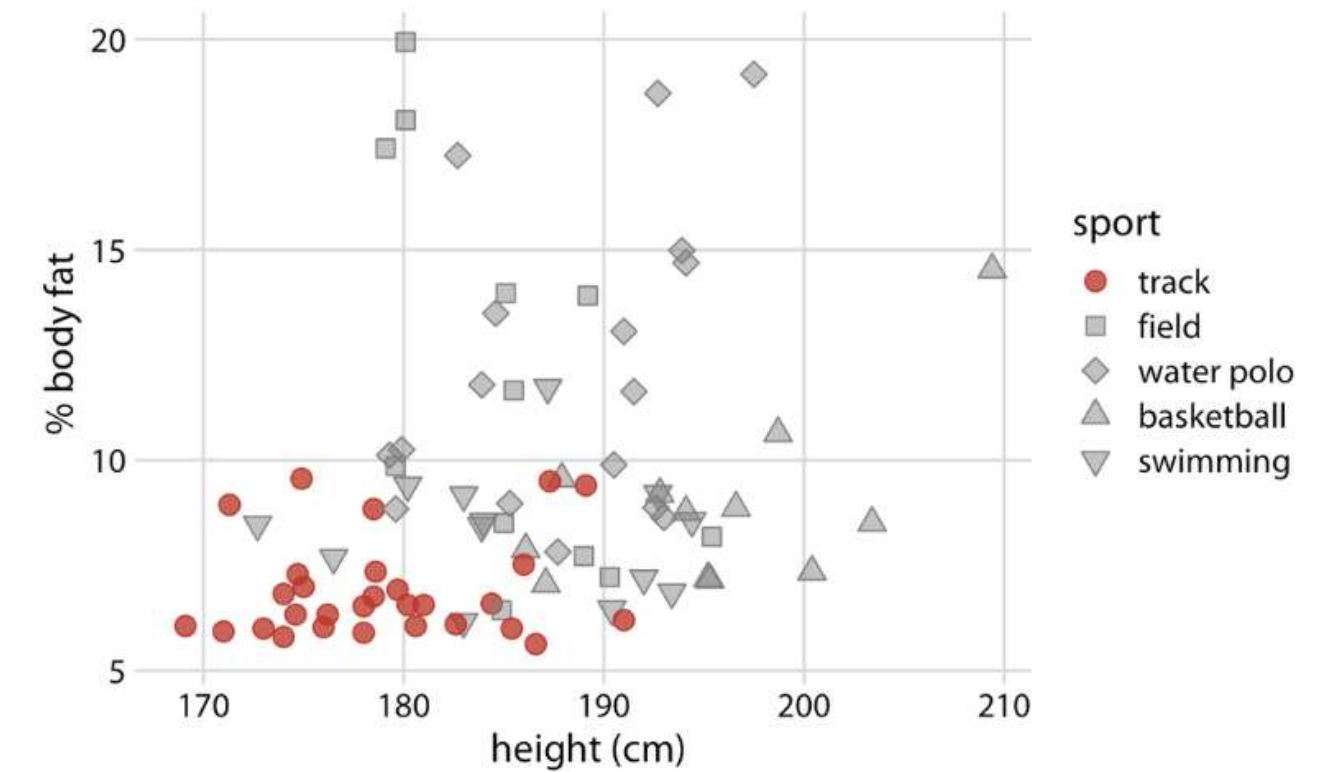
Color as a Tool to Highlight: Accent Scales

(Sec 4.3)

Color is an effective tool to highlight specific elements that carry the "key information" of the story. By using an accent color scale, we can emphasize relevant figure elements to ensure the reader identifies them immediately.

- Highlighting is required to prevent the reader from solving "visual puzzles". In complex datasets, the audience may not know where to look; color serves as a guide to the most important technical findings.
- Accent scales contain a set of subdued baseline colors (like grays or light pastels) paired with a matching set of stronger, more saturated colors. It is critical that baseline colors do not compete for attention.

Example: A diagnostic dashboard for a server farm where all "Healthy" servers are shown in subdued gray, but a "Critical" server is highlighted in vivid red to prompt immediate action.



Track athletes are among the shortest and leanest of male professional athletes participating in popular sports.

Visualization of Data, Understanding and Interpretation

Color as a Tool to Highlight: Accent Scales

(Sec 4.3)

The Narrative Role of Accents

- *Primary Function:* Direct the audience's focal point explicitly to specific data values, categories, or trends that carry the primary story.
- *Reducing Cognitive Search:* Rather than scanning a dense figure, the reader's eye is preattentively targeted, leading to faster insight comprehension.
- *Storytelling Contexts:* Critical for business presentations, lecture materials, executive summaries, and defense slide decks.

The Mechanics of Preattentive Contrast

How to guarantee highlight isolation:

- *Drab Baselines:* All background or context data must be colored in highly desaturated, drab, or completely neutral tones (like light gray or light slate).
- *Saturated Highlights:* The key highlighted element is colored in a vivid, highly saturated, or contrasting hue (PES Orange, Gold, or bold Crimson).
- *Eliminating Competitors:* If background colors are too saturated, they compete with the accent, destroying preattentive contrast.

Visualization of Data, Understanding and Interpretation

Methods for Designing Accent Color Scales

(Sec 4.3)

Method 1

Desaturate & Darken Base *Modifying Standard Scales*

- Take a balanced qualitative scale like Okabe Ito.
- Mute and lighten the context categories, reducing their saturation.
- Darken and heavily saturate only the active target category.

Method 2

Neutral Gray + Vivid Hue *The Cleanest Technical Palette*

- Strictly remove all color from non-essential context categories.
- Map all background bars, lines, or scatter points to a single neutral light gray.
 - Highlight the active category with a sharp, saturated primary color.

Method 3

Cartographic Presets *Using Expert-Designed Scales*

- Avoid manual generation by using professional preset scales.
 - ColorBrewer Accent Scale: Balanced cartographic preset with built-in subdued and strong pairings.
- Maintains accessibility metrics automatically.

Visualization of Data, Understanding and Interpretation

Case Study - U.S. State Growth Storytelling with Accent Colors

The Contrast Between Texas & Louisiana

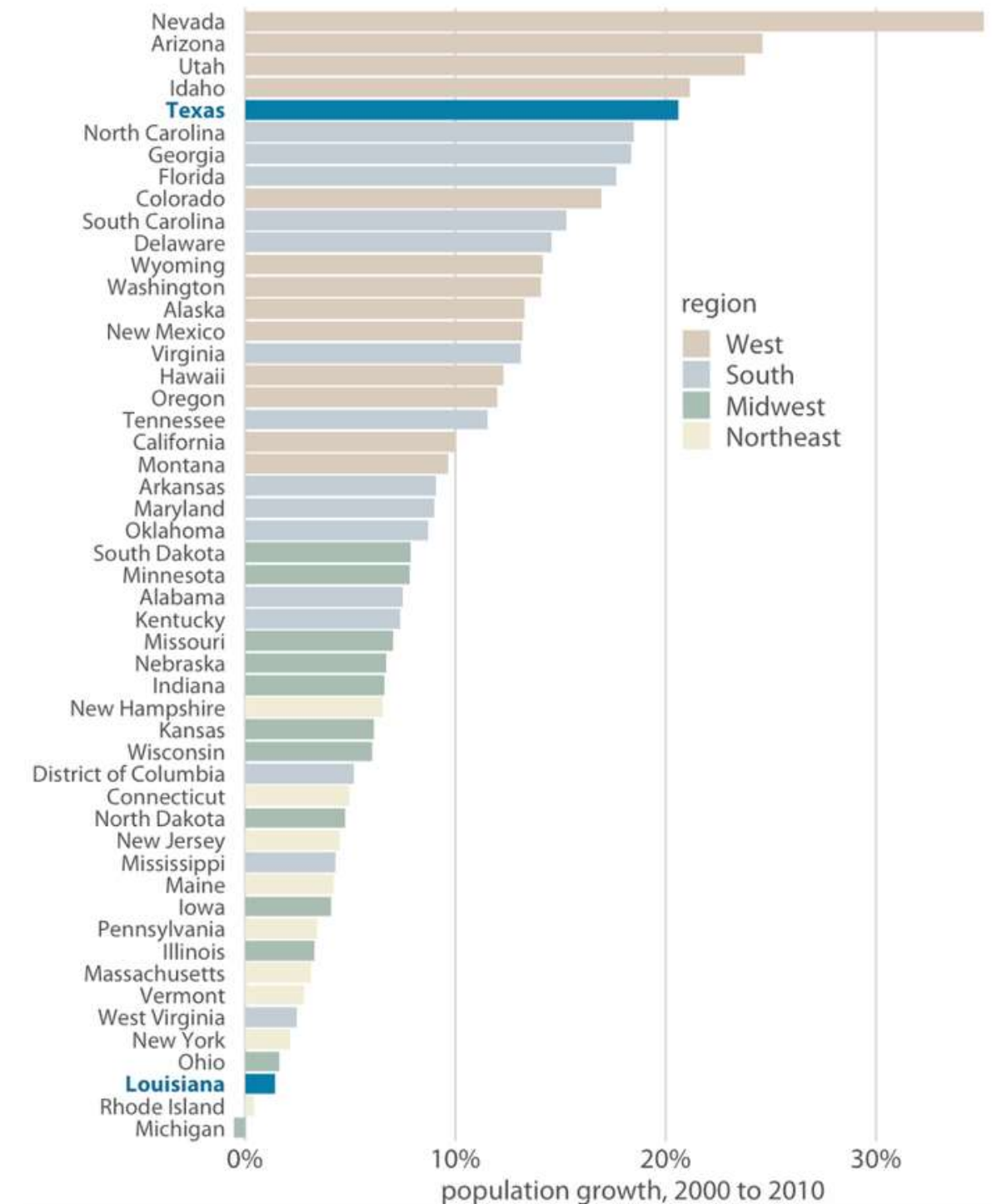
Case Study: Figure 4.8 (TB 1, Sec 4.3)

- The Analytical Problem: Both states are in the South, meaning they are colored in the same region color in qualitative mapping (Slide 6).
- The Missing Insight: Southern states are assumed to share a single trajectory, obscuring a major outlier story.
- Opposite Dynamics: Texas was the 5th fastest-growing U.S. state, while Louisiana was the 3rd slowest (2000-2010).

Applying the Accent Solution

Outcome: Structural context is preserved

- Muting the Backdrop: The other 48 states are colored in a highly desaturated, drab light blue-gray background.
- Vivid Highlights: Texas is colored in a saturated, dark crimson, while Louisiana is highlighted in bright orange.
- Instant Narrative: The eye is drawn to the extremes immediately, identifying the outlier comparison within milliseconds.



Visualization of Data, Understanding and Interpretation

Color Palettes : Univariate

- Categorical



- Ordered

→ Sequential

→ Diverging

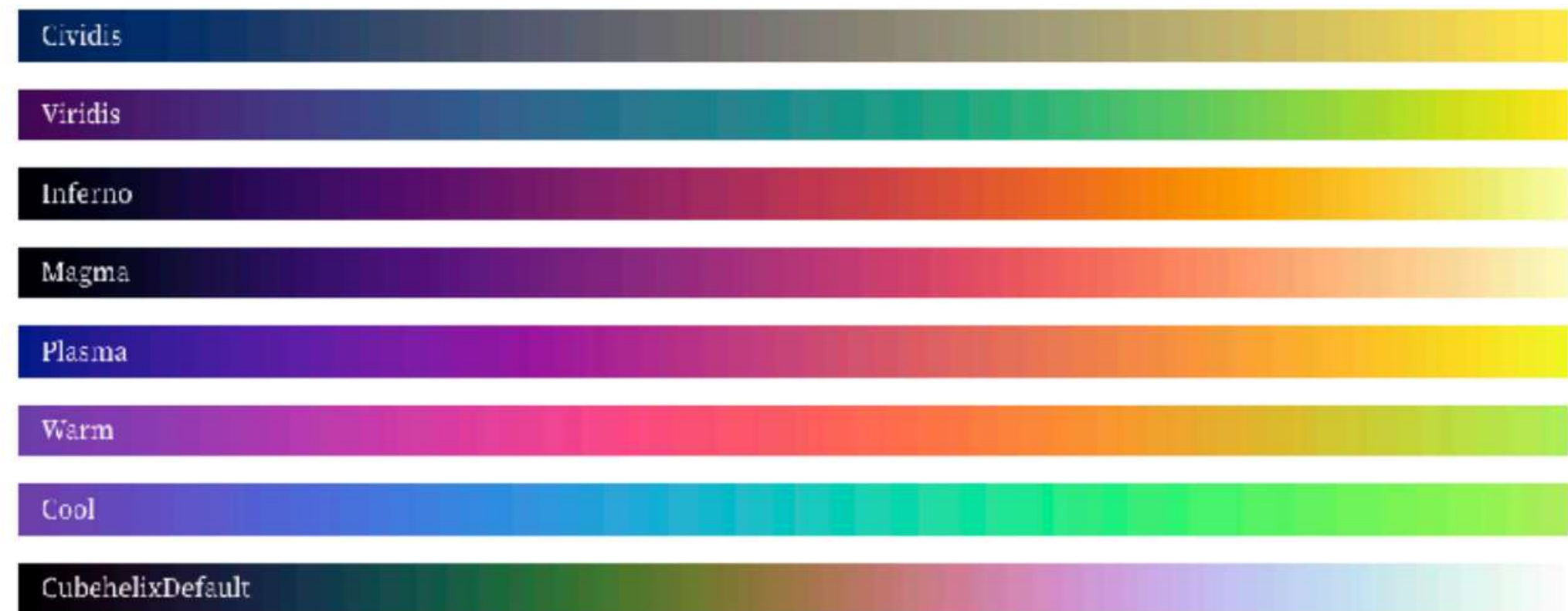


- Diverging

- Useful when data has meaningful "midpoint"
- Use neutral color for midpoint
 - white, yellow, grey
- Use saturated colors for endpoints

- Sequential

- Ramp luminance or saturation
- If multi-hue, good to order by luminance



diverging



sequential



Visualization of Data, Understanding and Interpretation

Avoiding the "Rainbow Scale" Pitfall

The rainbow scale is a popular but highly flawed color scale that runs through the entire color spectrum. While visually striking, it is frequently "bad" or "wrong" for technical data representation.

- Technical figures must use monotonic scales for sequential data. The rainbow scale violates this because it is effectively circular; the ends (dark red) are nearly the same color, meaning we don't instinctively perceive them as being maximally apart.
- The scale is non-monotonic in lightness. If converted to grayscale, it goes from medium-dark to light to very dark and back again. This causes the scale to highlight arbitrary features of the data while obscuring real trends.

rainbow scale



rainbow converted to grayscale

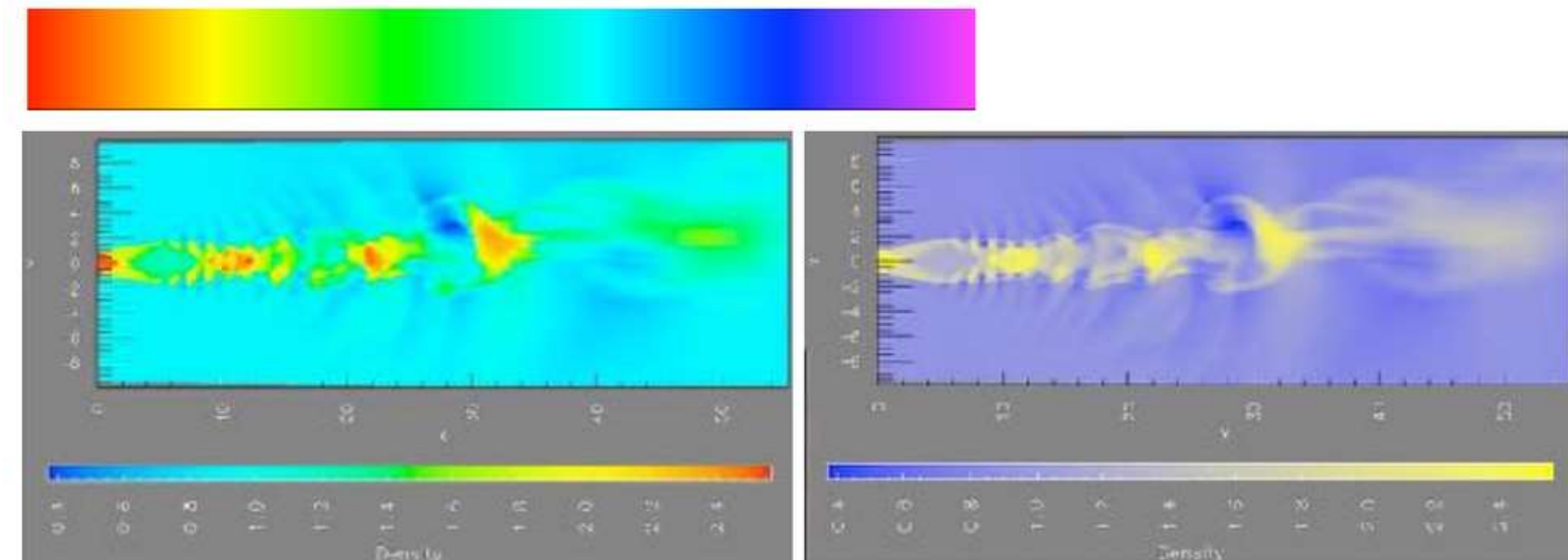


The rainbow colorscale is highly non-monotonic. This becomes clearly visible by converting the colors to gray values. From left to right, the scale goes from moderately dark to light to very dark and back to moderately dark. In addition, the changes in lightness are very non-uniform. The lightest part of the scale (corresponding to the colors yellow, light green, and cyan) takes up almost a third of the entire scale while the darkest part (corresponding to dark blue) is concentrated in a narrow region of the scale.

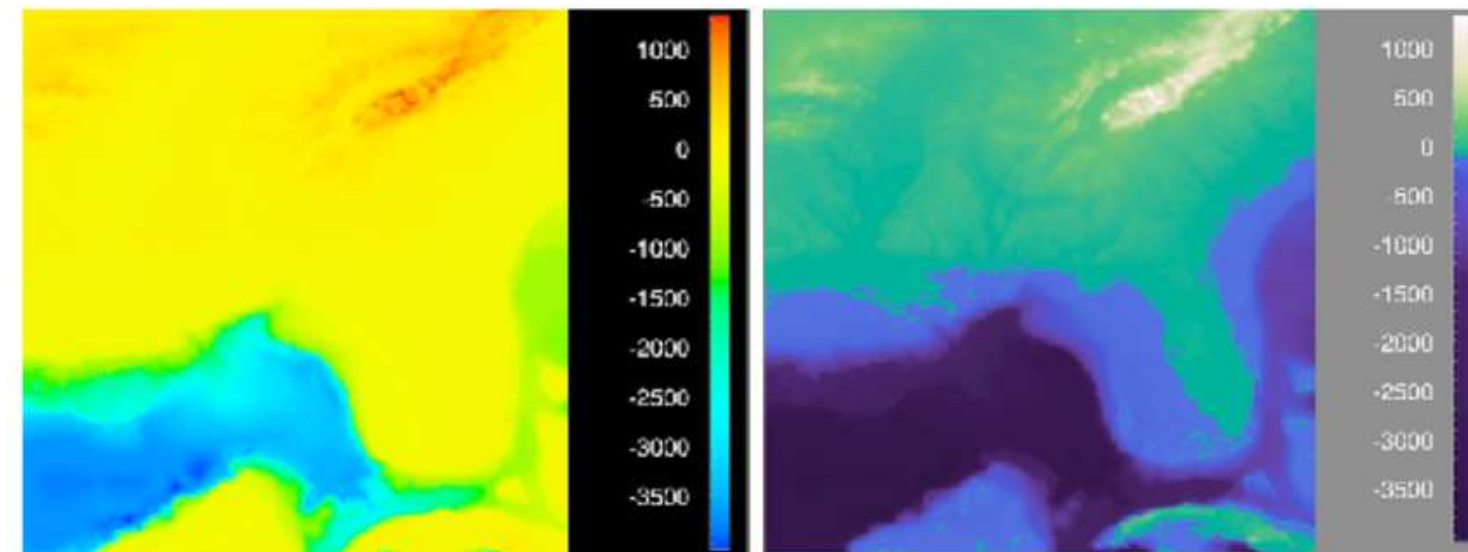
Visualization of Data, Understanding and Interpretation

Ordered Color: Rainbow - Poor Default

- Problems
 - Perceptually unordered
 - Perceptually nonlinear
- Benefits
 - Fine-grained structure visible and namable
- Alternatives
 - Large-scale structure: fewer hues
 - Fine structure: multiple hues with monotonically increasing luminance [eg viridis]
- Legit for categorical
 - Segmented saturated rainbow is good!



[A Rule-based Tool for Assisting Colormap Selection. Bergman, Rogowitz, and Treinish. Proc. IEEE Visualization (Vis), pp. 118–125, 1995.]



[Why Should Engineers Be Worried About Color? Treinish and Rogowitz 1998. <http://www.research.ibm.com/people/lloyd/color/color.HTM>]



[Transfer Functions in Direct Volume Rendering: Design, Interface, Interaction. Kindlmann. SIGGRAPH 2002 Course Notes]

Visualization of Data, Understanding and Interpretation

MCQs

1. Which of the following aesthetics is capable of representing both continuous (numerical) and discrete (categorical) data?

- A. Shape
- B. Line type
- C. Size
- D. None of the above

Visualization of Data, Understanding and Interpretation

MCQs



1. Which of the following aesthetics is capable of representing both continuous (numerical) and discrete (categorical) data?

- A. Shape
- B. Line type
- C. Size
- D. None of the above

Answer: C

2. Under the strict constraints of qualitative color scale design, what does "equivalent visual weight" mean?

- A. Every hue must have a different level of saturation to create maximum contrast.
- B. No single hue can stand out, look 'louder' than others, or be overly saturated due to equal luminance.
- C. The colors must transition sequentially from a pale shade to a dark shade of the same hue.
- D. The colors must follow a strict mathematical order based on their physical light wavelength.

2. Under the strict constraints of qualitative color scale design, what does "equivalent visual weight" mean?

- A. Every hue must have a different level of saturation to create maximum contrast.
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- D. The colors must follow a strict mathematical order based on their physical light wavelength.

Answer: B

3. Which qualitative color palette is considered the academic standard for Color Universal Design (CUD) because it is completely accessible to colorblind viewers?

- A. ColorBrewer Dark2
- B. ggplot2 Default Hue
- C. Viridis Scale
- D. Okabe Ito Scale

3. Which qualitative color palette is considered the academic standard for Color Universal Design (CUD) because it is completely accessible to colorblind viewers?

- A. ColorBrewer Dark2
- B. ggplot2 Default Hue
- C. Viridis Scale
- D. Okabe Ito Scale

Answer: D

4. What is a critical structural requirement for designing a balanced "diverging" quantitative color scale?

- A. It must map continuous values from dark to light monotonically to represent progression.
- B. It must stitch two sequential scales symmetrically at a common neutral midpoint, ensuring equal visual weight for equal deviations.
- C. It must use highly saturated background colors to compete with the central accent midpoint.
- D. It must run through the entire rainbow spectrum to maximize visual impact.

4. What is a critical structural requirement for designing a balanced "diverging" quantitative color scale?

- A. It must map continuous values from dark to light monotonically to represent progression.
- B. It must stitch two sequential scales symmetrically at a common neutral midpoint, ensuring equal visual weight for equal deviations.
- C. It must use highly saturated background colors to compete with the central accent midpoint.
- D. It must run through the entire rainbow spectrum to maximize visual impact.

Answer: B

5. Why is the popular "rainbow scale" considered mathematically flawed and "wrong" for representing sequential technical data?

- A. It is non-monotonic in lightness, which highlights arbitrary features while obscuring actual trends.
- B. It is effectively circular because its ends (dark red) are nearly the same color, violating the requirement of monotonic progression.
- C. Its changes in lightness are highly non-uniform across the spectrum.
- D. All of the above.

5. Why is the popular "rainbow scale" considered mathematically flawed and "wrong" for representing sequential technical data?

- A. It is non-monotonic in lightness, which highlights arbitrary features while obscuring actual trends.
- B. It is effectively circular because its ends (dark red) are nearly the same color, violating the requirement of monotonic progression.
- C. Its changes in lightness are highly non-uniform across the spectrum.
- D. All of the above.

Answer: D



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Thank You



Visualization of Data, Understanding and Interpretation

Unit 1 : Foundations of Visualization, Data Interpretation

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

Dr.Prajwala TR (Course Instructor)

Dept. of CSE, PES University, Bangalore

Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.

Visualization of Data, Understanding and Interpretation

Lecture Overview

- Introduction to visualization techniques categorized by analytical tasks, including how to select appropriate methods for comparing amounts, distributions, proportions, relationships, and uncertainty.
- Exploration of histograms for continuous data and how the selection of bin widths directly affects the visibility of overall patterns and data features.
- Understanding density plots and Kernel Density Estimation (KDE).
- Analyzing proportional visualization methods such as pie charts, stacked bar charts, and side-by-side bar charts to compare their effectiveness in displaying categories across different baselines.

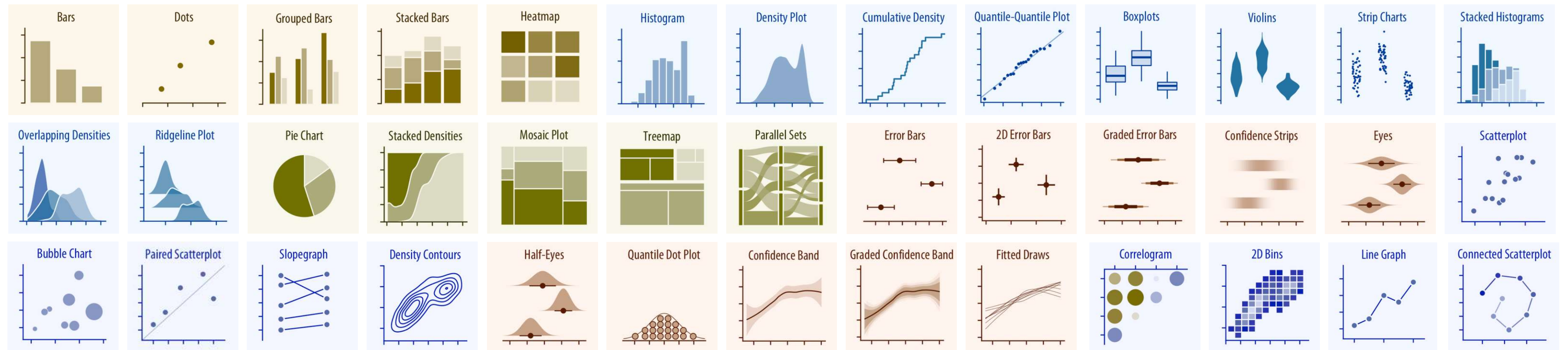
Visualization of Data, Understanding and Interpretation

Directory of visualizations

(TB 1, Ch5)

This chapter introduces visualization techniques for different analytical tasks:

- Amounts: Compare numerical values across categories
- Distributions: Understand how data values are spread
- Proportions: Show parts of a whole
- X-Y Relationships: Explore associations between variables
- Uncertainty: Communicate confidence and variability



Visualization of Data, Understanding and Interpretation

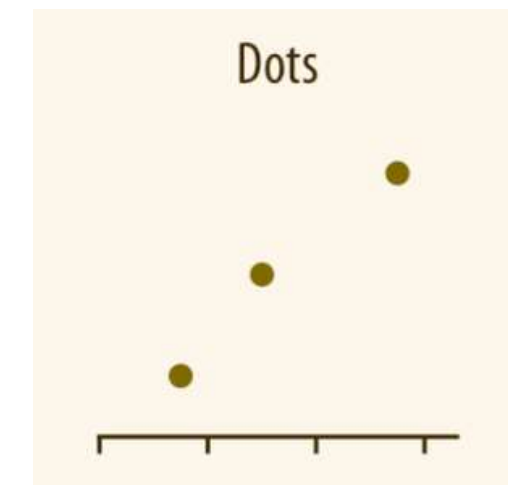
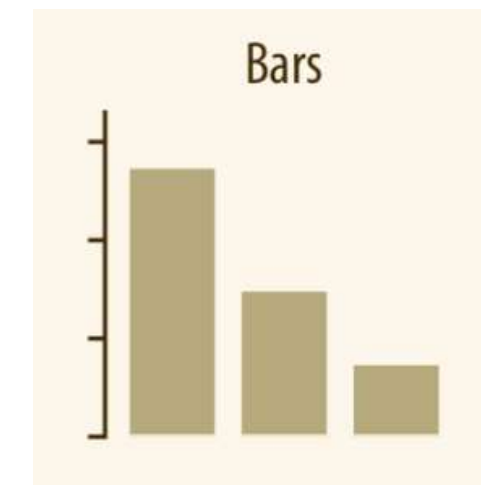
Visualizing Amounts

(Sec 5.1)

When to Use: To display and compare numerical values across categories.

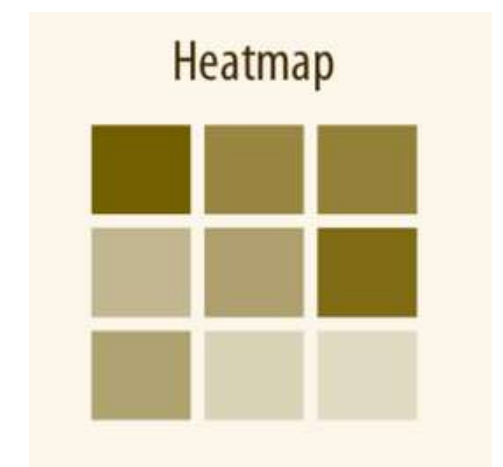
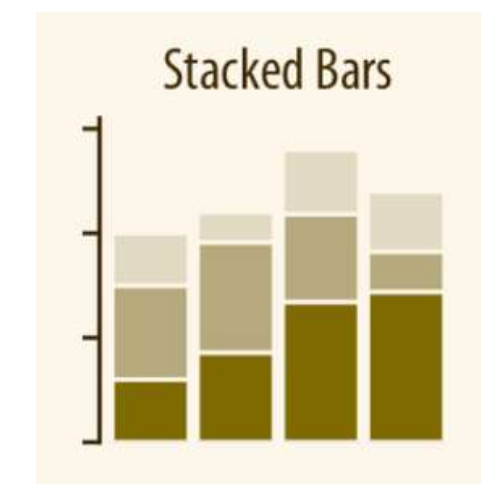
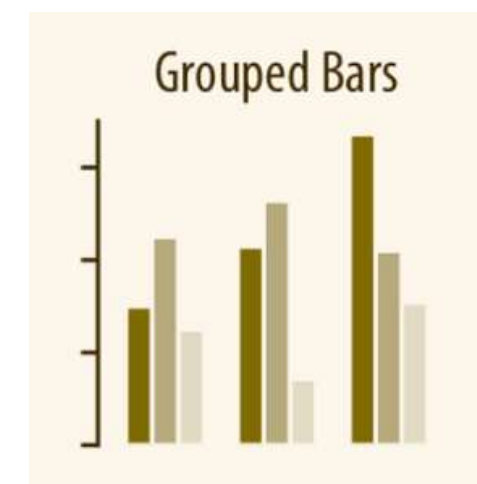
Common Charts:

- Bar charts: Best general-purpose comparison
- Dot plots: Cleaner alternative for simple comparisons
- Grouped bars: Compare multiple groups
- Stacked bars: Show totals + composition
- Heatmaps: Compare many values using color



Example:

- Comparing sales across products
- Student marks in different subjects



Visualization of Data, Understanding and Interpretation

Visualizing Distributions

(Sec 5.2)

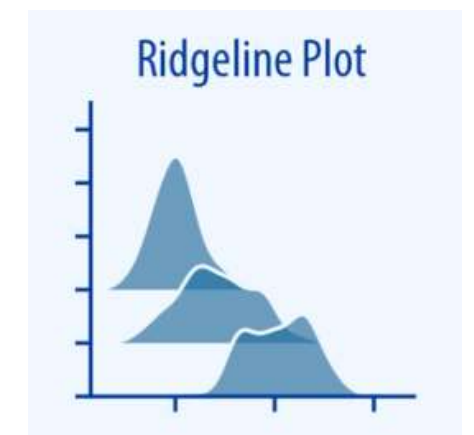
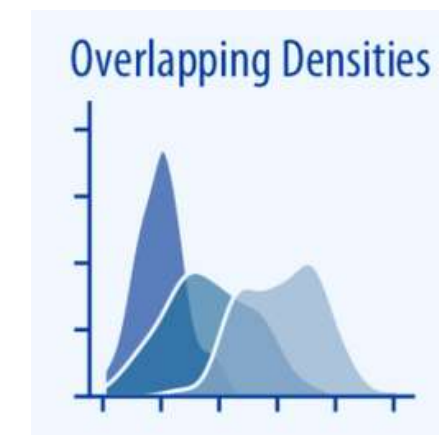
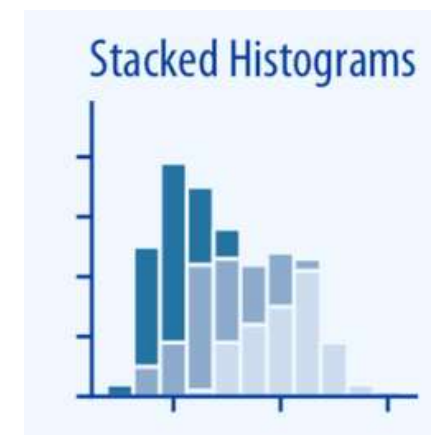
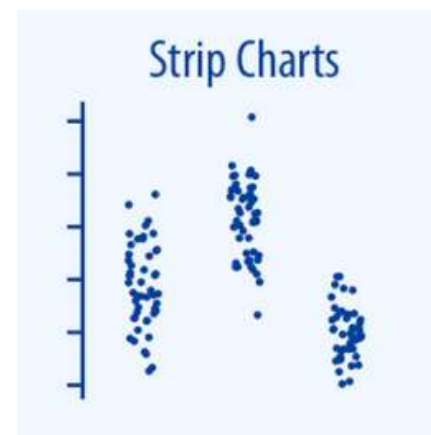
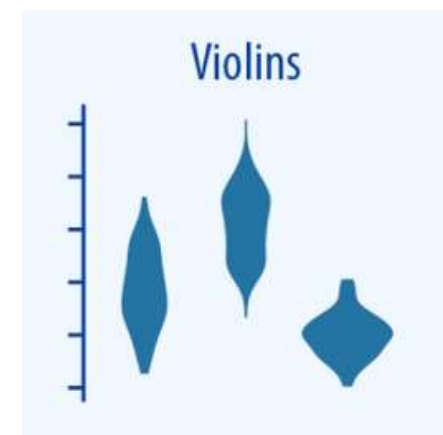
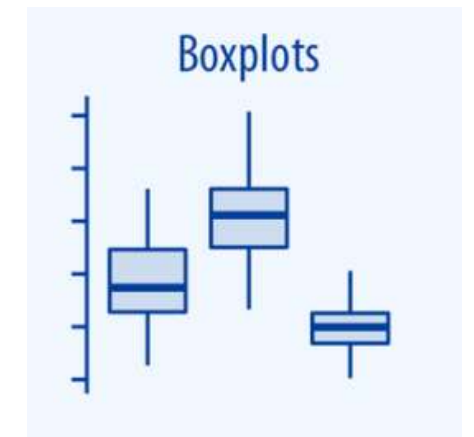
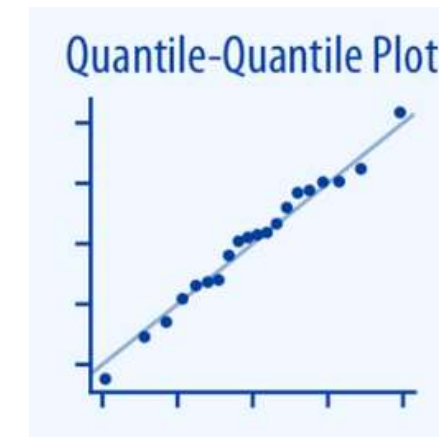
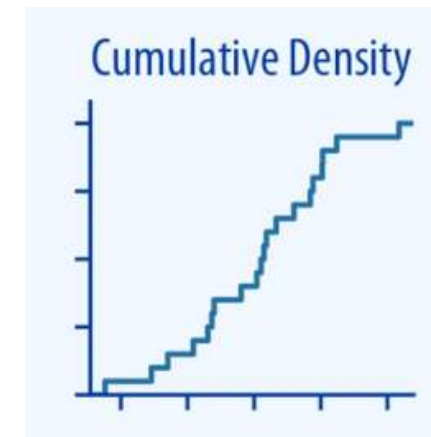
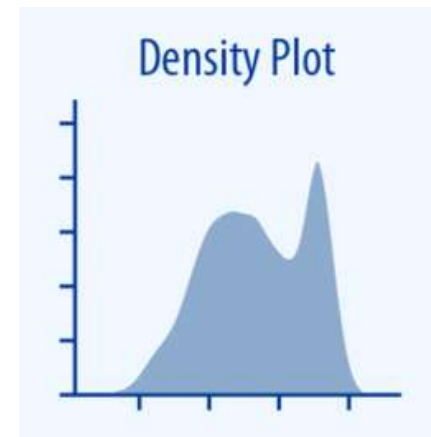
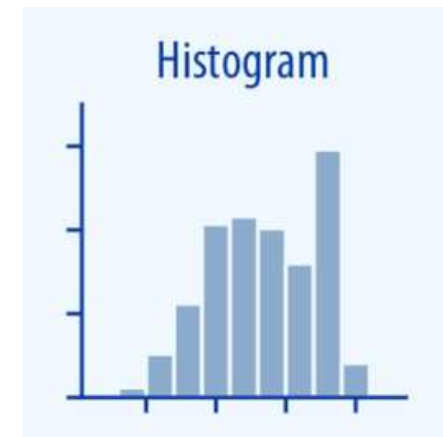
When to Use: To understand how data values are spread.

Common Charts:

- Histogram
- Density plot
- Box plot
- Violin plot
- Strip chart
- Ridgeline plot

What they reveal:

- Center
- Spread
- Skewness
- Outliers
- Multiple peaks



Example: Distribution of exam scores or customer ages.

Visualization of Data, Understanding and Interpretation

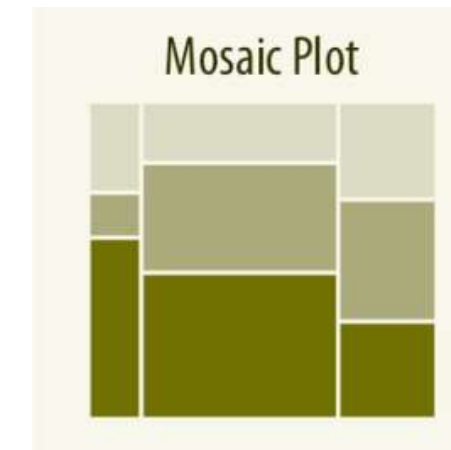
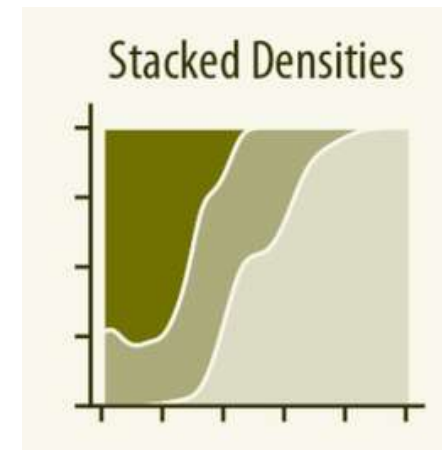
Visualizing Proportions

(Sec 5.3)

When to Use: To show parts of a whole.

Common Charts:

- Pie chart
- Bar chart
- Stacked bar chart
- Mosaic plot
- Treemap
- Parallel sets



Best Practice:

- Bar charts generally make comparisons easier than pie charts.
- Treemaps are useful for hierarchical proportions.

Example:

- Market share of companies
- Budget allocation
- Election vote share



Visualization of Data, Understanding and Interpretation

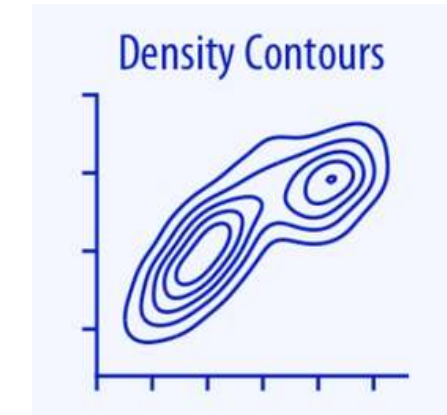
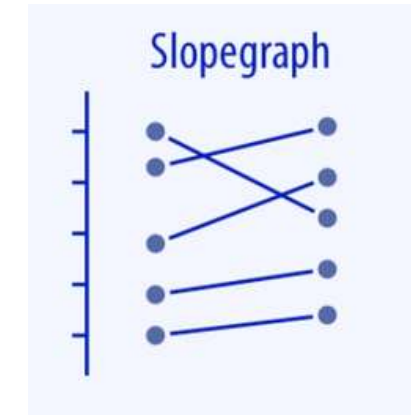
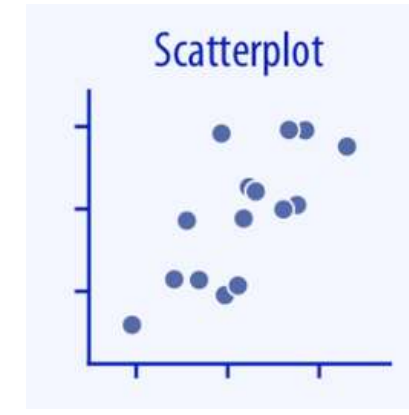
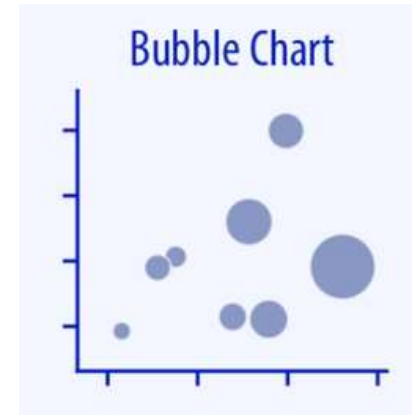
Visualizing X-Y Relationships

(Sec 5.4)

When to Use: To explore relationships between two or more quantitative variables.

Common Charts:

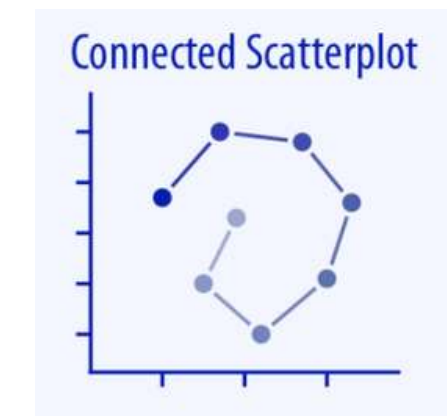
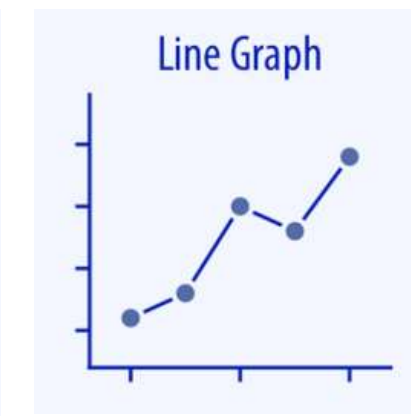
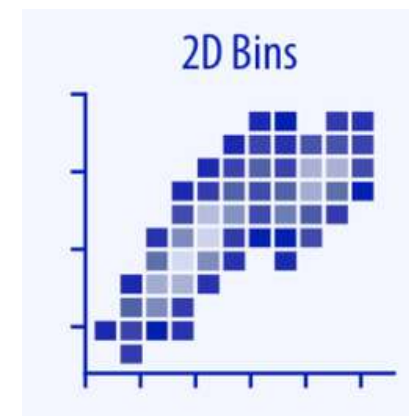
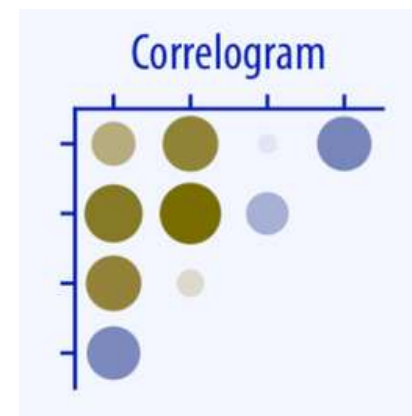
- Scatter plot
- Bubble chart
- Line graph
- Correlogram
- Smooth trend line



What they reveal: correlation, trends, clusters, patterns, outliers

Example:

- Height vs Weight
- Study hours vs Marks
- Temperature over time



Visualization of Data, Understanding and Interpretation

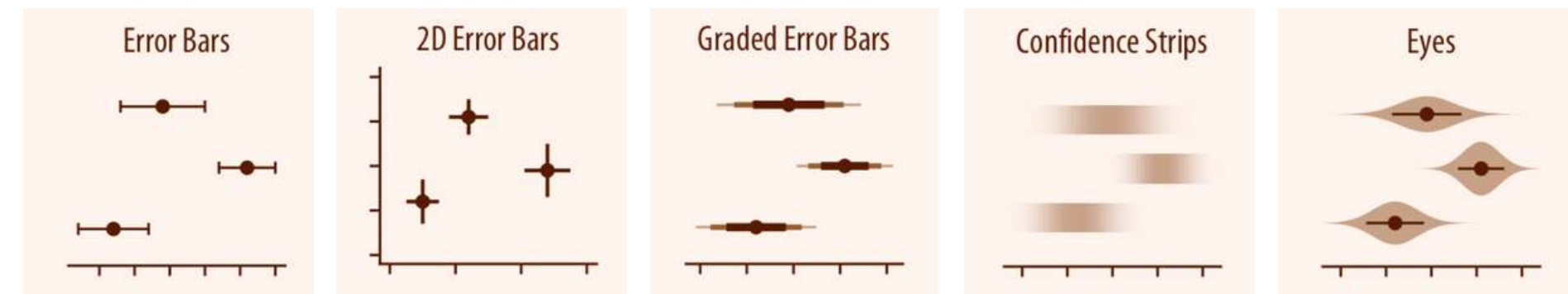
Visualizing Uncertainty

(Sec 5.6)

When to Use: To communicate the confidence or reliability of measurements and predictions.

Common Charts:

- Error bars
- Confidence intervals
- Confidence bands
- Quantile dot plots
- Half-eye plots
- Fitted draws

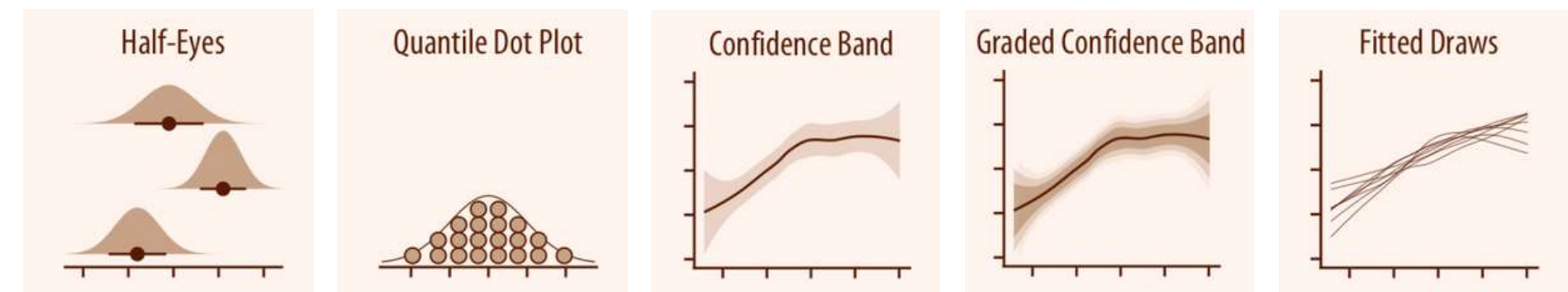


Why it matters:

- Showing only a single value can be misleading.
- Visualizing uncertainty helps viewers understand how reliable an estimate is.

Example:

- Weather forecast ranges
- Polling margins of error
- Machine learning prediction intervals



Visualization of Data, Understanding and Interpretation

Visualizing Distributions: In Depth

(TB 1, Ch7)



What is a distribution?
A distribution describes how the values of a variable are spread across a dataset.

- Why Visualize Distributions?**
- Understand how data is distributed.
 - Identify common value ranges, spread, and patterns.
 - Detect unusual observations or gaps in the data.
 - Support better data analysis and decision-making.

Example: Titanic Passenger Ages

- Dataset contains approximately 1300 passengers.
- Age is known for 756 passengers.
- *Goal:* Visualize the age distribution to understand how passengers are spread across different age groups.

Table 7.1: Numbers of passenger with known age on the Titanic.

Age range	Count	Age range	Count	Age range	Count
0–5	36	31–35	76	61–65	16
6–10	19	36–40	74	66–70	3
11–15	18	41–45	54	71–75	3
16–20	99	46–50	50		
21–25	139	51–55	26		
26–30	121	56–60	22		

Visualization of Data, Understanding and Interpretation

Histograms: Visualizing a Distribution

(Sec 7.1)

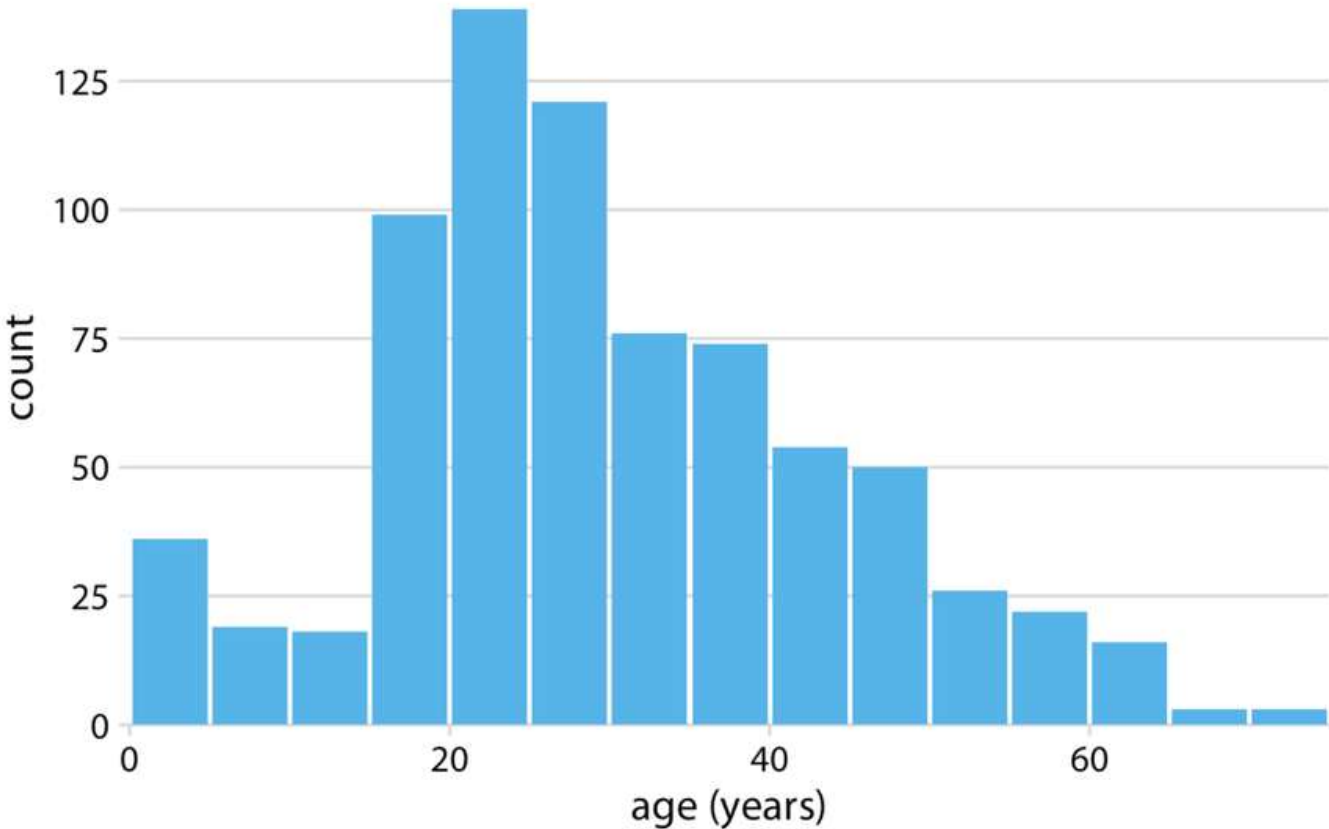
A histogram groups continuous data into bins (intervals) and displays:

- X-axis: Value ranges (Age)
- Y-axis: Number of observations (Count)
- Each bar represents the frequency within an interval.

Example: Age bins [0-5, 6-10, 11-15, ..., 71-75]
These counts are converted into bars to visualize the age distribution.

Table 7.1: Numbers of passenger with known age on the Titanic.

Age range	Count	Age range	Count	Age range	Count
0–5	36	31–35	76	61–65	16
6–10	19	36–40	74	66–70	3
11–15	18	41–45	54	71–75	3
16–20	99	46–50	50		
21–25	139	51–55	26		
26–30	121	56–60	22		



Visualization of Data, Understanding and Interpretation

Choosing the Right Bin Width

(Sec 7.1)



The appearance of a histogram depends on the chosen bin width.

Bin Width	Effect
Too Small	Very noisy, too many peaks
Appropriate	Shows overall trend very clearly
Too Large	Important patterns disappear

For the Titanic age distribution:

- 1-year bins: Too much detail; difficult to identify overall trends.
- 3–5-year bins: Clearly reveal the distribution while preserving important features.
- 15-year bins: Oversimplify the data and hide meaningful patterns.

💡 *Always try multiple bin widths before deciding on a histogram.*

Visualization of Data, Understanding and Interpretation

Choosing the Right Bin Width

(Sec 7.1)

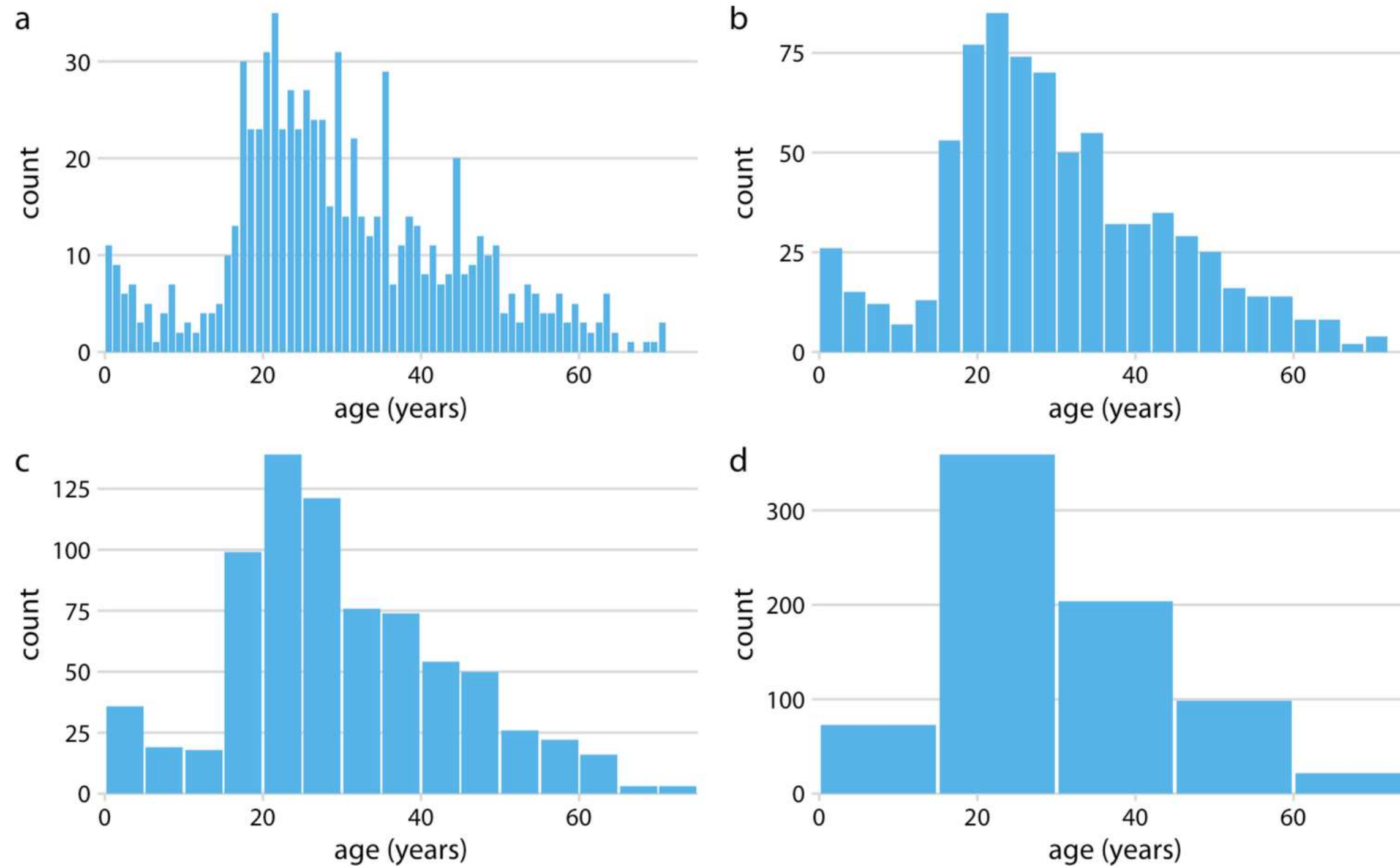


Figure 7.2: Histograms depend on the chosen bin width. Here, the same age distribution of Titanic passengers is shown with four different bin widths: (a) one year; (b) three years; (c) five years; (d) fifteen years.

What is a density plot?

A density plot estimates the underlying probability distribution using a smooth curve.

Instead of grouping data into bins (like a histogram), it estimates a *continuous probability density curve*.

Key Characteristics

- No fixed bins, produces a smooth representation of the data, useful for identifying the overall shape of a distribution.
- The area under the curve equals 1 (represents total probability).

Why use a Density Plot?

- Smoother than histograms.
- Easier to compare multiple distributions.
- Less affected by arbitrary bin boundaries.

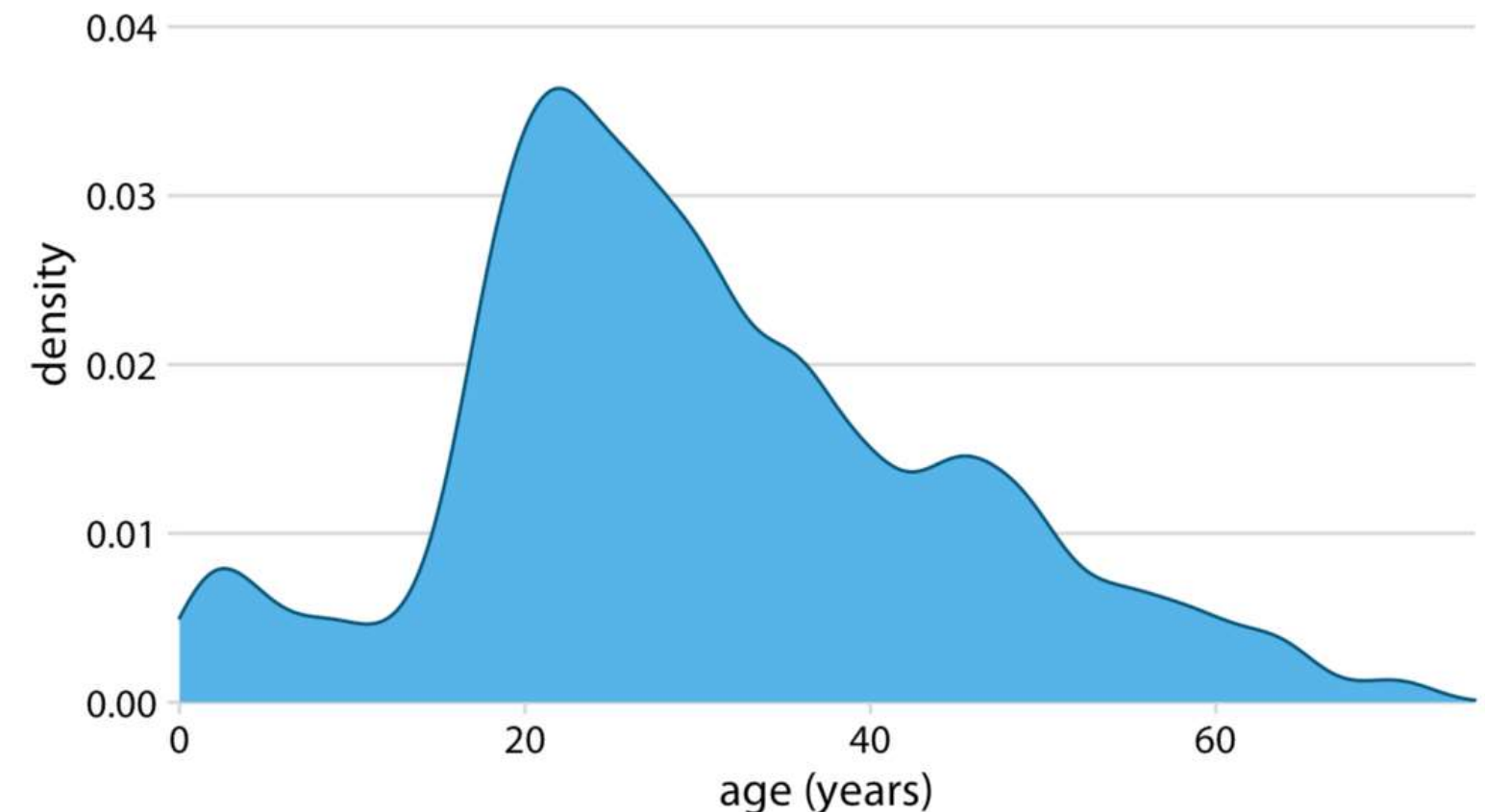


Figure 7.3: Kernel density estimate of the age distribution of passengers on the Titanic. The height of the curve is scaled such that the area under the curve equals one. The density estimate was performed with a Gaussian kernel and a bandwidth of 2.

💡 *A density plot doesn't show the exact number of observations in each interval. Instead, it estimates where data values are more or less likely to occur.*

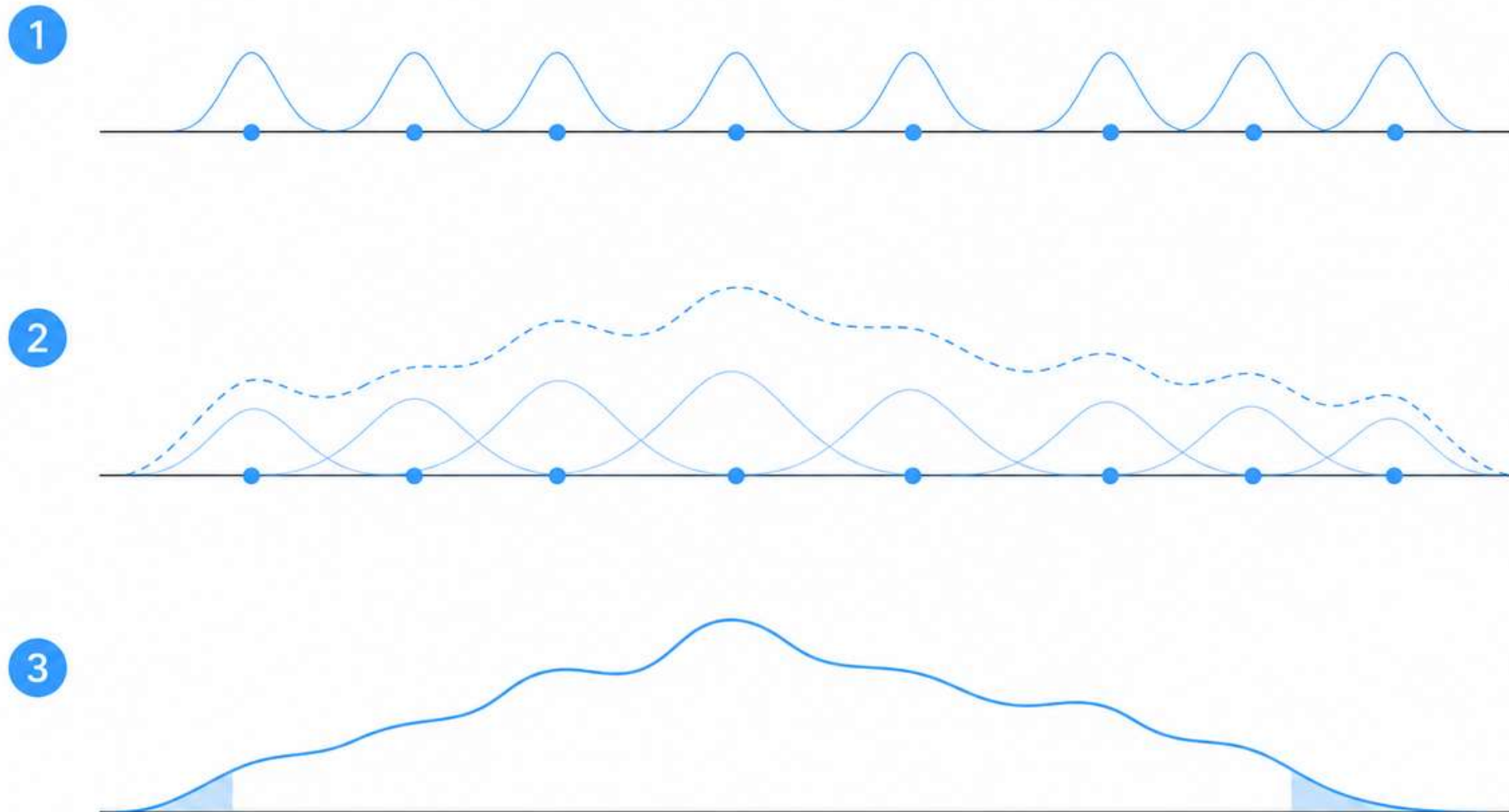
Visualization of Data, Understanding and Interpretation

Kernel Density Estimation

(Sec 7.1)

Kernel Density Estimation (KDE) builds the density curve in three simple steps:

1. Place a small smooth curve (kernel) on every data point.
2. Add all the kernels together.
3. Normalize the result so the total area under the curve equals 1.



Visualization of Data, Understanding and Interpretation

Using Gaussian Kernels for KDE (Sec 7.1)

A Gaussian kernel places a small *bell-shaped (normal distribution) curve* on each data point.

Why Gaussian?

- Produces smooth, continuous curves.
- Handles noise well.
- Most widely used kernel in practice.

Bandwidth

The bandwidth controls smoothness.

- small bandwidth: very wavy/noisy
- large bandwidth: too smooth, hides important features

The kernel also affects the curve shape:

- gaussian: smooth curves
- rectangular: step-like appearance

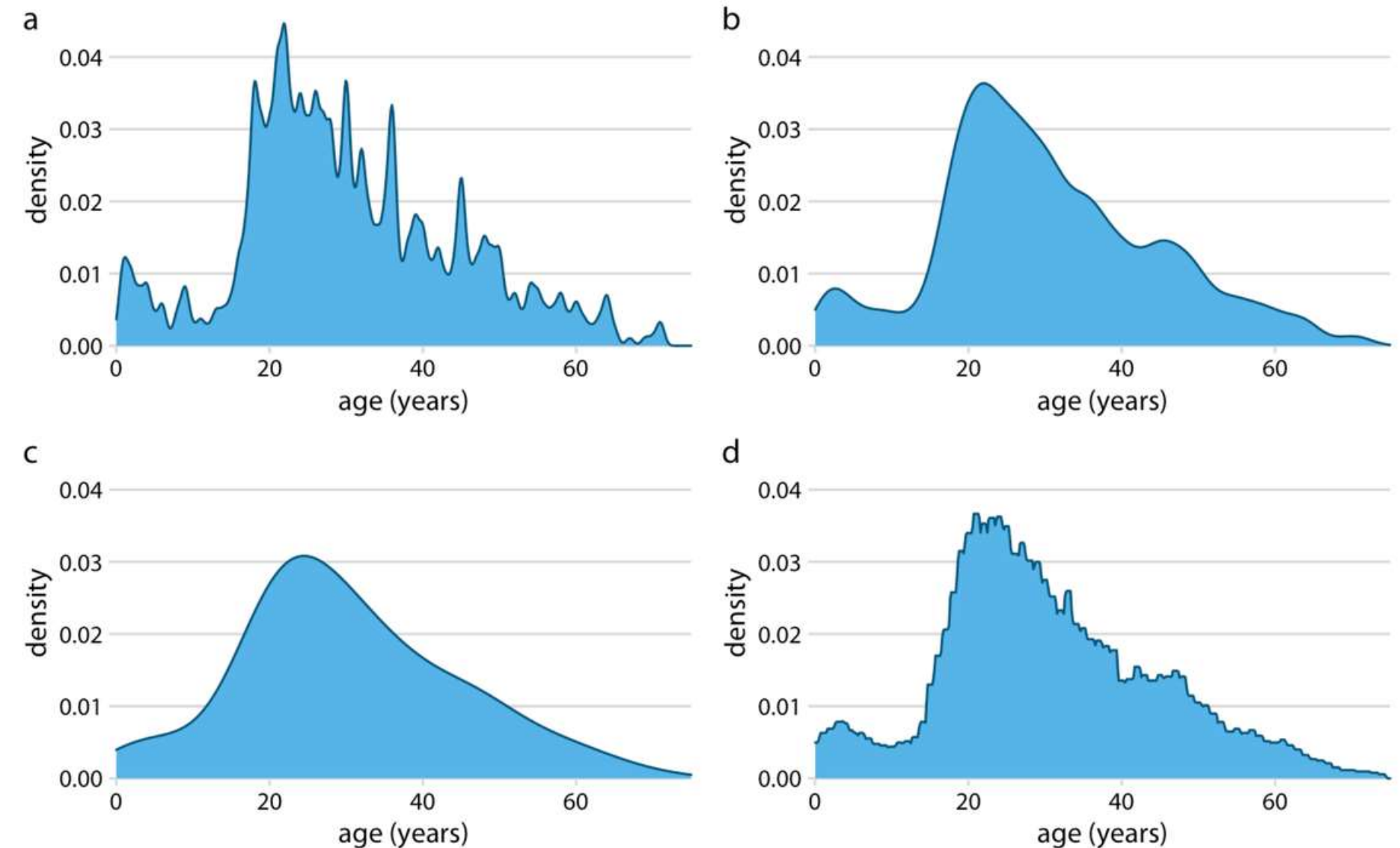


Figure 7.4: Kernel density estimates depend on the chosen kernel and bandwidth. Here, the same age distribution of Titanic passengers is shown for four different combinations of these parameters: (a) Gaussian kernel, bandwidth = 0.5; (b) Gaussian kernel, bandwidth = 2; (c) Gaussian kernel, bandwidth = 5; (d) Rectangular kernel, bandwidth = 2.

- 💡 *Bandwidth plays the same role in KDE that bin width plays in histograms.*
- 💡 *For larger datasets, kernel choice matters less than selecting a reasonable bandwidth.*

Visualization of Data, Understanding and Interpretation

Limitations & Best Practices (Sec 7.1)

Pitfalls of Density Plots

Density estimation may predict values where no data can exist.

Example:

For age data:

- Negative ages are impossible.
- A poorly constrained density estimate may extend below 0 years.

Best Practices

- Explore different bandwidths
- Ensure the curve does not extend into impossible values
- Validate that the visualization matches the actual data

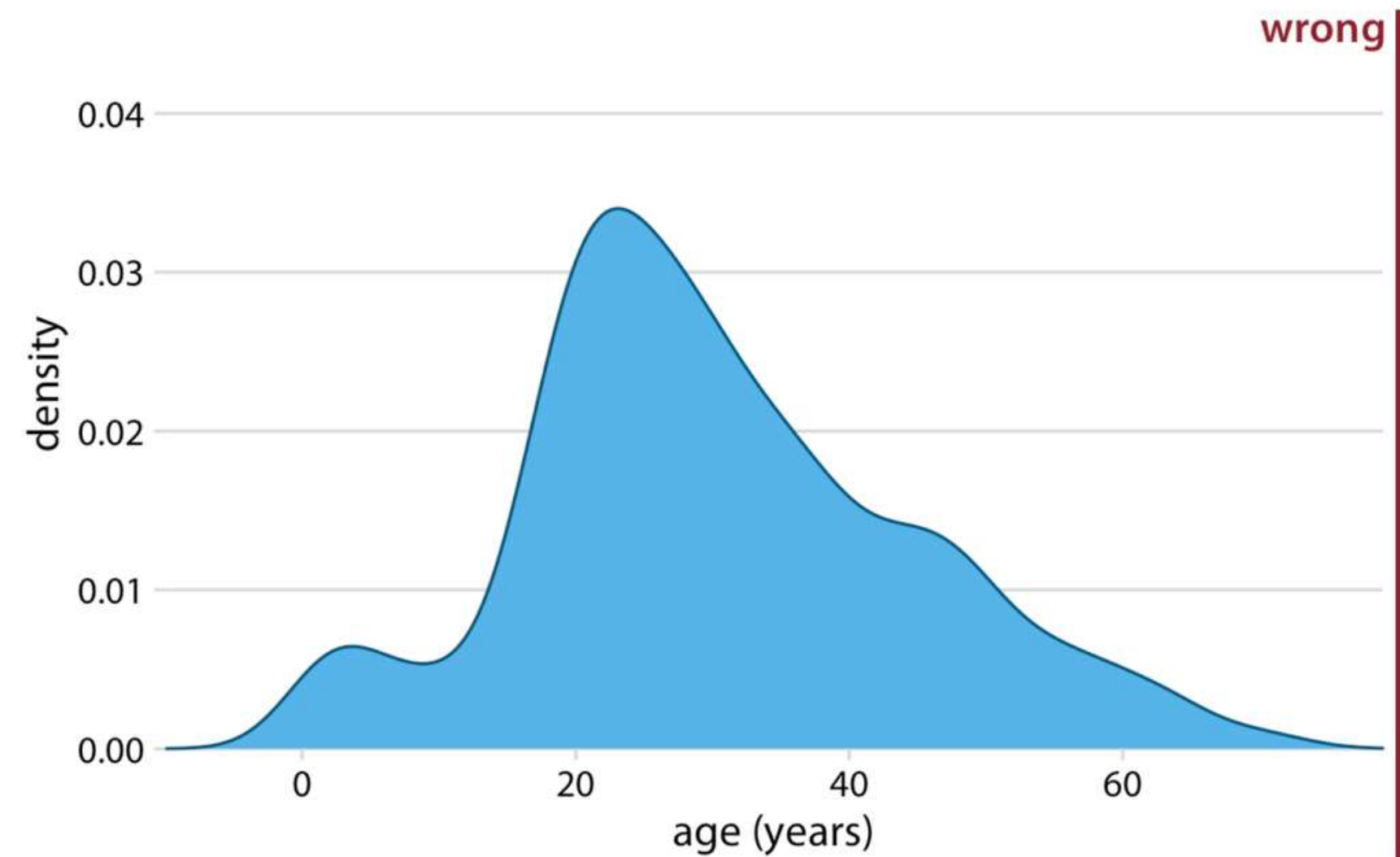


Figure 7.5: Kernel density estimates can extend the tails of the distribution into areas where no data exist and no data are even possible. Here, the density estimate has been allowed to extend into the negative age range. This is clearly nonsensical and should be avoided.

Visualization of Data, Understanding and Interpretation

Histogram vs Density Plot

(Sec 7.1)



Histogram	Density Plot
Uses bins	Uses a smooth curve
Shows counts/frequencies	Shows estimated probability density
Sensitive to bin width	Sensitive to bandwidth
Easy to interpret	Better for comparing multiple distributions

Visualization of Data, Understanding and Interpretation

Visualizing Proportions: In Depth

(TB 1, Ch10)



What are proportions?

- Proportions describe how a whole is divided into individual parts.
- Visualizing proportions helps communicate the relative contribution of each category to the total.
- Common applications include market share, election results, budget allocation, and population composition.

💡 *The most effective visualization depends on the dataset and the message that needs to be communicated*

Visualization	Primary Purpose
Pie Chart	Shows the contribution of each category to the whole
Side-by-side Bar Chart	Enables accurate comparison of category values
Stacked Bar Chart	Compares proportions across multiple groups or conditions

Visualization of Data, Understanding and Interpretation

Pie Charts & Bar Charts (Sec 10.1)

Pie Charts

Advantages:

- Pie charts clearly represent parts of a whole.
- They are effective for displaying simple fractions and a small number of categories.

Limitations:

- Comparing slices of similar size is difficult.
- They become difficult to interpret when many categories are included.

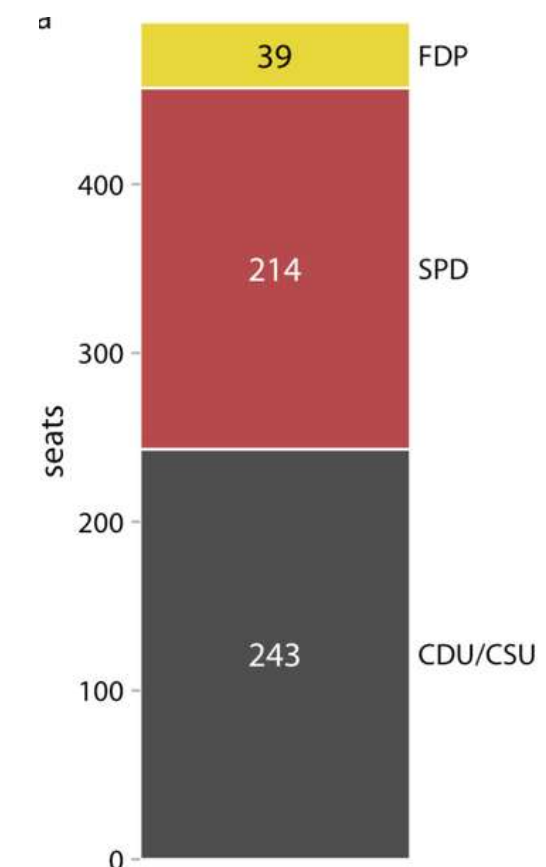
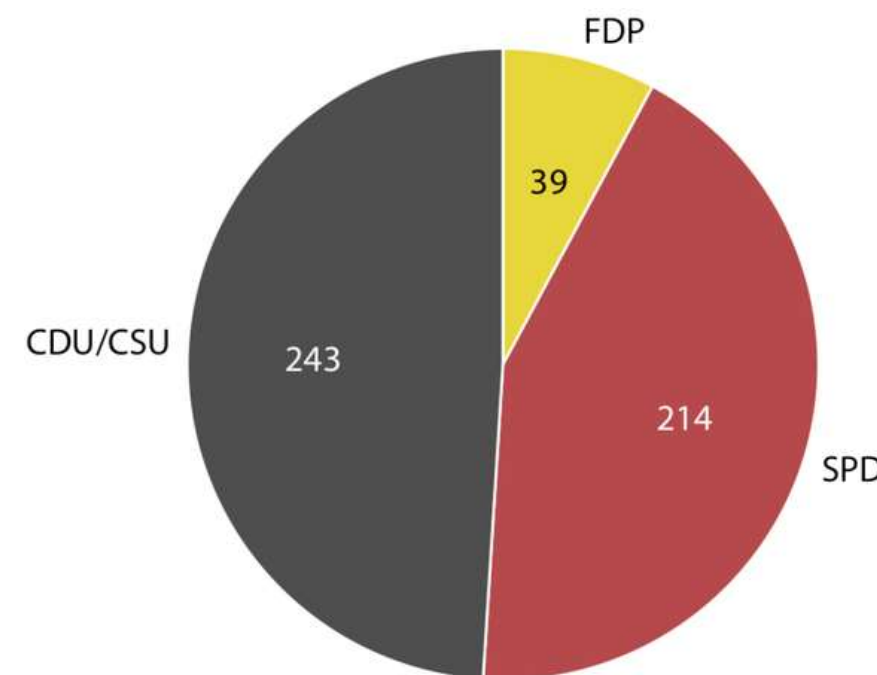
Stacked Bar Charts

Advantages:

- Stacked bar charts compare proportions across multiple groups or time periods.

Limitations:

- Individual segments are more difficult to compare because they do not share a common baseline.

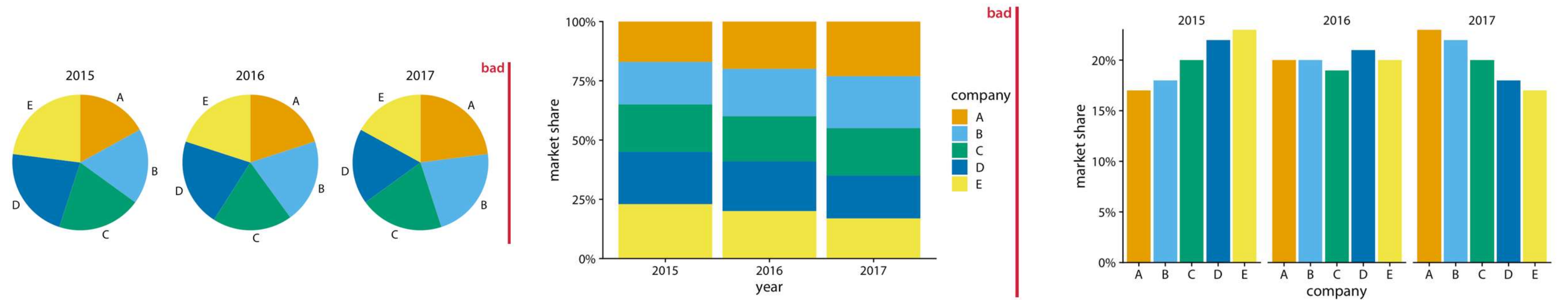


Visualization of Data, Understanding and Interpretation

Pie Charts & Bar Charts

(Sec 10.1)

Pie Charts	Stacked Bar Charts	Side-by-side Bar Charts
Make it difficult to compare category values across years.	Reveal overall trends but make middle segments difficult to compare because they lack a common baseline.	Place every category on the same baseline, making comparisons accurate and straightforward.



Visualization of Data, Understanding and Interpretation

MCQs

1. Which visualization technique groups continuous data into intervals (bins) to show the frequency of observations?

- A. Scatter Plot
- B. Density Plot
- C. Histogram
- D. Box Plot

Visualization of Data, Understanding and Interpretation

MCQs

1. Which visualization technique groups continuous data into intervals (bins) to show the frequency of observations?

- A. Scatter Plot
- B. Density Plot
- C. Histogram
- D. Box Plot

Answer: C

Visualization of Data, Understanding and Interpretation

MCQs

2. A histogram created using a very small bin width is most likely to:

- A. Hide important patterns by combining observations.
- B. Produce a smoother representation of the data.
- C. Show many small fluctuations, making the distribution appear noisy.
- D. Estimate the underlying probability density.

Visualization of Data, Understanding and Interpretation

MCQs

2. A histogram created using a very small bin width is most likely to:

- A. Hide important patterns by combining observations.
- B. Produce a smoother representation of the data.
- C. Show many small fluctuations, making the distribution appear noisy.
- D. Estimate the underlying probability density.

Answer: C

Visualization of Data, Understanding and Interpretation

MCQs

3. In Kernel Density Estimation (KDE), how is the final density curve constructed?

- A. By placing a kernel on each data point and combining their contributions.
- B. By fitting a single Gaussian curve to the entire dataset.
- C. By counting the number of observations in each bin.
- D. By connecting consecutive data points with straight lines.

Visualization of Data, Understanding and Interpretation

MCQs

3. In Kernel Density Estimation (KDE), how is the final density curve constructed?

- A. By placing a kernel on each data point and combining their contributions.
- B. By fitting a single Gaussian curve to the entire dataset.
- C. By counting the number of observations in each bin.
- D. By connecting consecutive data points with straight lines.

Answer: A

4. Which statement correctly describes the role of the bandwidth in a density plot?

- A. It determines the number of bins in the histogram.
- B. It controls the width of each kernel and the smoothness of the density estimate.
- C. It changes the total area under the density curve.
- D. It determines the number of observations included in the dataset.

4. Which statement correctly describes the role of the bandwidth in a density plot?

- A. It determines the number of bins in the histogram.
- B. It controls the width of each kernel and the smoothness of the density estimate.
- C. It changes the total area under the density curve.
- D. It determines the number of observations included in the dataset.

Answer: B

5. A density plot of passenger ages extends slightly below zero years. What is the most appropriate interpretation?

- A. The dataset contains passengers with negative ages.
- B. The Gaussian kernel is mathematically incorrect.
- C. Kernel Density Estimation can assign small probabilities to regions where no observations are possible.
- D. The histogram would produce the same result.

5. A density plot of passenger ages extends slightly below zero years. What is the most appropriate interpretation?

- A. The dataset contains passengers with negative ages.
- B. The Gaussian kernel is mathematically incorrect.
- C. Kernel Density Estimation can assign small probabilities to regions where no observations are possible.
- D. The histogram would produce the same result.

Answer: C



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Thank You



Visualization of Data, Understanding and Interpretation

Unit 1 : Foundations of Visualization, Data Interpretation

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

Dr.Prajwala TR (Course Instructor)

Dept. of CSE, PES University, Bangalore

Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.



Visualization of Data, Understanding and Interpretation

Lecture Overview

- Visualizing multi-variable associations using basic scatter plots, color-coded subgroups, bubble charts, and pairwise scatter plot matrices.
- Measuring relationship strength and direction with the linear correlation coefficient (r) and interpreting dense correlation patterns using correlograms.
- Designing effective time-series visualizations using guidelines for markers, line charts, and area charts, while avoiding clutter in multiple datasets.
- Applying and comparing data smoothing techniques; Moving Averages, LOESS, and Splines to reveal underlying trends by balancing noise, lag, and overfitting.

Visualization of Data, Understanding and Interpretation

Visualizing Associations with Scatter Plots (TB 1, Ch12)



Why Study Relationships Between Variables?

Many datasets contain multiple quantitative variables, and we often want to understand how they are related.

Example: height vs weight, age vs income, study hours vs exam score

Throughout this lecture, we'll use measurements from a dataset, including:

- Body Mass
- Head Length
- Skull Size
- Sex

Goal: Identify relationships between these variables.

Visualization of Data, Understanding and Interpretation

Scatter Plots: The Basics

(Sec 12.1)

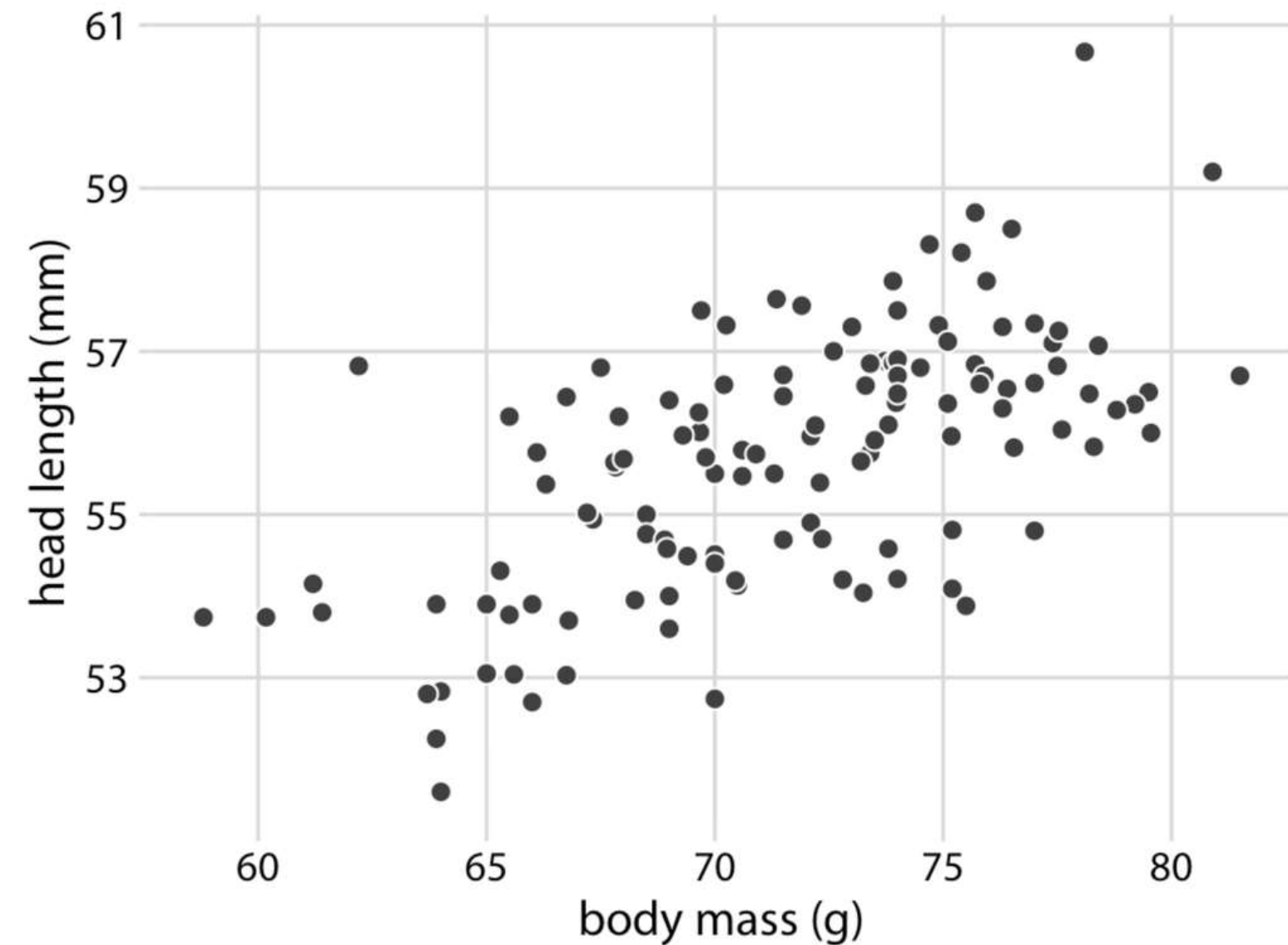
A scatter plot displays the relationship between two quantitative variables.

Components:

- X-axis: Independent variable
- Y-axis: Dependent variable
- Each dot represents one observation

What can scatter plots reveal?

- Positive or negative relationships
- Strength of association
- Clusters
- Outliers



Example: Body Mass (x-axis) vs Head Length (y-axis)

The points form a cloud showing that heavier birds generally have longer heads.

Visualization of Data, Understanding and Interpretation

Interpreting Scatter Plots

(Sec 12.1)

When analyzing a scatter plot, observe:

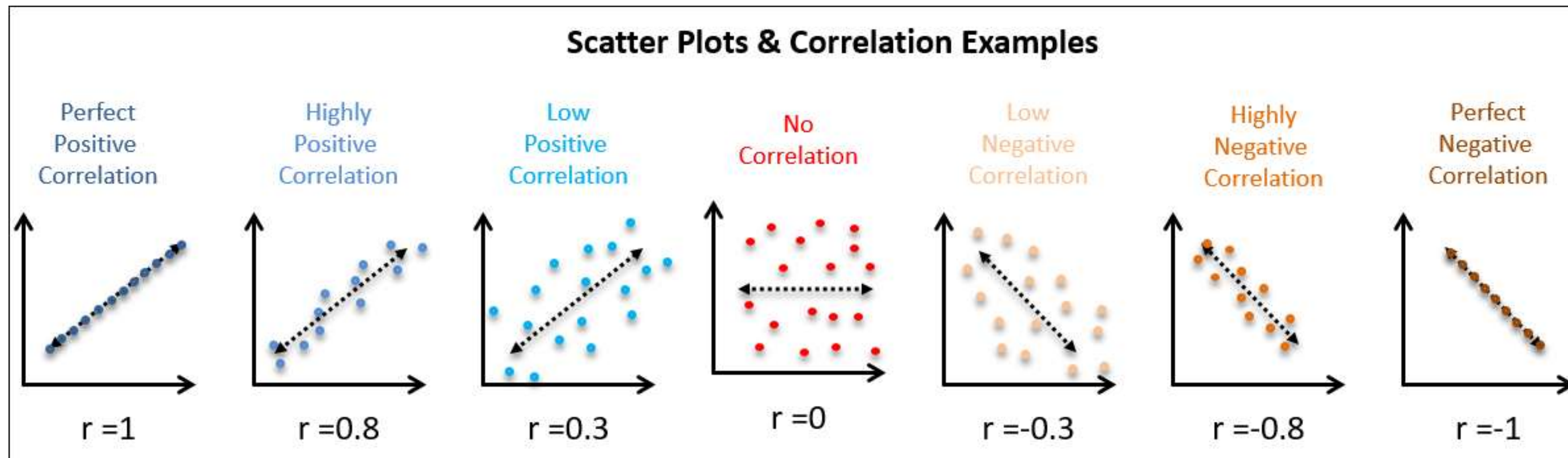
Direction: Positive, Negative, No relationship

Strength: Weak, Moderate, Strong

Patterns: Clusters, Outliers, Trends

Example observation from previous scatter plot:

Moderate positive relationship, larger body mass generally corresponds to larger head length, some variation still exists



Visualization of Data, Understanding and Interpretation

Adding a Third Variable Using Color (Sec 12.1)

Color Reveals Hidden Patterns

Scatter plots can encode an additional variable using color.

In this example:

- Orange: Female birds
- Blue: Male birds

What does color reveal?

- Males generally have longer heads
- Females tend to have lower body mass
- The overall trend is partly influenced by sex

Why use color?

- Compare groups
- Detect subgroup patterns
- Improve interpretation without clutter

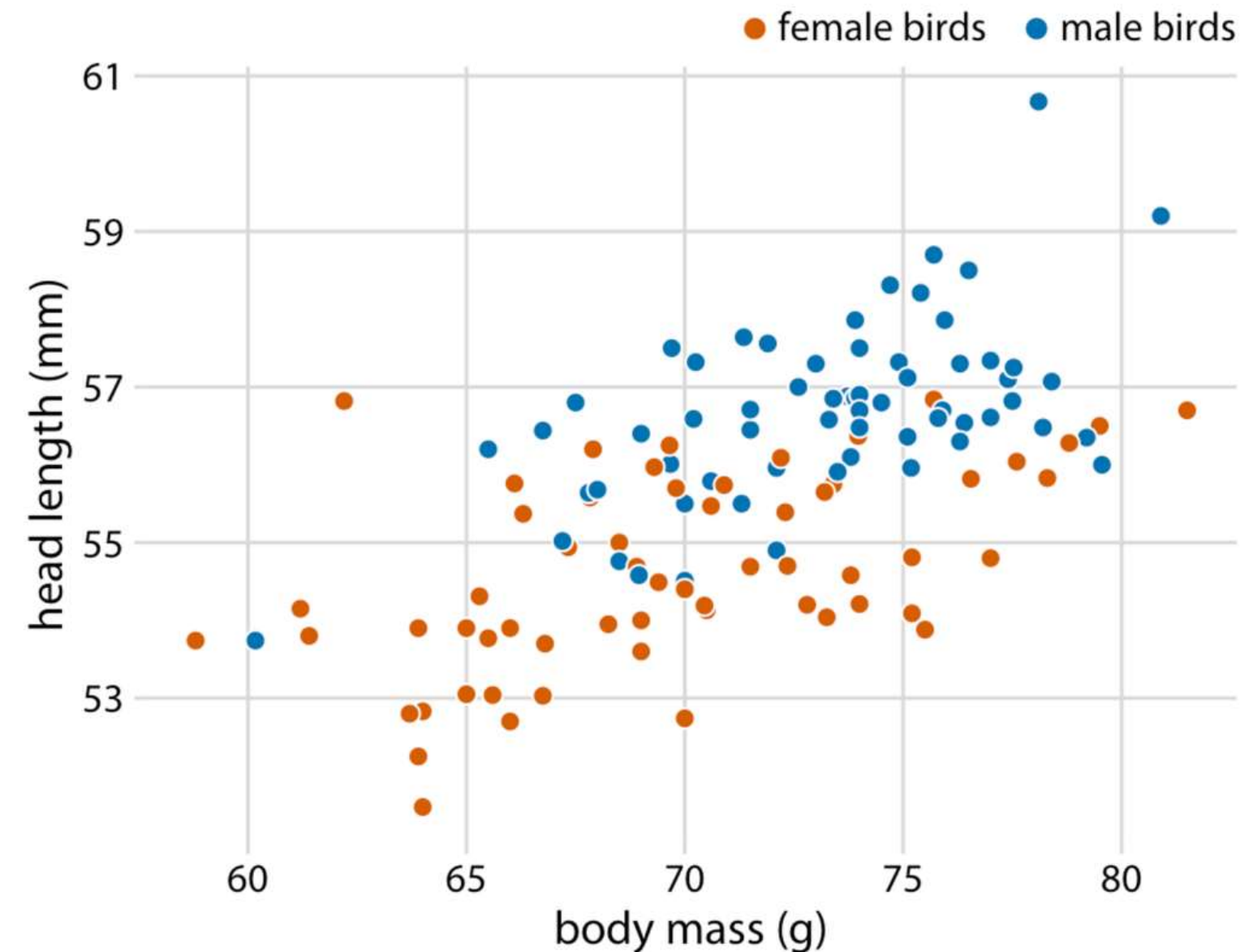


Figure 12.2: Head length versus body mass for 123 blue jays. The birds' sex is indicated by color. At the same body mass, male birds tend to have longer heads (and specifically, longer bills) than female birds. Data source: Keith Tarvin, Oberlin College

Visualization of Data, Understanding and Interpretation

Bubble Charts (Sec 12.1)

Representing Four Variables

When x-position, y-position, and color are already used, another variable can be represented using *marker size*. This allows us to display four variables simultaneously.

Example

- X-axis: Body Mass
- Y-axis: Head Length
- Color: Bird Sex
- Bubble Size: Skull Size

Limitations

- Size differences are harder to judge than position.
- Small differences in values are difficult to perceive.
- Can become visually cluttered.

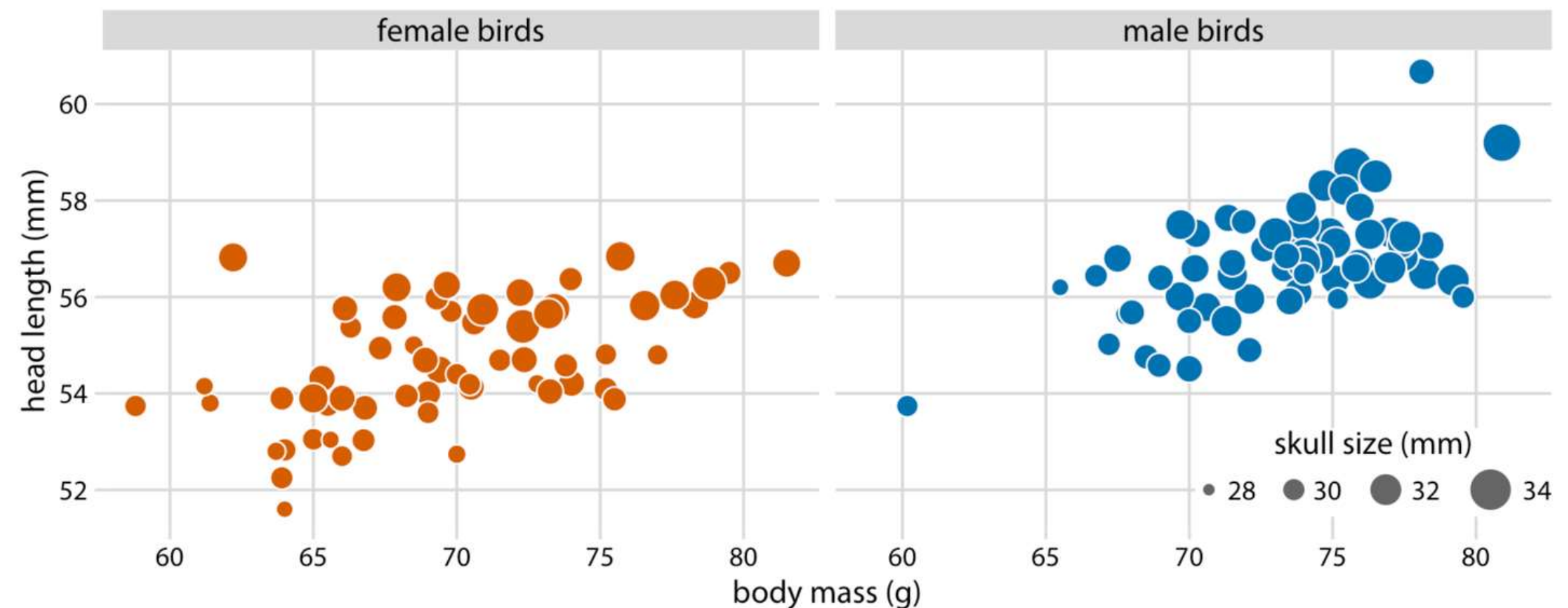


Figure 12.3: Head length versus body mass for 123 blue jays. The birds' sex is indicated by color, and the birds' skull size by symbol size. Head-length measurements include the length of the bill while skull-size measurements do not. Head length and skull size tend to be correlated, but there are some birds with unusually long or short bills given their skull size. Data source: Keith Tarvin, Oberlin College

Visualization of Data, Understanding and Interpretation

Scatter Plot Matrix (Sec 12.1)

Comparing Many Variables at Once

Instead of using bubble sizes, we can display all pairwise relationships using a scatter plot matrix.

Benefits

- Compare every pair of variables
- Easier to identify correlations
- More effective than bubble charts for multiple quantitative variables

From the example,

- Skull size and body mass show a clear relationship.
- Male and female birds form distinct groups.
- Relationships can be compared across all variables.

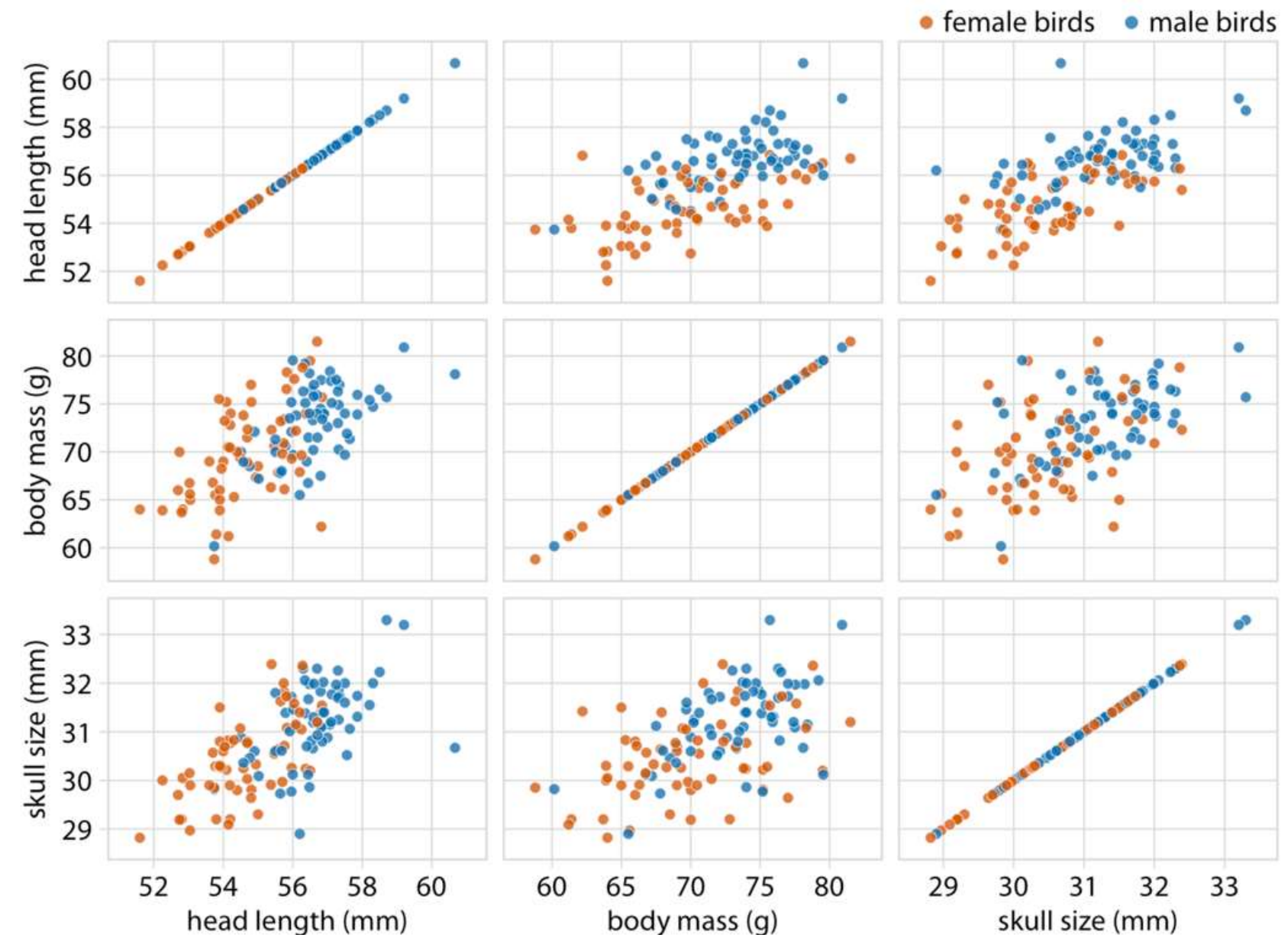


Figure 12.4: All-against-all scatter plot matrix of head length, body mass, and skull size, for 123 blue jays. This figure shows the exact same data as Figure 12.2. However, because we are better at judging position than symbol size, correlations between skull size and the other two variables are easier to perceive in the pairwise scatter plots than in Figure 12.2. Data source: Keith Tarvin, Oberlin College

Visualization of Data, Understanding and Interpretation

Choosing the Right Visualization

(Sec 12.1)



Visualization	Best Used For
Scatter Plot	Relationship between two variables
Colored Scatter Plot	Compare groups while showing relationships
Bubble Chart	Show four variables simultaneously
Scatter Plot Matrix	Compare many quantitative variables

Key Considerations:

- Position is easier to interpret than size.
- Avoid encoding too many variables in a single plot.
- Choose the simplest visualization that answers the question.

Why Correlograms?

When analyzing many quantitative variables, scatter plot matrices become crowded. Instead of visualizing every pair of variables, we can summarize their relationships using the *correlation coefficient* (r).

Correlation Coefficient (r)

- Measures the strength and direction of a linear relationship.
- Values range from -1 to $+1$.
- $+1$: perfect positive correlation, 0 : no correlation, -1 : perfect negative correlation

$$r = \frac{\sum_i (x_i - \bar{x}) (y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 (y_i - \bar{y})^2}}$$

Visualization of Data, Understanding and Interpretation

Correlation and Correlograms

(Sec 12.2)

Example:

Student	Hours Studied	Score
A	2	50
B	4	70
C	6	90

$$\bar{x} = \frac{2 + 4 + 6}{3} = 4$$

$$\bar{y} = \frac{50 + 70 + 90}{3} = 70$$

Student	x	y	(x- $\bar{x}$)	(y- $\bar{y}$)	(x- $\bar{x}$)(y- $\bar{y}$)	(x- $\bar{x}$)^2	(y- $\bar{y}$)^2
A	2	45	-2	-20	40	4	400
B	4	60	0	0	0	0	0
C	6	74	2	20	40	4	400
Total:					80	8	800

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$$
$$r = \frac{80}{\sqrt{8}\sqrt{800}} = \frac{80}{\sqrt{6400}}$$
$$= \frac{80}{80} = 1$$

Visualization of Data, Understanding and Interpretation

Correlation and Correlograms (Sec 12.2)

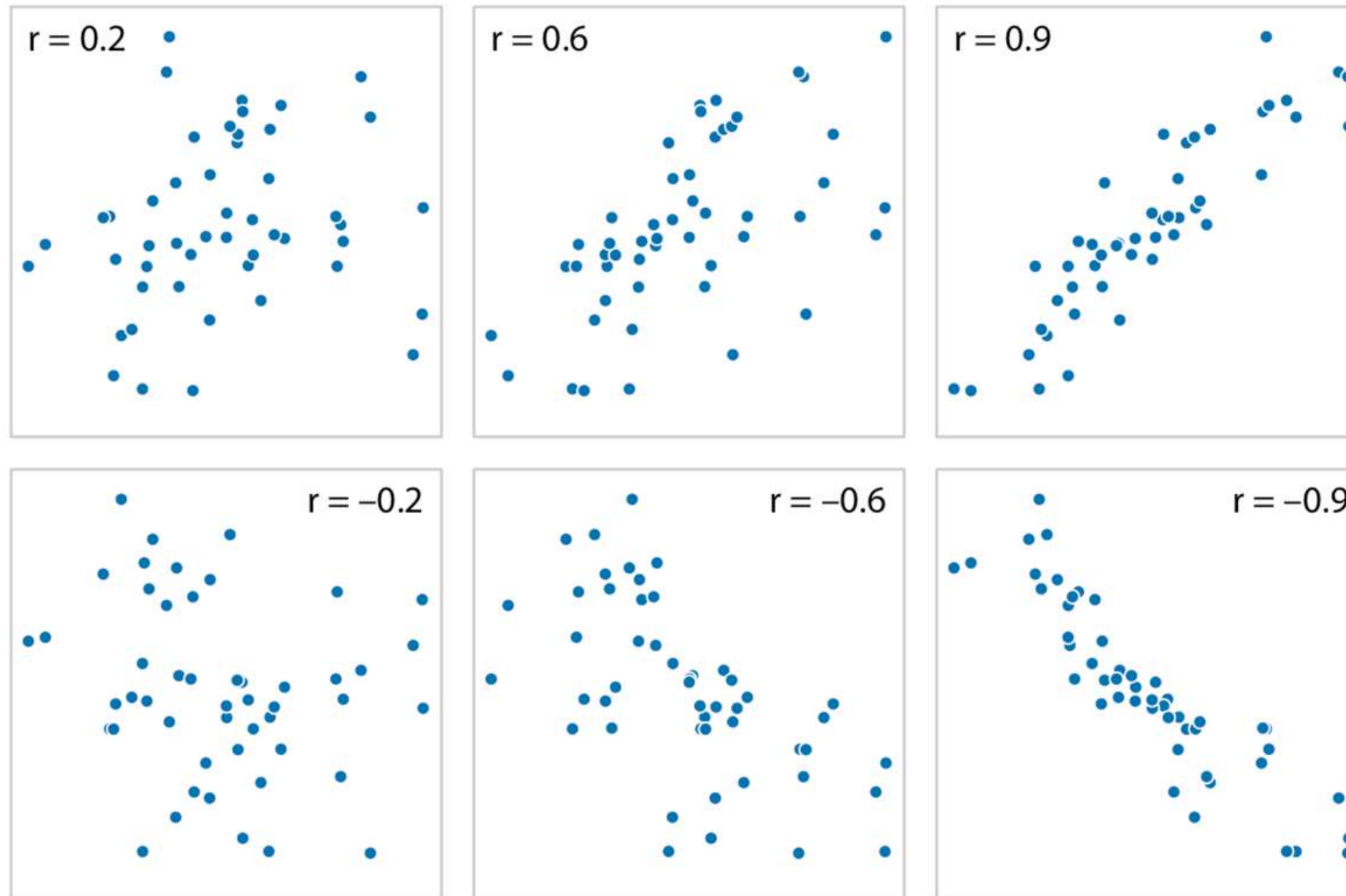


Figure 12.5: Examples of correlations of different magnitude and direction, with associated correlation coefficient r . In both rows, from left to right correlations go from weak to strong. In the top row the correlations are positive (larger values for one quantity are associated with larger values for the other) and in the bottom row they are negative (larger values for one quantity are associated with smaller values for the other). In all six panels, the sets of x and y values are identical, but the pairings between individual x and y values have been reshuffled to generate the specified correlation coefficients.

Visualization of Data, Understanding and Interpretation

Correlograms (Sec 12.2)

Visualizing Many Correlations

A correlogram summarizes pairwise correlations using color (and sometimes size).

Advantages

- Compact summary of many variables
- Easy to spot strong associations

Limitation

- Does not show the original data points.
- Correlations near zero can be difficult to distinguish.

Improved Design

- Represent correlation magnitude using circle size.

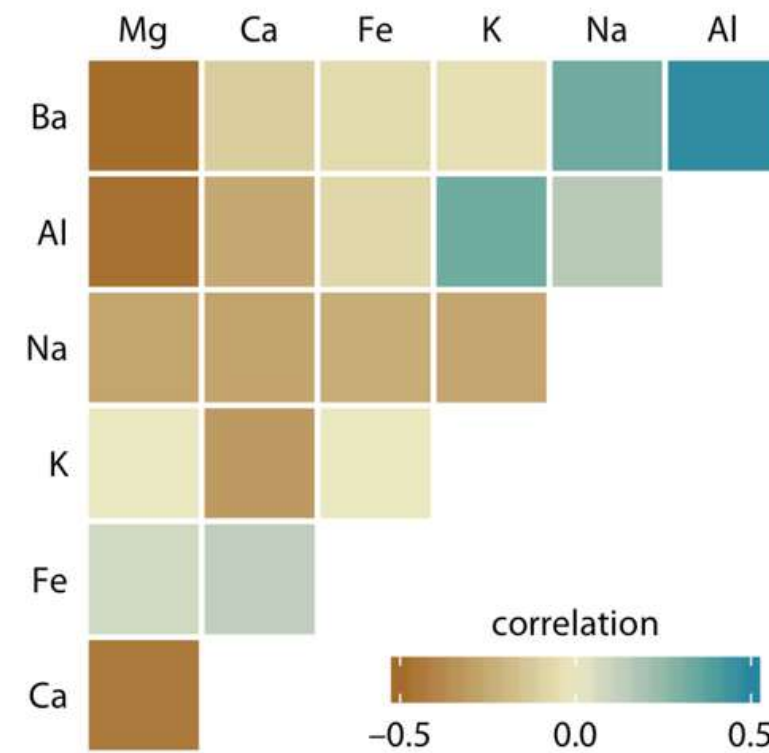


Figure 1:

Color indicates the direction and strength of correlation:

blue = positive, brown = negative, lighter shades = weaker relationships

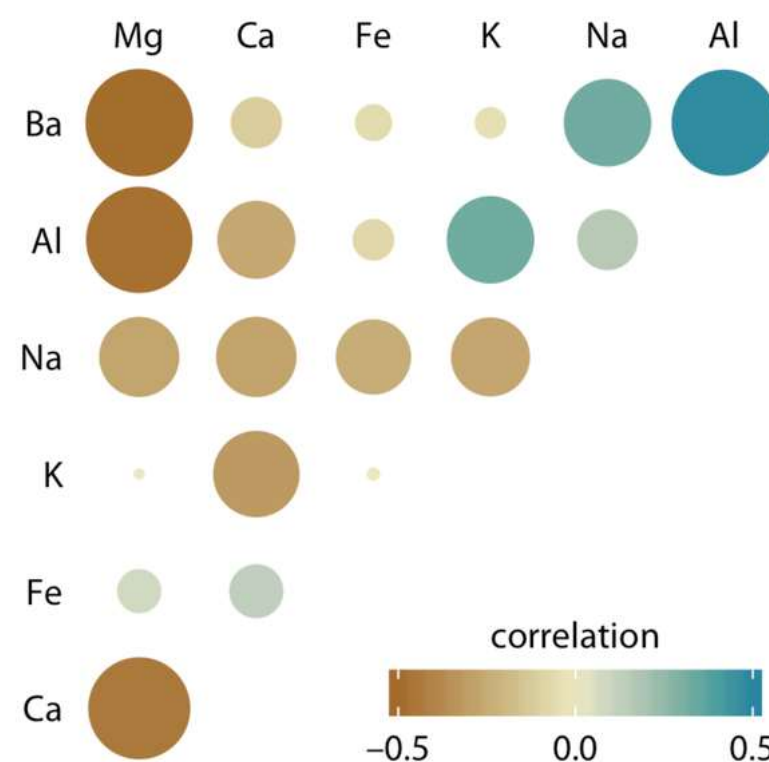


Figure 2:

Both color and circle size represent correlation. Larger circles indicate stronger correlations, while small circles highlight values close to zero.

Visualization of Data, Understanding and Interpretation

Visualizing Time Series: Line Graphs (TB 1, Ch13)



What is a Time Series?

A time series represents observations collected in chronological order. Since time provides a natural sequence, preserving this order is essential for revealing trends, patterns, and changes over time.

Why Use Line Graphs?

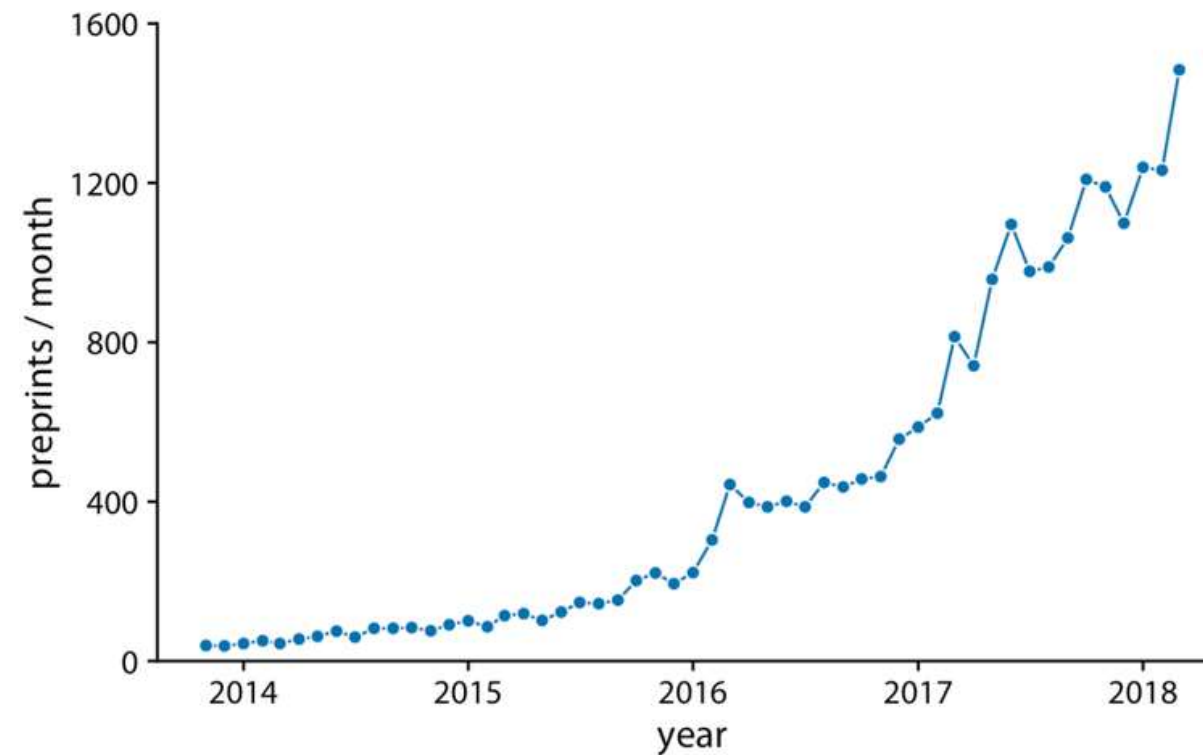
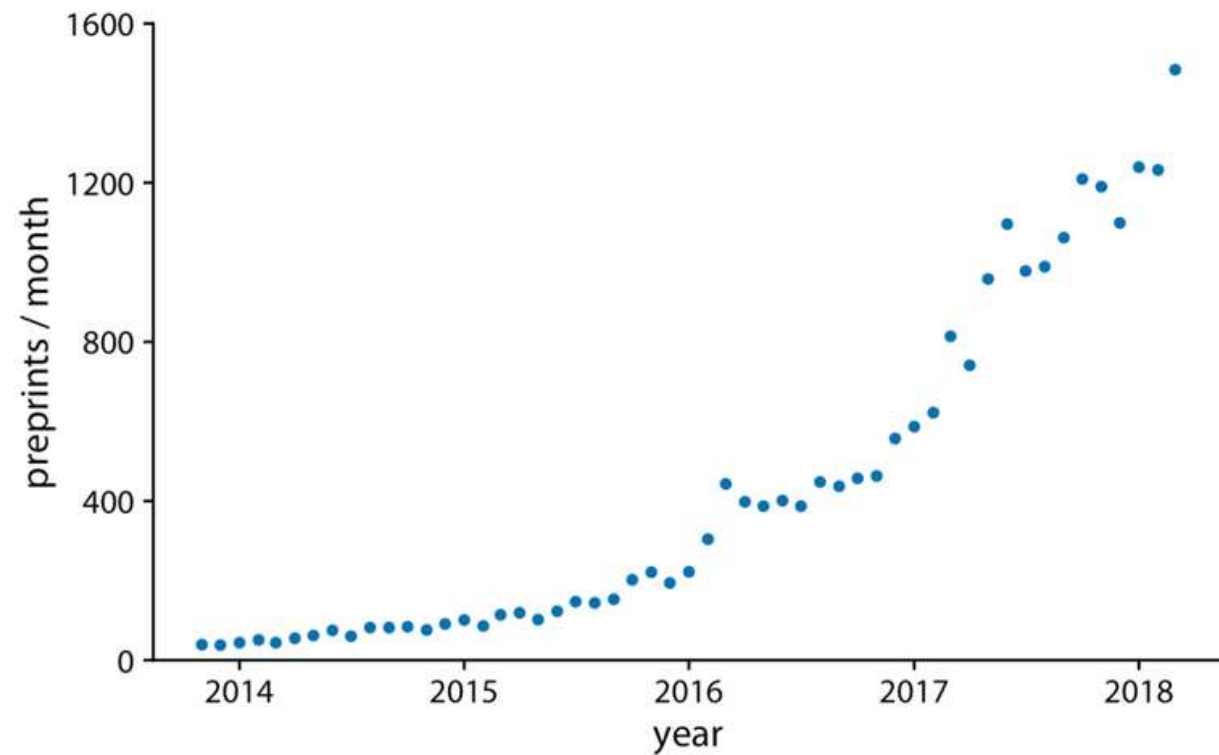
- Line graphs connect consecutive observations to emphasize temporal continuity.
- They make long-term trends, cycles, and sudden changes easier to recognize than disconnected points.
- Lines serve as visual guides that help viewers follow the progression of data over time.
- When many observations are recorded, markers can be removed to reduce clutter and focus attention on the overall trend.

Design Guidelines:

- Use a line with markers when individual observations are important.
- Use a line only when the dataset is dense and the overall trend is the primary focus.
- Use an area chart only when the y-axis begins at zero and emphasizing cumulative magnitude is appropriate.

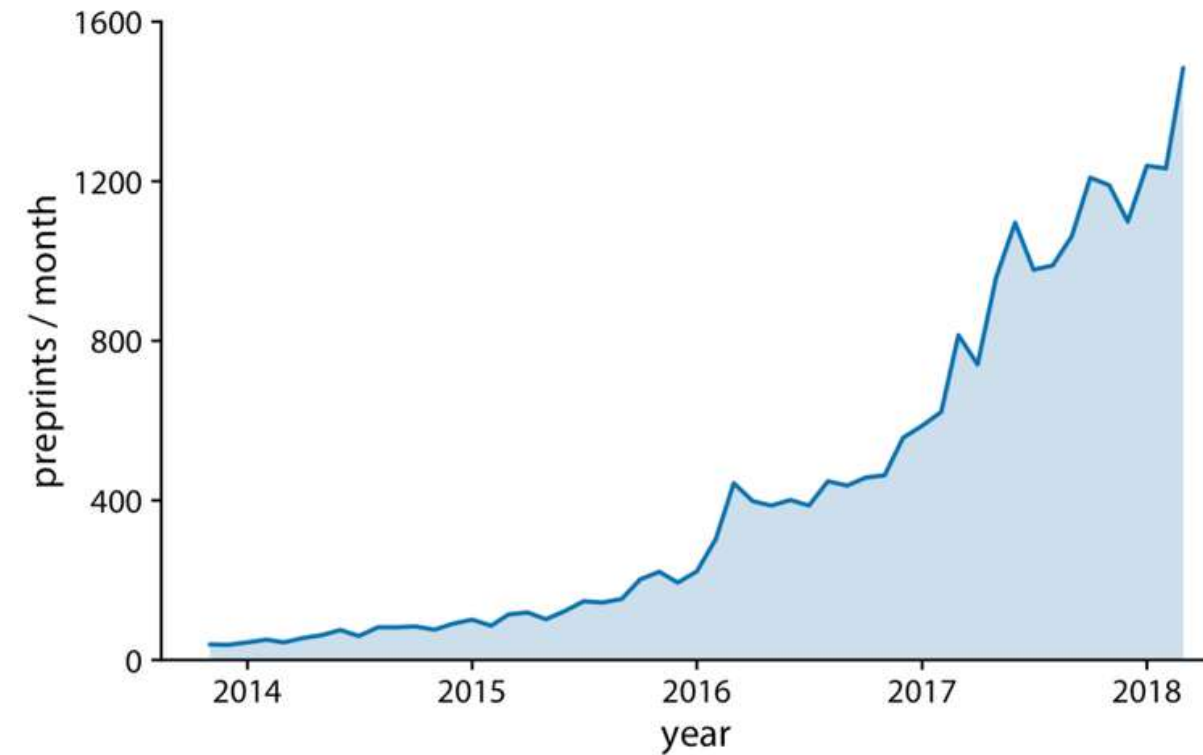
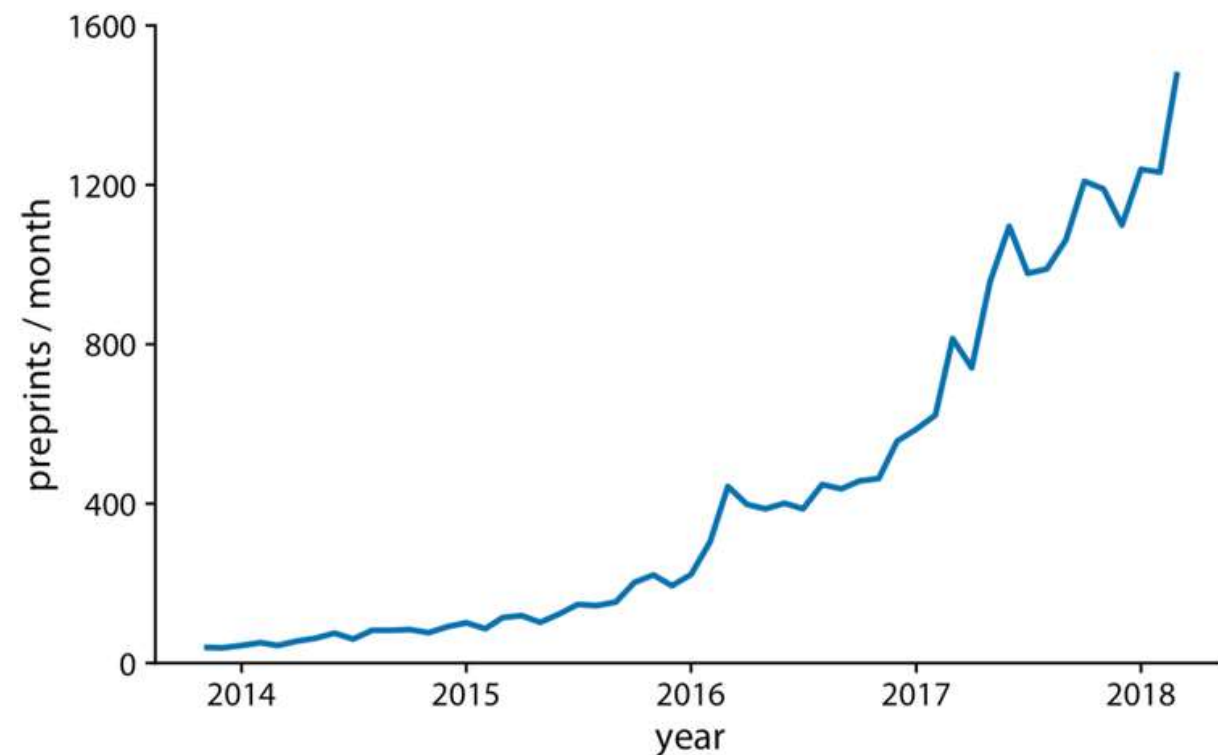
Visualization of Data, Understanding and Interpretation

Visualizing Time Series: Line Graphs (Sec 13.1)



(Top Left) Scatter Plot: Monthly observations shown as individual points. The temporal order exists but is not visually emphasized.

(Top Right) Line Graph with Markers: Connecting observations highlights the chronological order and makes temporal trends easier to follow while preserving individual data points.



(Bottom Left) Line Graph: Removing markers reduces visual clutter and shifts attention from individual observations to the overall trend.

(Bottom Right) Area Chart: Filling the area beneath the line further emphasizes the magnitude and overall trend. This visualization should only be used when the y-axis starts at zero.

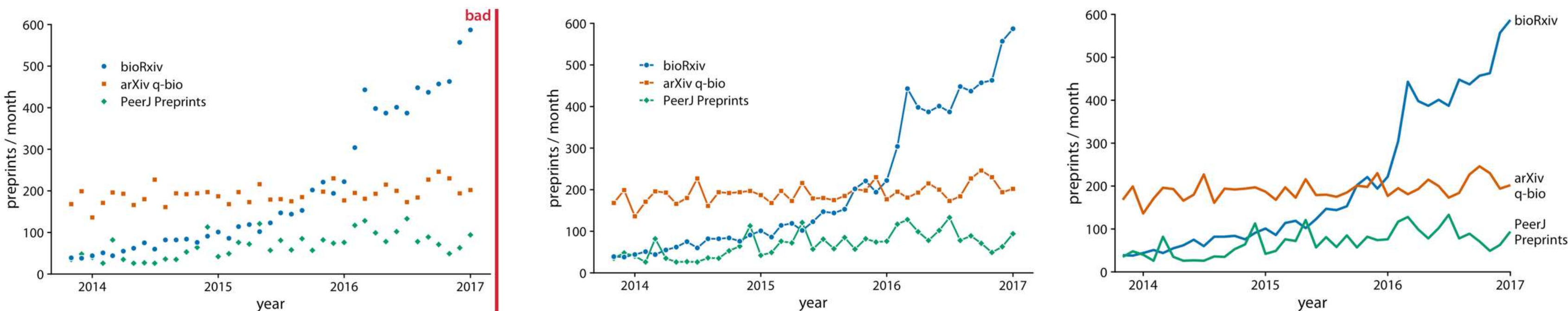
Visualization of Data, Understanding and Interpretation

Multiple Time Series

(Sec 13.2)

Comparing Multiple Time Series: Displaying several time series on one chart requires careful design so that each trend remains distinguishable.

Poor Practice	Recommended Practice
Plotting multiple series as scattered points creates visual clutter and makes trends difficult to follow.	Connect observations with lines so that each series can be tracked continuously over time.
Relying only on a legend forces viewers to repeatedly match colors with labels.	Label each line directly whenever possible to reduce cognitive effort and improve readability.



Visualization of Data, Understanding and Interpretation

Visualization of Trends

(TB 1, Ch14)

Why do we smooth data?

When datasets contain many observations, individual points often hide the overall pattern. Instead of focusing on every fluctuation, smoothing reveals the underlying trend.

Trend visualization helps us identify:

- Overall increase or decrease
- Turning points
- Long-term behavior
- Noise vs meaningful changes

Trend = Signal + Noise



Figure 14.1: Daily closing values of the Dow Jones Industrial Average for the year 2009. Data source: Yahoo! Finance

Instead of seeing every point individually, we estimate the signal beneath the noise.

💡 *Think about stock prices. They change every day. If we stare at every daily change, it's difficult to understand what actually happened during the year. A trend line summarizes thousands of tiny movements into one understandable pattern*

Visualization of Data, Understanding and Interpretation

Moving Average (Sec 14.1)

Moving Average Smoothing

The simplest smoothing technique is the Moving Average.

MA decreases random fluctuations, resulting in clearer long-term trend.

💡 *Instead of plotting today's value, plot the average of the nearby values*

Example: Compute 3-day Moving Average

Day	Price
1	100
2	104
3	98
4	102
5	96

$$\text{Days 1-3: } Average = \frac{100 + 104 + 98}{3} = \frac{302}{3} = 100.67$$

$$\text{Days 2-4: } Average = \frac{104 + 98 + 102}{3} = \frac{304}{3} = 101.33$$

$$\text{Days 3-5: } Average = \frac{98 + 102 + 96}{3} = \frac{296}{3} = 98.67$$

💡 Each time the window moves one day forward, the oldest value leaves, the newest value enters, and a new average is calculated. This sequence of averages forms the smoothed trend line.

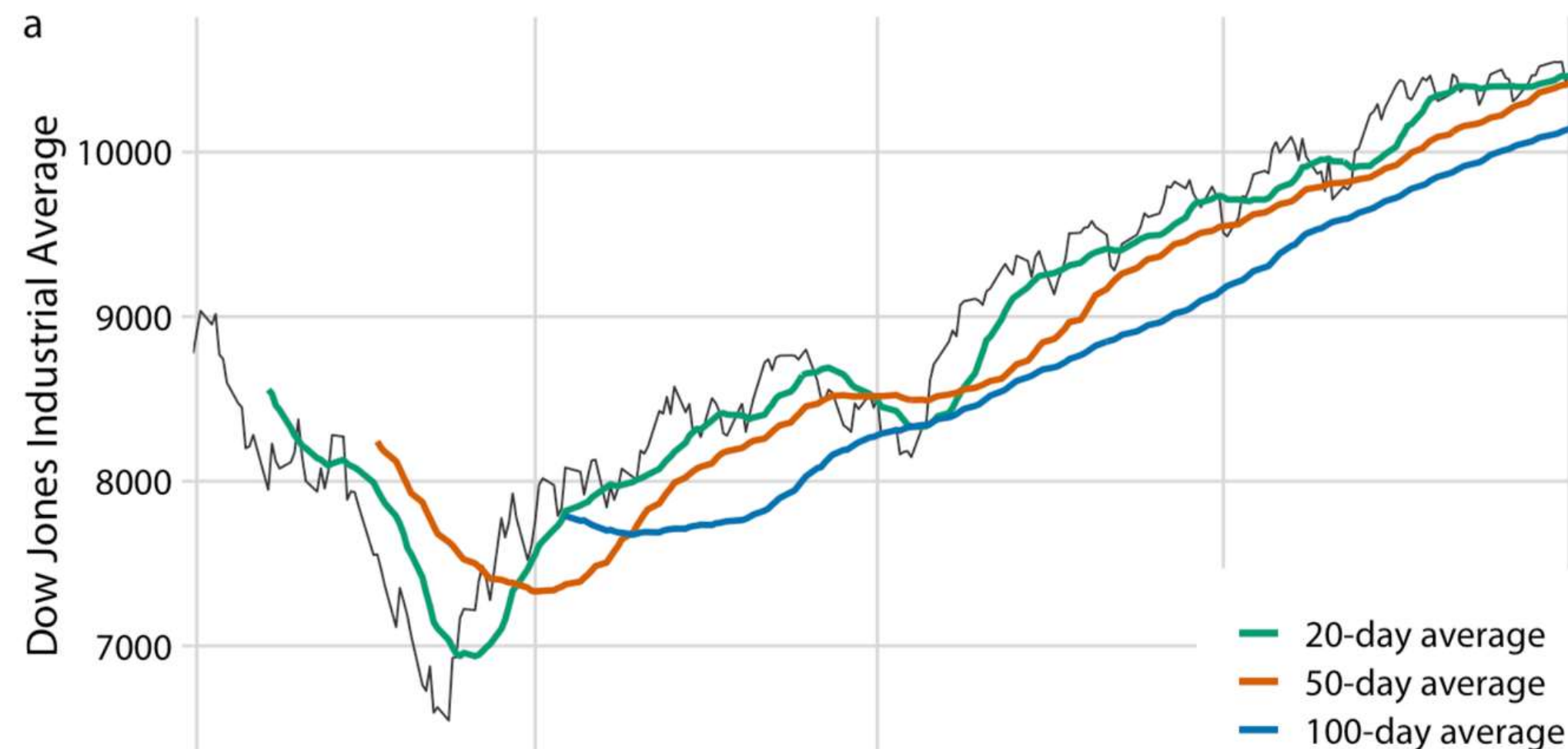
Visualization of Data, Understanding and Interpretation

Window Size (Sec 14.1)

Effect of Window Size

Different window sizes produce different levels of smoothing.

- smaller window size: responds quickly, more sensitive to fluctuations
- larger window size: less noise, smoother curve, misses rapid changes
- smaller window sizes help capture short-term trends, whereas larger window sizes help capture very smooth long-term trend



- Comparison of 20-, 50-, and 100-day moving averages.
- Each point on the colored curves represents the average of the previous 20, 50, or 100 days, reducing short-term fluctuations and highlighting the long-term trend.
- Larger windows produce smoother curves but respond more slowly to changes.

Visualization of Data, Understanding and Interpretation

Understanding Lag in Moving Averages

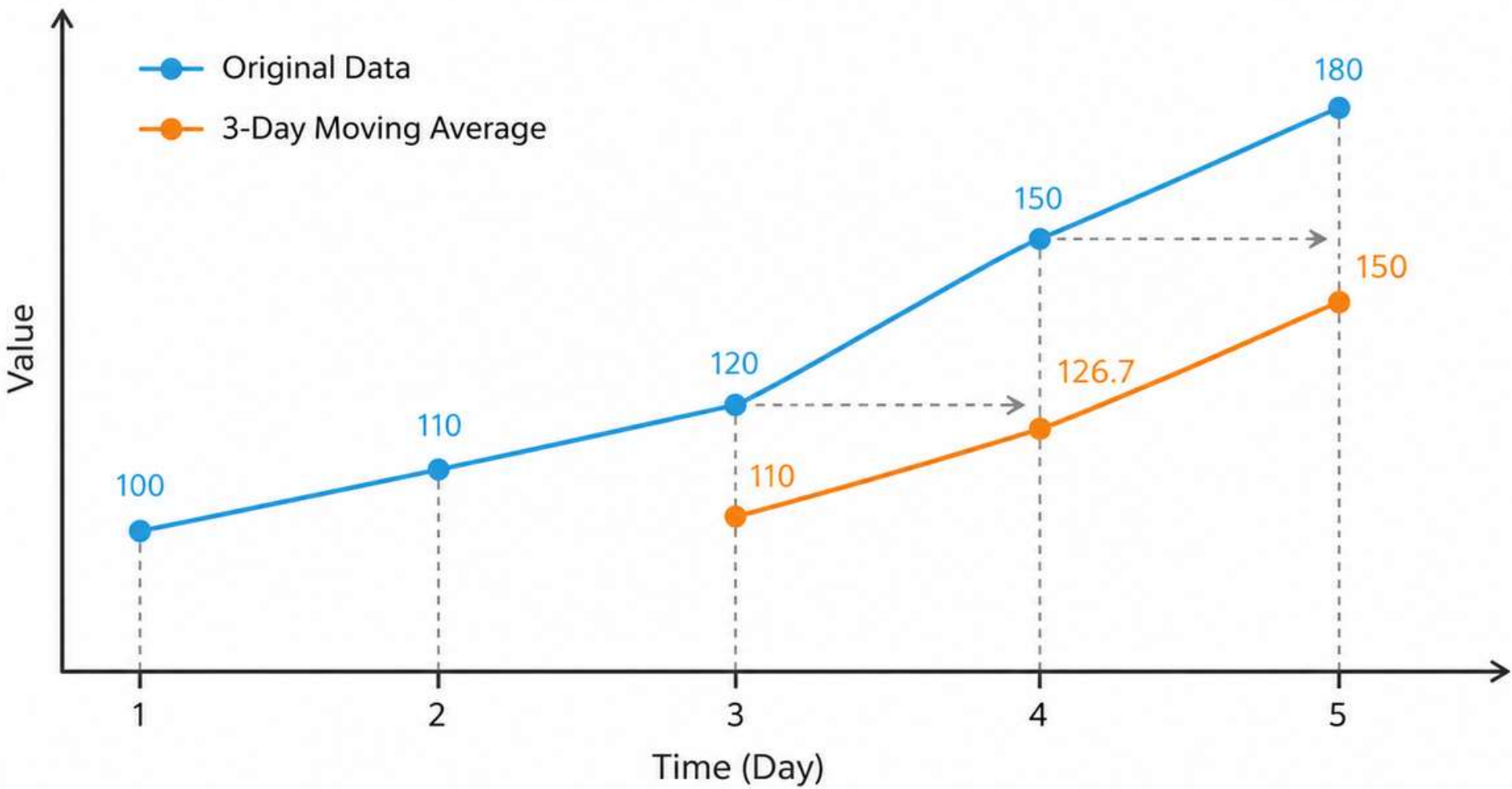
(Sec 14.1)

What is Lag?
Lag means the moving average reacts later than the original data because it summarizes past observations.

Example: 3-Day Moving Average

Suppose sales over five days are:

Day	1	2	3	4	5
Sales	100	110	120	150	180

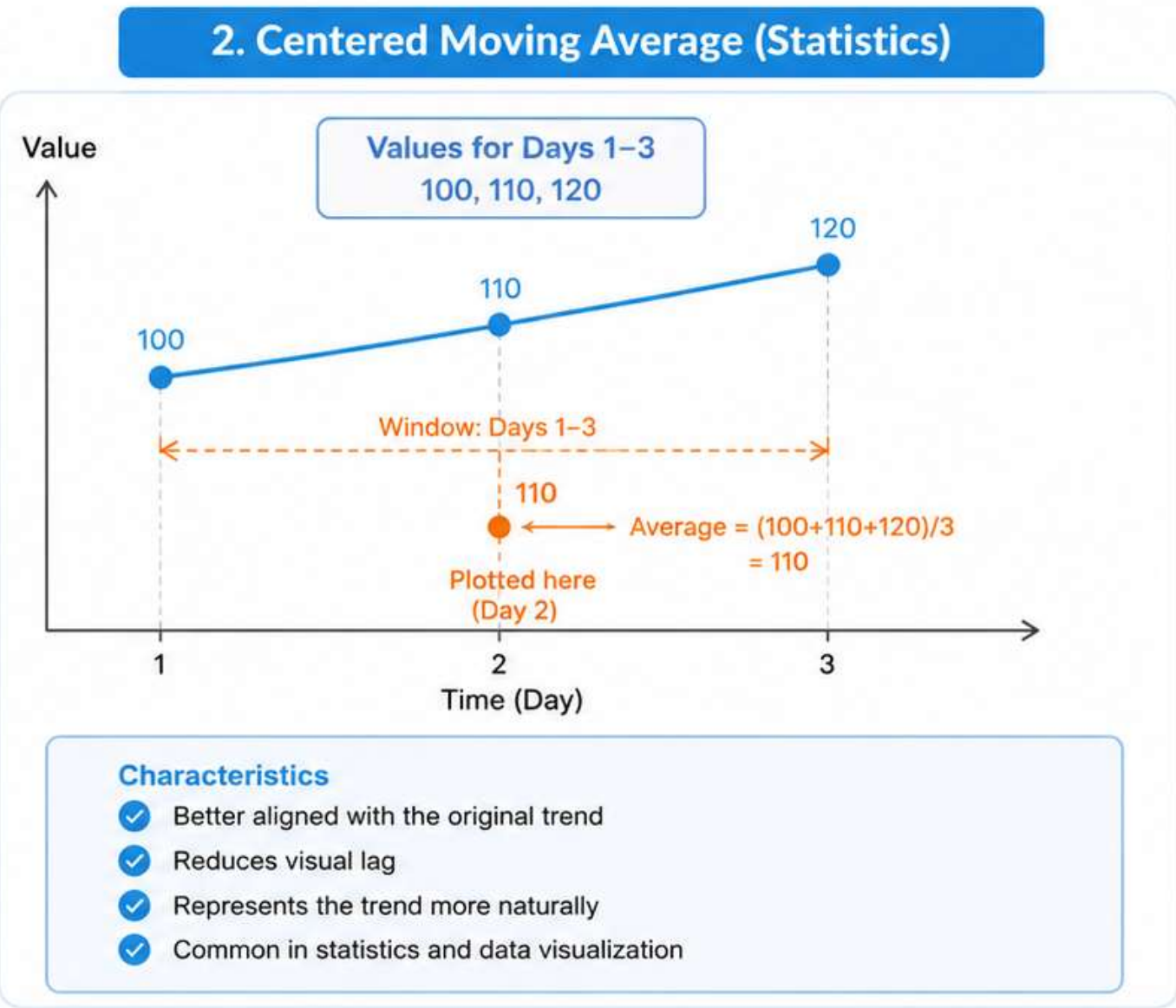
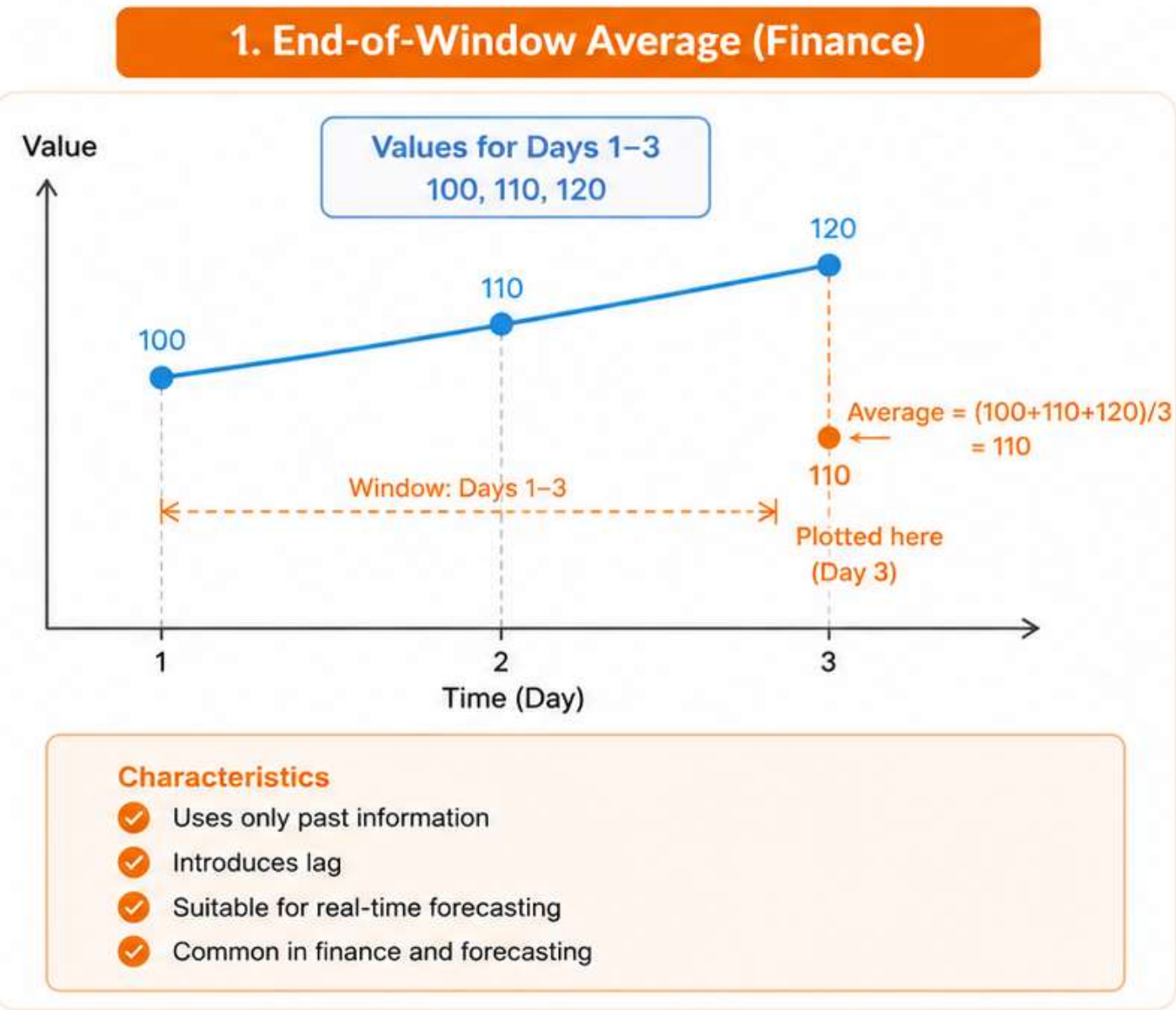


Visualization of Data, Understanding and Interpretation

End-of-Window vs Centered Moving Average (Sec 14.1)

Reducing the Effect of Lag

- The moving average value stays the same.
- The only difference is where we place that value on the timeline.



End-of-Window	Centered
Plot at the last day of the window	Plot at the middle day
Trend appears delayed	Trend aligns with the data

Visualization of Data, Understanding and Interpretation

Limitations of Moving Average (Sec 14.1)



Why isn't MA perfect?

Problem 1: Data near the beginning and end cannot be averaged

- Trend becomes shorter

Problem 2: Every observation receives equal weight

- Even an outlier entering or leaving the window changes the average noticeably.

Problem 3: Large windows may hide important events.

💡 Moving averages are easy to understand but can lag, lose information, and may oversmooth data.

Visualization of Data, Understanding and Interpretation

LOESS Smoothing

(Sec 14.1)

Locally Estimated Scatterplot Smoothing

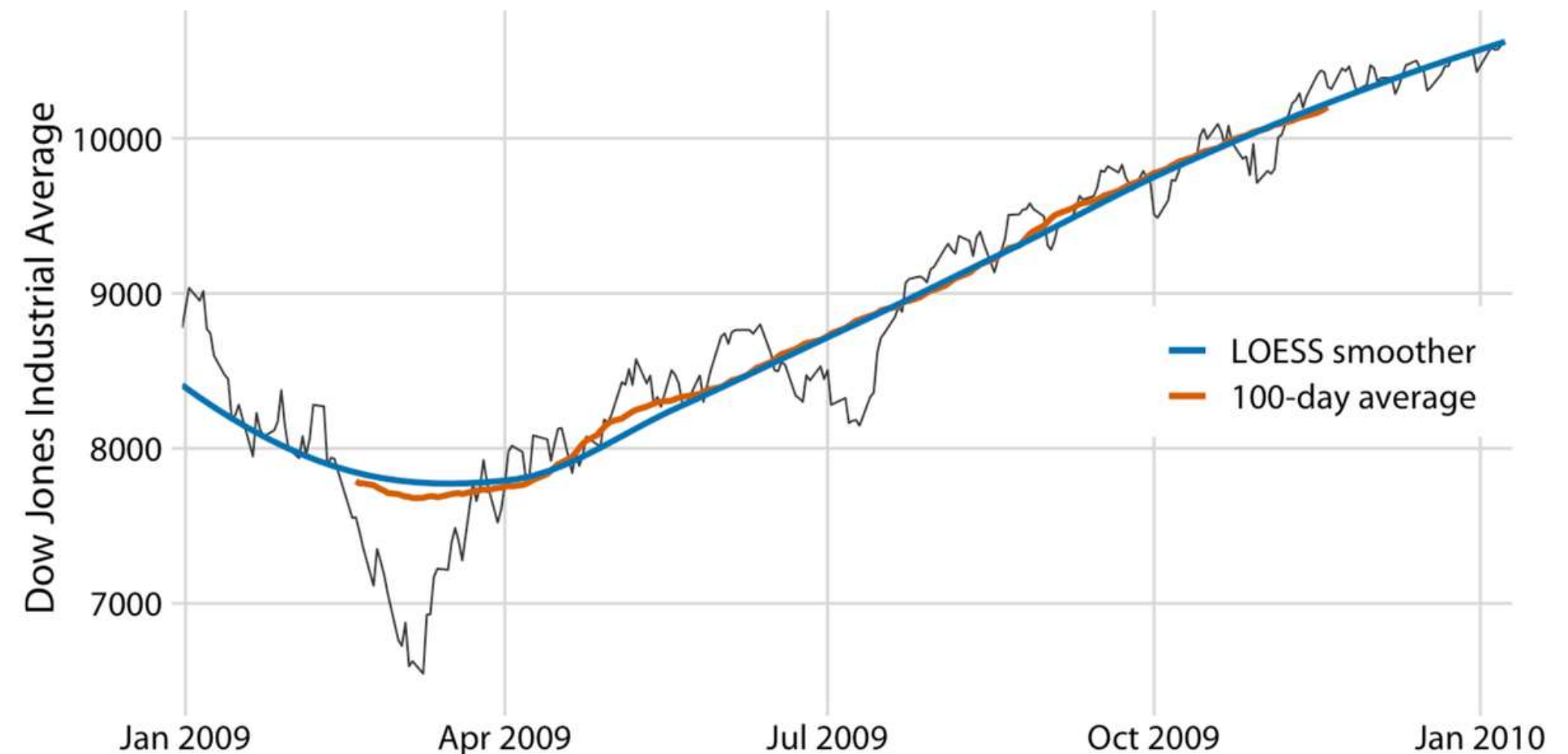
Instead of averaging values, LOESS fits small local regression curves across the data.

How it works:

1. Select nearby points
2. Give higher weight to closer points
3. Fit a small polynomial
4. Move to next location
5. Join all local fits smoothly

Advantages:

- Smoother than moving averages
- No missing endpoints
- Captures nonlinear relationships



Comparison of a 100-day Moving Average and LOESS smoothing. Both capture the overall trend, but LOESS produces a smoother curve without the edge truncation caused by fixed moving windows.

Visualization of Data, Understanding and Interpretation

LOESS Beyond Time Series

(Sec 14.1)

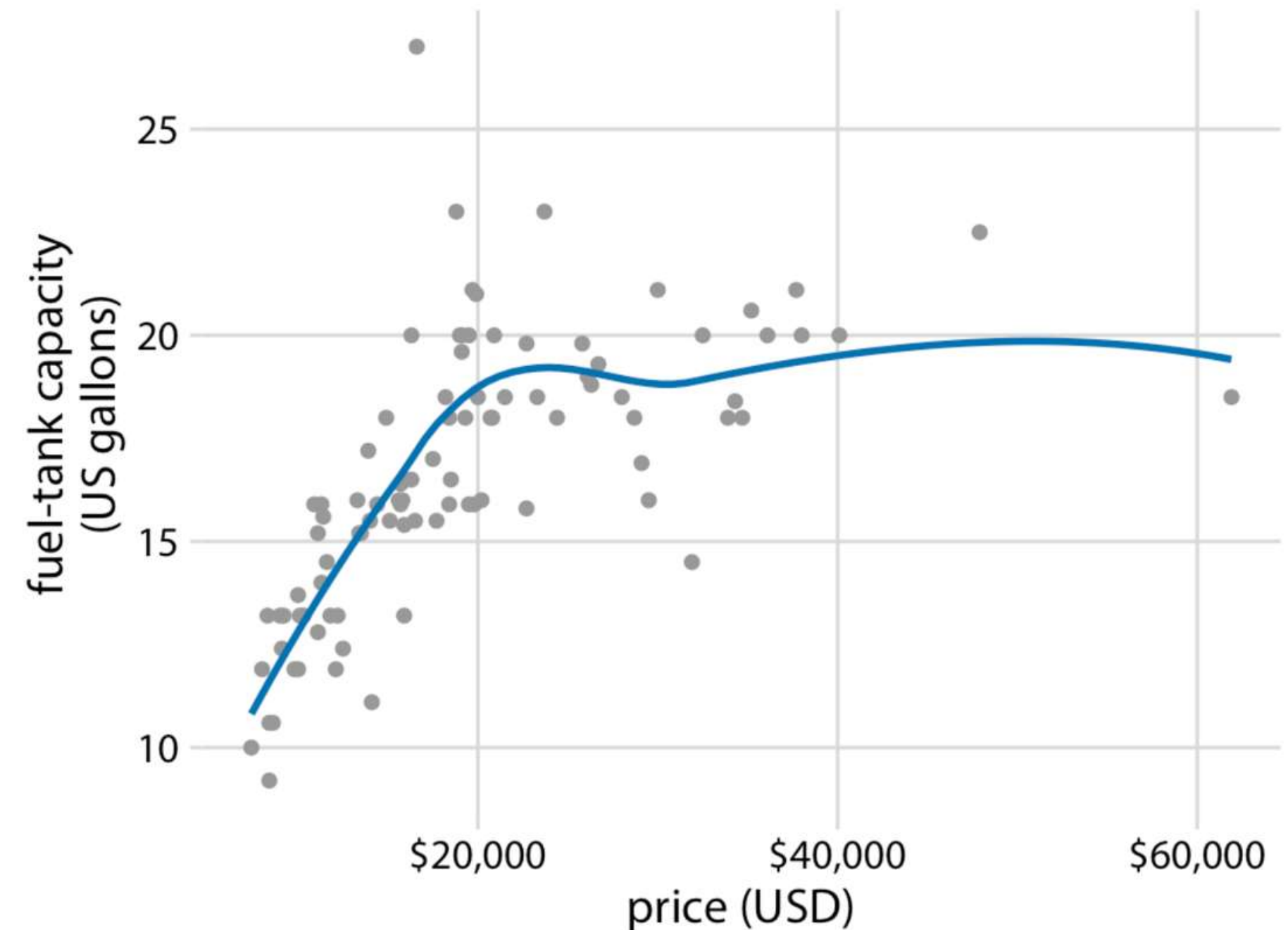
LOESS for Scatter Plots

LOESS works for any scatter plot, not just time series.

Example:

Relationship between Car Price vs Fuel Tank Capacity

LOESS smoothing applied to a scatter plot. It reveals the underlying relationship by reducing noise, showing that fuel-tank capacity increases with price up to about \$20,000, after which it levels off.



Visualization of Data, Understanding and Interpretation

Splines (Sec 14.1)

- Splines are another type of smoothing method.
- A spline is a smooth curve built from multiple polynomial pieces.
- These pieces meet at points called *knots*.

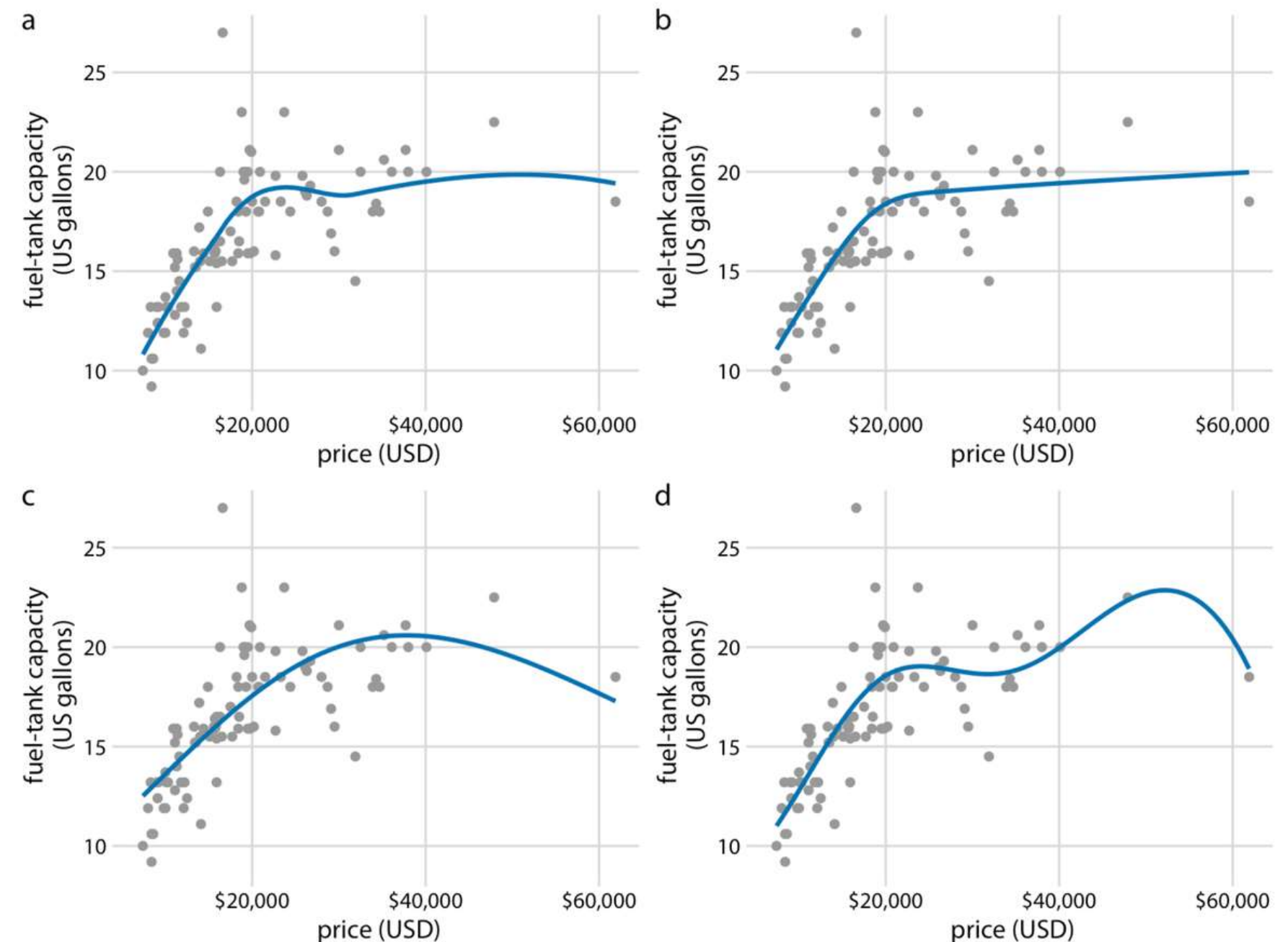
Advantages:

- Fast
- Flexible
- Handles large datasets efficiently

Challenges:

Different spline types and different numbers of knots produce different curves.

- Too many knots: overfitting
- Too few knots: underfitting



Different smoothing methods can produce noticeably different trend curves on the same dataset, especially near the edges of the data.

Moving Average Smoothing

The simplest smoothing technique is the Moving Average.

MA decreases random fluctuations, resulting in clearer long-term trend.

💡 *Instead of plotting today's value, plot the average of the nearby values*

Example: Compute 3-day Moving Average

Day	Price
1	100
2	104
3	98
4	102
5	96

$$\text{Days 1-3: } \textit{Average} = \frac{100 + 104 + 98}{3} = \frac{302}{3} = 100.67$$

$$\text{Days 2-4: } \textit{Average} = \frac{104 + 98 + 102}{3} = \frac{304}{3} = 101.33$$

$$\text{Days 3-5: } \textit{Average} = \frac{98 + 102 + 96}{3} = \frac{296}{3} = 98.67$$

💡 Each time the window moves one day forward, the oldest value leaves, the newest value enters, and a new average is calculated. This sequence of averages forms the smoothed trend line.

Visualization of Data, Understanding and Interpretation

Choosing the Right Smoother (Sec 14.1)

Method	Best Used For	Limitation
Moving Average	Simple trend visualization	Lag & edge effects
LOESS	Smooth nonlinear trends	Computationally slower
Splines	Flexible large datasets	Depends on knot selection

💡 *There is no universally best smoother*

Choose based on:

- Dataset size
- Amount of noise
- Desired level of detail
- Interpretation requirements

Different smoothing methods can produce different conclusions from the same data.

1. What is a major visual limitation of using a bubble chart to display a fourth quantitative variable via marker size?

- A. It cannot be used alongside color encoding for subgroups.
- B. Small differences in values are difficult for viewers to perceive accurately compared to spatial position.
- C. It automatically introduces temporal lag into the plotted data series.
- D. It is restricted only to categorical data on the primary axes.

1. What is a major visual limitation of using a bubble chart to display a fourth quantitative variable via marker size?

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- C. It automatically introduces temporal lag into the plotted data series.
- D. It is restricted only to categorical data on the primary axes.

Answer: B

Visualization of Data, Understanding and Interpretation

MCQs

2. A correlation coefficient of $r = -0.85$ indicates:

- A. A weak negative relationship
- B. No relationship between variables
- C. A strong negative relationship
- D. A perfect negative relationship

Visualization of Data, Understanding and Interpretation

MCQs

2. A correlation coefficient of $r = -0.85$ indicates:

- A. A weak negative relationship
- B. No relationship between variables
- C. A strong negative relationship
- D. A perfect negative relationship

Answer: C

3. A dataset contains 30 numerical features. You want to visualize the major patterns while retaining most of the information. Which approach is most appropriate?

- A. Create 435 scatter plots for every pair of variables.
- B. Use a pie chart.
- C. Convert all variables into categorical values.
- D. Apply PCA and visualize the first two principal components.

Visualization of Data, Understanding and Interpretation

MCQs



3. A dataset contains 30 numerical features. You want to visualize the major patterns while retaining most of the information. Which approach is most appropriate?

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Answer: D

Visualization of Data, Understanding and Interpretation

MCQs

4. Increasing the moving average window from 20 days to 100 days primarily results in:

- A. Higher sensitivity to daily changes
- B. More fluctuations in the trend
- C. Smoother trends but reduced responsiveness to short-term changes
- D. Elimination of all noise

Visualization of Data, Understanding and Interpretation

MCQs

4. Increasing the moving average window from 20 days to 100 days primarily results in:

- A. Higher sensitivity to daily changes
- B. More fluctuations in the trend
- C. Smoother trends but reduced responsiveness to short-term changes
- D. Elimination of all noise

Answer: C

5. Why does a moving average plotted at the end of each averaging window appear to lag behind the original data?

- A. It ignores recent observations.
- B. Each average is associated with the last observation in its window.
- C. It assigns unequal weights to observations.
- D. The averaging window overlaps incorrectly.

5. Why does a moving average plotted at the end of each averaging window appear to lag behind the original data?

- A. It ignores recent observations.
- B. Each average is associated with the last observation in its window.
- C. It assigns unequal weights to observations.
- D. The averaging window overlaps incorrectly.

Answer: B

Visualization of Data, Understanding and Interpretation

MCQs

6. In spline smoothing, increasing the number of knots generally makes the fitted curve:

- A. Less flexible
- B. More rigid
- C. More flexible but potentially prone to overfitting
- D. Identical to a moving average

Visualization of Data, Understanding and Interpretation

MCQs

6. In spline smoothing, increasing the number of knots generally makes the fitted curve:

- A. Less flexible
- B. More rigid
- C. More flexible but potentially prone to overfitting
- D. Identical to a moving average

Answer: C



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Thank You



Visualization of Data, Understanding and Interpretation

Unit 1 : Foundations of Visualization, Data Interpretation

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

Dr.Prajwala TR (Course Instructor)

Dept. of CSE, PES University, Bangalore

Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.



Visualization of Data, Understanding and Interpretation

Lecture Overview

- Fundamental principles of effective data visualization by understanding context, selecting appropriate visual displays, and matching chart types to the communication goal.
- Choosing and designing charts based on functional effectiveness and analytical questions rather than purely aesthetic preferences, ensuring that comparisons are simple, intuitive, and reduce cognitive load.
- Understanding how computers store images through the differences in scalability, resolution, and ideal use cases between bitmap (raster) and vector graphics.
- Evaluating the impact of image compression techniques, specifically why lossless formats preserve visual quality while lossy formats like JPEG degrade text, sharp edges, and line-based figures.
- Navigating the visualization workflow from exploratory analysis to final presentation, focusing on reproducibility, repeatability, and the critical need to separate underlying data content from design styling.

Visualization of Data, Understanding and Interpretation

Principles of Effective Data Visualization (TB 2, Ch1)

The Six Principles of Effective Data Visualization

- *Understand the Context*
Identify the audience, objective, and communication strategy before designing any visualization.
- *Choose an Appropriate Visual Display*
Select the chart type that best communicates the intended insight.
- *Eliminate Clutter*
Remove unnecessary elements that distract from the message.
- *Focus Attention Where You Want It*
Use visual cues such as color, size, and annotation to direct the audience's attention.
- *Think Like a Designer*
Apply design principles to improve clarity, readability, and aesthetics.
- *Tell a Story*
Organize visuals into a compelling narrative that leads the audience to meaningful conclusions.

Visualization of Data, Understanding and Interpretation

Principles of Effective Data Visualization (Ch1)



The STORYTELLING with DATA PROCESS

STEP 1: UNDERSTAND the CONTEXT

WHO is your audience?
WHAT do you need them to do?
HOW will data help make your point?

ARTICULATE
your BIG IDEA



CREATE a
STORYBOARD



→ brainstorm
→ edit
→ get feedback

STEP 4: DRAW ATTENTION where you WANT it

Use POSITION, SIZE, and COLOR to
FOCUS your AUDIENCE'S ATTENTION

Use the
"where are
your eyes
drawn?"
test



CLOSE
EYES...



then
OPEN
EYES



WHAT do
you SEE FIRST?

STEP 2: CHOOSE an APPROPRIATE VISUAL

DRAW IT!
then... CREATE it in
your TOOL



iterate and
look at your data
different ways

SEEK FEEDBACK
from others



STEP 5: THINK LIKE a DESIGNER

FIRST FUNCTION →
think about what you want your
audience to DO with the data



ANALYZE
fine detail



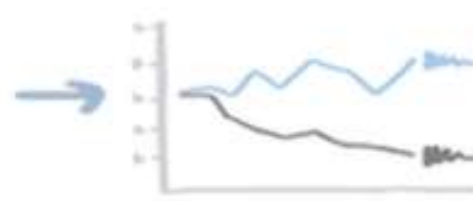
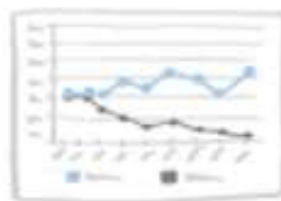
COMMUNICATE
key trends

SECOND FORM ✨
create a visualization that
will make this easy

→ Affordances
→ Accessibility
→ Aesthetics
→ Acceptance

STEP 3: ELIMINATE CLUTTER

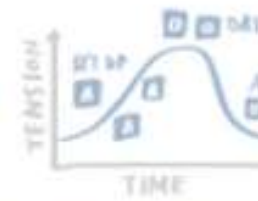
IDENTIFY UNNECESSARY
ELEMENTS and REMOVE THEM



→ Leverage white space
→ Align elements
→ Avoid diagonal components

STEP 6: TELL a STORY

RETURN to your STORYBOARD



Where and how
will data fit into
the story?

Use the narrative arc to plan
your story and form a pithy, repeatable
phrase to help your message stick

Visualization of Data, Understanding and Interpretation

Understanding the Context Before Visualizing Data (Ch1)



Good data visualization begins with understanding the problem, not by choosing a chart.

- Before creating any visualization, ask:
- Who is the audience?
 - What do they need to know or do?
 - How will data help communicate the message?

Exploratory Analysis	Explanatory Analysis
Explores data to discover patterns	Communicates a clear insight
Multiple hypotheses are tested	One key message is emphasized
Used by the analyst	Designed for the audience

Visualization of Data, Understanding and Interpretation

What is Data Preprocessing?

Data preprocessing is the process of converting raw, messy data into clean, structured, and analysis-ready data.

- Real-world datasets often contain:
 - Missing values
 - Duplicate records
 - Incorrect or inconsistent entries
 - Outliers and noise
 - Different data formats
- Visualizing raw data can produce misleading graphs and incorrect conclusions.
- Clean data enables visualizations that are:
 - Accurate
 - Easy to interpret
 - Reliable for decision-making

Key Idea: Garbage In → Garbage Out

Poor-quality data leads to poor-quality visualizations and wrong insights. Proper preprocessing ensures that charts represent the true story hidden in the data.

Visualization of Data, Understanding and Interpretation

Data Preprocessing Steps

Steps Before Creating Visualizations

- Data Assessment Understand the dataset structure.
- Identify missing values, duplicates, incorrect formats, and outliers.
- Data Cleaning Remove duplicate records.
- Handle missing values (delete or fill).
- Correct inconsistent entries (e.g., "NY", "New York", "new york").
- Data Transformation Convert data into a suitable format.
- Standardize units (kg, cm, ₹, \$).
- Convert dates and categorical values into consistent formats.
- Data Reduction Remove unnecessary columns or records.
- Aggregate or sample large datasets for faster visualization.

Outcome:

A clean dataset that produces meaningful and trustworthy charts instead of misleading ones.

Visualization of Data, Understanding and Interpretation

Why Preprocessing Matters



Raw Dataset
Missing values
Duplicate records
Mixed date format
Different city names ("NY", "I
Extreme errors (Age =

Impact on Visualization

- Without preprocessing: Misleading charts and incorrect insights.
- With preprocessing: Accurate visualizations, clearer patterns, and better decision-making.

Takeaway:

Data preprocessing is the foundation of every effective visualization. Spending time cleaning data often improves the quality of insights more than choosing a different chart type.

Visualization of Data, Understanding and Interpretation

The Three Questions Every Visualization Must Answer (Ch1)



1. Who is your audience?

- Identify a specific audience instead of trying to address everyone.
- Tailor the visualization to the audience's background, goals, and decision-making needs.

2. What do you want them to know or do?

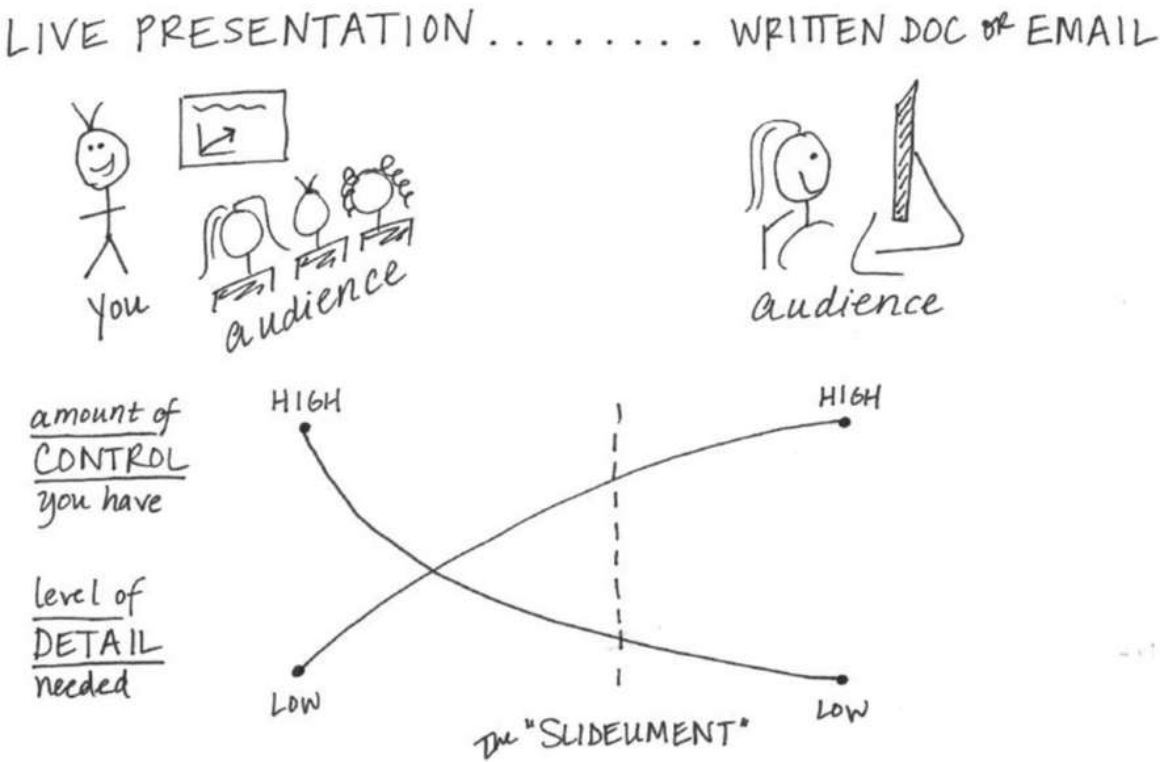
- Every visualization should support a clear action or decision.
- End with a recommendation or prompt rather than simply presenting data.

3. How will you communicate?

Choose the communication format based on the audience.

Live Presentation	Report
Presenter guides the audience	Communicates a clear insight
Less text, more visuals	More explanation and detail

💡 *Data is evidence, not the story itself. Use it to support your message!*



Visualization of Data, Understanding and Interpretation

Planning Before Designing (Ch1)

The 3-Minute Story

Summarize the entire message in about three minutes.

- Clarifies the main narrative.
- Helps communicate effectively even if presentation time changes.

The Big Idea

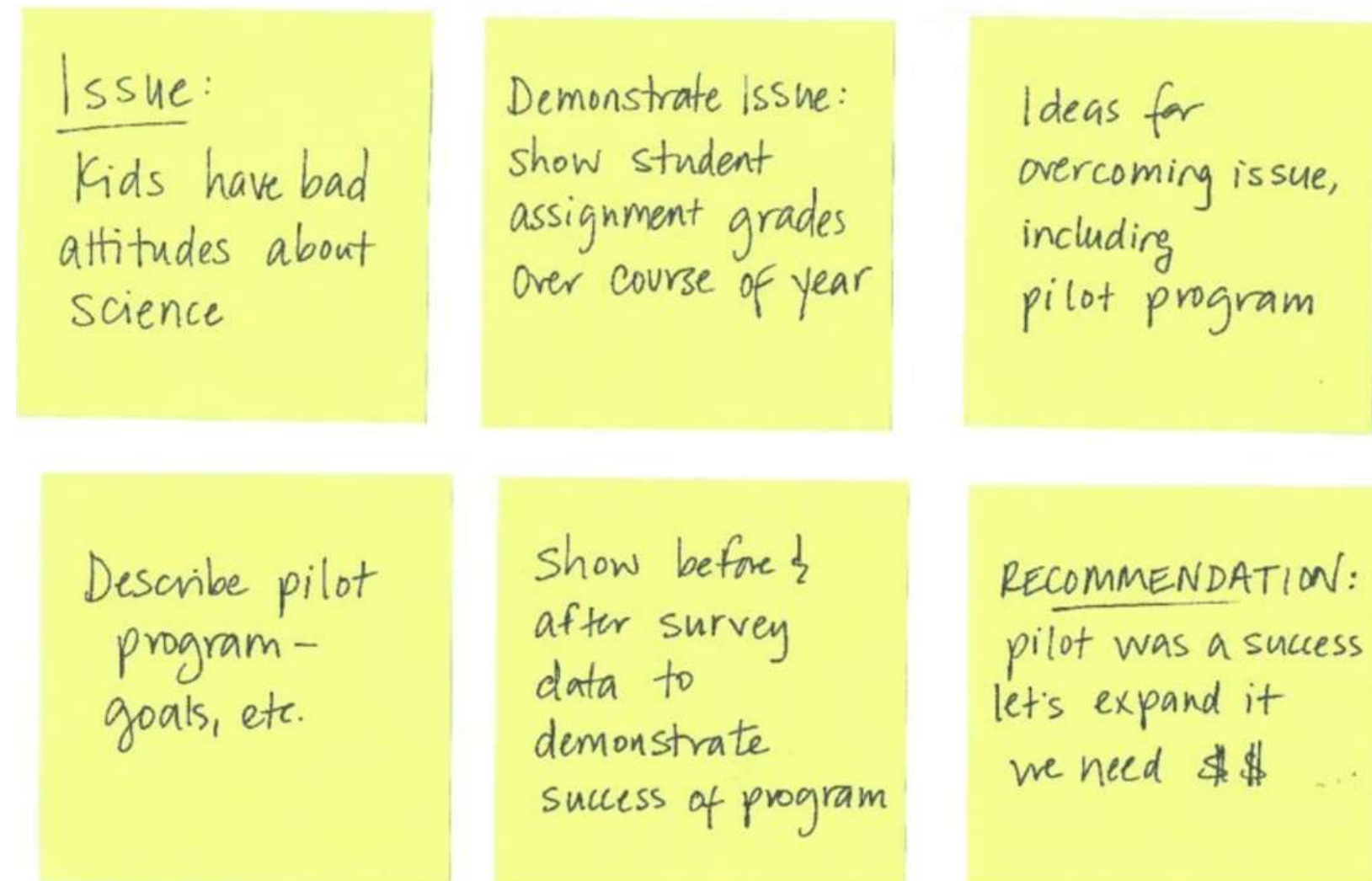
Reduce the message to one complete sentence that:

- Expresses your main point.
- Explains why it matters.
- States what is at stake.

Storyboarding

Plan the communication before opening presentation software.

- Arrange ideas using paper, sticky notes, or a whiteboard.
- Organize the narrative before creating visuals.
- Modify ideas easily without becoming attached to slide designs.



Visualization of Data, Understanding and Interpretation

Choosing an Appropriate Visual Display (Ch2)



Why does choosing the right visualization matter?

Even a valuable insight can be misunderstood if it is displayed using an inappropriate chart.

An effective visualization should:

- Match the type of data being presented.
- Highlight patterns, comparisons, relationships, or trends.
- Reduce cognitive effort for the audience.
- Communicate the intended message accurately and efficiently.

What we have already learned

- We have explored how different chart types communicate different kinds of information, including: comparison charts, distribution charts, relationship charts, composition charts, time-series visualizations.
- Next, we will learn what makes a chart perceptually effective, choosing the right chart, image formats, etc.

Visualization of Data, Understanding and Interpretation

What Makes a Chart Perceptually Effective? (Ch2)



A chart is perceptually effective when it allows viewers to interpret data accurately, quickly, and with minimal mental effort.

An effective chart should:

- Emphasize the key message rather than decoration.
- Match the chart type to the data being displayed.
- Make comparisons simple and intuitive.
- Reduce unnecessary cognitive load.

💡 *Choose the visualization that makes the answer easiest to see, not the most visually attractive one*

Visualization of Data, Understanding and Interpretation

Choosing the Right Chart

(Ch2)



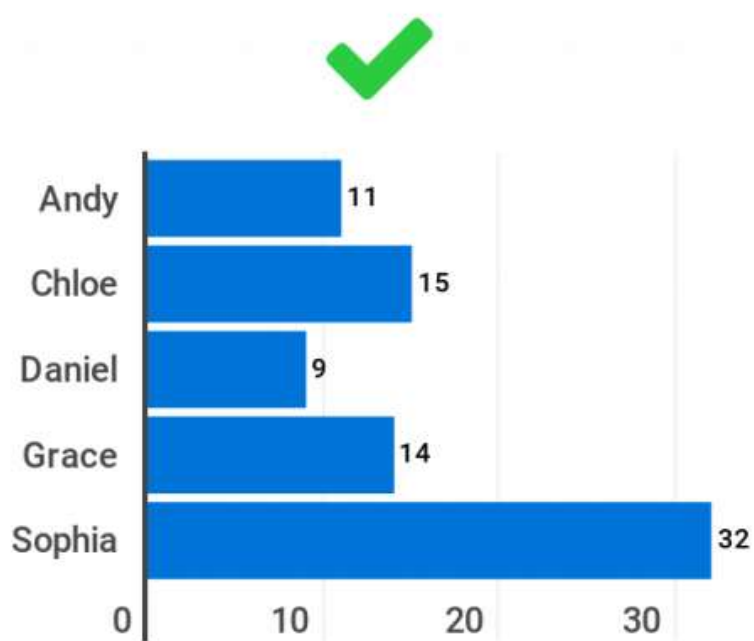
Different chart types communicate different relationships. While we have understood the use of various charts previously, here are a few recommendations:

Purpose	Recommended Chart
Compare categories	Bar Chart
Show trends	Line Chart
Show relationships	Scatter Plot
Show distributions	Histogram / Box Plot
Show proportions	Stacked Bar (limited categories)

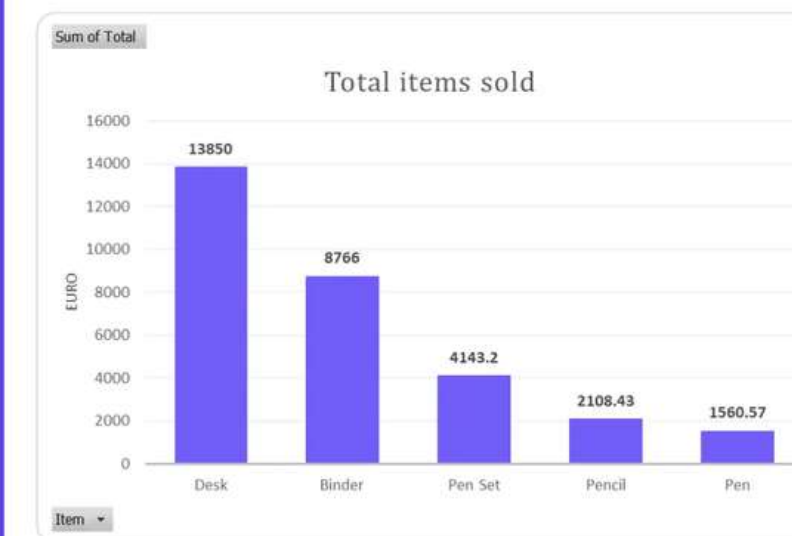
Avoid choosing charts based only on appearance.
Instead, first identify what question the data should answer.

Visualization of Data, Understanding and Interpretation

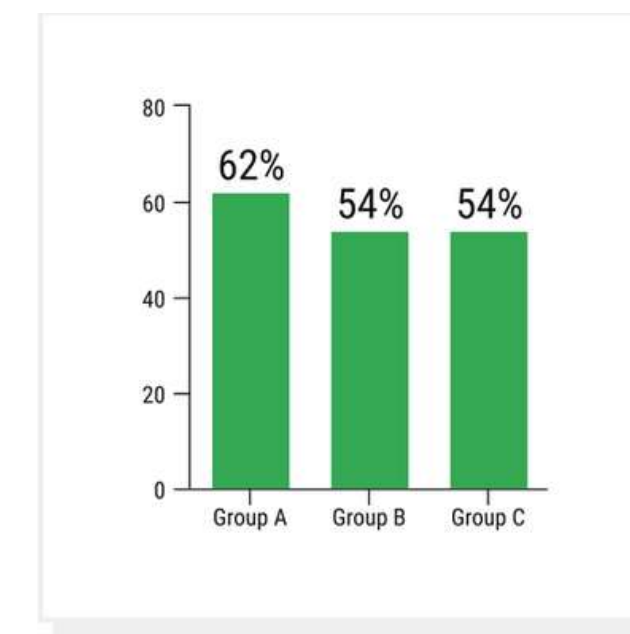
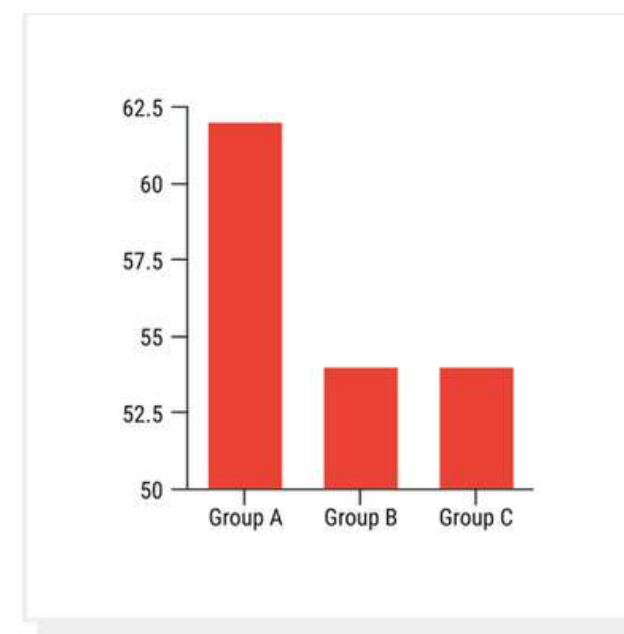
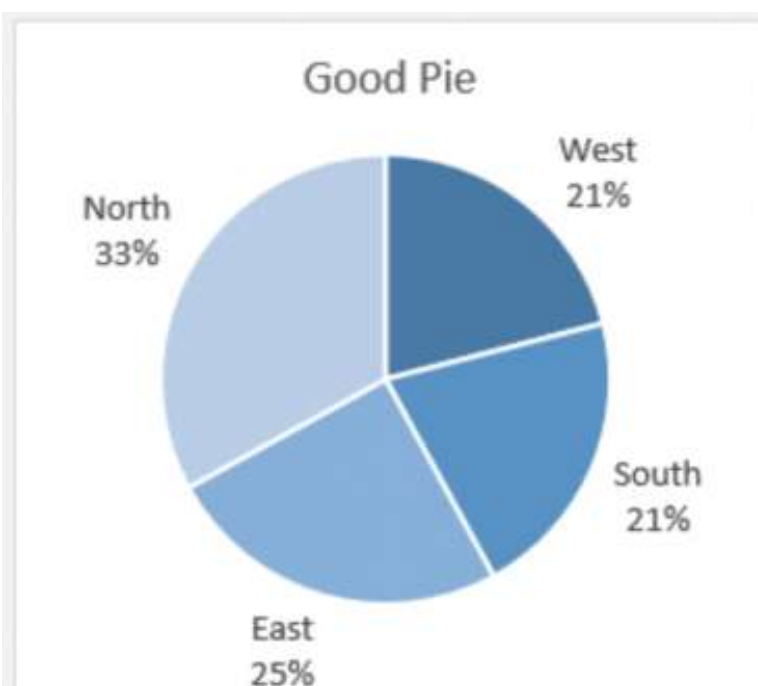
Choosing the Right Chart (Ch2)



Cluttered chart
(bad data visualization example)



Clean chart
(good data visualization example)



Visualization of Data, Understanding and Interpretation

Why Image Formats Matter? (TB 1, Ch27)

Why should data scientists care?

A visualization is only as good as the file used to save it.

Choosing the correct format affects:

- Image quality
- File size
- Scalability
- Ease of editing
- Print vs web usage

💡 *Which image format should you use for charts, reports, presentations, and publications?*

Visualization → SaveImage → Quality, Size, Scalability, Compatibility

Visualization of Data, Understanding and Interpretation

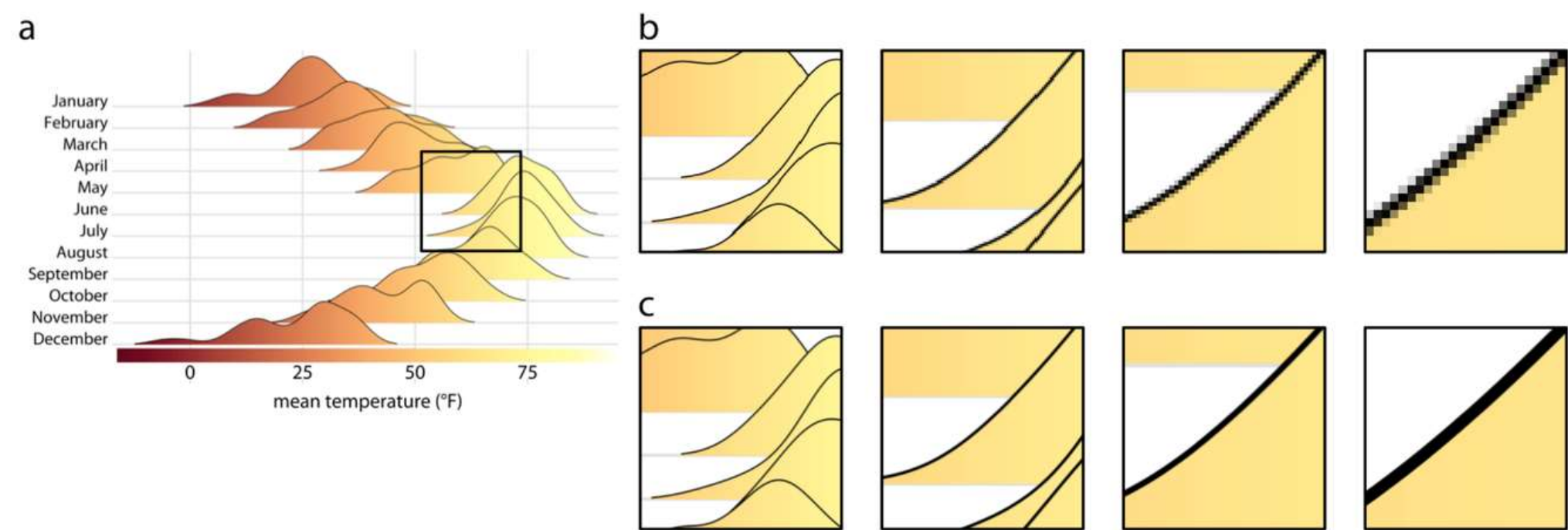
Bitmap vs Vector Graphics

(Sec 27.1)

Two Ways Computers Store Images:

Bitmap (Raster)	Vector
Made of pixels	Made of mathematical shapes
Fixed resolution	Resolution independent
Can become pixelated	Always remains sharp
Best for photos	Best for charts, icons, diagrams

💡 *Bitmap stores pixels, vector stores instructions for drawing shapes*



💡 *Zooming reveals that bitmap images become pixelated, while vector graphics remain sharp at any magnification*

Visualization of Data, Understanding and Interpretation

Common Image File Formats

(Sec 27.1)



Which format should you use?

Format	Type	Best Use
PDF	Vector	Publications, reports
SVG	Vector	Websites
PNG	Bitmap	Charts, Screenshots
JPEG	Bitmap	Photographs
TIFF	Bitmap	Printing
GIF	Bitmap	Simple Animations

Compression is the process of reducing an image's file size by removing or encoding data more efficiently while trying to preserve its visual quality.

Lossless Compression:

- No information removed
- Original image can be perfectly reconstructed
- Larger files
- Examples: PNG, TIFF

Lossy Compression:

- Removes "less noticeable" details
- Smaller file size
- Permanent quality loss
- Example: JPEG

Visualization of Data, Understanding and Interpretation

Why JPEG is Bad for Charts

(Sec 27.2)

JPEG Introduces Compression Artifacts

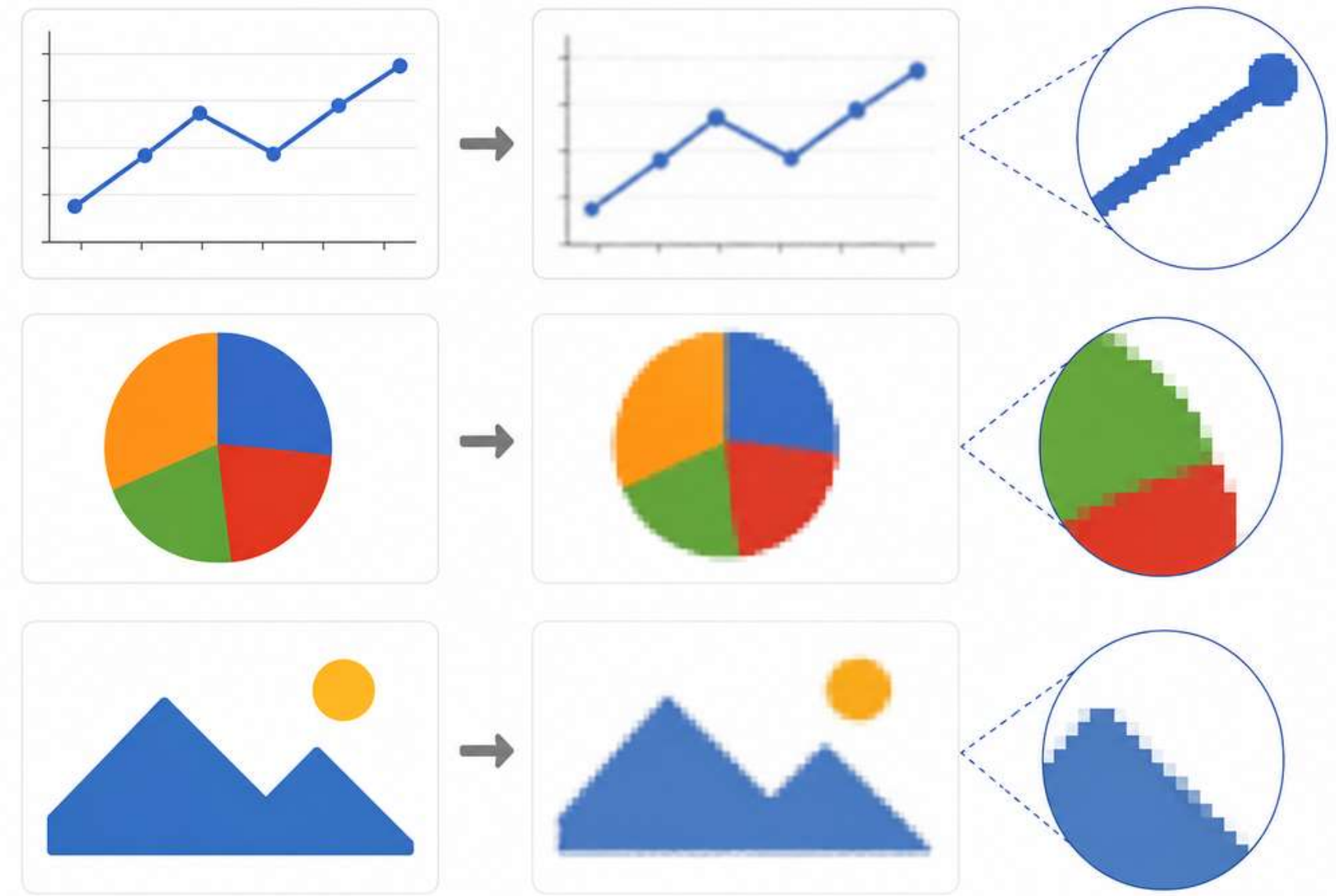
Photographs compress well because colors change gradually.

Charts and text contain:

- Sharp edges
- Fine lines
- Labels

JPEG creates:

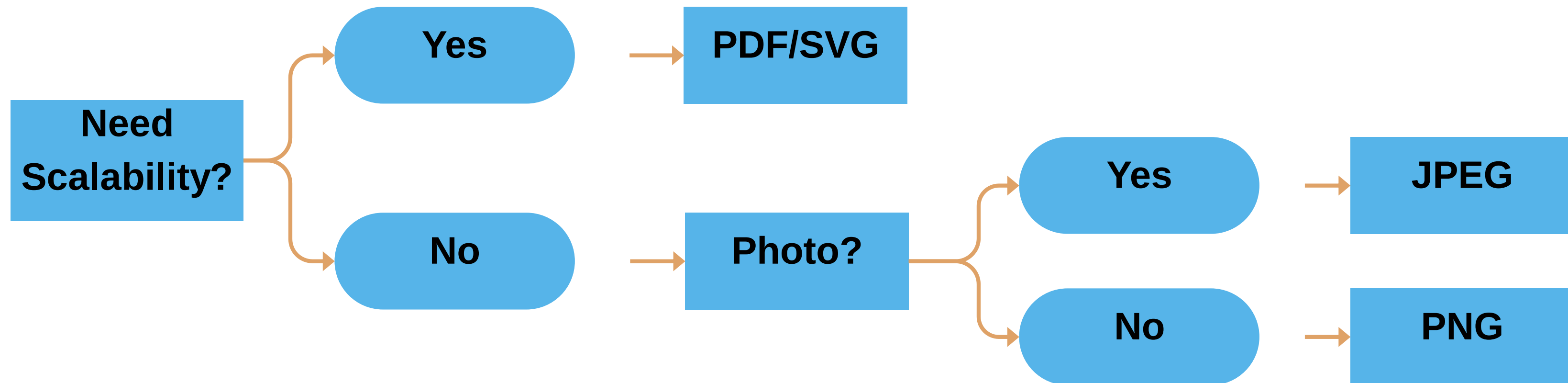
- Blurry text
- Pixelated gradients
- Compression artifacts



Visualization of Data, Understanding and Interpretation

Choosing the Right Format: Best Practices (Sec 27.2)

- Keep the original high-quality version.
- Convert only when necessary.
- Avoid repeatedly converting between formats.



Visualization of Data, Understanding and Interpretation

Choosing Visualization Software

(TB 1, Ch28)



A visualization tool should make it easy to:

- Explore data
- Produce publication-quality figures
- Modify designs
- Reproduce results later

Reproducibility: Someone else can recreate the same visualization using the same data.

Repeatability: The exact same figure (including styling) can be regenerated.

Why does it matter?

Interactive tools often make reproducibility harder.

Visualization of Data, Understanding and Interpretation

Data Exploration vs Data Presentation

(Sec 28.2)



Data Exploration	Data Presentation
Discover patterns	Communicate findings
Fast iteration	Polished appearance
Try many plots	Final visualization
Speed matters	Design matters



Visualization of Data, Understanding and Interpretation

Separate Content from Design (Sec 28.3)

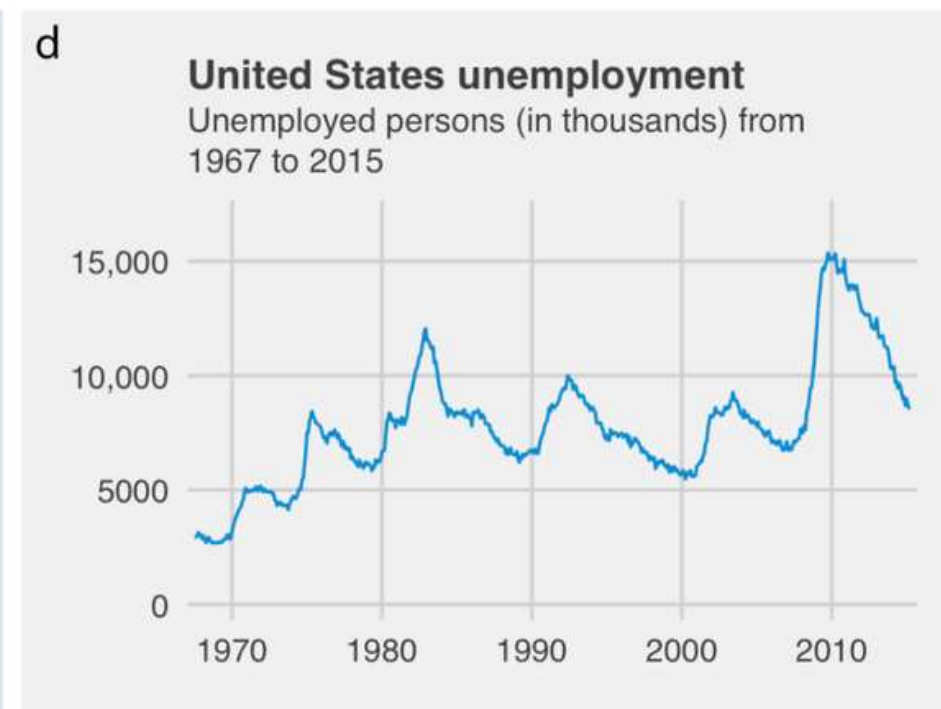
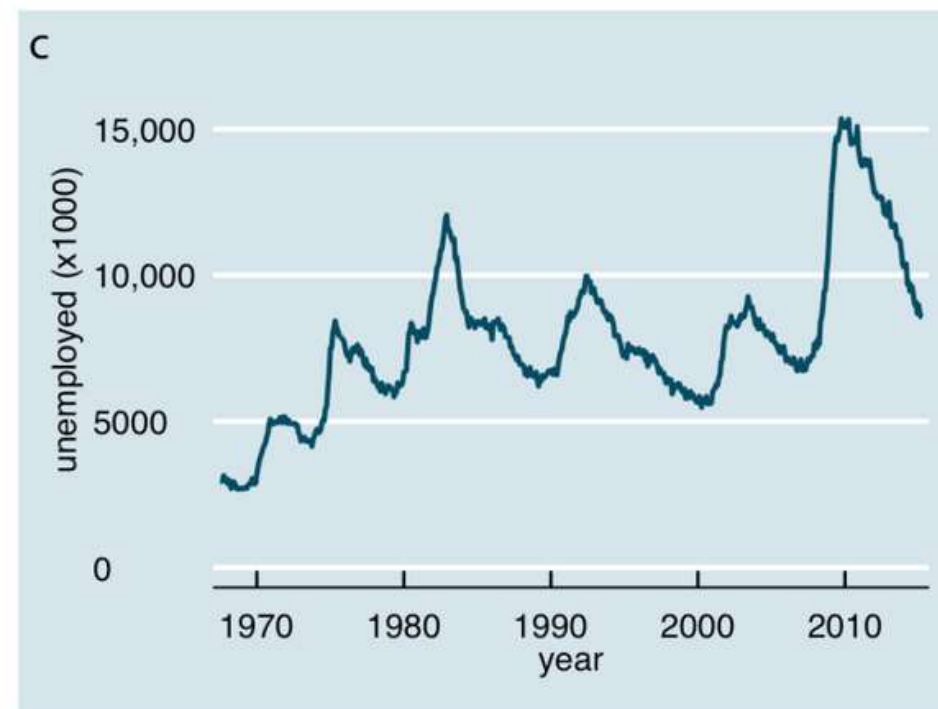
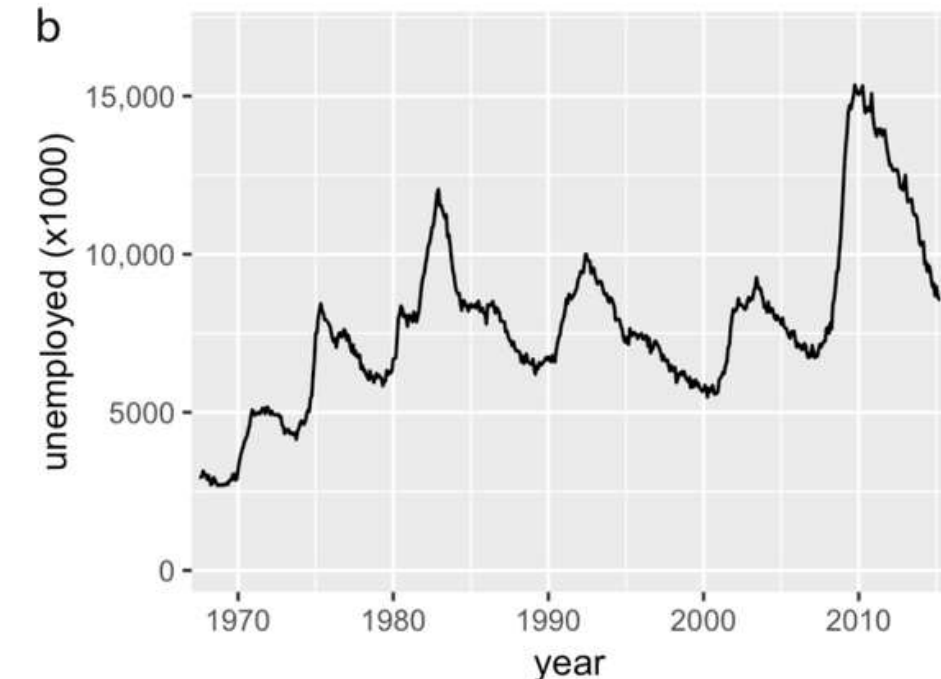
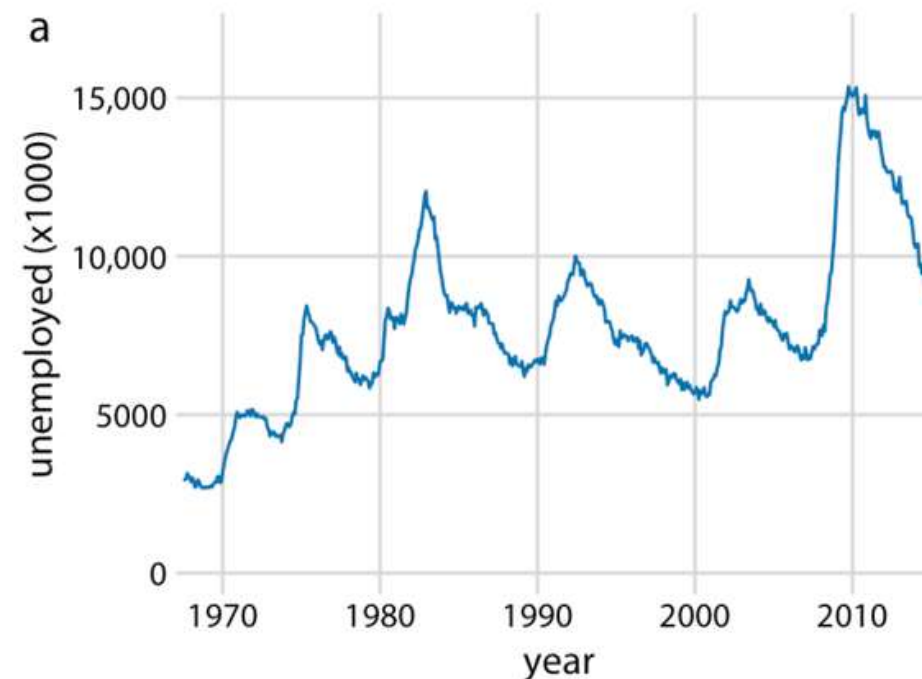
Content:

- Data
- Variables
- Axes
- Plot type

Design:

- Colors
- Fonts
- Gridlines
- Themes
- Labels

💡 *Changing a theme should not change the data*



The same data can look very different simply by changing the visual theme, while the underlying information remains unchanged.

Visualization of Data, Understanding and Interpretation

MCQs

1. Which image format is best for charts and screenshots?

- A. JPEG
- B. PNG
- C. GIF
- D. RAW

Visualization of Data, Understanding and Interpretation

MCQs

1. Which image format is best for charts and screenshots?

- A. JPEG
- B. PNG
- C. GIF
- D. RAW

Answer: B

Visualization of Data, Understanding and Interpretation

MCQs

2. Vector graphics differ from bitmap graphics because they:

- A. Store pixels
- B. Use mathematical shapes
- C. Always have smaller file sizes
- D. Are only used online

Visualization of Data, Understanding and Interpretation

MCQs

2. Vector graphics differ from bitmap graphics because they:

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Answer: B

Visualization of Data, Understanding and Interpretation

MCQs

3. Which stage focuses on discovering patterns in data?

- A. Data presentation
- B. Data exploration
- C. Image compression
- D. Publication

Visualization of Data, Understanding and Interpretation

MCQs

3. Which stage focuses on discovering patterns in data?

- A. Data presentation
- B. Data exploration
- C. Image compression
- D. Publication

Answer: B

4. A public health analyst has completed an extensive analysis on vaccination rates. The Minister of Health has requested a 2-minute briefing before a cabinet meeting. Which approach is most consistent with Knafllic's recommendations?

- A. Present the most statistically significant findings first, followed by supporting charts and appendices.
- B. Begin with the big idea, support it using only the evidence needed for the decision, and reserve detailed analysis for questions.
- C. Show the exploratory dashboard so the audience can independently evaluate every possible interpretation.
- D. Summarize the analysis using one chart per hypothesis that was tested to demonstrate analytical rigor.

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- D. Summarize the analysis using one chart per hypothesis that was tested to demonstrate analytical rigor.

Answer: B

Visualization of Data, Understanding and Interpretation

MCQs

5. When separating content from design in a chart, which of the following elements belongs to the "Design" category rather than "Content"?

- A. Plot type
- B. Gridlines
- C. Variables
- D. Axes

Visualization of Data, Understanding and Interpretation

MCQs

5. When separating content from design in a chart, which of the following elements belongs to the "Design" category rather than "Content"?

- A. Plot type
- B. Gridlines
- C. Variables
- D. Axes

Answer: B



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Visualization of Data, Understanding and Interpretation

Unit 1 : Foundations of Visualization, Data Interpretation

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

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Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.

Visualization of Data, Understanding and Interpretation

Lecture Overview

- Understand what visual clutter is and why reducing cognitive load improves communication.
- Learn practical decluttering techniques using alignment, white space, and contrast.
- Apply a systematic decluttering process to transform busy charts into clear, audience-focused visualizations.
- Understand how preattentive attributes guide visual attention before conscious thought.
- Learn how size, color, and position create visual hierarchy and emphasize key insights.
- Apply attention-guiding techniques to make visualizations faster and easier to interpret.

Visualization of Data, Understanding and Interpretation

Eliminating Clutter

(TB 2, Ch3)

So far, we have learned to:

- Start with the context by understanding the audience and defining the communication goal.
- Select an appropriate visual display that best represents the data and supports the intended message.

Choosing the right chart is only the beginning. Even an appropriate chart can fail if unnecessary visual elements distract from the message.

In this section, we focus on eliminating clutter—removing anything that does not improve understanding, so the audience can interpret the visualization quickly and effortlessly.



1. Understand the Context

Know your Who-What-How and the Big Idea



2. Choose an Effective Visual

Use the best type of visual(s) for your data and message



3. Identify & Remove Clutter

Apply the Gestalt Principles, alignment, white spaces, contrast and visual order



4. Direct the Audience's Focus

Guide people's attention with preattentive attributes (size, color, position, texts)



5. Think like a Designer

Achieve your goals with great design (affordance, accessibility, aesthetics, acceptance)



6. Tell a Story

Use the proven 3-part narrative to tell a story with your data

Visualization of Data, Understanding and Interpretation

Sources of Visual Clutter

(Ch3)

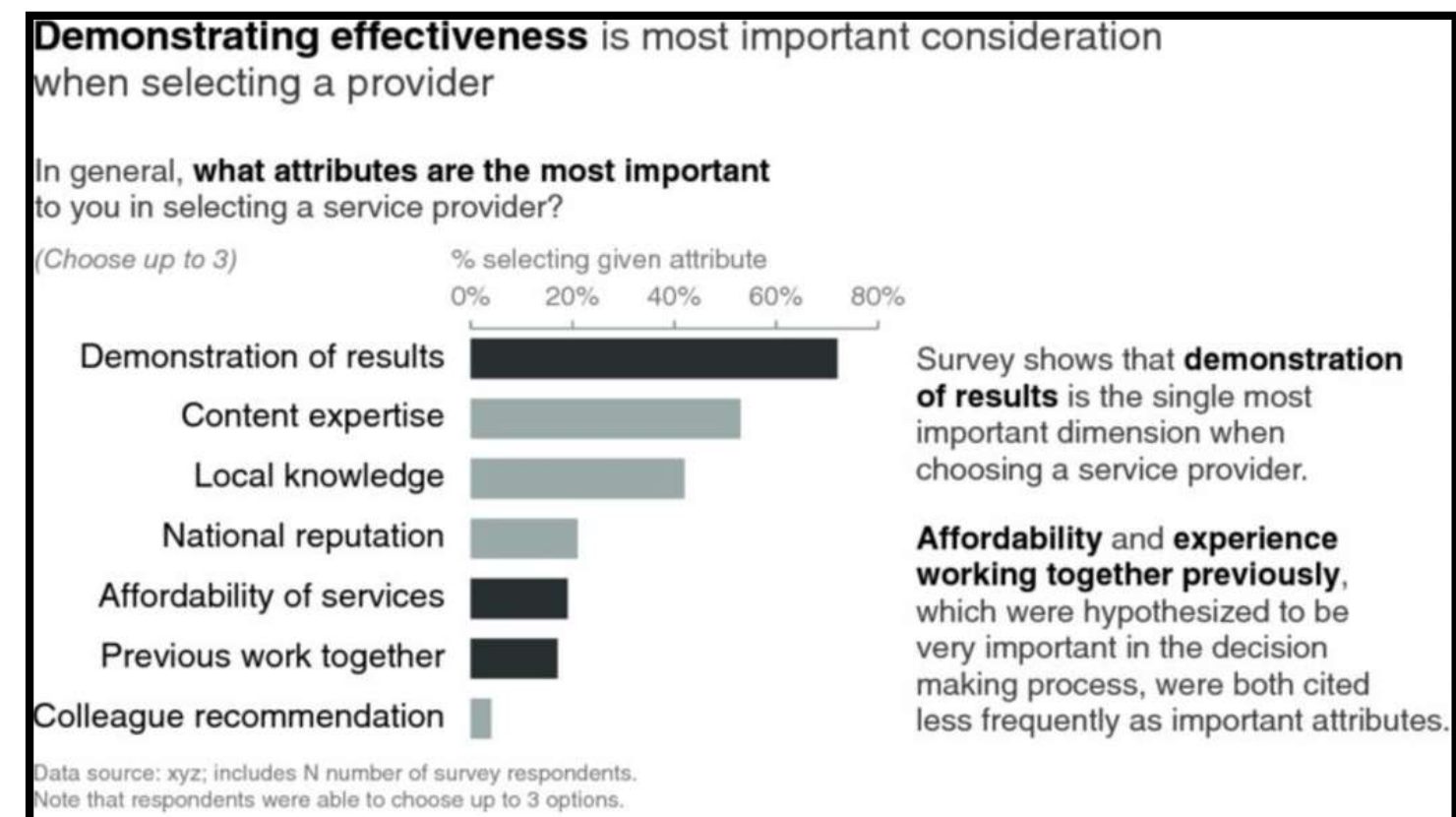
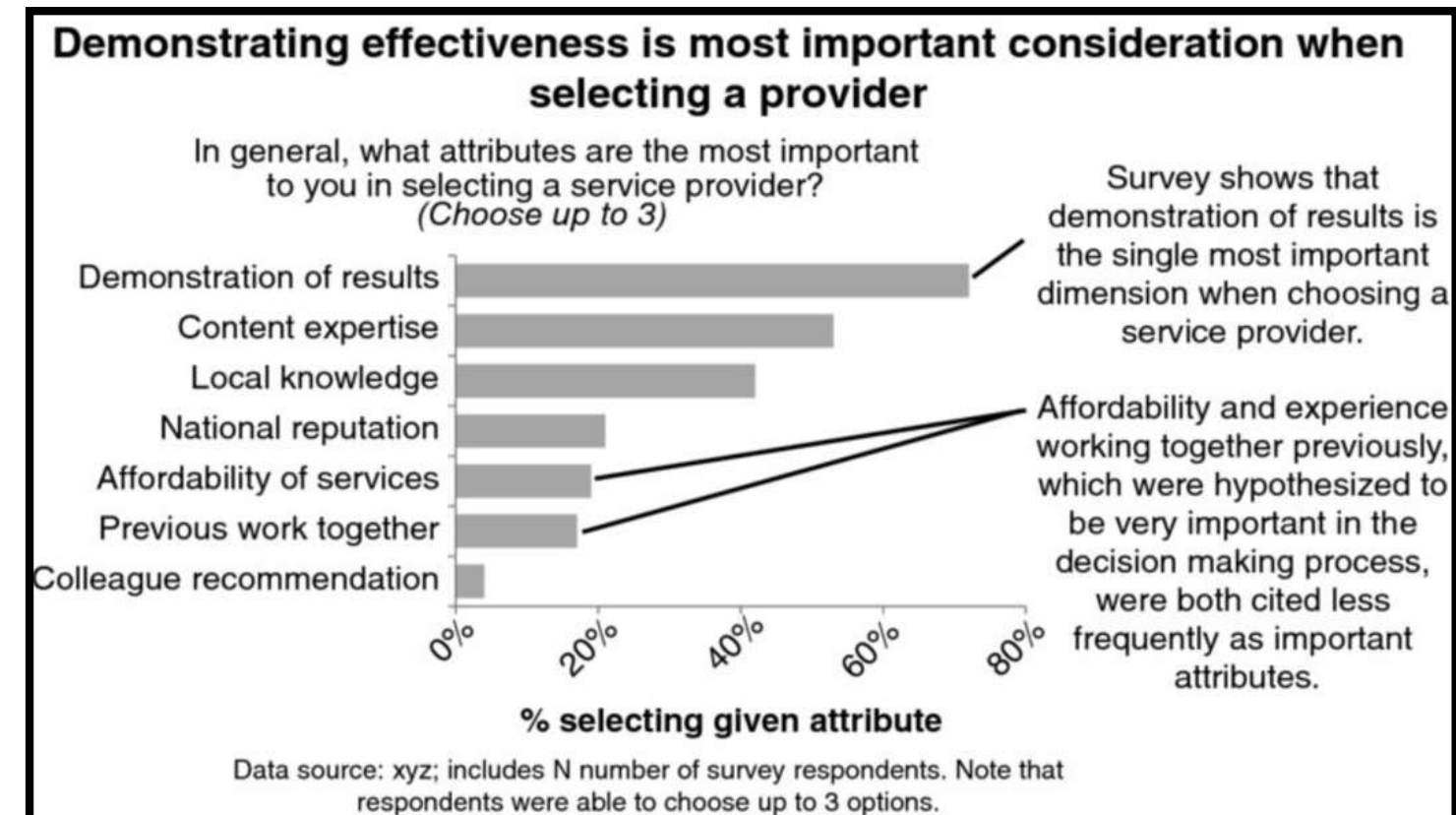
Why does clutter matter?

Visual clutter forces the audience to spend more effort understanding a chart instead of understanding the data.

Common Sources of Clutter

- Poor alignment creates a disorganized appearance.
- Little or no white space makes information feel crowded.
- Excessive use of colors, shapes, and formatting reduces visual contrast.
- Diagonal text and connector lines are harder to read than horizontal layouts.
- Too many elements compete equally for attention.

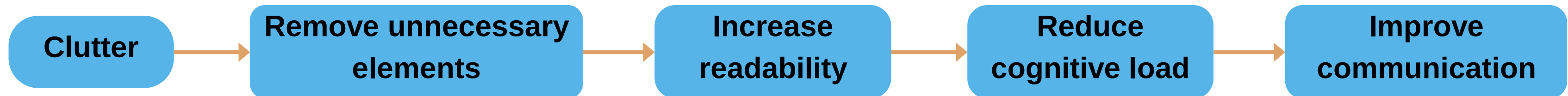
! *Every element should justify its existence by improving understanding*



Practical Techniques for Cleaner Visualizations

- Use alignment to create clean horizontal and vertical structure.
- Preserve white space to separate information and emphasize important content.
- Use contrast strategically- highlight only what matters instead of emphasizing everything.
- Avoid unnecessary decorations, including chart borders, heavy gridlines, redundant markers, and diagonal labels.
- Label data directly whenever possible to eliminate unnecessary legends.

💡 *Reducing clutter decreases cognitive load and makes important patterns easier to recognize*



Visualization of Data, Understanding and Interpretation

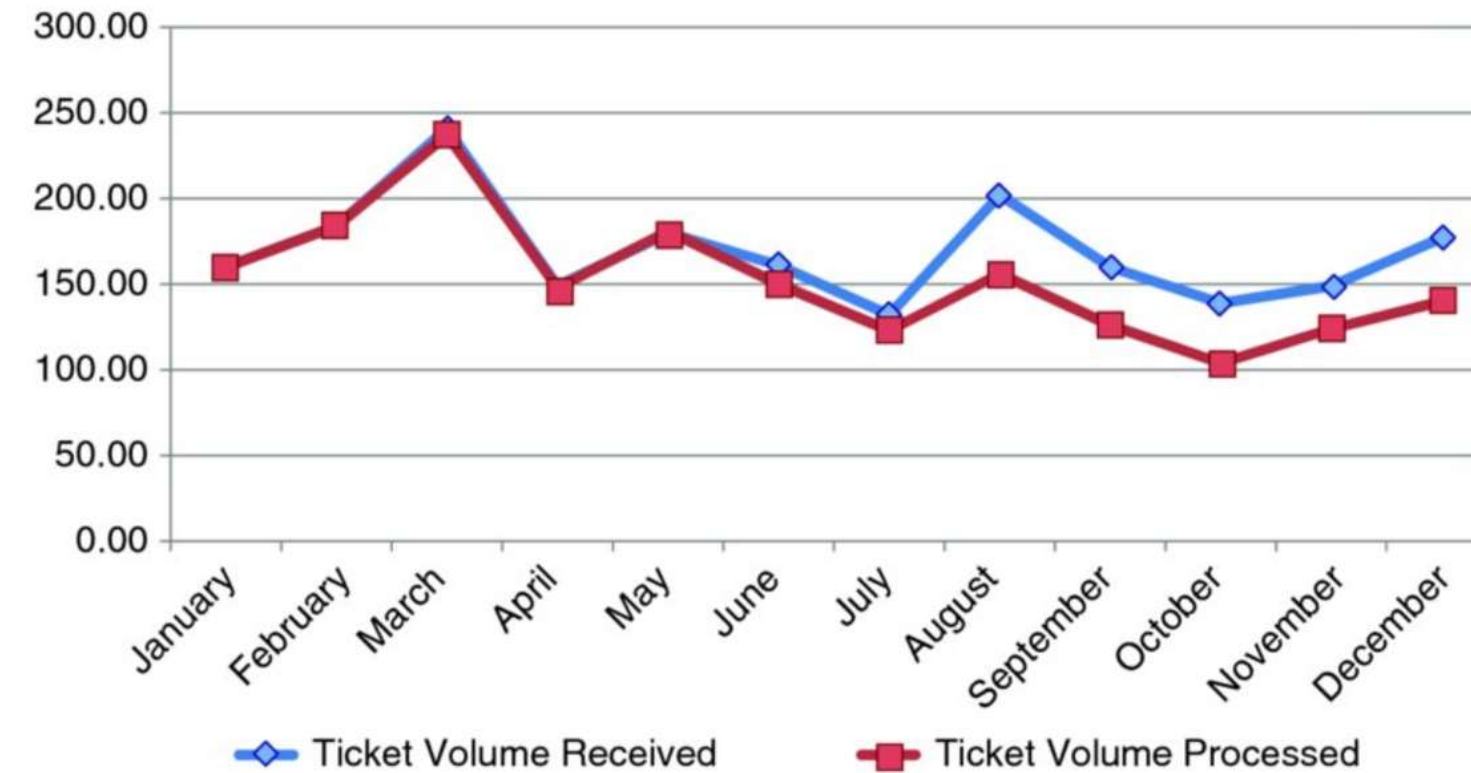
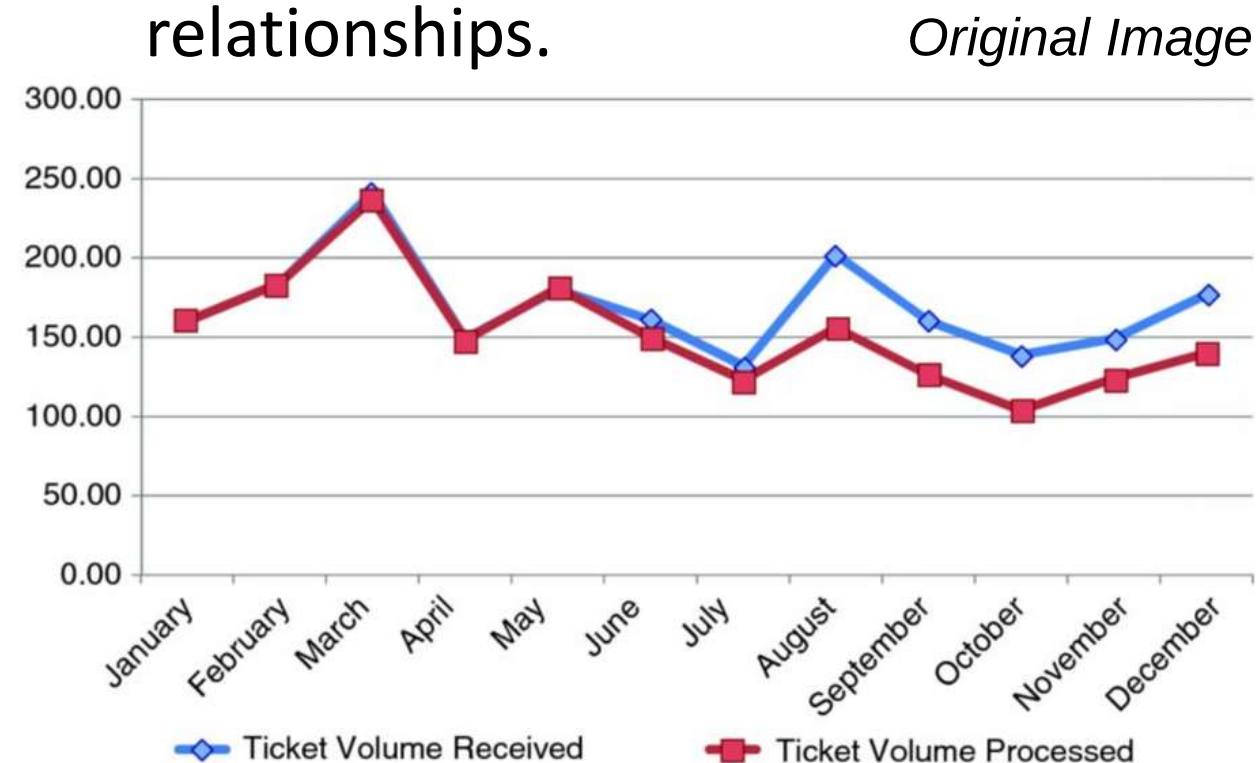
Decluttering in Practice

(Ch3)

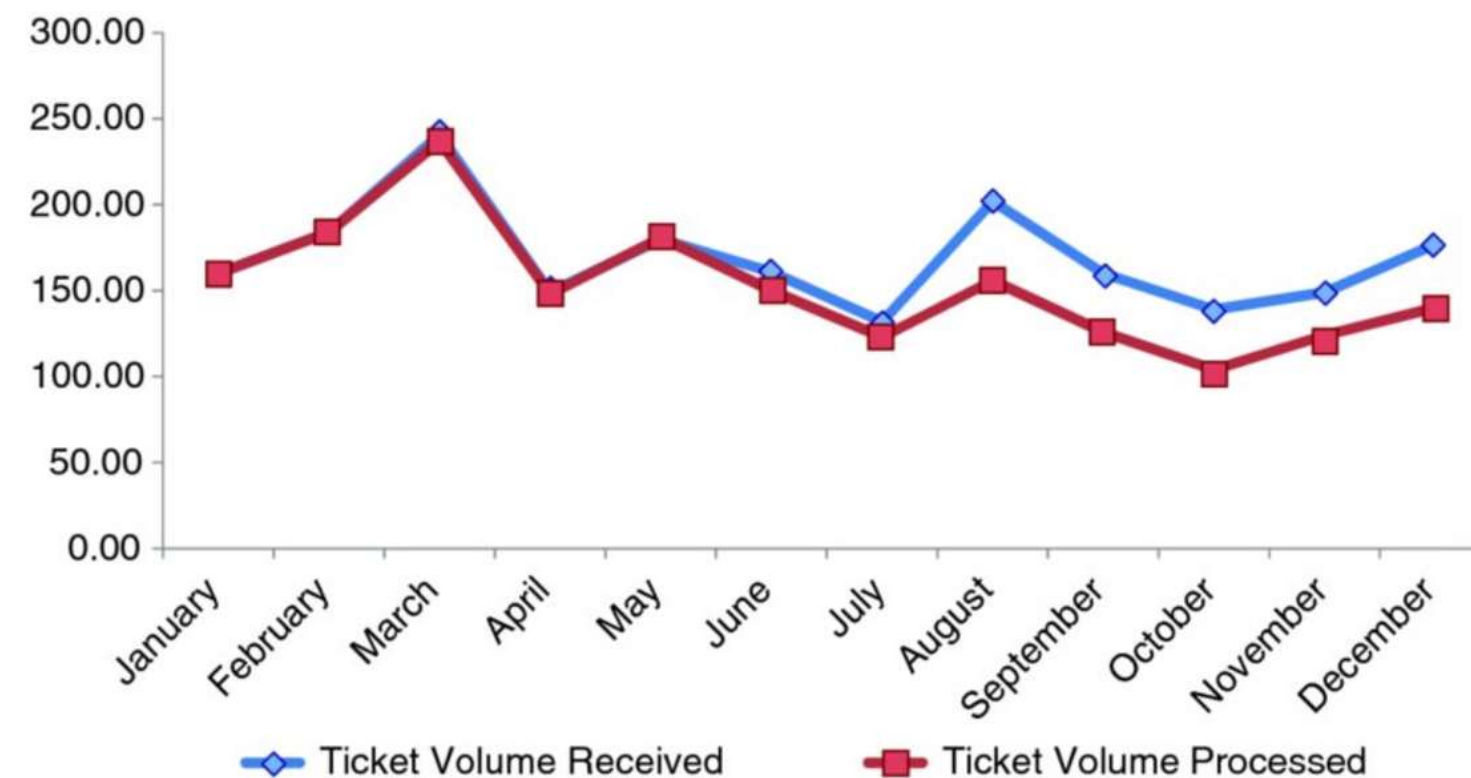
Walkthrough Example:

The example demonstrates a step-by-step decluttering process improving a line chart through six simple changes:

1. Remove chart borders.
2. Remove or lighten gridlines.
3. Remove unnecessary data markers.
4. Simplify axis labels.
5. Replace legends with direct labels.
6. Use consistent colors to reinforce relationships.



(1) Removing Chart Borders

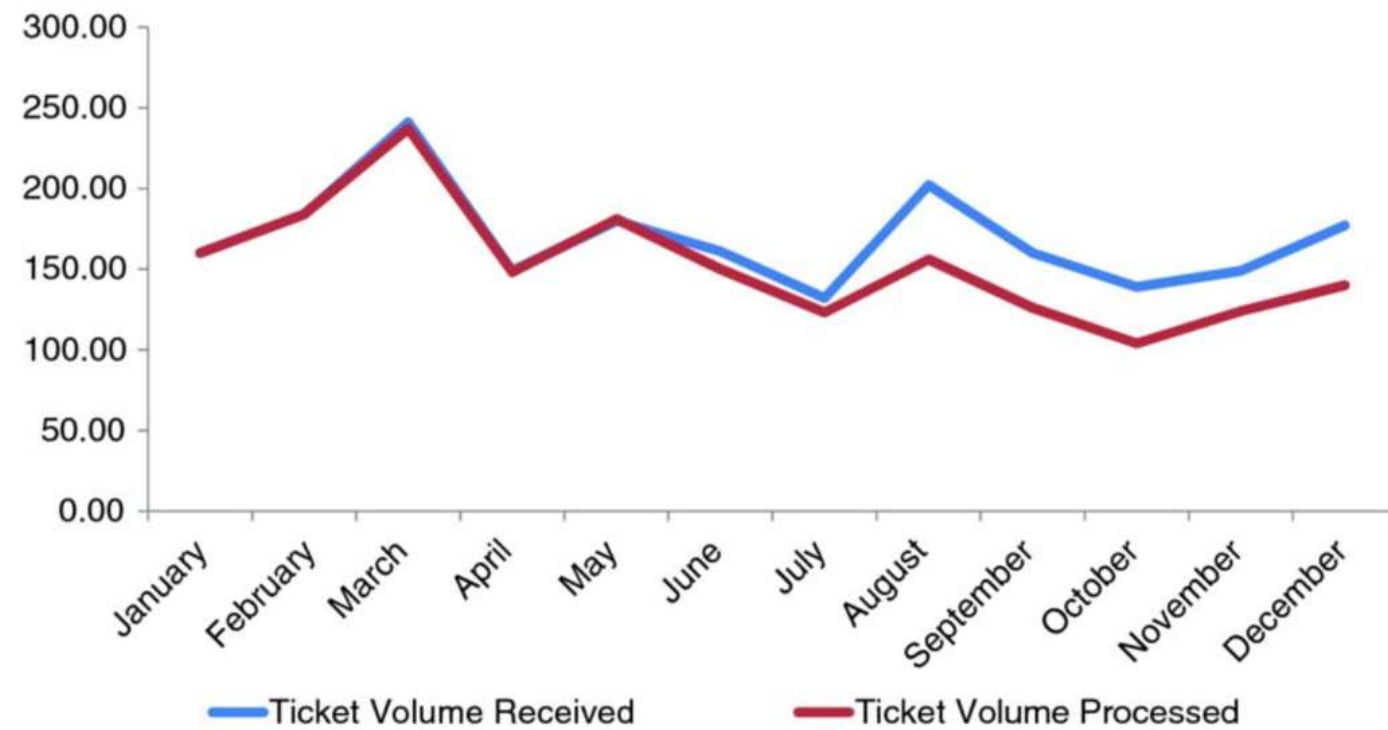


(2) Removing Gridlines

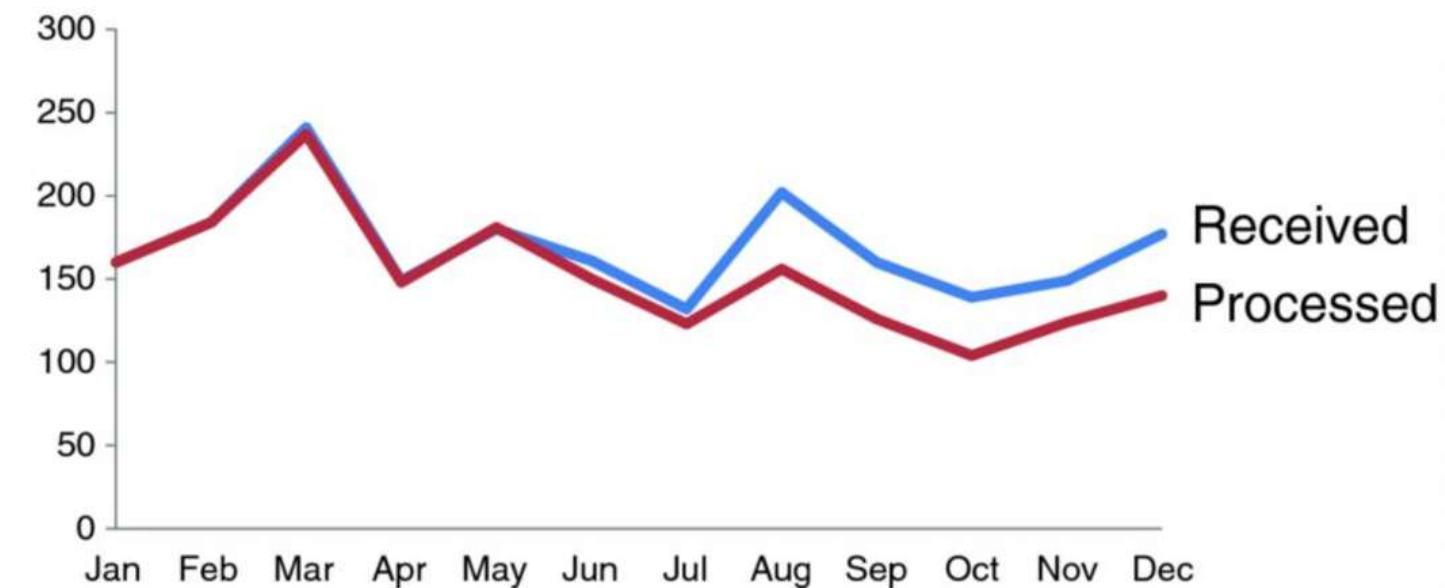
Visualization of Data, Understanding and Interpretation

Decluttering in Practice

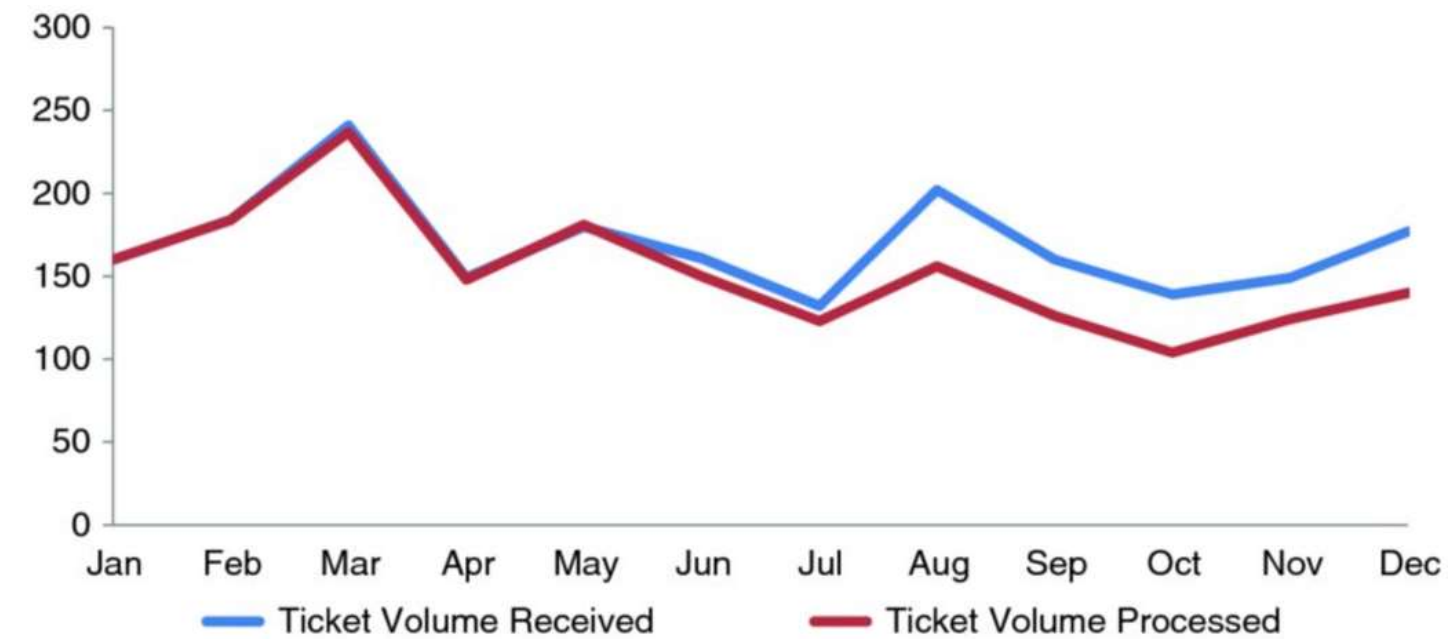
(Ch3)



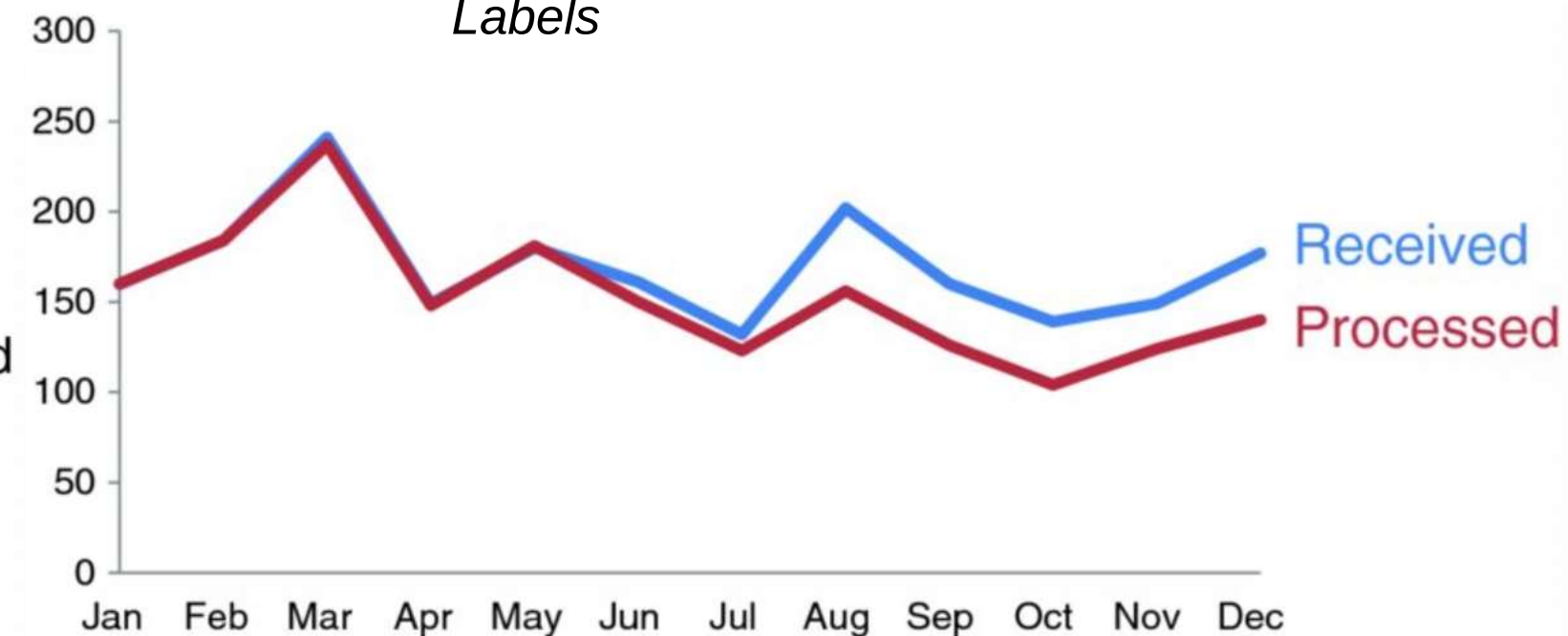
(3) Removing Data Markers



(5) Labelling Data Directly



(4) Cleaning Axis Labels



(6) Leveraging Consistent Colour

Visualization of Data, Understanding and Interpretation

Focus Attention Where You Want It

(Ch4)



So far, we have learned how to:

- Understand the purpose of the visualization by considering the audience and context.
- Choose an appropriate chart to communicate the intended message.
- Remove unnecessary clutter that distracts from the data.

What comes next?

A visualization can be accurate and uncluttered, yet still fail to communicate its message if viewers do not know where to look first.

The next principle focuses on guiding attention intentionally. By using visual emphasis strategically, we can highlight the most important information, establish a clear viewing order, and make the message easier to understand at a glance.

What are Preattentive Attributes?

- Preattentive attributes are visual properties that the brain processes almost instantly, before conscious attention is required.
- They leverage our iconic memory, allowing viewers to detect important differences immediately.
- Common preattentive attributes- color (hue & intensity), size, position, shape, orientation, motion, line length & width

Why are they important?

- Direct attention immediately.
- Reduce search time.
- Create visual hierarchy.
- Improve communication efficiency.



Orientation



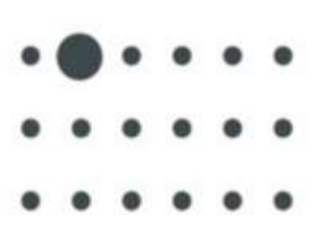
Shape



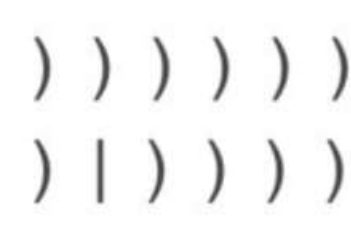
Line length



Line width



Size



Curvature



Added marks



Enclosure

*Preattentive
Attributes*



Hue



Intensity



Spatial position



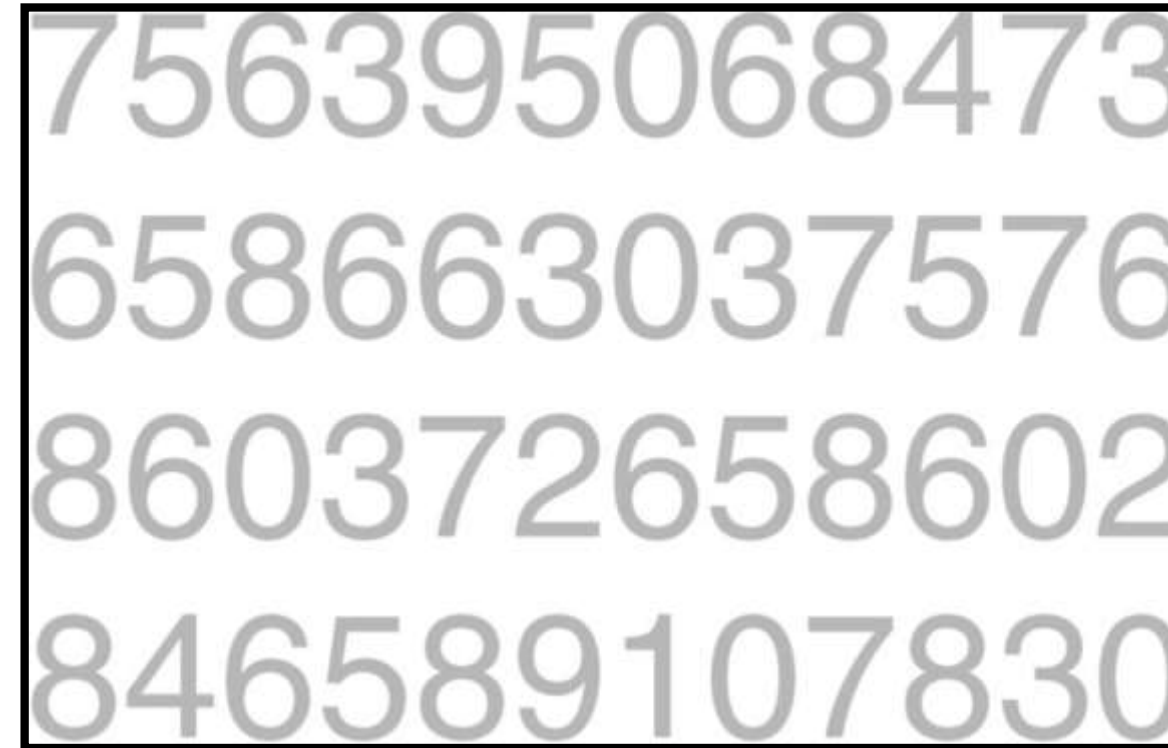
Motion

Visualization of Data, Understanding and Interpretation

Preattentive Attributes: Directing Attention Instantly (Ch4)

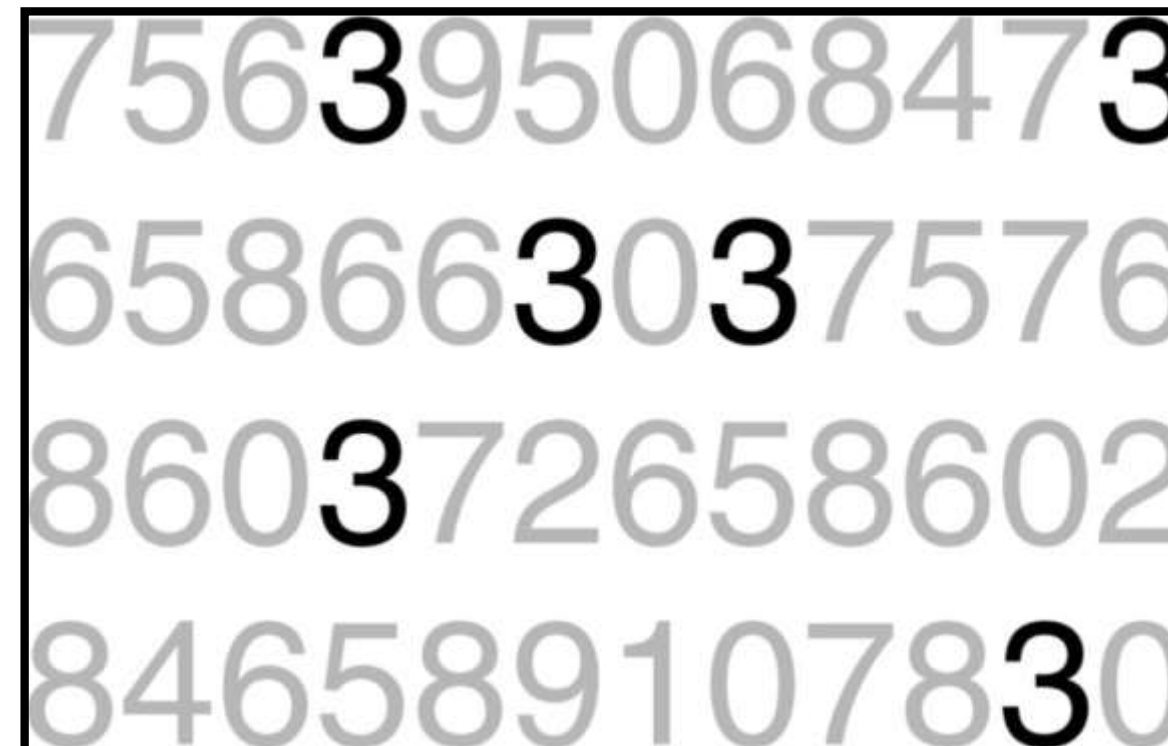
Example:

- Task: Count the 3s in the example
- In the first figure, without visual cues, viewers must consciously search for the target information.
- Identifying the six occurrences of the number 3 requires sequential scanning and greater cognitive effort.
- In the second figure, highlighting the 3s using colour makes them immediately stand out.
- The brain detects the highlighted elements almost instantly, without deliberate searching.



7	5	6	3	9	5	0	6	8	4	7	3
6	5	8	6	6	3	0	3	7	5	7	6
8	6	0	3	7	2	6	5	8	6	0	2
8	4	6	5	8	9	1	0	7	8	3	0

*Figure
1*



7	5	6	3	9	5	0	6	8	4	7	3
6	5	8	6	6	3	0	3	7	5	7	6
8	6	0	3	7	2	6	5	8	6	0	2
8	4	6	5	8	9	1	0	7	8	3	0

*Figure
2*

Visualization of Data, Understanding and Interpretation

Creating Visual Hierarchy

(Ch4)

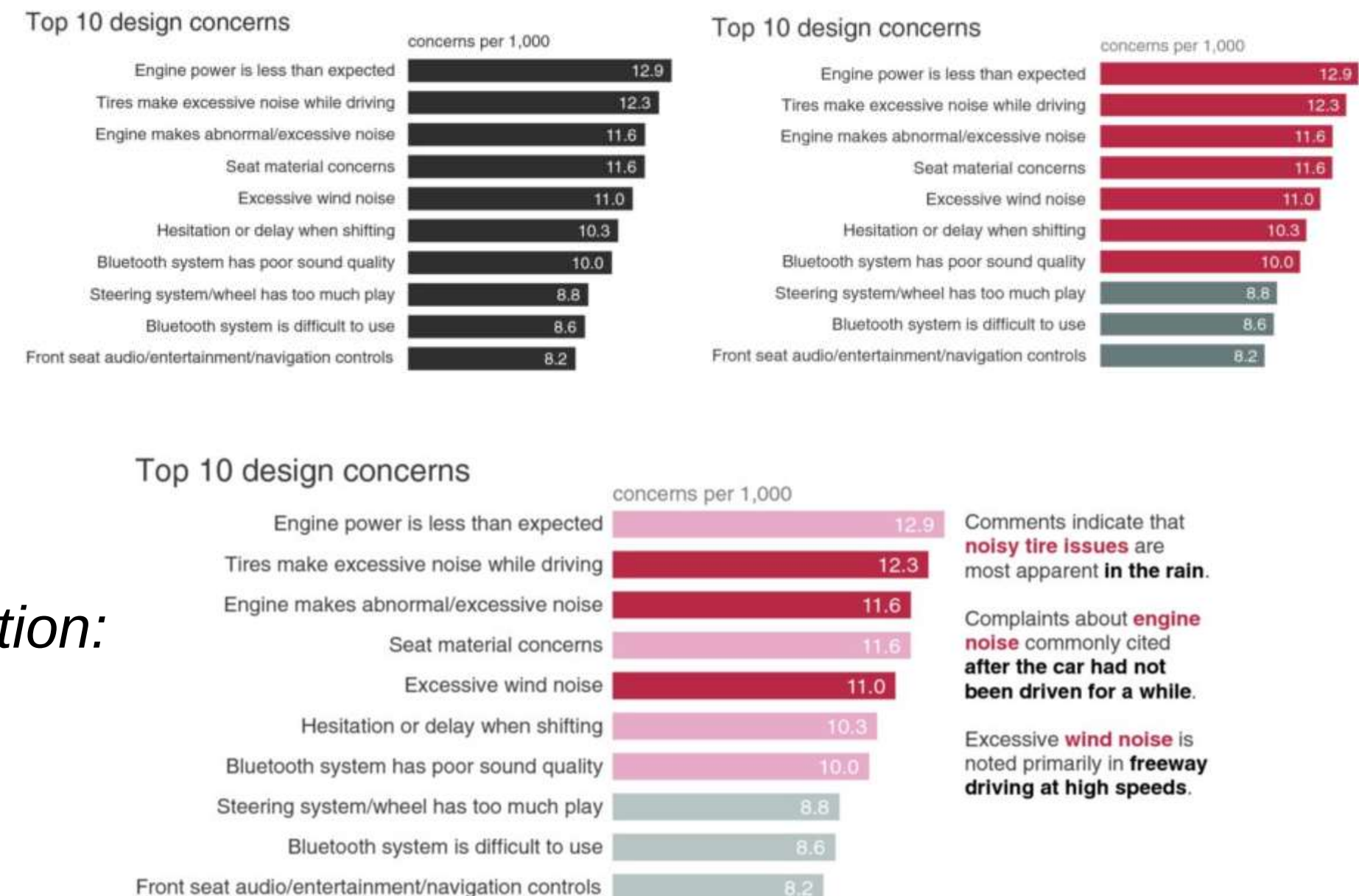
Visual Hierarchy

- Not every element should receive equal attention.
- Use stronger visual emphasis for important information and reduce emphasis for supporting details.

Effective techniques

- Highlight one important element using color.
- Use larger size for important content.
- Keep supporting elements in muted grey.
- Combine multiple preattentive attributes sparingly.
- Guide viewers from overview to detail.

💡 *Key Idea: A good visualization answers the question:
"Where should the audience look first?"*



Visualization of Data, Understanding and Interpretation

Using Size, Color & Position Strategically

(Ch4)

Three Powerful Attention Tools

Size

- Larger objects naturally appear more important.
- Equal importance should be represented with similar sizes.

Color

- Use one bold color sparingly.
- Keep remaining elements neutral.
- Maintain color consistency throughout the presentation.
- Design with colorblind users in mind.

Position

- Audiences naturally scan from top-left in a zigzag ("Z") pattern.
- Place the key message where viewers look first.

*Using Colour
Sparingly*

Country Level Sales Rank Top 5 Drugs

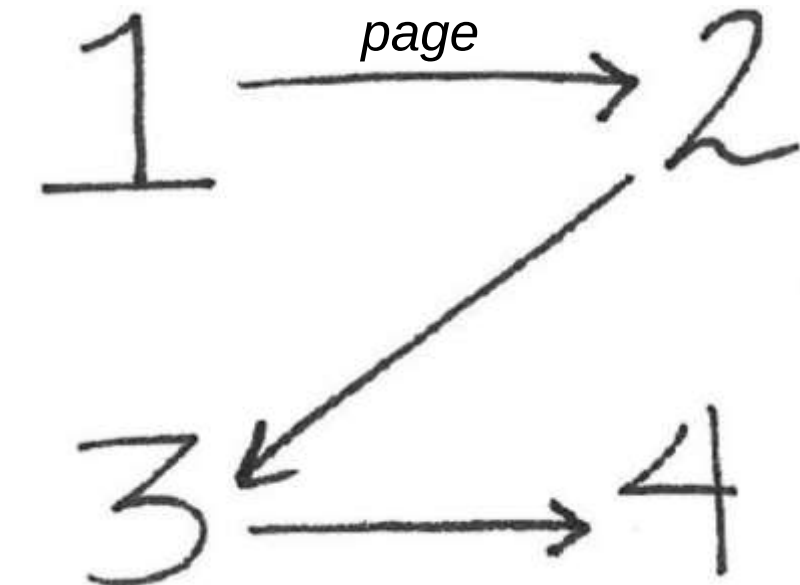
Rainbow distribution in color indicates sales rank in given country from #1 (red) to #10 or higher (dark purple)

Country	A	B	C	D	E
AUS	1	2	3	6	7
BRA	1	3	4	5	6
CAN	2	3	6	12	8
CHI	1	2	8	4	7
FRA	3	2	4	8	10
GER	3	1	6	5	4
IND	4	1	8	10	5
ITA	2	4	10	9	8
MEX	1	5	4	6	3
RUS	4	3	7	9	12
SPA	2	3	4	5	11
TUR	7	2	3	4	8
UK	1	2	3	6	7
US	1	2	4	3	5

Top 5 drugs: country-level sales rank

		RANK				
		1	2	3	4	5+
COUNTRY DRUG	A	B	C	D	E	
Australia	1	2	3	6	7	
Brazil	1	3	4	5	6	
Canada	2	3	6	12	8	
China	1	2	8	4	7	
France	3	2	4	8	10	
Germany	3	1	6	5	4	
India	4	1	8	10	5	
Italy	2	4	10	9	8	
Mexico	1	5	4	6	3	
Russia	4	3	7	9	12	
Spain	2	3	4	5	11	
Turkey	7	2	3	4	8	
United Kingdom	1	2	3	6	7	
United States	1	2	4	3	5	

The zigzag "z" of taking in information on a screen or page



Visualization of Data, Understanding and Interpretation

Guiding Attention Without Adding Clutter

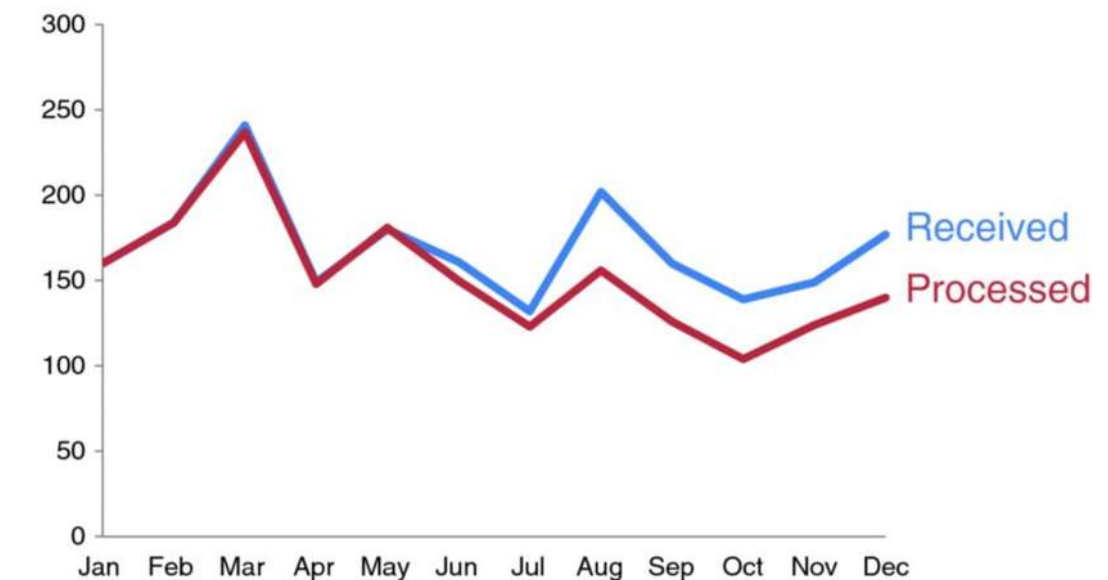
(Ch4)

Strategic Emphasis

Attention should be guided using minimal but meaningful cues.

Examples include:

- Highlighting only critical data labels.
- Using bold lines only where comparison matters.
- Pushing less important information into the background.
- Adding annotations only when they improve interpretation.



First, push everything to the background



Make the data stand out



Too many data labels feels cluttered



Data labels used sparingly help draw attention

Visualization of Data, Understanding and Interpretation

MCQs

1. A visualization contains aligned elements, direct labels, generous white space, and minimal gridlines. However, every data series is displayed in a different bright color with equal visual prominence. Which design principle is still being violated?

- A. White space
- B. Alignment
- C. Strategic use of contrast
- D. Direct labeling

Visualization of Data, Understanding and Interpretation

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Answer: C

2. A designer removes all gridlines, borders, legends, and axis labels from a chart to eliminate clutter. The resulting visualization becomes difficult to interpret. According to *Storytelling with Data*, what best explains this outcome?

- A. Decluttering requires maximizing white space.
- B. Elements should be removed only if they do not contribute to interpretation or communication.
- C. Every chart should retain all default formatting.
- D. Direct labeling should never replace legends.

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Answer: B

Visualization of Data, Understanding and Interpretation

MCQs

3. A dashboard uses five equally bright colors to highlight five key performance indicators, claiming that each metric is equally important. According to *Storytelling with Data*, what is the most likely consequence?

- A. The dashboard becomes more objective because no metric is prioritized.
- B. The audience processes all metrics equally quickly.
- C. The effectiveness of preattentive attributes is reduced because nothing stands out.
- D. Equal emphasis increases visual hierarchy.

Visualization of Data, Understanding and Interpretation

MCQs

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- D. Equal emphasis increases visual hierarchy.

Answer: C



4. A presenter intentionally places a secondary statistic at the top-left corner of a slide while positioning the primary conclusion near the bottom-right. Which principle is being violated?

- A. Gestalt proximity
- B. Position as a preattentive attribute
- C. Redundant encoding
- D. Direct labeling

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Answer: B

5. Which modification best increases visual hierarchy without introducing unnecessary clutter?

- A. Assigning a different bright color to every category.
- B. Increasing the font size of all labels equally.
- C. Displaying all data markers and values for every observation.
- D. Highlighting the key series in a bold color while muting all remaining elements in grey.

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Answer: D



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